



Transportation Management (TM)

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Planning

In planning, freight documents and transportation units are created on the basis of business documents, for example, forwarding orders, from which freight units have been created, and by assigning capacities. Various constraints are taken into account, for example, requested delivery dates.

Integration

The component **Planning** of Transportation Management (TM) is integrated with the following other TM components:

Component	Integration Type
Forwarding Order Management	The system creates freight units from forwarding orders. These freight units form the basis for planning.
Freight Order Management	Freight documents and transportation units are the result of the planning.
Charge Calculation	You can have the system calculate the transportation charges for your freight orders and freight bookings.
Master Data	You need to create the required master data as a prerequisite for planning.

Features

- Freight units

Planning is based on freight units. For more information, see [Creation and Editing of Freight Units](#).

- Unified package building

Unified package building allows you to package your products. For more information, see [Unified Package Building in TM](#).

- Profiles and settings

By defining profiles and settings, you can determine which objects are available for planning and you can influence the planning results. For more information, see [Profiles and Settings](#).

- Manual planning and automatic planning

Manual and automatic planning (sometimes also referred to as interactive planning) offer you a wide variety of planning options. For more information, see [Manual Planning](#) and [Automatic Planning](#).

- Scheduling

You can determine the start and end dates for a set of activities. For more information, see [Scheduling](#).

- Carrier selection

You can assign a suitable carrier to your subcontractable business documents (for example, freight orders). For more information, see [Carrier Selection](#).

Creation and Editing of Freight Units

Use

The system automatically creates freight units from the following business documents:

- Forwarding orders
- Order-based transportation requirements
- Delivery-based transportation requirements
- Other order-based business documents, such as sales orders or purchase orders from Sales and Distribution (SD) or Materials Management (MM) or outbound delivery orders from Extended Warehouse Management (EWM)
- Other delivery-based business documents, such as inbound and outbound deliveries
- JIT calls

The system can split and merge the product quantities of the requirements when it creates freight units based on the freight unit building rules used.

The creation of freight units in turn affects planning. The larger the freight units are, the simpler the planning activities in VSR optimization. If the freight units are smaller, there may well be a higher theoretical potential for optimization, but in practice, planning in VSR optimization is made more difficult due to the higher number of freight units. To achieve the best planning result, you are advised to aim somewhere in the middle.

❖ Example

You have a truck with a capacity of 20 tons. You are planning to transport 20 tons of flour in 1 kg packages. Neither extreme is advisable here for the reasons given above. You should not create a single freight unit with a weight of 20 tons, nor should you create 20,000 freight units with a weight of 1 kg each.

Prerequisites

- Business documents exist.
- If you would like the system to automatically create freight units when business documents are saved, you have specified this in Customizing:
 - In the case of forwarding orders, you have selected the checkbox for automatic freight unit building in the forwarding order type. For more information, see Customizing for [Transportation Management](#) under [►Forwarding Order Management ► Forwarding Order ► Define Forwarding Order Types ►](#).
 - In the case of order-based business documents, such as sales orders or purchase orders, you have selected the checkbox for automatic freight unit building in the logistics integration profile. For more information, see Customizing for [Transportation Management](#) under [►Integration ► Logistics Integration ► Internal TM Component Integration ► Define Logistics Integration Profile for SD, MM, and LE ►](#).
- If required, you have extended the standard checks that the system executes to decide whether a freight unit or transportation unit must be rebuilt during business document maintenance. For more information, see Customizing for [Transportation Management](#) under [►Business Add-Ins \(BADIs\) for Transportation Management ► Forwarding Order Management ► Freight Unit Building ► BAdI: Extension of Checks for Triggering FU Building from TRQ ►](#).
- You have defined freight unit types in Customizing. For more information, see Customizing for [Transportation Management](#) under [►Planning ► Freight Unit ► Define Freight Unit Types ►](#).
- You have defined freight unit building rules (FUB rules) and assigned freight unit types to them (see [Freight Unit Building Rule](#)).

- Optionally, you have defined incompatibilities (see [Incompatibilities](#)).
- If goods have been entered in the freight document as missing, you can specify in Customizing that the system automatically splits the associated freight unit into a freight unit containing the actual quantity and a freight unit that has the cargo receipt status **Missing Cargo**. You can then assign the freight unit for the missing cargo to a different freight document so that the goods can be redelivered at a later date. For more information, see Customizing for **Transportation Management** under **Freight Order Management > Define Discrepancy Profile**.
- If you want to use normalized load consumption quantities (normalized quantities, NLQ) and additional normalized load consumption quantities (additional normalized quantities, ALQ) as split criteria, you have made the following settings:
 - You have defined an alternative quantity for the unit of measure of the normalized quantity in your material master or you have specified an individual quantity per product or reference product in the **Define Normalized Quantity** app.
 - You have defined a condition with the condition type /SCMTMS/NLQ_UOM.
 - You have defined the conversion between the units of measure of normalized and additional normalized quantities in the Customizing activity **Define Conversion Between NLQ and ALQ UoM**.
 - Optionally, you have defined the relevance of units of measure for normalized and additional normalized quantities in the Customizing activity **Define UoM Relevance of NLQs and ALQs**.
 - In your freight unit building rule, you have defined normalized or additional normalized quantities or both as the planning quantity.

For more information, see [Use of Normalized Quantities and Capacities](#).

Features

Automatic Creation of Freight Units

First, the system groups all business document items under consideration of the following parameters:

- Hard constraints (such as source location and destination location, carrier, and pick-up and delivery dates, see [Hard Constraints](#))
- Business document attributes (such as delivery group) and packaging information
- Incompatibilities
- FUB rules

Here the system takes into account freight units that already exist for these business documents.

The system then creates one or more freight units from these groups. If you have defined a split quantity and have selected the checkbox for the item split, the system takes these entries into account when creating multiple freight units.

The business document item attribute **Delivery Group** is used to consolidate all items with the same delivery group into one freight unit. It is also possible to consolidate items with different delivery groups into the same freight unit based on the above grouping parameters.

In the case of automatically creating freight units, you can specify an FUB rule. If you do not specify an FUB rule, the system automatically determines the FUB rule. For more information, see [Determination of Freight Unit Building Rules](#). In the FUB rule you can enter, for example, that freight orders or transportation units are created automatically instead of freight units (see [Transportation Unit](#)) (shortcut scenario). For more information, see [Freight Unit Building Rule](#) and [Direct Creation of Freight Orders](#).

When freight units are built, the main cargo items in the forwarding documents can have different categories as defined by the shipping type. Items with different categories form a compatibility group, which means that these items cannot be grouped into one freight unit or transportation unit.

A distinction is made between the following groups:

- Package, product, and schedule line
- Container
- Railcar
- Trailer

❖ Example

A forwarding order has a container, package, and railcar as its main cargo items. All of these items have a different compatibility group, which means that the system will create three freight units or transportation units.

You can also create freight units in the background (see [Forwarding Order Management Preparation](#)). However, we recommend that you use automatic creation.

You can use strategies to create freight units in accordance with your own requirements. SAP delivers the standard strategy FUB_AUTO for freight unit building. For more information, see [Process Controller Configuration](#).

The system automatically creates, for example, freight units when you save your forwarding order. For more information, see [Editing of a Forwarding Order](#).

Creation of Freight Units Based on Normalized and Additional Normalized Quantities

You can have the system build freight units based on normalized and additional normalized quantities as split criteria. You can define several units of measure for normalized and additional normalized quantities as the planning quantity in your freight unit building rule. The system determines the relevant unit of measure to be used based on the condition that you have defined.

For more information, see [Use of Normalized Quantities and Capacities](#).

Manual Splitting and Merging of Freight Units

You can manually split or merge the freight units that the system has created automatically. You can do this on the user interface of the freight unit, for example. After you have selected **Split/Merge**, the system proceeds as follows:

- When freight units are split, the target freight unit is the one that takes on the product items (or partial quantities of these items) that are split off and that is created during the split.
- When merging freight units, the target freight unit is the existing freight unit that takes on the product items (or partial quantities of these items) that are split off.
- If you select **Adopt Stops**, the system copies the stops from the existing freight unit. The system also copies the planning results (routing and assignments to freight orders) from the existing freight unit if it has already been planned.
- If you want to prevent the freight units from being changed after a manual split or merge by the automatic creation of freight units, select **Fix Source Freight Unit** or **Fix Target Freight Unit**.

In the case of product items, you can split off or merge partial quantities. When doing so, you must specify the required quantity. You can split or merge package items and container items if they do not have any subitems, for example, products.

i Note

- If empty freight units are created during splitting, the system does not immediately delete them. You can fill them again. If they are still empty when you save, the system deletes them.
- You can split a freight unit several times. If your freight unit contains, for example, 100 tons, you can enter 10 tons as the split quantity and 5 as the number of item splits. The system splits the freight unit into 5 freight units, each containing 10 tons. The original freight unit then still contains 50 tons.
- You can still split freight units manually after you have started executing the transportation (once the cargo receipt status has been set to **On Hand**). Note, however, that the forwarding order is no longer adjusted accordingly.
- If a cargo item has subitems that are not assigned to a forwarding order, you can also split the associated freight units manually (so that you can assign a damaged item to a different freight document, for example).

For manual splitting and merging, the system checks the following parameters:

- Source location, destination location, Incoterms, and dates/times

The system can only split or merge freight units if these parameters match the source freight unit and target freight unit.

- Incompatibilities for which you have specified in the **Manual Violation** field that the incompatibility must not be violated during manual planning
- Partial quantities

You are not allowed to split off or merge partial quantities for package items or container items.

These “hard” checks are mandatory.

The function for splitting and merging is available on the following user interfaces:

- Transportation cockpit
- User interface for displaying freight units
- Freight unit table in the planning overview

Related Information

[Consideration of Equipment Data During FU or TU Creation](#)

[Time Windows](#)

[Integration of Next Generation Just-In-Time](#)

Hard Constraints

The input for the creation of freight units is taken from the predecessor documents. Freight unit building tries to consolidate the items of the predecessor documents as much as possible taking into account freight unit building rules. Further, certain attributes on item and header level of the predecessor documents may prevent this consolidation and lead to the creation of separate freight units for the predecessor documents and their items. These attributes are considered as hard constraints and may apply to both order-based and delivery-based business documents.

The following list contains the main hard constraints using the sales order as an example:

Hard Constraints

Freight Unit Hard Constraint	Constraint Level	Predecessor Document Type	Example: Sales Order
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Freight Unit Hard Constraint	Constraint Level	Predecessor Document Type	Example: Sales Order
Carrier	Header/item	Order-based, delivery-based, forwarding order	Forwarding agent/carrier
Incoterms	Item	Order-based, delivery-based, forwarding order	Incoterms attributes
Service level (shipping condition)	Item	Order-based, delivery-based, forwarding order	Shipping condition
Shipping type	Header	Forwarding order	n/a
Transportation mode	Item	Order-based, delivery-based, forwarding order	Mode of transport of shipping type
Pick-up and delivery window specification	Item	Order-based, delivery-based	Delivery date rule
Unloading point	Item	Order-based	Unloading point
Warehouse-related information	Item	Order-based, delivery-based	n/a
No ADR Exemption Allowed indicator	Item	Order-based, delivery-based, forwarding order	Determined by dangerous goods check
Empty provisioning information	Item	Forwarding order	n/a
Block	Item	Order-based, delivery-based, forwarding order	Based on delivery block Customizing
Legal Control License Indicator	Item	Order-based, delivery-based	Legal Control License indicator
Source location	Item	Order-based, delivery-based, forwarding order	Shipping point (alternatively: plant)
Destination location	Item	Order-based, delivery-based, forwarding order	Ship-to party (alternatively: buyer)
Acceptable start date	Item	Order-based, delivery-based, forwarding order	Pick-up date
Requested start date	Item	Order-based, delivery-based, forwarding order	Delivery date
Logical system	Header	Order-based, delivery-based, forwarding order	System in which sales order was created
Base document type code	Item	Order-based, delivery-based, forwarding order	Sales order
Base document type code of successor document	Header	Order-based	Outbound delivery, for example
Order combination indicator	Header	Order-based	Order combination
Fixed date/time indicator	Item	Order-based	Delivery date and quantity fixed
Product transportation group	Item	Order-based	Product-specific attributes

Determination of Freight Unit Building Rules

Use

When creating freight units, the system determines which freight unit building rule (FUB rule) it is to take into account for a business document item. To find out which business documents are supported, see [Creation and Editing of Freight Units](#).

Prerequisites

- Optionally, you have defined FUB rules. For more information, see [Freight Unit Building Rule](#).
- You have defined conditions with the condition type /SCMTMS/FUBR. For more information, see [Condition](#).

Features

To determine FUB rules, the system does the following:

1. The system checks whether you have specified a condition in the business document type, for example, forwarding order type. There can be only one condition per business document type.
2. If the system does not find a condition or if this condition does not return a result, the system checks whether you have specified a standard FUB rule in the business document type.
3. If you have not specified one, the system uses default settings.

i Note

You can see which FUB rule the system has used in the freight unit.

Freight Unit Building Rule

Use

The system takes into account freight unit building rules (FUB rules) when creating and editing freight units (see [Creation and Editing of Freight Units](#)).

Features

Freight Unit Building Strategy

In the FUB rule, you define how the system is to consolidate freight units. To do this, you can use various freight unit building strategies (FUB strategies):

- Consolidate as much as possible

In this case, the items come from different business documents, for example, forwarding orders. The system groups the items in freight units.

i Note

In addition to forwarding orders, other possible business documents are forwarding quotations, order-based transportation requirements, and delivery-based transportation requirements.

- Consolidate per item

In this case, the system generates one freight unit per business document item.

- Consolidate per request (compatible parts)

In this case, the system groups all items of a business document to form a freight unit.

For all three FUB strategies, the system takes into account hard constraints and incompatibilities (see [Incompatibilities](#)).

i Note

The FUB strategy also applies to the creation of transportation units and freight orders (see below under [Document Type](#) and [Automatic Creation of Freight Orders](#)).

Planning and Split Quantities

In the FUB rule, you enter planning quantities (for example, gross weight and gross volume) with the corresponding units of measure. The system copies the units of measure to the freight units that it creates. The units of measure form the basis for checking the capacity of resources. You define the maximum capacity of a freight unit using the split quantity.

i Note

- If you create equipment-based freight units or transportation units, the system automatically fills the planning and split quantities from Customizing for the equipment. For more information, see below under [Equipment-Based Freight Units and Transportation Units](#).
- You can also create freight unit building rules without indicating a unit of measure. For more information, see [UoM Setting for Quantity and Alternative Quantity](#) below.

❖ Example

You want to transport 5 tons of cement, but you can only transport it in batches of 500 kg. In this case, you define a split quantity of 500 kg. The system generates 10 freight units.

If the system cannot find the planning unit in the business document, it converts the unit using the values that you have defined in the product master.

❖ Example

You want to transport 5 tons of cement, but you can only transport it in batches of 0.5 m³. In the product master, you have defined that 0.8 m³ of cement is equivalent to 1 ton of cement. The system then converts 5 tons of cement into 4 m³ and generates 8 freight units.

In addition to the split quantity, you can also specify a rounding value. In this case, the system calculates the freight unit quantity as a multiple of the rounding value.

❖ Example

You define the following values:

Product	1 computer = 1 piece = 0.06 m³
Split quantity	1 m³

Rounding value	1 piece
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Since 1 m³ divided by 0.06 m³ makes 16.67 m³, with this split quantity, the freight unit can contain a maximum of 16 computers.

If you enter 3 pieces as the rounding value, for example, because you can only pack three computers in a box, the freight unit can contain a maximum of 15 computers, since 15 is a multiple of 3 and is less than 16.67.

In the FUB rule, you can use the item split checkbox to control whether items are allowed to be split if the split quantity is exceeded.

❖ Example

You have defined the following values:

Freight unit building strategy	Consolidate as much as possible
Critical quantity	Gross weight
Item split allowed	No
Split quantity	20 tons

You want to schedule four items with a total weight of 21 tons. Items 1 to 3 each contain 5 tons and item 4 contains 6 tons. Taking into account your settings, the system creates two freight units:

- Freight unit 1 with items 1 to 3
- Freight unit 2 with item 4

Freight Unit Update Strategy

You can specify whether it is possible to move items from one freight unit to another when freight units are updated. You do so by defining the freight unit update strategy in the freight unit building rule.

❖ Example

There are two cargo items in a forwarding order. The system creates one freight unit for each item. One freight unit is planned (FU1) and the other freight unit is not planned (FU2). Item 1 belongs to FU1 and item 2 belongs to FU2.

If item 1 is deleted from the forwarding order, the following occurs:

- In the standard system (that is, **Item Swap Allowed (Standard Behavior)** is selected in the freight unit building rule), item 2 is moved to FU1 and FU2 is canceled.
- If you have selected **Item Swap Not Allowed** in the freight unit building rule, item 2 stays in FU2 and FU1 is canceled.

UoM Setting for Quantity and Alternative Quantity

You can indicate which units of measure are allowed for the quantity and alternative quantity that are to be used as planning quantity. You can either allow dimensionless units of measure only, such as EACH, Pieces, and PAL, or units of measure of all dimensions. You can also indicate that no unit of measure is required. The system then uses the units of measure defined in the predecessor document items. In this case, splitting of freight units is not supported.

Other planning quantities are not affected by this setting.

i Note

Quantities or alternative quantities that have a unit of measure with a dimension are ignored by vehicle scheduling and routing (VSR) optimization, transportation proposal, and load consolidation in the respective requirement documents, such as freight units.

Document Type

You can control which document type is to be created by the freight unit builder, that is, freight unit type, transportation unit type, or freight order type. You can either enter a document type directly or enter a condition for determining the document type in the advanced settings. You define freight unit types, transportation unit types, and freight order types in Customizing. For more information, see Customizing for [Transportation Management](#) under the following:

- [Planning](#) > [Freight Unit](#) > [Define Freight Unit Types](#) >
- [Planning](#) > [Transportation Unit](#) > [Define Transportation Unit Types](#) >
- [Freight Order Management](#) > [Freight Order](#) > [Define Freight Order Types](#) >

If the system determines a freight order type as the document type, it automatically generates freight orders without generating freight units or transportation units (see below under [Automatic Creation of Freight Orders](#)).

Default Business Document Categories

During freight unit building, the system can create different categories of transportation units. If you do not specify a freight document type in the freight unit building rule (either directly or using a condition), the system creates the following business document categories:

Item Category	Business Document Category
Product / package	Freight unit
Package	Package unit as an instance of the transportation unit with TU category "Package"
Container	Container unit as an instance of the transportation unit with TU category "Container"
Railcar	Railcar unit as an instance of the transportation unit with TU category "Railcar"
Trailer	Trailer unit as an instance of the transportation unit with TU category "Trailer"

In this case, the system reads the default document type from Customizing. You must therefore ensure that a default type has been defined in the Customizing activities [Define Freight Unit Types](#) and [Define Transportation Unit Types](#).

If you want the system to create "Freight unit" instead of "Container unit" as the business document category for container items, you must specify the document type of the freight unit in the freight unit building rule (either directly or via a condition).

Equipment-Based Freight Units and Transportation Units

If you enter an equipment type and group in the freight unit building rule, the freight unit builder creates equipment-based freight units or transportation units. Depending on the equipment type, the freight unit builder creates transportation units of the category container, railcar, or trailer.

The system automatically derives the planning quantities, units of measure, and split quantities from Customizing for the equipment (Customizing activity [Define Equipment Groups and Equipment Types](#)). You can delete or change these quantities in the freight unit building rule. This enables you to create transportation units or freight orders (shortcut scenario) with multiple items (see [Example: Freight Unit Building for Railcar Units](#)).

If you want the system to create “Freight unit” instead of “Container unit” as the business document category for container items, you must specify the document type of the freight unit in the freight unit building rule (either directly or via a condition). We recommend that you enter it directly, rather than using a condition.

For more information, see [Consideration of Equipment Data During FU or TU Creation](#).

Automatic Creation of Freight Orders

If you want a freight order to be created automatically instead of a freight unit (shortcut scenario), enter a freight order type as the document type or a condition for determining this freight order type. This function is also supported for transportation units (see [Transportation Unit](#)).

❖ Example

You specify in a condition that if the quantity exceeds 20 tons, freight order type 1 is to be used; otherwise freight unit type 1 is to be used.

This function is suitable, for example, for the following cases:

- In general you want to send small orders that are below one of the upper limits you have defined with an express delivery company.
- You always have to transport orders as a full load due to their size.

Process Controller Strategies

You can use strategies to define the way in which freight units are created in accordance with your own requirements. SAP delivers the standard strategy FUB_AUTO for freight unit building. For more information, see [Process Controller Configuration](#).

Unified Package Building

On the tab page [Advanced Settings](#) you can specify if during freight unit building the system creates freight units, freight orders, or package units with a complete hierarchy of package and product items based on the information from the order documents. Moreover, the system determines the exact position of each product in the package and displays these positions in the load plan.

Alternatively, you can define that the system calculates only a rough estimation of the number of packages that can be built. This estimation is displayed in the header data of a freight unit on the [General Data](#) tab page. It can also be used by the VSR optimizer during planning.

As a prerequisite, you must have entered a unified package building profile. For the determination and display of the exact product positions, you must have selected [Optimizer-Based Mixed Package Building](#) as the mixed package building mode in your unified package building profile.

Whether the system creates a freight unit, freight order, or package unit during freight unit building depends on which document type you enter on the [General Data](#) tab page of the freight unit building rule. If you want the system to create a complete package hierarchy, you can choose a document type of the categories **FU**, **FO** (Freight Order), or **TU** (Transportation Unit). If you use a TU type, the category must be **Package** (that means package unit). We recommend to use a TU type with category **Package** instead of an FU type.

You can also specify the maximum number of top-level packages that a freight unit, freight order, or package unit may contain when the system creates them during freight unit building. The system considers this number when it creates a complete package hierarchy.

Further, you can specify that a separate freight unit, freight order, or package unit is to be created for unpackaged items, that means, one common business document is created for all unpackaged items that contains no other items. The system considers this checkbox when it creates a complete package hierarchy. If you use the package builder as your unified package building engine, you must select the **Only Full Packages in PB Result** checkbox in your unified package building profile as a prerequisite. In this case, the system builds full packages for a product and the remaining quantities of this product stay unpackaged.

For more information, see [Unified Package Building in TM](#).

Activities

You can define FUB rules by opening the app **Create Freight Unit Building Rule** from the launchpad.

More Information

[Determination of Freight Unit Building Rules](#)

[Definition of Dimensions and Units of Measurement](#)

Example: Freight Unit Building for Railcar Units

Use

There are different ways of creating railcar units using the freight unit builder, as shown in the examples below.

i Note

The examples cover railcar units only, but you can also use the freight unit builder to create containers and trailers.

Prerequisites

- In the freight unit building (FUB) rule, you have specified the FUB strategy **Consolidate per Request (Compatible Parts)**. You define FUB rules by using the **Create Freight Unit Building Rule** app, which is available from the Launchpad.
- To enable the system to create multi-item railcar units, you have defined item types with multi-items in Customizing for **Transportation Management** under **Freight Order Management** > **Define Item Types for Freight Order Management**.
- You have assigned the item types with multi-items to the transportation unit type for the railcar unit in Customizing for **Transportation Management** under **Planning** > **Transportation Unit** > **Define Transportation Unit Types**. If you use the shortcut scenario, you have assigned them to the freight order type in Customizing for **Transportation Management** under **Freight Order Management** > **Freight Order** > **Define Freight Order Types**.

Note that if you do not assign the item types, the system creates only single items for the required number of railcars or containers. It does not create a multi-item with subitems.

Activities

The following examples are based on a forwarding order with a product item that weighs 500 tons and has a volume of 200 m³.

Example 1: Equipment-Based Transportation Units (TUs) as Single-Item Railcar Units

You have configured the freight unit building rule as follows:

Freight Unit Building Rule	Value
Split Quantity	100 tons
Document Type	Not specified
Equipment or Vehicle Group / Equipment or Vehicle Type	Railcar (maximum payload 100 tons, cubic capacity 50 m ³)

Result of freight unit building: Five railcar units, each with a capacity of 100 tons and a cubic capacity of 50 m³

Example 2: Equipment-Based TU as Multi-Item Railcar Unit

You have configured the freight unit building rule as follows:

Freight Unit Building Rule	Value
Split Quantity	0
Document Type	Not specified
Equipment or Vehicle Group / Equipment or Vehicle Type	Railcar (maximum payload 100 tons, cubic capacity 50 m ³)

Result of freight unit building: One multi-item railcar unit with a capacity of 500 tons and a cubic capacity of 200 m³. The railcar unit has five subitems, each of which represents one railcar.

Example 3: Equipment-Based TU as Multi-Item Rail Freight Order (Shortcut Scenario)

You have configured the freight unit building rule as follows:

Freight Unit Building Rule	Value
Split Quantity	0
Document Type	Freight order type
Equipment or Vehicle Group / Equipment or Vehicle Type	Railcar (maximum payload 100 tons, cubic capacity 50 m ³)

Result of freight unit building: One freight order with a capacity of 500 tons and a cubic capacity of 200 m³. The freight order has five subitems, each of which represents one railcar.

i Note

The system calculates the required number of railcar or container items based on the maximum payload and cubic capacity of the equipment or vehicle type. The total product quantity is consumed by the railcars or containers created, for example, as follows:

- The vehicle type has a maximum payload of 100 tons and a cubic capacity of 50 m³. The product (coal) has a total weight of 450 tons and a volume of 200 m³. The required number of railcars is therefore 5.
- The vehicle type has a maximum payload of 100 tons and a cubic capacity of 50 m³. The product (cotton) has a total weight of 200 tons and a volume of 220 m³. The required number of railcars is therefore also 5.

Unified Package Building in TM

You can use this function to create packages.

Use

In Transportation Management (TM) you can have the system create packages as follows:

- When freight units are created

This type of unified package building takes place **early** in the process if the system creates freight units based on order documents (for example, forwarding orders or sales orders).

When freight units are created, you can have freight orders or package units created instead of freight units. You make this setting in the freight unit building rule (see "Prerequisites").

- Based on capacity documents

This type of unified package building takes place **late** in the process if capacity documents exist already. You can start from the user interface of an individual capacity document or from the transportation cockpit. It is offered for road freight orders, trailer units, container units (**Road** transportation mode only), and package units.

- During the automatic creation of package units

You can create package units based on unified package building for freight units.

Before you build packages, you can check whether your settings are consistent. For more information, see [Consistency Check of Package Building Constraints](#).

You can use the built packages in load planning. For more information about load planning, see [Load Planning](#).

You can simulate unified package building. For more information, see [Test Cockpits for Unified Package Building](#).

i Note

Several Business Add-Ins (BADIs) are available for unified package building. For more information, see Customizing for Transportation Management under [Business Add-Ins \(BADIs\) for Transportation Management](#) [Planning](#).

Prerequisites

Unified Package Building Using the Package Builder

If you want to build packages using the package building engine (package builder), you must have made the required settings. For more information, see [Package Building](#).

i Note

Some of the SCM Basis transactions are available on the SAP Easy Access screen or in Customizing for TM:

- Customizing activity for defining unified package building profiles:

[Planning](#) [Unified Package Building](#) [Define Unified Package Building Profile](#)

- Customizing activity for defining product orientation profiles:

[Master Data](#) [Product](#) [Define Product Orientation Profile](#)

- Customizing activity for defining profiles for the package building optimizer:

►► [Planning](#) ► [Unified Package Building](#) ► [Package Building](#) ► [Define Profile for Package Building Optimizer](#) ►

- Customizing activity for defining rule profiles for the package building optimizer:

►► [Planning](#) ► [Unified Package Building](#) ► [Package Building](#) ► [Define Rule Profile for Package Building Optimizer](#) ►

- SAP Easy Access transaction for defining products:

►► [Logistics](#) ► [Transportation Management](#) ► [Master Data](#) ► [Material](#) ► [MM01 - Create](#) ► For information about defining product master data for package building, see SAP Note [2842396](#) .

- SAP Easy Access transaction for checking the consistency of your settings:

►► [Logistics](#) ► [Transportation Management](#) ► [Administration](#) ► [Unified Package Building](#) ► [Package Building](#) ► [Check Consistency of Package Building Constraints](#) ►

Unified Package Building During Freight Unit Building

You have made the following settings in TM:

- You have defined a freight unit building rule with the following settings:
 - You have entered a unified package building profile (see [Unified Package Building Profile](#)).
 - You have specified whether the system is to create freight units, freight orders, or package units.
 - You have specified whether the system is to create freight units, freight orders, or package units with a complete hierarchy of package and product items based on the information from the order documents or only a rough estimation of the number of packages that can be built. Note that an estimation of the number of packages is supported only for freight units.
 - Optional: You have specified the maximum number of top-level packages that a freight unit, freight order, or package unit may contain. The system takes this number into account only if you have specified that a complete hierarchy of package and product items is to be created (see above).

For more information, see [Freight Unit Building Rule](#).

- You have entered your freight unit building rule in one of the following places:
 - In the document type of the order document, such as in the forwarding order type in Customizing for TM if you are working with forwarding orders
 - In the logistics integration profile if you are working with the “Internal TM Component Integration” (see [Logistics Integration Profile for SD, MM, and LE](#) and [Internal TM Component Integration](#))

Unified Package Building Based on Capacity Documents

You have made the following settings in TM:

- You have defined a subprofile [Settings for Unified Package Building](#) in your planning profile and entered your unified package building profile in it (see [Planning Profile](#)).
- You have assigned your planning profile to the document type of your capacity documents (for example, the freight order type in Customizing for TM). You have also activated unified package building in the document type. There you can also specify whether you want the system to update the load plan automatically if you change items again after unified package building.

Unified Package Building During Automatic Creation of Package Units

For more information, see [Use of Package Units in Planning](#) ("Automatic Creation of Package Units" section).

Features

Unified Package Building During Freight Unit Building

Depending on your settings in the freight unit building rule, the system creates either a complete hierarchy of package and product items (that exist only locally in the document) or only a rough estimation of the number of packages that can be built. This estimated number of packages is displayed in the header data of the freight unit and can be used later during VSR optimization.

Unified Package Building Based on Capacity Documents

If you select **Build Packages** in a capacity document or in the transportation cockpit, the system creates a complete hierarchy of package and product items that exist only locally in the document. In doing so, the system does not change packages that originate from preceding documents, such as forwarding orders.

If you select several capacity documents in the transportation cockpit, the system creates the packages for each document only, and not for all documents.

You can use **Remove Packages** to remove the local packages. This action does not remove packages that originate from preceding documents such as forwarding orders.

The results of unified package building are displayed on the **Items** tab page in the **Load Plan** hierarchy view of the capacity document as well as in the load plan hierarchy in the transportation cockpit. You also get a 3D visualization of the package. For more information, see [Load Plan](#).

You can also build packages in the background on the basis of capacity documents. For more information, see [Unified Package Building](#).

To improve performance, you can perform unified package building for multiple capacity documents in parallel. To do this, you have to define a parallel processing profile and assign it to your planning profile (in the subprofile **Settings for Unified Package Building**). For more information, see [Definition of Parallel Processing Profiles](#).

Unified Package Building During Automatic Creation of Package Units

For more information, see [Use of Package Units in Planning](#) ("Automatic Creation of Package Units" section).

Related Information

[Transportation Planning with Mixed Packages and Deconsolidation](#)

[Unified Package Building in Integration Based on EWM TUs](#)

Freight Unit

Definition

A set of goods that are transported together across the entire transportation chain. A freight unit can include transportation constraints for transportation planning.

Use

The freight unit is used to merge items in the following business documents that are to be transported together:

- Forwarding order
- Order-based transportation requirement
- Delivery-based transportation requirement

The freight unit is the smallest unit that can be transported. The business document data and the freight unit building rules form the basis for building freight units.

You define the most important settings for the freight unit in the freight unit type. For more information, see Customizing for [Transportation Management](#) under [►Planning►Freight Unit►Define Freight Unit Types►](#).

After the freight units have been created automatically, you can display them on the user interface for the freight unit. You can also split and merge freight units there (see [Creation and Editing of Freight Units](#)). Furthermore, you can also split or merge freight unit stages on this user interface and in the transportation cockpit. When splitting a stage, you manually insert a transshipment location. When merging stages, transshipment locations are deleted automatically.

You can create a freight document for a freight unit. For more information, see [Creation and Editing of Freight Orders](#) and [Creation and Changing of Freight Bookings](#).

You can create a house bill of lading or a house air waybill for a freight unit (see [Building and Printing of House Bills of Lading and House Air Waybills](#)).

You can use SAP Event Management to track and monitor freight unit events (see [Tracking of Freight Units](#)).

You can use the change controller (see [Change Controller](#)) to check whether freight units were changed and how the system reacts to these changes.

Structure

The freight unit data is taken automatically from the business document. However, you can still change some of the data. You can access the following data from the freight unit, for example:

- Items

The freight unit items can stem from different business documents, such as forwarding orders, but a freight unit item is assigned to only one business document item, such as a forwarding order item.

- Locations

You can enter one-time locations at stage level. The system stores these new locations and adjusts stages in the document. It also adjusts the locations in the assigned execution documents.

i Note

However, you should continue to make changes to existing locations in the master data.

- Transportation stages

The system displays loading and unloading times, transshipment locations, and means of transport, for example.

- Quantities and deviations

You can display actual quantities and deviations, for example.

- Transportation constraints

The system copies this data about the transportation stages from the business document or from the transportation stages that were created manually or by the system during planning. Depending on how the stage was created, you can make changes to them. For more information, see [Transportation Stages and Dates/Times in the Freight Unit](#).

- Document flow

The system displays the corresponding business documents with the relevant status information.

- Dangerous goods information (see [Considering Dangerous Goods](#))
- Status information and blocking information (see [Statuses of Business Documents](#) and [Blocking Information](#))
- Instructions (see [Management of Instructions](#))
- Change documents (see [Change Tracking](#))
- CO₂ emissions (see [Calculation of CO₂ Emissions](#))

More Information

[Freight Unit Building Rule](#)

[Determination of Direct Shipment Options](#)

[Interaction Between Organizational Units](#)

[Nature of Goods](#)

[Deviating Business Partner Addresses](#)

Transportation Stages and Dates/Times in the Freight Unit

Use

Transportation stages and dates/times are copied from the forwarding order to the freight unit. Only the dates/times are copied from order-based and delivery-based transportation requirements.

The following section describes special features related to transportation stages and dates/times in the freight unit.

Features

Transportation Stages

If a freight unit contains transportation stages because they were predefined in the business document, transshipment must take place. In this case, you cannot delete the related transportation stops at a later point in time.

If transportation stages result from manual planning or VSR optimization for a freight unit, these transportation stages reflect the transshipment. In this case, you can change the related transportation stops at a later time, for example, delete or add further transportation stops. You can only delete if the transportation stage has not been planned (that is, you have to cancel the assignment to a freight order or a freight booking for the transportation stage).

❖ Example

For a forwarding order, goods must be transported from A to Z. The freight unit generated from the forwarding order also goes from A to Z. In planning you refine this, since it is to be a multimodal transport. This results in the freight unit being transported

from A to Z via B. These transportation stages are mapped to the freight unit accordingly. This applies for manual planning and for VSR optimization.

Dates/Times

After the freight units are generated, the dates/times of the business document are available in the freight unit (in the transportation stages). Windows for pickup and delivery are also copied to the freight unit from the business document.

You can overwrite windows for pickup and delivery. To do this, you must define a condition (see [Condition](#)) and enter it in the freight unit type in Customizing or enter a corresponding rule in the freight unit type. The system then calculates the window when generating the freight unit and displays it in the freight unit.

You cannot change dates/times that are copied from the business document. These dates/times can only be changed using the condition you defined for your window.

Interaction Between Organizational Units

Use

You can use this function to model how organizational units interact with each other, for example, how a sales organization interacts with a planning and execution organization. This interaction is especially important if you have to transport certain goods in a specific transportation stage, for example, dangerous goods.

Features

The sales organization is responsible for entering the order, for example, but is not allowed to create freight bookings for specific transportation stages or send them to a carrier. It can therefore propose how to transport the goods (by specifying a route and a schedule, and assigning a freight order or a freight booking) and sets the organization interaction status to transfer the affected stages to the planning and execution organization to be checked. The planning and execution organization checks the proposal and the transportation stage details in the transportation cockpit. It can then confirm the data exactly as proposed in the forwarding order, change data such as the departure and then confirm the proposal, or reject the proposal outright. The status for the stage in the forwarding order then changes to **Confirmed**, **Confirmed with Deviations**, or **Rejected** respectively.

i Note

The functions for confirming or rejecting proposals are not displayed as standard, and so the planning and execution organization must first select an appropriate layout set in the transportation cockpit. For more information, see [Use of Profile and Layout Sets](#).

The planning and execution organization can also create a freight booking or a freight order for the transportation stage in the transportation cockpit, or assign the transportation stage to a document that already exists. In this case, the system sets the organization interaction status automatically from **To Be Checked** to **Confirmed**. The sales organization can then confirm the transportation stages with the sold-to party.

Furthermore, the planning and execution organization can select an existing freight booking in addition to the route and schedule and confirm the details for the carrier.

Even if a stage has already been confirmed, the sales organization can still make changes to the order. In this case, the system sets the status of the stage to **Changed After Confirmation**.

If the planning and execution organization rejects a stage, the sales organization can submit a new proposal by changing the status of the stage back to **To Be Checked**.

i Note

If the planning and execution organization rejects unconfirmed stages that are assigned to a freight booking via an auto-confirmation profile these unconfirmed stages are automatically unplanned.

If the sales organization wants to cancel a stage, this must also be approved by the planning and execution organization. The sales organization therefore sets the status of the relevant stage in the forwarding order to **Cancellation Requested**. In the transportation cockpit, the planning and execution organization chooses **Confirm** for the stage and the status in the forwarding order subsequently changes to **Cancellation Approved**.

The sales organization can also remove the default route from a stage in the forwarding order by selecting the stage and choosing **Schedule > Unassign**.

If the sales organization wants to delete the stages of the default route from the actual route, it can do so by selecting one of the relevant stages and choosing **Delete Stage**. The system then replaces the stages containing the default route by the original stages.

Automatically Setting the Organization Interaction Status

The sales organization can set the organization interaction status manually from **Confirmed** or **Rejected** to **To Be Checked, Not Yet Requested**, or **Cancellation Requested**.

You define criteria for automatic confirmation in a profile, which the system determines based on the freight booking type. These criteria can include, for example, the security status, volume, or weight. If a transportation stage meets the criteria defined in the profile, the stage can be confirmed automatically (for example, if the volume and weight defined in the profile are less than the volume and weight defined in the transportation stage).

You can use the Business Add-In **BAdI: Change Organization Interaction Status** (/SCMTMS/TOR_ORG_INT_STATUS) to control the following cases:

- A transportation stage can be confirmed automatically (that is, the organization interaction status is set automatically from **To Be Checked** to **Confirmed**), such as when the forwarding order does not contain any dangerous goods.
- A confirmed transportation stage must be rechecked (that is, the organization interaction status is set automatically from **Confirmed** or **Rejected** to **To Be Checked**), for example, when quantities are changed.

Aggregated Organization Interaction Status

The **Statuses** tab page of the forwarding order shows the aggregated organization interaction status of the forwarding order. This is an overall status that is based on the statuses of the individual stages.

If all of the stages have the same status, this status is shown as the aggregated organization interaction status (unless all stages have the status **Confirmed with Deviations**, in which case the aggregated organization interaction status is set to **Confirmed**). If all of the stages have a status that is either **To Be Checked** or **Changed After Confirmation**, the aggregated organization interaction status is set to **Changed After Confirmation**.

In cases where the status of one stage differs from all of the others, the system sets the aggregated organization interaction status based on the following rules, which it applies in the sequence specified:

Sequence	Status of at Least One Stage	Aggregated Status
1	Cancellation Approved	Cancellation Partially Approved
2	Cancellation Requested	Cancellation Partially Requested

Sequence	Status of at Least One Stage	Aggregated Status
3	Rejected	Partially Rejected
4	Confirmed	Partially Confirmed
5	Confirmed with Deviations	Partially Confirmed
6	Changed After Confirmation	Changed After Confirmation
7	To Be Checked	To Be Checked
8	Not Yet Requested	Not Yet Requested

More Information

[Auto-Confirmation Profile](#)

[Coordination Between Customer Service Agent and Gateway Agent](#)

Consideration of Equipment Data During FU or TU Creation

Use

i Note

In the following documentation, the term "equipment" refers to equipment and vehicles, unless otherwise stated.

You can use this function to take into account equipment groups and equipment types as well as the physical properties of your equipment when creating freight units (FUs) and transportation units (TUs). During freight unit building, the system consolidates the business document items into containers, railcar units, or trailer units and displays them as freight units or transportation units. For these FUs and TUs, the system copies the physical properties as well as the equipment type and equipment group that you defined in Customizing.

i Note

If you later change the equipment type in Customizing (for example, the physical properties), you must trigger an update in the freight unit building rule based on the equipment type. The system then updates the planning-relevant quantities in the freight unit building rule based on the changed equipment type.

Prerequisites

- Business documents (forwarding orders, order-based or delivery-based transportation requirements) exist.
- You have defined equipment groups, equipment types, and physical properties in Customizing for [Transportation Management](#). For more information, see [► Master Data ► Resources ► Define Equipment Groups and Equipment Types ►](#).

Features

Automatic Creation of Freight Units and Transportation Units

You can influence the automatic creation of freight units and transportation units by defining equipment, that is, an equipment group and an equipment type, in your freight unit building rule.

If you want to use the equipment type and the equipment group contained in the business document item (except for container item), you must make a customer-specific implementation. For more information, see SAP Note [1552874](#) .

The system automatically derives the planning quantities, units of measure, and split quantities from Customizing for the equipment. It uses the maximum payload weight as the maximum gross weight and the cubic capacity as the maximum volume. All of the other physical properties that you have defined in Customizing are not considered by the system when you create freight units and transportation units.

If required, you can change the planning quantities and split quantities. This enables you to create transportation units or freight orders (shortcut scenario) with multiple items (see [Example: Freight Unit Building for Railcar Units](#)).

If the capacity is exceeded, the system splits a business document item according to the standard process, provided that the freight unit building rule and the item type allow a split.

The system only considers equipment for the business document items of the type **Package Item** and **Product Item** (schedule line). The system does not consider equipment for container items, railcar items, and trailer items. If the system determines a freight unit building rule for these types of item, and you have specified in this freight unit building rule that the system is to copy the equipment from the business document item, an error message appears and the creation of the freight units or transportation units is canceled.

Manual Splitting and Merging of Freight Units

You also have the option of using manual splitting and merging to create empty containers based on equipment. Here you specify the following:

- Source location and destination location
- Freight unit building rule, equipment group, and equipment type

You have the following options:

- You enter a freight unit building rule in which you have defined an equipment group and an equipment type.
- You enter the equipment group and the equipment type.

In this case, the system uses the standard freight unit building rule.

- You enter a freight unit building rule as well as an equipment group and an equipment type.

If the freight unit building rule contains a different equipment group and equipment type, the system overwrites the values in the freight unit building rule with the values you enter here.

The system creates a freight unit that represents the container. You can then move freight units into this container.

Alternatively, you can mark freight units and merge them in a container. You enter a freight unit building rule as well as an equipment group and an equipment type. The system creates a container and assigns the freight units to this container.

The system deletes empty containers when you save.

Example

You have a forwarding order with the following two items:

Item	Quantity	Gross Weight
Item 10	10 Pallets	10,000 kg
Item 20	9 Pallets	9,000 kg

If you use equipment type CNT_EXAMPLE, the system creates the following item:

Item	Equipment Type	Gross Weight	Tare Weight
Item 10 (1 container)	CNT_EXAMPLE	22,000 kg	3,000 kg

This item has two subitems that correspond to the items of the forwarding order.

Related Information

[Equipment Groups and Equipment Types](#)

Statuses of Business Documents

Use

This section contains the list of statuses that the following business documents can have:

- Freight order (see [Freight Order](#))
- Freight booking (see [Freight Booking](#))
- Freight unit (see [Freight Unit](#))
- Transportation unit (see [Transportation Unit](#))
- Service order (see [Service Order](#))
- Consignment order (see [Consignment Order](#))

Features

The following table lists the statuses that business documents can have.

Status	Status Values	Business Documents That Can Have This Status
Life cycle status	New	Freight order
	In Process	Freight booking
	Completed	Freight unit
	Canceled	Transportation unit
		Service order
Driver assignment status	Not Relevant	Freight order
	Underassigned	

Status	Status Values	Business Documents That Can Have This Status
	Fully Assigned Overassigned	
Planning status	Not planned Partially Planned Planned Not Ready for Planning Not Published	All plannable business documents, such as freight units or transportation units
Subcontracting status	No Subcontracting Result In Tendering Carrier Assigned Sent Changed After Sending	All subcontractable business documents
Confirmation status	No Confirmation Yet Invalid Confirmation Rejected Confirmed Confirmed with Changes Update Sent, No Confirmation Yet Document Changed After Confirmation Document Changed After Rejection	All subcontractable business documents
Tendering status	Not Published Published Review Required Completed	Freight order (not supported for rail freight order)
Booking confirmation status	Not Sent to Carrier Sent to Carrier Rejected by Carrier Confirmed by Carrier Canceled by Shipper Partially Confirmed by Carrier Shipping Instruction Sent to Carrier Acknowledged by Carrier, Confirmation Is Pending Confirmed by Carrier with Deviations	Freight booking

Status	Status Values	Business Documents That Can Have This Status
	Canceled by Shipper and Acknowledged by Carrier	
Cargo receipt status	Picked Up Missing Cargo On Hand Shipped Delivered Delivery Failed	Freight order Freight booking Freight unit Transportation unit
Execution status	Not Relevant Not Started/MAWB Open (Air) In Execution Executed Interrupted Canceled Ready for Transportation Execution/MAWB Finalized (Air) Not Ready for Transportation Execution/MAWB Not Finalized (Air) Loading in Process Capacity Planning Finished	Freight order Freight booking Freight unit Transportation unit Consignment Order
Handling execution status	Not Determined Cargo Ready for Loading Arrived (only on stop level) Not Unloaded (only on item level) Partially Unloaded (only on stop level) Unloaded Not Loaded Partially Loaded (only on stop level) Loaded Departed (only on stop level) Not Coupled (only on item level and only for freight orders, trailer units and railcar units) Coupled (only on item level and only for freight orders, trailer units and railcar units) Not Uncoupled (only on item level and only for freight orders, trailer units and railcar units)	Freight order Freight booking Transportation unit Consignment order

Status	Status Values	Business Documents That Can Have This Status
	Uncoupled (only on item level and only for freight orders, trailer units and railcar units) Checked In (only on stop level) Checked Out (only on stop level)	
Cargo execution status	Not Determined Cargo Not Unloaded Cargo Partially Unloaded Cargo Unloaded Cargo Not Loaded Cargo Partially Loaded Cargo Loaded Not Uncoupled Cargo Not Unloaded/Not Uncoupled Cargo Partially Unloaded/Not Uncoupled Cargo Unloaded/Not Uncoupled Partially Uncoupled Cargo Not Unloaded/Partially Uncoupled Cargo Partially Unloaded/Partially Uncoupled Cargo Unloaded/Partially Uncoupled Uncoupled Cargo Not Unloaded/Uncoupled Cargo Partially Unloaded/Uncoupled Cargo Unloaded/Uncoupled Not Coupled Cargo Not Loaded/Not Coupled Cargo Partially Loaded/Not Coupled Cargo Loaded/Not Coupled Partially Coupled Cargo Not Loaded/Partially Coupled Cargo Partially Loaded/Partially Coupled Cargo Loaded/Partially Coupled Coupled Cargo Not Loaded/Coupled Cargo Partially Loaded/Coupled	Freight order Freight booking Transportation unit

Status	Status Values	Business Documents That Can Have This Status
	Cargo Loaded/Coupled	
Load plan status/unload plan status (on stop and item level)	Not Planned Not Relevant Invalidated Planned Finalized	Freight order Freight booking Transportation unit Consignment order
Load plan item status/unload plan item status (on item level)	Not Planned Not Relevant Invalidated Planned Finalized	Freight order Freight booking Transportation unit Consignment order
Transmission-to-EWM status	Not Sent Partially Sent (only on stop level) Sent Partially Received Received Cancellation Requested Cancellation of TM Document Requested (only on stop level) Cancellation Confirmed	Freight order Freight booking Transportation unit
Publishing status	Published Not Published	Freight booking (air)
Waybill/bill of lading finalized status	Not Finalized Finalized Partially Finalized (only for freight bookings)	Freight unit Freight booking
Shipped-on-board status	Not Shipped on Board Shipped on Board	Freight booking (ocean)
Uplift confirmation status	Uplift Not Confirmed Uplift Confirmed	Freight booking (air)
Master-bill-of-lading status	Master Bill of Lading Not Received Master Bill of Lading Received	Freight booking (ocean)
Manifest status	Manifest Not Created Manifest Created	Freight order Freight booking

Status	Status Values	Business Documents That Can Have This Status
Delivery status	Not Delivered Partially Delivered Delivered	Freight unit, transportation unit, and freight order if freight order or transportation unit is automatically created instead of freight unit (see “Automatic Creation of Freight Orders” in Freight Unit Building Rule)
Customs status	N/A	For more information, see Global Trade .
Invoicing status	Not Invoiced Partially Invoiced Completely Invoiced Over-Invoiced Not Up-to-Date	Freight order Freight unit Service order
Transmission-to-ERP status	New — Not Yet Transmitted Initial Transmission Started, Not Yet Confirmed by ERP Transmission Successful Retransmitted after Changes in TM, Not Yet Confirmed by ERP	Freight order Freight Booking
Schedule Data Status	Data Is Up-to-Date Data Is Not Up-to-Date	Freight order Freight booking
Document check status	Last Check with Errors Last Check Without Errors or Warnings Last Check with Warnings Not Yet Checked	Freight Order Freight Booking
Cross-document check status	Last Cross-Document Check with Green Traffic Light Last Cross-Document Check with Yellow Traffic Light Not Yet Checked	Freight Order Freight Booking
Archiving Status	Not Archived Archiving in Process Archived	Freight order Freight booking Freight unit Transportation unit

Fixing of Business Documents

- You can unfix the business document and then change it.

- The fixing only prevents the change of already existing planning results, that is, a fixed subcontractable document is allowed to be subcontracted until a carrier is assigned. If a fixed document is already assigned to a carrier, no more change is allowed in the planning results.
- In VSR optimization, a fixed freight order must not be deleted and assigned freight units must not be unassigned.
- For manual planning, you can specify in your planning profile whether you are allowed to change a fixed freight order or not ([Consider Fixing Status](#)).

You can specify in Customizing for the business document type (that is, freight order type and freight booking type) that the business document is to be fixed automatically each time you save it. In case of one-step planning, the system saves the business documents only at the very end of the planning process, the fixing takes place at the very end accordingly.

More Information

- For more information about the statuses listed above, see [Explanation of Statuses of Business Documents](#).
- For information about statuses in Advanced Shipping and Receiving, see [Statuses in Advanced Shipping and Receiving](#).
- For information about statuses in the integration of a TM system with SAP GTS, see [Statuses in the Integration with SAP GTS \(Inbound Process\)](#) and [Statuses in the Integration with SAP GTS \(Outbound Process\)](#).

Explanation of Statuses of Business Documents

Use

This section contains an explanation about how the various statuses of business documents in Freight Order Management are set and what effects these statuses have.

Features

The following statuses are available:

Life Cycle Status

This status has the following values:

- **New**

The business document has been created, but hasn't been considered in planning and execution yet.

- **In Process**

Planning, execution, subcontracting, or invoicing was started.

You can specify in Customizing for the business document type that the life cycle status is to be set to **In Process** immediately after creation of the business document (that is, even if none of the mentioned processes were started yet). This setting is required for business documents resulting from a planning process.

- **Complete**

The transportation execution is finished and the invoicing status is set to **Completely Invoiced** (or no invoice is expected). This status is set automatically by the system. You can set this status manually if the business document has the life cycle status **In Process**.

- **Canceled**

Planning and execution of transportation is canceled. You can set this status manually at any time in the planning process by selecting the **Cancel Document** pushbutton.

Driver Assignment Status

This status is set automatically by the system on header level when you assign drivers to your entire freight order or individual transportation stages of a freight order. This status is based on the number of required drivers that you indicate in the freight order and the number of drivers that you assign to the freight order.

The status has the following values:

- **Not Relevant**

The freight order is not relevant for driver planning.

- **Underassigned**

At least one transportation stage of the freight order has too few drivers assigned to it.

- **Fully Assigned**

The required number of drivers has been assigned to all transportation stages.

- **Overassigned**

All transportation stages of the freight order have too many drivers assigned to them.

Planning Status

This status relates to the different planning states of the individual transportation stages. The planning status is set automatically by the system and is determined based on the planning status of the individual stages.

The status has the following values:

- **Not Planned**

No transportation stage is planned.

- **Partially Planned**

At least one transportation stage is planned.

- **Planned**

All transportation stages are planned.

- **Not Ready for Planning**

No transportation stage is ready for planning.

- **Not Published**

No transportation stage is published.

The planning status can relate to an individual transportation stage (not only to a business document). In this case, the planning status can have the following values:

- **Unplanned**

The transportation stage hasn't been planned yet.

- **Planned**

The transportation stage has been planned.

- **Planned, Conflict Between FU and Freight Order/Booking Times**

The transportation stage has been planned, but there exist conflicts with time constraints regarding the predecessor or the successor transportation stage.

- **Not Ready for Planning**

The transportation stage is not ready for planning because a location is missing. The system automatically sets a planning block.

Subcontracting Status

This status is set automatically by the system and is relevant for subcontractable business documents. You define the subcontracting relevance in Customizing for the business document type.

The status has the following values:

- **No Subcontracting Result**

The carrier hasn't been determined for the business document.

- **In Tendering**

The carrier hasn't been determined, but a tendering process is already in process.

- **Carrier Assigned**

The carrier was assigned.

- **Sent**

The business document was sent to the assigned carrier.

The value **Carrier Assigned** is a prerequisite for this value.

- **Changed After Sending**

A relevant change was made to the business document after it was sent to the assigned carrier. Relevant changes are:

- Quantity changes
- Changes to dates/times
- Changes to source locations, intermediate stops, and destination locations
- Changes to the transportation service level

For more information, see [Change Controller](#).

Confirmation Status

This status is either set automatically by the system or manually by the user and is relevant for subcontractable business documents. You define the subcontracting relevance in Customizing for the business document type.

This status can only be set automatically by the system if the subcontracting status has the value **Sent**.

The status has the following values:

- **No Confirmation Yet**

The carrier hasn't sent a confirmation yet.

- **Invalid Confirmation Received**

The confirmation sent by the carrier cannot be interpreted.

- **Rejected**

The carrier rejected the business document.

- **Confirmed**

The carrier confirmed the business document.

- **Confirmed with Changes**

The carrier confirmed the business document with some changes.

- **Update Sent, No Confirmation Yet**

The business document was changed and an update was sent but the carrier hasn't sent a confirmation yet.

- **Document Changed After Confirmation**

The business document was changed after having been confirmed by the carrier.

- **Document Changed After Rejection**

The business document was changed after having been rejected by the carrier.

Tendering Status

This status is set automatically by the system and is relevant for subcontractable business documents that are tendered, that is, the subcontracting status has the value **In Tendering**. You define the tendering relevance in Customizing for the business document type.

The status has the following values:

- **Not Published**

The user or the system has started tendering for a freight order, but no requests for quotation to carriers have been published yet.

- **Published**

The user or the system has started tendering for a freight order, and has published one or several requests for quotation to carriers.

- **Review Required**

Review is required by tendering manager.

- **Completed**

Tendering is completed.

There are also the following statuses:

- Freight Request for Quotation: Life Cycle Status
 - **Not Sent**

The freight request for quotation has been created in the tendering plan, but it has not been published yet.

- **Open**

The freight request for quotation has been published and sent to a carrier. Freight quotations can be submitted for this freight request for quotation.

- **Closed**

The freight request for quotation has been published and sent to a carrier, but no more freight quotations can be submitted for it.

- **Omitted**

The tendering process was completed before the freight request for quotation was sent. In other words, an omitted freight request for quotation has never been published.

- **Freight Quotation: Life Cycle Status**

- **Draft**

The freight quotation has been created in the tendering worklist for carriers, but it has not been submitted. It is not considered in the corresponding tendering process.

- **Sent**

The freight quotation has been created and submitted for a freight request for quotation. It is considered in the corresponding tendering process and the evaluation of submitted freight quotations.

- **Tendering Step: Life Cycle Status**

- **Not Started**

The tendering step has been created in a tendering plan, but not started yet. All freight requests for quotations assigned to this step are in status **Not Sent**.

- **In Process**

At least one freight request for quotation assigned to this step has been published and is in status **Open**.

- **Completed**

At least one freight request for quotation assigned to this step had been sent to a carrier. No more freight quotations can be submitted to the assigned freight requests for quotations.

- **Omitted**

The tendering process was completed before the tendering step started.

Booking Confirmation Status

This status is either set automatically by the system or manually by the user.

The status has the following values:

- **Not Sent to Carrier**

The freight booking hasn't been sent to the carrier.

- **Sent to Carrier**

The freight booking has been sent to the carrier.

- **Rejected by Carrier**

The freight booking has been rejected by the carrier.

- **Confirmed by Carrier**

The freight booking has been confirmed by the carrier.

- **Canceled by Shipper**

The freight booking has been canceled by the shipper.

- **Partially Confirmed by Carrier**

The freight booking has been confirmed by the carrier with quantity changes.

- **Shipping Instruction Sent to Carrier**

A shipping instruction has been sent to the carrier for the freight booking.

- **Acknowledged by Carrier, Confirmation Is Pending**

The carrier has received the freight booking but has not confirmed it yet.

- **Confirmed by Carrier with Deviations**

The carrier has confirmed the freight booking with changes.

- **Canceled by Shipper and Acknowledged by Carrier**

The shipper has canceled the freight booking and the carrier has received this cancellation.

Cargo Receipt Status

This status is set automatically by the system. It cannot be set manually.

The status has the following values:

- **Picked Up**

The cargo has been picked-up and is now in transit.

- **Missing Cargo**

The cargo has not yet been received at all or has only been partially received. This status value is set when the actual quantity (in pieces) is lower than the planned quantity. In this case, the system executes a freight unit (FU) split. This means that it updates the existing FU and its predecessor documents with the actual quantity, and creates a new FU for the missing pieces.

- **On Hand**

The cargo has been received. This status is set automatically when actual quantities are reported. After this status has been set, only the actual quantities in the freight order or freight booking are taken into account. You can change these quantities in the freight order or freight booking. If changes are subsequently made in the forwarding order, they are not copied to the freight order, freight booking, or freight unit.

- **Shipped**

The cargo has been shipped. This status value is used in the direct integration of Transportation Management (TM) and Extended Warehouse Management (EWM) in SAP S/4HANA or SAP Extended Warehouse Management (SAP EWM).

- **Delivered**

The cargo has been delivered.

- **Delivery Failed**

The cargo item could not be delivered.

After you remove a freight unit from a freight order, the system allows you to keep the items without reference to the predecessor document in the freight order if the cargo receipt status of the items is at least **On Hand**. In this case, the cargo receipt status is set to **Delivery Failed** for these items. As a prerequisite, you must have selected the **Enable Additional Execution Checks** checkbox in the Customizing activity for defining freight order types.

❖ Example

You have planned a freight unit on the freight order and execution of the freight order has started. During execution, not all cargo could be delivered to the ship-to party, for example. Therefore, you have removed the freight unit from the freight order. If you have selected this checkbox, the cargo items without reference to the freight unit will remain in the freight order and the cargo receipt status of the items is set to **Delivery Failed**. If you have not selected the **Enable Additional Execution Checks** checkbox in the Customizing activity for defining freight order types, such cargo items are removed from the freight order.

i Note

You have also other options to handle a transport when the delivery has failed and to set the status of a transportation stage to **Delivery Failed**. For more information, see [Handling of Failed Deliveries](#).

Execution Status

This status is set automatically by the system for business documents that are subject to tracking. You can specify the tracking relevance in Customizing for the business document type. For more information, see [Execution Tracking](#).

You can also set this status manually.

The status has the following values:

- **Not Started/MAWB Open (Air)**

Execution hasn't started yet.

- **Capacity Planning Finished**

Capacity planning was finished.

- **Loading in Process**

Loading was started.

- **Ready for Transportation Execution/MAWB Finalized (Air)**

Transportation execution can be started, that is, loading was finished or a carrier accepted an order.

i Note

- A business document with status **MAWB Finalized** can no longer be planned manually or automatically.
- In Customizing, you can enable the automatic derivation of **Ready for Transportation Execution/MAWB Finalized (Air)**. This triggers when either the first execution-relevant event is reported or the first stop's execution status in the freight document is updated. For more information, see Customizing for **Transportation Management** under ► **Freight Order Management** ► **Execution Settings** ► **Define Event Profiles** ►

- **Not Ready for Transportation Execution/MAWB Not Finalized (Air)**

Transportation execution cannot be started yet because either the check for readiness of execution failed or you set the status manually to this value.

- **In Execution**

Execution was started.

You can specify in Customizing for the business document type that the business document is set immediately to **In Execution** after creation.

- **Executed**

Execution has finished.

If you have specified in Customizing that your business document type is relevant for execution tracking, we recommend that you specify a last expected event. If this event is reported, the execution status is set automatically to **Executed**.

- **Interrupted**

Execution has been interrupted.

- **Canceled**

Execution was canceled.

- **Not Relevant**

You can specify in Customizing for the business document type that the business document is not relevant for execution tracking. The system automatically sets this status. The status does not change anymore.

Handling Execution Status

This status is either set automatically by the system or manually by the user.

For more information about how the handling execution status is set by event reporting, see [Execution of Freight Documents](#)

You can set the handling execution status on the following levels:

- On item level for cargo items at the current stop

i Note

A cargo item can be a product, a package, or a container.

For example, this status can indicate that a package has been loaded into a container.

You can also set this status on item level for passive vehicle resource items. However, this is only possible for the status values **Coupled**, **Uncoupled**, **Not Coupled**, and **Not Uncoupled**.

- On stop level

On this level, this status indicates the progress of the execution activities at a stop. It refers to a container (capacity item or cargo item) or a cargo item.

The handling execution status is related to the cargo execution status. If all cargo items have been loaded into the container, for example, the system automatically sets the cargo execution status of the container to **Cargo Loaded**. Moreover, the handling execution status on item level and the handling execution status on stop level are related to each other.

In addition, the execution status on stop level considers the handling execution status. For example, the execution status on stop level is only set to a green traffic light if the handling execution status has been set to **Unloaded**.

For air freight bookings, the handling execution status is also displayed on the **Operations** tab page and indicates the logistical execution status. It contains the following:

- Location ID (location code)
- Status at last location at which freight units were picked up or delivered
- Date/time when the status was set

For ocean freight bookings and freight orders, this information is displayed on the **Statuses** tab page.

For example, if a freight order for pick-up is assigned to the air freight booking and you change the status of this freight order from **Arrived** to **Unloaded**, the logistical execution status of the air freight booking will indicate **Unloaded**.

The handling execution status **on item level** has the following values:

- **Not Determined**

The system sets this initial status value after you have created the business document. It indicates that no execution activities are to be performed for the cargo item at the current stop.

- **Not Unloaded**

The cargo item hasn't been unloaded at the current stop.

- **Unloaded**

The cargo item has been unloaded at the current stop.

In the inbound part of the Advanced Shipping and Receiving process, you can use the handling execution status option **Set Items to Unloaded** to manually set the status of the items that are unloaded from the truck to **Unloaded**.

- **Not Loaded**

The cargo item hasn't been loaded at the current stop.

- **Loaded**

The cargo item has been loaded at the current stop.

In the outbound part of the Advanced Shipping and Receiving process, you can use the handling execution status option **Set Items to Loaded** to manually set the status of the items at the warehouse loading stop to **Loaded**.

- **Not Coupled**

The trailer unit or railcar unit hasn't been coupled at the current stop.

- **Coupled**

The trailer unit or railcar unit has been coupled at the current stop.

- **Not Uncoupled**

The trailer unit or railcar unit hasn't been uncoupled at the current stop.

- **Uncoupled**

The trailer unit or railcar unit has been uncoupled at the current stop.

The handling execution status **on stop level** has the following values:

- **Not Determined**

The system sets this initial status value after you have created the business document. It indicates that no execution activities have been performed for the containers at a stop.

If you change the status from **Arrived** to **Set to Not Arrived**, the status is displayed again.

- **Cargo Ready for Loading**

The goods are ready to be loaded and you have set the warehouse processing status to **Completed**. You can initiate customs processes and have the good cleared before the truck's arrival and check in. Any items whose status is not set to **Completed** at this point are removed from the freight order.

To revert the status, select **Cargo Not Ready for Loading**. The customs invoice and export declaration must not have been requested.

- **Arrived**

The goods have arrived.

The **Arrival at Destination** event is sent to all relevant document in TM automatically.

- **Checked In**

The business documents of the goods that have arrived at the gate match the documents in the system. The truck is allowed to drive to the yard to unload or load the goods.

If you set the **Checked In** status without first setting the **Arrived** status, then the system sets the **Arrived** status automatically.

The system automatically reports the **Check-In** event to the relevant document in TM. The system also automatically reports the **Arrival at Destination** event if the **Checked In** status is set before the **Arrived** status.

In the inbound part of the Advanced Shipping and Receiving process, if you don't have an explicit check-in process at your company gate, you can directly set the warehouse processing status to **Ready for Warehouse Processing** at the warehouse unloading stop level. The system will automatically set the **Checked In** status. When you set the **Checked In** status manually or the system sets it automatically, all the inbound deliveries for that stop get the status **In Yard** (for EWM deliveries) or **In Plant** (for Logistics Execution (LE) deliveries).

When you set the **Checked In** status for a stop in a road freight order or consignment order that is relevant for a customs process in the inbound part of Advanced Shipping and Receiving, the system sends a request for unloading permission to a SAP Global Trade Services (SAP GTS) system that is integrated with a TM system. The TM system also sets an execution block for the business document.

To revert the status, select **Not Checked In**. The system sets the handling execution status back to **Arrived**.

If a road freight order or consignment order that is relevant for a customs process in Advanced Shipping and Receiving has an execution block due to a pending unloading permission, the block is not reversed when you reverse the **Checked In** status.

- **Unloaded**

The cargo items have been unloaded.

In the inbound part of the Advanced Shipping and Receiving process, when unloading is completed, you can use the handling execution status option **Set Stop to Unloading Completed** to set the status of the warehouse unloading stop to **Unloaded**.

- **Not Unloaded**

The truck has arrived at the destination location of the items to be unloaded, but the unloading has not been completed.

- **Not Loaded**

The cargo items have not been loaded.

- **Loaded**

The cargo items have been loaded.

In the outbound part of the Advanced Shipping and Receiving process, after you have finished loading the truck, you can use the handling execution status option **Set Stop to Loading Completed** to set the status of the warehouse loading stop to **Loaded**. After this status has been set, no further items can be loaded onto the truck, and the system automatically removes all the items that do not have the status **Loaded** from the road freight order.

- **Checked Out**

The goods have been unloaded from the truck or loaded onto the truck. The truck is checked to make sure that, for example, the doors are closed. The documents of the truck are also checked.

The system automatically reports the **Checkout** event in the assigned freight units and in the business document.

In the Advanced Shipping and Receiving process, if you set the **Checked Out** status without setting the **Departed from Door** status, the system automatically sets the **Departed from Door** status.

If the export declaration is missing or incomplete for a road freight order or consignment order that is relevant for a customs process in the outbound part of Advanced Shipping and Receiving, the system sets a block. You cannot set the handling execution status of the stop to **Checked Out** until the block is resolved.

To revert the status, select **Set to Not Checked Out**. The system sets the handling execution status back to **Unloaded** or **Loaded**.

- **Departed**

This status value indicates that the truck has departed.

If you set the source stop of the main carriage to **Departed**, the vessel is loaded and has departed (**Shipped on Board**).

The **Departure** event is automatically sent to all documents in TM. Similarly, if you report a **Departure** event, the system automatically sets the handling execution status at stop level to **Departed**.

To revert the status, select **Set to Not Departed**. The system sets the handling execution status back to **Checked Out**.

For more information about the handling execution status in rail freight orders, railcar units, and container units, see [Rail Freight Order](#).

Cargo Execution Status

This status is either set automatically by the system or manually by the user. It indicates the execution status of a container (capacity item or cargo item) at the current stop, for example a container has been fully loaded at the source stop of the main carriage.

The status has the following values:

- **Not Determined**

The system sets this initial status value after you have created the business document. It indicates that no execution activities are to be performed for the container at the current stop.

- **Cargo Not Unloaded**

The container hasn't been unloaded at the current stop.

- **Cargo Partially Unloaded**

The container has been partially unloaded at the current stop.

- **Cargo Unloaded**

The container has been unloaded at the current stop.

- **Cargo Not Loaded**

The container hasn't been loaded at the current stop.

- **Cargo Partially Loaded**

The container has been partially loaded at the current stop.

- **Cargo Loaded**

The container has been loaded at the current stop.

Furthermore the status contains values that show for a truck not only the status of loading and unloading (as described above), but also whether all its assigned trailers are coupled or uncoupled (or only some of them). For example, you have the status values **Cargo Loaded/Coupled** or **Cargo Partially Loaded/Partially Coupled**. For a complete list of all status values, see [Statuses of Business Documents](#).

For more information about the cargo execution status in rail freight orders, railcar units, and container units, see [Rail Freight Order](#).

Load Plan Status and Unload Plan Status

You can set this status to indicate that goods are ready for loading or unloading at a stop. You can set the status on header, stop, and item level of a freight document with the following effects:

- Header level: The system searches for the next relevant stop for loading or unloading of the goods, sets the status on stop level, and sends a loading or unloading request (LDAP request) to EWM or SAP EWM for the relevant stop.
- Stop level: The system sends an LDAP request to EWM or SAP EWM for this stop.
- Item level: The system does not send an LDAP request.

You can set the load plan or unload plan status independently from the handling execution status. By setting this status, for example, you notify the next relevant stop for loading or unloading goods that the truck will arrive soon. The next relevant stop is usually the next stop after the current stop that has been modelled in the location master data as a warehouse. The current stop of execution and the handling execution status are only changed when EWM or SAP EWM sends an LDAP notification to TM, when you change the status manually, or when an execution event is reported, for example, by SAP Event Management.

The status has the following values:

- **Not Planned**

The plan hasn't been started. This is the initial status value.

- **Not Relevant**

The plan is not relevant for this stop or item.

- **Invalidated**

The plan has changed (for example, items have been added or deleted); you have to set the status to **Finalized** again on stop level to send an LDAP request to EWM or SAP EWM.

- **Planned**

The plan is in a preliminary state.

- **Finalized**

The plan is complete. The system automatically sends an LDAP request to EWM or SAP EWM, when the status value is set on header or stop level of the freight document.

The load plan status for referenced capacity items (package items, container items, trailer items, and railcar items) is displayed in the assigned successor document and can be set there as well. For example, if you have a road freight order with an assigned trailer unit, you can set the load plan status for the referenced trailer item hierarchy in the road freight order directly.

A status change can be triggered by the following actions:

- You can set the status manually.
- You can perform unified package building.
- You can perform load planning.

In freight bookings and freight orders for pick-up or delivery, the load plan status of the referenced container item hierarchy is displayed but you can't change it.

Load Plan Item Status and Unload Plan Item Status

This status shows whether the plan has changed at item level. It enables you to establish, for example, whether new items have been added to or deleted from the plan. If the plan has changed, the load or unload plan item status is set to **Invalidated**.

The status has the same values as the load or unload plan status.

Note that you can set this status for items or packages of a business document in the context of load planning and packaging. For more information, see [Transportation Planning with Mixed Packages and Deconsolidation](#). In this context, an additional status value is available for the load plan item status:

- **Packaging not Relevant**

The item does not need to be packaged.

Transmission-to-EWM Status

This status describes the message exchange with an external warehouse management system, for example, SAP EWM. The status is automatically set when the `LoadingAppointmentRequest` message is sent to SAP EWM or the `LoadingAppointmentNotification` message is received from SAP EWM. The system can set it on item level or on stop level. On stop level, the system sets it separately for an inbound or an outbound stop. On item level, the status refers to the inbound or outbound stop respectively.

The status on **item level** has the following values:

- **Not Sent**

The item hasn't been sent at the current stop.

- **Sent**

The item has been sent at the current stop.

- **Partially Received**

A partial quantity has been received at the current stop and further updates are to be expected. The status won't change until the truck departs, even if all items have already been loaded.

- **Received**

The item has been received at the current stop and the truck has departed.

- **Cancellation Requested**

A cancellation has been requested for the business document at the current stop.

- **Cancellation Confirmed**

The cancellation for the business document has been confirmed at the current stop.

The status on **stop level** has the following values:

- **Not Sent**

No items have been sent.

- **Partially Sent**

Some of the items have been sent.

- **Sent**

All items have been sent.

- **Partially Received**

At least one of the items has been received or partially received.

- **Received**

All items have been received.

- **Cancellation Requested**

A cancellation has been requested for all items.

- **Cancellation of TM Document Requested**

A cancellation of a freight order or freight booking has been requested.

- **Cancellation Confirmed**

The cancellation for the items has been confirmed.

Publishing Status

This status controls whether a business document can be used by other users, for example, other organizational units.

The status has the following values:

- **Not Published**

The business document has not yet been published and cannot be used by other users.

- **Published**

The business document has been published and can be used by other users.

Waybill/Bill of Lading Finalized Status

You set this status for freight units via your air forwarding orders.

This status has the following values:

- **Not Finalized**

The waybill or bill of lading hasn't yet been finalized.

- **Finalized**

The waybill or bill of lading has been finalized.

- **Partially Finalized** (only for freight bookings)

This status is set for a freight booking if its freight units have different waybill or bill of lading statuses, for example, if one freight unit is finalized and another freight unit is not finalized.

Shipped-on-Board Status

You set this status manually after the vessel has departed the source stop of the main carriage. The shipped-on-board date must not lie in the future.

As a prerequisite, the execution status is to be set to **Ready for Transportation Execution**.

The status has the following values:

- **Not Shipped on Board**

The vessel hasn't departed yet.

- **Shipped on Board**

The vessel has departed.

The system automatically sets the handling execution status of the source stop of the main carriage and all previous outbound stops to **Departed**.

Uplift Confirmation Status

You set this status manually after the aircraft has departed the source stop of the main carriage. The uplift date must not be in the future.

As a prerequisite, the execution status is to be set to **MAWB Finalized**.

The status has the following values:

- **Uplift Not Confirmed**

The aircraft hasn't departed yet.

- **Uplift Confirmed**

The aircraft has departed.

The system automatically sets the handling execution status of the source stop of the main carriage and all previous outbound stops to **Departed**.

Master-Bill-of-Lading Status

You set this status manually after a master bill of lading has been created for an ocean freight booking. If you want to change the master bill of lading number after having set this status, you first have to reset this status.

The status has the following values:

- **Master Bill of Lading Not Received**

No master bill of lading has been received yet.

- **Master Bill of Lading Received**

A master bill of lading has been received.

Manifest Status

You set this status manually after loading has been finished.

The status has the following values:

- **Manifest Not Created**

No manifest has been created yet.

- **Manifest Created**

A manifest has been created.

Delivery Status

This status is set automatically by the system. It is derived from the cargo receipt status on item level:

- If you set the first item to **Delivered**, the system automatically sets the delivery status of the freight unit to **Partially Delivered**.
- If you set the last item to **Delivered**, the system automatically sets the delivery status of the freight unit to **Delivered**.

The status is also automatically set to **Delivered**, if the **Proof of Delivery** event is reported.

The status has the following values:

- **Not Delivered**

The freight unit hasn't been delivered.

- **Partially Delivered**

The freight unit has been partially delivered.

- **Delivered**

The freight unit has been delivered.

Transmission-to-ERP Status

This status is only relevant if you use transportation data from an ERP system and if you want to create ERP shipments based on the planned freight order.

The transmission-to-ERP status is set automatically by the system when the data exchange is processed.

As a prerequisite, you have set the shipment creation relevance in Customizing to [Shipment Creation in SAP ERP](#) for the business document type (freight order type and freight booking type).

The status has the following values:

- **New - Not Yet Transmitted**

Business document is new, and no data was transmitted to the ERP system yet.

- **Initial Transmission Started, Not Yet Confirmed by ERP**

Initial transmission of business document information to the ERP system has been started, but no confirmation has been received yet from the ERP system.

- **Transmission Successful**

Business document information was successfully transmitted to the ERP system.

- **Retransmitted after Changes in TM, Not Yet Confirmed by ERP**

Retransmission of business document information to the ERP system has been started after business document has been changed. New confirmation has not been received yet from the ERP system.

Invoicing Status

This status is only relevant if an invoice is expected from the carrier for a subcontractable business document.

The status is set automatically by the system once a cost calculation has been performed.

The status has the following values:

- **Not Invoiced**

No invoicing request was created.

- **Partially Invoiced**

Partial invoicing request was created.

- **Completely Invoiced**

Invoicing request was created for the total costs.

- **Over-Invoiced**

Invoicing request was created for additional costs.

- **Not Up-to-Date**

Invoicing request was created for a settlement document to accommodate the changes in the final amount of any charge item of an order.

Once a freight settlement document is created either manually or in background processing, the invoicing status can change to [Partially Invoiced](#), [Completely Invoiced](#), [Over-Invoiced](#), or [Not Up-to-Date](#). The system determines the invoicing status based on the calculated costs in the order and the invoiced values in the freight settlement document. An invoice is completed if the invoiced values equal the calculated costs. An order is partially invoiced if the invoiced values are less than the calculated costs. An invoice is over-invoiced if the invoiced values are more than the calculated costs.

Schedule Data Status

This status is set automatically if a referenced schedule has been changed. For more information, see [Use of Schedules](#).

The status has the following values:

- **Data Is Up-to-Date**

The referenced data is up-to-date.

- **Data Is Not Up-to-Date**

A referenced schedule has been changed and the referenced data is not up-to-date.

Document Check Status

This status is set automatically, If you start the check manually and save the business document. The status has the following values:

- **Last Check with Errors**

The last check was performed with errors.

- **Last Check Without Errors or Warnings**

The last check was performed without errors or warnings.

- **Last Check with Warnings**

The last check was performed with warnings.

- **Not Yet Checked**

The check hasn't been performed yet.

Cross-Document Check Status

The system checks if there are time conflicts between a business document and its predecessor or successor business documents. This check is performed automatically and the status set automatically when you save a business document. You can also start the check manually. In this case, the status is also set automatically.

Examples:

- A freight booking arrives too late for the subsequent freight order.
- Transportation times do not lie within the time windows for pickup and delivery.

The status has the following values:

- **Last Cross-Document Check with Green Traffic Light**

The check was performed without errors.

- **Last Cross-Document Check with Yellow Traffic Light**

The check was performed with warnings.

- **Not Yet Checked**

The check hasn't been performed yet.

Archiving Status

The status has the following values:

- **Not Archived**

The business document hasn't yet been archived.

- **Archiving in Process**

The business document has been selected for archiving and processed by the archiving pre-processing.

- **Archived**

The business document has been written to the archive. It can still be accessed in display mode.

For more information about archiving, see [Archiving in Transportation Management \(TM\)](#)

More Information

- For information about statuses in Advanced Shipping and Receiving, see [Statuses in Advanced Shipping and Receiving](#).
- For information about statuses in the integration of a TM system with SAP GTS, see [Statuses in the Integration with SAP GTS \(Inbound Process\)](#) and [Statuses in the Integration with SAP GTS \(Outbound Process\)](#).

Blocking Information of Freight Units and Capacity Documents

Use

You can set planning, execution, and invoicing blocks for business documents, such as freight units, freight documents, and transportation units. The blocks can be set as follows:

- Automatically, for example if a dangerous goods check indicates an error
- Automatically, by being copied from a predecessor document
- Manually

Prerequisites

You have defined block reasons in Customizing. For more information, see Customizing for [Transportation Management](#) under ► [Basic Functions](#) ► [General Settings](#) ► [Define Block Reason Codes](#) ►.

You have defined a block profile in Customizing. You can define tolerances for time conflicts in this profile and specify which blocks are copied to which specific successor documents.

Also, you have assigned the profile in Customizing to the document types (freight unit type, freight order type, freight booking type, and transportation unit type). For more information, see Customizing for [Transportation Management](#) under ► [Freight Order Management](#) ► [Define Block Profile](#) ►.

Features

You can display all the automatic blocks on the [Block Information](#) tab page. Furthermore, you can set or remove blocks manually:

- Manual setting of blocks

When you set blocks manually, you can select a block reason from the values defined in Customizing. You can also set, for example, multiple planning or execution blocks and choose different block reasons. All the block reasons are displayed both in the aggregated view of the blocks next to the block status and in the successor documents.

You can set manual blocks for an entire business document or only for individual document items or stages.

- Automatic setting of blocks

If blocks were set in a predecessor document (this means in a forwarding order or freight unit), blocks are set by the system. Automatic blocks can also be created after checking dangerous goods and approvals as well as during integration with SAP Extended Warehouse Management, for example. The following detailed information is displayed on the **Block Information** tab page:

- The predecessor document that caused the block
 - You can use a link to navigate to this document.
- The type and reason for the block
- The date on which the block was created and the person responsible

When you view the block details, the system displays more information about a block, such as why a forwarding order is still incomplete for planning or the item for which there is a discrepancy.

- Removal of blocks

When you remove a block, you can enter a comment. The system also logs and displays who removed the block and when. Note that you require the authorization to remove blocks.

If you have created a block profile in Customizing and you have defined tolerances for time conflicts, the system ignores these conflicts when it sets planning blocks. The system can also ignore consecutive time conflicts that arise after an initial time conflict has been ignored. For example, a system update can have led to a worsening of the conflict. Time conflicts can arise if a requested pick-up or delivery date/time is exceeded.

For freight orders and freight bookings, only the execution block is copied from the freight unit. The invoicing block is set automatically or manually. It cannot be copied from a predecessor document.

More Information

[Blocking Information of a Forwarding Order](#)

Handling of External Freight Units

In Transportation Management (TM), you can also plan freight units that originate from an external system (for example from TM in SAP S/4HANA Cloud Public Edition). However, restrictions apply to these freight units.

Freight units can be replicated from an external system to TM in SAP S/4HANA, where they can be planned and sent back to the external system. For example, freight units can be created in TM in SAP S/4HANA Cloud based on sales orders and sent to SAP S/4HANA for planning. The freight orders to which the freight units were assigned are sent back again to SAP S/4HANA Cloud Public Edition after planning. All changes to the freight orders in SAP S/4HANA are sent to SAP S/4HANA Cloud Public Edition until accruals have been posted there for the freight orders. Changes to the freight units in SAP S/4HANA Cloud Public Edition are also sent continuously to SAP S/4HANA.

To ensure that changes to the freight units in TM are not overwritten by updates from the original system, most of the fields of the freight units cannot be edited and certain actions cannot be performed. This should also ensure that the freight units still fulfill the requirements of the external system after planning in TM. Examples of permitted and not permitted changes and actions are detailed below.

Possible Changes and Actions

When you plan external freight units in TM, you can change the following data and perform the following actions:

- Adjusting times and dates
- Assigning freight units to freight orders

Certain actions, such as assigning freight units to other capacity documents, assigning freight units to trailers, or creating package units for a freight unit, are basically possible but might cause problems in the external system if this system does not support the corresponding function.

Changes and Actions That Are Not Permitted

You **cannot** change the following data or perform the following actions for freight units from an external system:

- Changing quantities
- Freight unit building
- Splitting and merging freight units
- Changing, splitting, and merging stages
- Canceling freight units
- Building house bills of lading

Identification of External Freight Units

The **Origin of Freight Unit** field shows in your personal worklist (POWL) as well as on the user interface of a freight unit if a freight unit originates from an external system (**External**) or was created in TM in SAP S/4HANA (**Internal**). In addition, the system displays how the freight units were created (for example, during freight unit building or in the export or import process).

Furthermore, the original freight unit is displayed in the document flow.

Unified Package Building

Unified package building is a **generic packaging layer** that allows the creation of packages.

The input for unified package building are items, such as products. The result is a possible packaging hierarchy consisting of packages and auxiliary packaging materials. This packaging hierarchy is saved in the corresponding business documents in the application from which you have called unified package building.

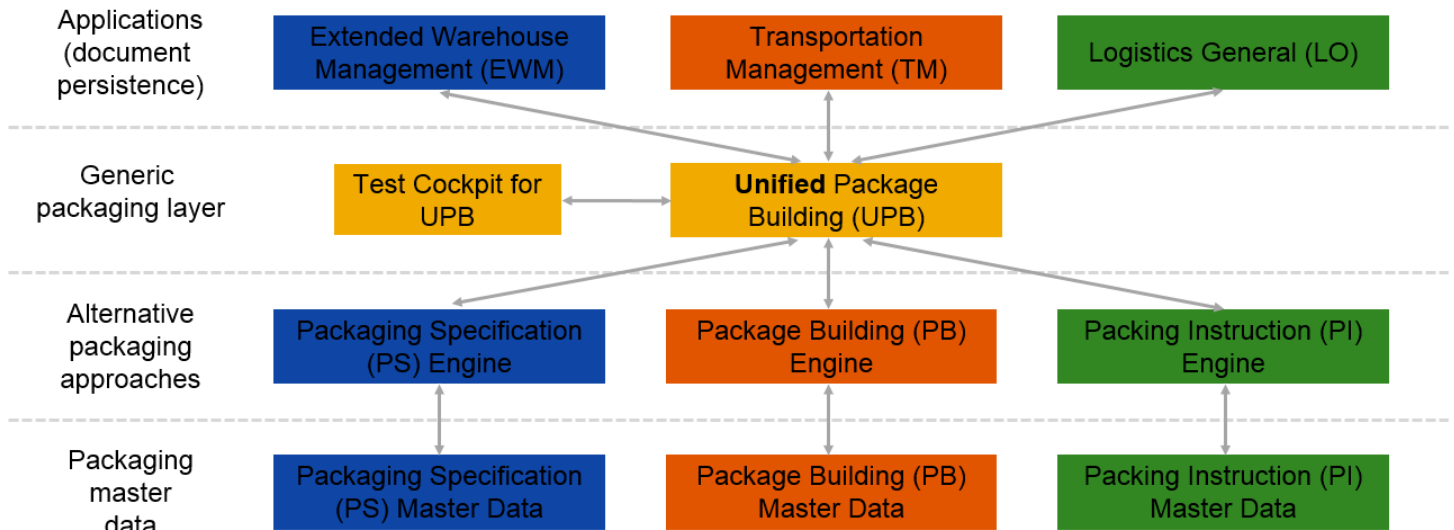
Unified package building supports alternative packaging approaches. These packaging approaches are called **unified package building engines** and process different sets of **packaging master data** and Customizing settings. The following unified package building engines are supported:

- Package building with the package builder
- Packing instruction
- Packaging specification

For more information, see [Unified Package Building Engines](#).

You can freely choose, which of these unified package building engines you want to use. You specify this in the **unified package building profile**. Depending on the unified package building engine you select, you can define specific settings. Further, you can specify generic settings for unified package building. For more information, see [Unified Package Building Profile](#).

The following graphic illustrates the different components and their relationship as explained above:



Unified package building offers several execution modes:

- The default execution mode is the building of packages. The system uses the best set of packaging master data or Customizing settings it can find.
- Further, the system provides the option to determine, which packing instructions or packaging specifications (**unified package building rules**) are available for certain products.
- Optionally, you can start the building of packages with specific unified package building rules.

Note however, that the last two execution modes are only available in certain processes. For more information, see SAP Note [3076865](#).

You can test the different execution modes independent from an application in the test cockpits for unified package building. For more information, see [Test Cockpits for Unified Package Building](#).

i Note

You can use the Business Add-In **BAdI: Unified Package Building** (/SCMB/EX_PB_UNIFIED) to adjust both the items to be packaged and the result items after packaging.

Unified package building is used in the following components of SAP S/4HANA:

- Transportation Management (TM)

For more information, see [Unified Package Building in TM](#).

- Extended Warehouse Management (EWM)

For more information, see [Unified Package Building in Extended Warehouse Management](#).

- Logistics General (LO)

For more information, see [Unified Package Building in Logistics General \(LO\)](#).

Unified Package Building in TM

You can use this function to create packages.

Use

In Transportation Management (TM) you can have the system create packages as follows:

- When freight units are created

This type of unified package building takes place **early** in the process if the system creates freight units based on order documents (for example, forwarding orders or sales orders).

When freight units are created, you can have freight orders or package units created instead of freight units. You make this setting in the freight unit building rule (see "Prerequisites").

- Based on capacity documents

This type of unified package building takes place **late** in the process if capacity documents exist already. You can start from the user interface of an individual capacity document or from the transportation cockpit. It is offered for road freight orders, trailer units, container units (**Road** transportation mode only), and package units.

- During the automatic creation of package units

You can create package units based on unified package building for freight units.

Before you build packages, you can check whether your settings are consistent. For more information, see [Consistency Check of Package Building Constraints](#).

You can use the built packages in load planning. For more information about load planning, see [Load Planning](#).

You can simulate unified package building. For more information, see [Test Cockpits for Unified Package Building](#).

i Note

Several Business Add-Ins (BADIs) are available for unified package building. For more information, see Customizing for Transportation Management under [Business Add-Ins \(BADIs\) for Transportation Management](#) [Planning](#).

Prerequisites

Unified Package Building Using the Package Builder

If you want to build packages using the package building engine (package builder), you must have made the required settings. For more information, see [Package Building](#).

i Note

Some of the SCM Basis transactions are available on the SAP Easy Access screen or in Customizing for TM:

- Customizing activity for defining unified package building profiles:

[Planning](#) [Unified Package Building](#) [Define Unified Package Building Profile](#)

- Customizing activity for defining product orientation profiles:

[Master Data](#) [Product](#) [Define Product Orientation Profile](#)

- Customizing activity for defining profiles for the package building optimizer:

►► [Planning](#) ► [Unified Package Building](#) ► [Package Building](#) ► [Define Profile for Package Building Optimizer](#) ►

- Customizing activity for defining rule profiles for the package building optimizer:

►► [Planning](#) ► [Unified Package Building](#) ► [Package Building](#) ► [Define Rule Profile for Package Building Optimizer](#) ►

- SAP Easy Access transaction for defining products:

►► [Logistics](#) ► [Transportation Management](#) ► [Master Data](#) ► [Material](#) ► [MM01 - Create](#) ► For information about defining product master data for package building, see SAP Note [2842396](#) .

- SAP Easy Access transaction for checking the consistency of your settings:

►► [Logistics](#) ► [Transportation Management](#) ► [Administration](#) ► [Unified Package Building](#) ► [Package Building](#) ► [Check Consistency of Package Building Constraints](#) ►

Unified Package Building During Freight Unit Building

You have made the following settings in TM:

- You have defined a freight unit building rule with the following settings:
 - You have entered a unified package building profile (see [Unified Package Building Profile](#)).
 - You have specified whether the system is to create freight units, freight orders, or package units.
 - You have specified whether the system is to create freight units, freight orders, or package units with a complete hierarchy of package and product items based on the information from the order documents or only a rough estimation of the number of packages that can be built. Note that an estimation of the number of packages is supported only for freight units.
 - Optional: You have specified the maximum number of top-level packages that a freight unit, freight order, or package unit may contain. The system takes this number into account only if you have specified that a complete hierarchy of package and product items is to be created (see above).

For more information, see [Freight Unit Building Rule](#).

- You have entered your freight unit building rule in one of the following places:
 - In the document type of the order document, such as in the forwarding order type in Customizing for TM if you are working with forwarding orders
 - In the logistics integration profile if you are working with the “Internal TM Component Integration” (see [Logistics Integration Profile for SD, MM, and LE](#) and [Internal TM Component Integration](#))

Unified Package Building Based on Capacity Documents

You have made the following settings in TM:

- You have defined a subprofile [Settings for Unified Package Building](#) in your planning profile and entered your unified package building profile in it (see [Planning Profile](#)).
- You have assigned your planning profile to the document type of your capacity documents (for example, the freight order type in Customizing for TM). You have also activated unified package building in the document type. There you can also specify whether you want the system to update the load plan automatically if you change items again after unified package building.

Unified Package Building During Automatic Creation of Package Units

For more information, see [Use of Package Units in Planning](#) ("Automatic Creation of Package Units" section).

Features

Unified Package Building During Freight Unit Building

Depending on your settings in the freight unit building rule, the system creates either a complete hierarchy of package and product items (that exist only locally in the document) or only a rough estimation of the number of packages that can be built. This estimated number of packages is displayed in the header data of the freight unit and can be used later during VSR optimization.

Unified Package Building Based on Capacity Documents

If you select **Build Packages** in a capacity document or in the transportation cockpit, the system creates a complete hierarchy of package and product items that exist only locally in the document. In doing so, the system does not change packages that originate from preceding documents, such as forwarding orders.

If you select several capacity documents in the transportation cockpit, the system creates the packages for each document only, and not for all documents.

You can use **Remove Packages** to remove the local packages. This action does not remove packages that originate from preceding documents such as forwarding orders.

The results of unified package building are displayed on the **Items** tab page in the **Load Plan** hierarchy view of the capacity document as well as in the load plan hierarchy in the transportation cockpit. You also get a 3D visualization of the package. For more information, see [Load Plan](#).

You can also build packages in the background on the basis of capacity documents. For more information, see [Unified Package Building](#).

To improve performance, you can perform unified package building for multiple capacity documents in parallel. To do this, you have to define a parallel processing profile and assign it to your planning profile (in the subprofile **Settings for Unified Package Building**). For more information, see [Definition of Parallel Processing Profiles](#).

Unified Package Building During Automatic Creation of Package Units

For more information, see [Use of Package Units in Planning](#) ("Automatic Creation of Package Units" section).

Related Information

[Transportation Planning with Mixed Packages and Deconsolidation](#)

[Unified Package Building in Integration Based on EWM TUs](#)

Unified Package Building Profile

The unified package building profile is a collection of parameters with which you control the creation of packages.

In the unified package building profile, you can define generic settings for unified package building (UPB):

- You can control UPB in a way that items belonging to different locations or containing different products, for example, must not be packaged together.
- You can indicate that auxiliary packaging materials are treated as separate items in the result of UPB. For more information, see [Defining Auxiliary Packaging Materials](#).

- You can activate the application log and indicate the logging severity.
- You can choose, which UPB engine the system is to use. The following engines are available:
 - Package building with the package builder
 - Packing instruction
 - Packaging specification

For more information, see [Unified Package Building Engines](#).

i Note

- In the test cockpits for UPB, you can see, which packing instruction or packaging specification was used. This information is provided in the **Unified Package Building Rule** field. The UPB rule is an abstraction of the used packing instruction or packaging specification, which both work based on rules. If you use package building as your UPB engine, this field is empty because package building does not work based on rules.
- The determination of UPB rules, that is, which packing instructions or packaging specifications are available for certain products, is only supported in certain processes. For more information, see SAP Note [3076865](#) . This SAP Note also contains information about other technical restrictions.

In the following, the settings for these UPB engines are described:

Settings for the Package Builder

You can make **general settings**, such as a product relationship profile. You can also indicate, for example, that remaining quantities of a product stay unpackaged and you can activate package level aggregation.

Under the **mixed package building settings**, you can indicate whether the system is to create mixed packages or not and you can specify various attributes for **mixed package building**.

Under the **grouping settings**, you can indicate a location hierarchy for grouping and specify whether products belonging to different location groups must not be packaged together (hard constraint) or are not to be packaged together if possible (soft constraint). A location group could be all products to be delivered to a certain area or region, for example. In addition, you can specify utilization thresholds for products to be delivered to certain soft location groups or business partners. Based on these thresholds, the system checks if the utilization of a package is sufficient related to its weight, height, or volume so that a separate package can be built. If the utilization is not sufficient, the products are consolidated with products for other soft location groups or business partners.

Under the **layer settings**, you can specify for **layer-based mixed package building** that no mixed layers are built, for example, or that the system is not allowed to create crowns.

Further settings are available for the embedded layer-based mixed package building (ABAP-based heuristic), which can also be disabled. In this case, layer-based mixed package building is based on the package building optimizer (if you select optimizer-based mixed package building as your mixed package building mode (see below)) or no layers are built (if you select volume-based mixed package building as your mixed package building mode).

Finally, under the **mode settings**, you can specify the mixed package building mode, that is, whether packages are to be built based on volume (**volume-based mixed package building**) or whether the exact positions of the products are to be determined (**optimizer-based mixed package building**). For optimizer-based mixed package building, you can make further settings. For example, you can indicate how the products will be arranged. Further, you can disable stacking, grouping, and overhang. And finally, by indicating a profile and a rule profile for the package building optimizer, you can let the system take into account further technical settings and predefined rules.

For more information, see the system documentation.

i Note

- You do not have to define any of these parameters. The package building will then run in standard mode.
- Some of these parameters are also available in the [Change Package Type Assignment](#) app. However, the parameters in the UPB profile are global settings and are taken into account only if you have not defined more specific values in the [Change Package Type Assignment](#) app.

To define a UPB profile, enter transaction code /SCMB/UPB_PROFILE in the command field.

For more information, see [Package Building](#).

Settings for the Packing Instruction

You indicate a packing transaction profile. For more information, see [Packing with Packing Instructions](#).

i Note

The following parameters of the packing transaction profile are not taken into account during UPB:

- [Packing Instruction Selection Dialog](#)
- [Start Automatically](#)
- [Display Mode](#)

Further, you can make UPB-specific settings, such as the relevant packaging level. With this setting, you can control whether the system is to determine the innermost packaging material or the outermost packaging material (default value) when determining UPB rules.

Settings for the Packaging Specification

You indicate a determination procedure. For more information, see [Packaging Specification](#).

Further, you can make UPB-specific settings. You can specify whether the system is to consider all relevant levels of the packaging specification irrespective whether you have selected the [HU Creation](#) checkbox for a level in the packaging specification or not. This setting can influence which of the packaging materials is determined as the innermost or the outermost packaging material. By indicating the relevant packaging level, you specify whether the system is to determine the innermost packaging material or the outermost packaging material (default value) when determining UPB rules.

Related Information

[Defining Package Type Assignments](#)

[Rule Profile for the Package Building Optimizer](#)

[Optimizer-Based Mixed Package Building](#)

[Reference Products for Package Building](#)

[Test Cockpits for Unified Package Building](#)

Defining Auxiliary Packaging Materials

Unified package building takes into account auxiliary packaging materials, such as filling material or slip sheets.

Depending on the unified package building engine you use, different auxiliary packaging materials are taken into account.

Prerequisites

If you want auxiliary packaging materials to be treated and displayed as separate items in the item hierarchy, you have selected the **Include Auxiliary Packaging Materials in UPB Result** checkbox in the unified package building profile. If you do not select this checkbox, the weight and volume of auxiliary packaging materials are still taken into account during unified package building.

Package Builder

If you perform unified package building using the package builder, the system takes into account the following auxiliary packaging materials:

- Packaging materials per layer

These packaging materials are used if a separate packaging material is required for each product in a mixed package. During package building, the system considers the weight and height of these additional packaging materials when calculating the total weight and height of the package.

The system takes these packaging materials into account when building mixed packages based on layers.

- Separation materials

Separation materials are product-specific packaging materials that are placed on top of a product in a mixed package, for example, slip sheets. During package building, the system considers the weight and the dimensions of this separation material when checking the limits of the mixed package. You can indicate that a separation material is mandatory.

The system takes separation materials into account when building mixed packages based on layer and volume.

You define these packaging materials in the **Change Package Type Assignment** app. For more information, see [Defining Package Type Assignments](#).

Packing Instruction and Packaging Specification

If you perform unified package building using the packing instruction or packaging specification, the system takes into account the auxiliary packaging materials that you defined in the packing instruction or packaging specification.

For more information, see [Packing with Packing Instructions](#) and [Packaging Specification](#).

Related Information

[Unified Package Building Profile](#)

Unified Package Building Engines

Unified package building (UPB) engines are alternative packaging approaches supported by unified package building. They use different processing logics to create packages based on different sets of master data and Customizing settings.

The following unified package building engines are supported:

- Package building with the package builder

For more information, see [Package Building](#).

- Packing instruction

For more information, see [Packing Instruction](#).

- Packaging specification

For more information, see [Packaging Specification](#).

i Note

In the test cockpits for unified package building, you can see, which packing instruction or packaging specification was used. This information is provided in the **Unified Package Building Rule** field. The unified package building rule is an abstraction of the used packing instruction or packaging specification, which both work based on rules. If you use package building as your UPB engine, this field is empty because package building does not work based on rules.

i Note

You can enhance these UPB engines with the following Business Add-Ins (BAIs):

- **BAI: Enhancements for Package Building** (/SCMB/EX_PB_ENGINE_PB)
- **BAI: Enhancements for Packing Instruction** (/SCMB/EX_PB_ENGINE_PI)
- **BAI: Enhancements for Packaging Specification** (/SCMB/EX_PB_ENGINE_PS)

Related Information

[Test Cockpits for Unified Package Building](#)

Package Building

You can use this function to create packages out of products.

Use

Package building is performed with the Supply Chain Execution (SCE) package builder in SCM Basis. In the unified package building profile, you need to select package building with the package builder as the unified package building engine.

During package building, the system bundles products into packages based on specific constraints, while also keeping the total number of packages to a minimum. The system can also specify the exact position and orientation of each product item.

You can define these constraints for different combinations of product, business partner, equipment, and location. If several constraints apply, the system always uses the most specific constraint. For more information, see [Determination of Package Type Assignments](#).

You can use the constraints that you define within this function to create packages in Transportation Management (TM) and in Extended Warehouse Management (EWM).

For more information about creating packages in TM, see <http://help.sap.com/s4hana> under **Product Assistance > Line of Business > Supply Chain > Delivery and Transportation > Transportation Management (TM) > Planning > Unified Package**

For more information about creating packages in EWM, see <http://help.sap.com/s4hana> under ► **Product Assistance** ► **Line of Business** ► **Supply Chain** ► **Warehousing** ► **Extended Warehouse Management (EWM)** ► **Goods Issue** ► **Cartonization Planning** ► **Configuring Package Building** ►

Prerequisites

- You have defined product master data. For more information, see [Defining Product-Specific Master Data](#).
- You have defined package type assignments. For more information see [Defining Package Type Assignments](#).
- You have defined a unified package building profile and indicated **Package Building** as the unified package building engine you want to use. For more information, see [Unified Package Building Profile](#).
- If you want to use mixed package building, you have optionally defined a product relationship profile in the **Change Product Relationship Profile** app. In this profile, you can define incompatibilities and stacking rules. Moreover, you can define the sequence in which the products are to be packaged.
- If you want to use optimizer-based mixed package building, you have done the following:
 - You have optionally defined a profile for the package building optimizer. This profile contains technical settings like runtimes and automatic runtime regulation as well as dump and trace levels. To define this profile, enter view /SCMB/V_PB_OPT in transaction SM30.
 - You have optionally defined a rule profile for the package building optimizer. This profile allows you to select predefined rules and define values and priorities. To define this profile, enter view cluster /SCMB/VC_PB_OPTRP in transaction SM34. For more information, see [Rule Profile for the Package Building Optimizer](#).
 - You have optionally defined a product orientation profile. In this profile, you define how a product can be positioned in a package. To define this profile, enter view /SCMB/V_MAT_ORNT in transaction SM30.

Features

When creating packages, the system can split the quantity of a product using the following criteria:

- Quantity per package
- Maximum height of the package
- Maximum weight of the package

If different products are to be packaged, the system considers the lowest maximum height and the lowest maximum weight for a product or a packaging material. For example, if each product has a different maximum height, the system uses the lowest maximum height.

The system creates mixed packages based on one of the following consolidation types:

- Volume
- Layer definitions
- Incompatibilities, stacking rules, and product orientations (based on these values, the system determines the exact positions of each product item in the package (optimizer-based mixed package building))

The result of package building is different types of packages:

- Packages that contain only a single type of product

- Mixed packages that contain several different products
- Packages that consist of several layers

The layers within a package can also contain different products, which results in mixed layers.

- Nested packages, that is a package in another package

For example, you can plan 24 cans in one box and 100 boxes in ten layers on one pallet.

- Full packages or incomplete packages

Incomplete packages provide room for further consolidation.

- Flat or uneven packages

Flat packages are a prerequisite for stackability.

These package types can be combined. For example, the result of package building can be an incomplete, uneven mixed package with mixed layers.

You can simulate the package building. For more information, see [Test Cockpit for Unified Package Building](#).

You can check whether you have defined your package building constraints in a consistent way. For more information, see [Consistency Check of Package Building Constraints](#).

For more information about best practices and the processing logic, see SAP Note [2581421](#) .

i Note

You can use the Business Add-In **BAdI: Package Building** (/SCMB/EX_PACKAGE_BUILDER) to change the standard settings of package building.

Example

You have 105 laptops and 117 mini hi-fi systems. 90 laptops and 80 mini hi-fi systems form a complete package, while 10 laptops and 16 mini hi-fi systems form one layer. If you allow mixed packages to be created with mixed layers, the system plans to put the remaining 15 laptops and 37 mini hi-fi systems in a mixed package with one layer consisting of 10 laptops, two layers each consisting of 16 mini hi-fi systems, and one mixed layer consisting of 5 laptops and 5 mini hi-fi systems.

Related Information

[Optimizer-Based Mixed Package Building](#)

[Unified Package Building](#)

Defining Product-Specific Master Data

In the master data of a product, you define product-specific attributes that the system considers when creating packages.

Procedure

- To create a product, in the command field enter **MM01**.
- On the **►Additional Data►Units of measure►** view, define the following:
 - Dimensions (length, width, height) of the product

- Weight of the product
- Definitions for full packages, that is, the product quantity that fits into one package as well as the height and weight of the package
- Definitions for full layers, that is, the product quantity that fits into one layer as well as the height and weight of the layer
- Maximum stacking factor (see [Stacking Factor](#))
- Maximum top load, that is, the maximum weight that you can place on top of a full package of this product
- A conversion factor between the base unit of measure and the unit of measure for normalized quantity (see [Determination of Normalized Load Consumption Quantities](#)).

Layer definitions are typically used for mixed packages. If you do not define any layer definition for your product, the system creates packages based on the available volume of the packaging material.

You can define multiple layer and package definitions for a product. To do so, you need to define packaging-specific units of measure in Customizing for SAP NetWeaver under ► **General settings** ► **Check Units of Measurement** ►. In the **Change Package Type Assignment** app, you can indicate in which cases these different layer and package definitions are to be used. For example, you can use different layer definitions for different business partners.

- On the **Sales: General/Plant** view, define the following:
 - Maximum level (that is the filling level for volume-based mixed package building)
 - Stackability factor (see [Stacking Factor](#))
 - Allowed packaging volume (for packaging materials only)

It is used to calculate the maximum height:

$$\text{Max. height} = \text{all. pack. volume} / ((\text{max.}) \text{ length} * (\text{max.}) \text{ width})$$

The maximum height is needed to calculate the capacity volume, which is the main volume considered during package building:

$$\text{Capacity volume for closed packages} = (\text{max.}) \text{ height} * (\text{max.}) \text{ length} * (\text{max.}) \text{ width}$$

$$\text{Capacity volume for open packages} = (\text{max. height} - \text{height of packaging material}) * (\text{max.}) \text{ length} * (\text{max.}) \text{ width}$$

In addition, the existing maximum height values defined in the material master data and the **Change Package Type Assignment** app need to be considered and the minimum is taken into account.

- Closed (closed packaging material)
- On the **WM Packaging** view, define settings for the packaging material. Note that these settings are only relevant if the physical attributes of the packaging material are to be taken into account during package building. In this case, you need to define a product for the packaging material and define the following settings:
 - Maximum height and weight

Note that these are global limits that apply for your entire company, for example. You can set lower limits, for example on a product-specific basis, in the **Change Package Type Assignment** app.
- On the **WM Packaging** view, define settings for package building:
 - A reference product for package building
 - A product orientation profile
 - The shape of the product
 - Thresholds for overhang, and absolute height

Related Information

[Defining Package Type Assignments](#)

Stacking Factor

The stacking factor defines how products and packages, for example pallets, can be stacked.

The following logic applies:

- A stacking factor of zero or an initial field means that the product or package can be stacked an infinite number of times on itself.
- A stacking factor of one means that the product or package can't be stacked on itself.
- A stacking factor greater than one, such as six, means that the product or package can be stacked up to six times on itself.

On the **Additional Data** > **Units of measure** view of the product master data, you can define a maximum stacking factor per base unit of measure. If, for example, you define a maximum stacking factor of six for the alternative unit of measure of the product (for example **PC** (Piece) or **EA** (Each)), it means that the product can be stacked up to six times on itself. This is taken into account during optimizer-based mixed package building. Optimizer-based mixed package building also takes into account maximum stacking factors that you have defined for reference products.

The maximum stacking factor that you define for the alternative unit of measure of the packaging material (for example **PAL** (Pallet)) is determined in package building and taken into account during load planning and load consolidation. If you want the packaging material to be marked as stackable, you need to specify a maximum stacking factor of zero or greater than one.

For closed packaging, only the stackability of the packaging material is relevant, the stackability of the products within the packaging material does not matter. For example, a pallet cage is considered as the outer packaging and thus the system checks the maximum stacking factor that you defined for the pallet cage and takes it into account during load planning and load consolidation. If a product is packaged in a carton, the stackability of the carton is relevant for optimizer-based mixed package building since the carton will be packaged further.

If an open package, such as a pallet, contains layers only (no matter if these layers consist of the same product or different products), the system takes into account and applies the maximum stacking factor (**Units of measure** view) that you have defined for full packages of the contained products or the stackability factor (**Sales: General/Plant** view) that you have defined on product header level. This is valid for layer-based mixed package building (embedded and based on optimizer-based mixed package building).

i Note

- If you have defined a stackability factor, the maximum stacking factor is not considered anymore.
- An open package is only stackable if all contained products are stackable.

Defining Package Type Assignments

By defining package attributes, for example the packaging material, you specify what kind of packages the system is to create for your products.

Context

You have the following options:

- You define package attributes for multiple different combinations of product, business partner, location, contained material, and equipment.
- You define package attributes for multiple different combinations of business partner, location, and equipment.
- You define an exclusive package for a business partner.
- You define alternative packaging materials. The following approaches are available:
 - You define the sequence in which the alternative packaging materials are to be used. This allows you to determine the packaging material based on the product quantity.
 - You define groups of packaging materials, assign packaging materials to each group, and define constraints for each packaging material. This allows you to determine the packaging material based on costs. This approach is only available for optimizer-based mixed package building. For more information, see [Cost-Based Selection of Packaging Materials](#).

i Note

If you define more specific attributes for a specific product (first option), the system considers these attributes and ignores the more global attributes (second option).

You can define the following package attributes:

- Packaging material and package type
- Limits to be applied, for example, the maximum height and weight of the package

Note that the lowest limits (that is, the maximum height of the (mixed) package, the maximum weight of the (mixed) package, and the maximum number of products) of all affected product items are considered.

- Product-specific attributes, for example, a separation material or if mixed packages and mixed layers are to be created

In the **Packaging Material** field, you enter the product ID of the packaging material that you defined in the product master data. This packaging material is used for full packages and mixed packages unless you have specified an additional product ID in the **Packaging Material (Mixed)** field. In this case, all remaining product items are packaged in mixed packages using this additional packaging material.

The package type can be either the packaging material, or the unit of measure, or both. You can enter the packaging material, or the package type, or both:

- If you only enter the packaging material and not the package type, the system uses the base unit of measure that you defined in the product master data.
- If you only enter the package type and not the packaging material, the weight and dimensions that you defined in the product master data are not considered. The system only considers limits that you specify in this transaction.
- If you enter both a packaging material and a package type, the system overrides the base unit of measure that you defined in the product master data with the package type that you specify in this transaction.

If you want the system to create layers, you have to indicate a unit of measure for layers. If you do not indicate such a unit of measure, the system creates packages based on your settings in the unified package building profile (either volume-based package building or optimizer-based mixed package building).

Some of these attributes can also be defined in the unified package building profile. However, the parameters in the unified package building profile are global parameters and are taken into account only if you have not defined more specific attributes here.

For more information about the individual attributes, see the system documentation.

For more information about best practices and the processing logic, see SAP Note [2581421](#) .

To define package attributes, you need to do the following:

Procedure

1. Open the **Change Package Type Assignment** app from the launchpad.
2. Define the package attributes.

Note that you have to define attributes on the **Product-BP-Location** tab page. The other tab pages are optional.

Related Information

[Unified Package Building Profile](#)

[Determination of Normalized Load Consumption Quantities](#)

Determination of Package Type Assignments

During package building, the system determines package type assignments to decide how to best package a product.

The system always uses the most specific value that it can find for a parameter.

To determine this value, the system does the following:

1. The system checks if you have defined a value for a specific parameter on the **Product-BP-Location** tab page. On this tab page, you define product-specific package attributes for different combinations of key fields, that is, product, business partner, location, contained material, as well as equipment type and group.
2. If the system does not find a value or if you selected **Undefined** for the parameter on the **Product-BP-Location** tab page, the system looks for a value on the **BP-Location** tab page. On this tab page, you define product-independent package attributes for different combinations of key fields, that is, business partner, location, as well as equipment type and group.
3. If the system does not find a value for the parameter on the **BP-Location** tab page either, it uses the initial value (0).

If you also selected **Undefined** for the parameter on the **BP-Location** tab page, the system checks if you made a global setting for the parameter in the unified package building profile.

4. If the system has found a package type assignment in one of the previous steps, it tries to find alternative packaging materials for the corresponding key fields by applying either the sequence defined on the **Alternative Packaging Materials Sequences** tab page or the groups defined on the **Alternative Packaging Materials Groups** tab page or both. Alternative packaging materials groups are only supported for optimizer-based mixed package building. For more information, see [Cost-Based Selection of Packaging Materials](#).

Note that the system can only determine alternative packaging materials, if you have deselected the **No Mixed Packages** checkbox in the unified package building profile and if you have defined consistent physical dimensions and their corresponding conversion factors in the material master.

5. On the **BP** tab page, the system checks if you indicated that an exclusive package is to be built for the business partner.

→ Recommendation

Please use placeholder (*) for the key fields very carefully. For products and locations, we recommend that you use reference products for package building and location hierarchies instead. For the contained material, you can't use placeholders.

- If several products are to be packaged in the same way, we recommend that you define reference products for package building. You define reference products for package building in the product master data.
- If the same attributes apply for a large number of locations, we recommend that you define a location hierarchy in the unified package building profile.

i Note

- The key fields **Equipment Type** and **Equipment Group** allow you to indicate the resource to which the product is assigned.

You can use these fields if the way a product is packaged depends on the resource with which it is going to be transported. For example, if you need to transport frozen goods, placing them on a pallet would be fine if the resource is a refrigerated truck. If the resource is a regular truck, however, you would need to package the goods into a refrigerated compartment.

Related Information

[Defining Product-Specific Master Data](#)

[Defining Package Type Assignments](#)

[Unified Package Building Profile](#)

Determination of Normalized Load Consumption Quantities

During package building, the system determines normalized load consumption quantities (normalized quantities, NLQ) to get accurate values regarding the load consumption of products and unit of measures (UoM) in top-level packages (that is, the outer packaging). Based on these values, vehicle scheduling and routing optimization is able to choose the most suitable resources later in the process.

To determine the normalized quantity for each product and UoM, the system does the following:

1. The system first checks if a value is available for each product and UoM in the **Normalized Quantity** field of the **Define Normalized Quantity** app. These values can have the following source:
 - You have entered the values manually.
 - You have determined NLQs with the **Determine Normalized Quantity Based on Randomized Data** app. The results are copied in this field automatically. For more information, see [NLQ Determination Based on Randomized Data](#).
 - You have determined NLQs with the **Determine Normalized Quantity Based on Documents** app. The results are copied in this field automatically. For more information, see [NLQ Determination Based on Documents](#).
 - You have uploaded a spreadsheet that contains the values.
2. If the system finds an NLQ for a product and UoM, it multiplies this value with the planned quantity indicated in the freight unit item for this product.

The system saves the results for all products in the freight unit items. In addition, it displays the sum of the normalized quantities of all items on header level of the freight unit.

3. If the system does not find an NLQ in this field, it checks the product master data. If it finds a conversion factor between the base UoM and the UoM for normalized quantity for a product and UoM, it uses this conversion factor to determine the normalized quantity for this product.
4. If you have not defined conversion factors in the product master data, the system does not determine normalized quantities during package building.

i Note

- The system only determines normalized quantities if you have defined a UoM for normalized quantity in your unified package building profile.
- In step 2 and 3, if the system finds an NLQ for a product and UoM, it uses this value. If it does not find an NLQ, it continues to search from bottom to top in the product hierarchy, starting with the reference product at the lowest level.

Related Information

[Reference Products for Package Building](#)

[Use of Normalized Quantities and Capacities](#)

NLQ Determination Based on Randomized Data

Use

You can use this report to determine a value for the normalized load consumption quantity (normalized quantity, NLQ) per product and unit of measure (UoM) by using a statistical approach. This value indicates how much space (footprint) a single product consumes in a top-level package (that is, the outer packaging).

The report determines this value in the following way:

1. Based on your selection, the report creates SPBUs (that is, units created by separate package building) with a random number of product items that contain a random quantity of products. Each SPBU represents a transportation demand, that is, a set of goods that are transported together across the entire transportation chain within a specific period of time (for example, one delivery day).
2. The report creates one CPBU (that is, a unit created by capacity-document-based package building) with an infinite capacity for all product items of these SPBUs. The CPBU represents a transportation order.
3. The report performs package building based on this CPBU and builds the best possible packages.
4. Based on the package building results, the report calculates a value for the normalized quantity for each product.
5. The report repeats step 1-3 as many times as you have indicated (number of calculation runs).
6. By evaluating the values calculated during these calculation runs, the report determines a unique weighted value for the normalized quantity per product.

Prerequisites

- You have defined dimensions for your products.
- We recommend that you check the consistency of your package building constraints before you run this report. For more information, see [Consistency Check of Package Building Constraints](#).

For more information about prerequisites for package building, see [Package Building](#).

Features

Selection

You need to indicate the following:

- A UoM for the normalized quantity
- A unified package building profile
- The number of calculation runs
- The level on which normalized quantities are to be determined (on product level or reference product level, or both)

We recommend that products belonging to the same reference product have the same dimension.

- The number of SPBUs, that is, the number of transportation demands (average and standard deviation)
- The number of items to be created per SPBU (average and standard deviation)
- The update mode, that is whether the system is to save the results or run the report in simulation mode (default setting)
- The input mode, that is whether the report is to run based on the input options or based on a spreadsheet upload
- At least one input option. For each input option, you can indicate the following:
 - A product or reference product
 - The absolute frequency of the product (the relative frequency is determined by the system)
 - A product quantity per item (average and standard deviation)

By indicating more than one input options, you can influence the weighting of specific products related to one SPBU.

When you execute the report, the NLQ determination screen is displayed:

- On the left, you can change the input data, for example, you can exclude certain products or change the quantities, and then execute or schedule the NLQ determination.
- On the right, the normalized quantities are displayed.

Outputs

The report determines a value for the normalized quantity for each product, which is saved in the **Normalized Quantity** field of the **Define Normalized Quantity** app, if you have indicated that the system is to save the results.

If you have executed the report in simulation mode, you can save the results by selecting **Update Selected Normalized Quantities**.

If the **Define Normalized Quantity** app already contains entries for combinations of product number and UoM for the normalized quantity, these entries are overwritten.

Activities

To start the report, open the **Determine Normalized Quantity Based on Randomized Data** app from the launchpad.

On the selection screen, choose **►Program ►Execute in Background ►** to execute the report in the background. Note however, that this is not supported if the input is based on spreadsheet upload. You need to execute the report based on spreadsheet upload

if you have more than five input options. On the NLQ determination screen, you can then schedule the NLQ determination for execution in the background.

We recommend that you check afterwards whether the normalized quantities in your business documents are up to date. For more information, see [Check Normalized Quantities in Business Documents](#).

Related Information

[Determination of Normalized Load Consumption Quantities](#)

[Unified Package Building Profile](#)

[Defining Package Type Assignments](#)

Optimizer-Based Mixed Package Building

You can use this function to create mixed packages for products.

Prerequisites

- You have installed the package building optimizer. For more information, see SAP Note [1686826](#) .
- In your unified package building profile in Customizing, you have selected **Optimizer-Based Mixed Package Building** in the **Mixed Package Building Mode** field. By doing so, additional fields are made editable, which allow you to make further settings. To define a unified package building profile, enter view /SCMB/V_PB_PROFX in transaction SM30.

Use

The system creates packages using the package building optimizer with the selected consolidation mode. By default, the consolidation mode **Detailed Consolidation** is applied, which determines the exact positions of the items in the package. If you use alternative packaging materials groups, you can select consolidation mode **Capacity-Based Consolidation**. For more information, see [Cost-Based Selection of Packaging Materials](#).

The package building optimizer takes into account the following constraints:

- Unified package building profile (settings for optimizer-based mixed packaging)
- Package type assignments

Note that the lowest limits (that is, the the maximum height of a (mixed) package, the maximum weight of a (mixed) package, and the maximum number of products) of all affected product items are considered.

- Product-specific master data (settings for package building)
- Product orientation profile
- Product relationship profile
- Profile for the package building optimizer
- Rule profile for the package building optimizer

Related Information

[Package Building](#)

[Defining Product-Specific Master Data](#)

Cost-Based Selection of Packaging Materials

The package building optimizer automatically selects the most cost-effective packaging material combination for each set of items that needs to be packaged based on packaging material costs.

Prerequisites

- You have defined alternative packaging materials groups on the [Alternative Packaging Materials Groups](#) view of the [Change Package Type Assignment](#) app. For each group, you have indicated the consolidation mode you want the package building optimizer to apply (detailed consolidation or capacity-based consolidation). Further, you have defined packaging materials for each group on the [Packaging Materials](#) view. For each packaging material, you can specify a package type, a layer unit of measure, as well as weight limits and height limits (for open packaging materials only).

The fixed cost that you define in this app is a dimensionless number and serves as the basis for selecting the best packaging material combination. If you don't provide fixed cost, it will be determined automatically using the following formula:

Fixed cost = square root of the volume (in cm³, decimal numbers are rounded up)

The system calculates the volume using the dimensions defined in the material master data.

For open packaging materials, if you enter a maximum height and a unit of measure in this app, and this maximum height is less than the maximum height in the material master data, the system uses your specified maximum height to calculate the volume. If you don't define a maximum height in either this app or the material master data, the system uses a standard height of 150 centimeters.

- On the [Product-BP-Location](#) view of the [Change Package Type Assignment](#) app, you have indicated the group IDs in the [Alternative Packaging Materials Group ID](#) field for the products for which you want to use cost-based packaging material selection.

Note that if you specify a group on this view, other packaging materials (mixed) and their details that you defined on this view are ignored.

- If you want the package building optimizer to consider incompatibilities, you have defined these in your product relationship profile. To do so, open the [Change Product Relationship Profile](#) app from the launchpad.

Features

The package building optimizer first evaluates a group of allowed packaging materials. Each packaging material has its own cost, layer unit of measure, weight limit, and height limit. Based on the consolidation mode that you indicated for the group, the package building optimizer evaluates the group in one of the following ways: It either determines detailed x, y, and z positions and considers various constraints. This is called detailed consolidation and is the default mode. Or it checks whether the product fits into the packaging material based on the total volume and total weight of the product. This is called capacity-based consolidation.

The package building optimizer manages both balanced and unbalanced multi-level packaging scenarios. It consolidates cartonized and non-cartonized products on pallets of various sizes. It can combine products with predefined packaging materials and those packaged using cost-based packaging material selection in the same package.

Cost-based packaging material selection supports packaging-material-specific layer definitions. For more information about layer definition, see [Unified Package Building Profile](#).

You can use **BAdI: Optimizer-Based Mixed Package Building (Postprocessing)** to modify the fixed cost, for example to consider parcel freight costs.

i Note

This function is supported in optimizer-based mixed package building only.

Related Information

[Defining Package Type Assignments](#)

[Determination of Package Type Assignments](#)

Reference Products for Package Building

You can define reference products to simplify the process of defining package building attributes.

If you have a group of products that need the same packaging attributes, you can define the attributes once for a reference product and then assign the reference product to each product of the group. By doing so, all of the products inherit the attributes of the reference product.

The following attributes can be defined for reference products:

- Definitions for full packages
- Definitions for full layers
- Package type assignments
- Incompatibilities
- Stacking rules
- Consolidation sequence

i Note

You define full packages and full layers in the product master data. You define incompatibilities, stacking rules, and the consolidation sequence in the product relationship profile.

When creating packages, the system first tries to create full packages with one product. If this is not possible, the system adds other products from the group with the same reference product.

You can define reference products on more than one level.

i Note

If you want to change the reference product for multiple products, we recommend to use the test cockpit for unified package building. For more information, see [Test Cockpit for Unified Package Building](#).

Related Information

[Defining Package Type Assignments](#)

[Defining Product-Specific Master Data](#)

[Unified Package Building Profile](#)

Rule Profile for the Package Building Optimizer

SAP offers a number of predefined rules that you can use for optimizer-based mixed package building. For each rule, you can indicate a value and a level. Indicating levels allows you to define a hierarchy between several rules.

Rule Number	Rule Description	Explanation
1110	Minimize height difference on capacity (in mm)	With this rule, you can minimize the height differences of the surface of the package. This objective is defined as standard deviation of the heights of the different cartonized or non-cartonized products that form the surface of the package, weighted by the sizes of these individual product surfaces. You can indicate the deviation from the standard deviation that the package building optimizer is to reach. This value is considered as a threshold, that means, all height differences above are penalized. It is always measured in millimeter.
3010	Minimize distance of center of mass from capacity center	With this rule, you can minimize the distance between the center of mass of the load (that is, the products) and the center of a package, for example, a pallet. You can indicate a threshold value that defines the maximum deviation above which the package building optimizer penalizes distances.
3020	Minimize height of center of mass on capacity	With this rule, you can minimize the height distance between the center of mass of the load (that is, the products) and the bottom side of a package, for example a pallet. You can indicate a threshold value that defines the maximum deviation above which the package building optimizer penalizes height distances.

Note that these rules only apply to top-level packages, that is, to the outer packaging.

Consistency Check of Package Building Constraints

With this report, you can check whether you have defined your package building constraints in a consistent way.

Use

Package building takes into account a wide range of constraints, such as master data and various profiles. When package building is complete, you might wonder why the system has built packages in a certain way. This report lets you check your constraints to discover any potential inconsistencies.

A number of individual checks are available. They are grouped in predefined categories, such as:

This is custom documentation. For more information, please visit [SAP Help Portal](#).

- Products
- Package type assignments
- Normalized quantities

Integration

By selecting **Open Ref. Mat. Hierarchy (Up)** (Open Reference Material Hierarchy (Up)) in the check results area, you can directly open the test cockpit for unified package building and analyze the product hierarchy. For more information, see [Test Cockpit for Unified Package Building](#).

Features

Selection

To run the report, you need to select the following:

- A product, a reference product, or a range of products
- One or several check categories

Moreover, you can specify how the output should look. You can indicate whether the report is to focus on products or checks and whether successful check results are to be displayed as well. Note that you can change these output settings on the consistency checks screen after you have executed the checks.

When you execute the report, the consistency checks screen is displayed:

- On the left, you can select the individual checks and execute them.
- On the right, the check results are displayed.

Output

The check results are displayed according to your output settings. A red check result indicates a critical inconsistency that might lead to a package building run failure. A yellow check result indicates that the system might not build the packages as you expect it.

From the check results area, you can directly open the following:

- Product master
- Test cockpit for unified package building (see **Integration**)
- Package type assignment transaction

Activities

Enter transaction code `/SCMB/CHECK_PB_CONS` in the command field to start the report `/SCMB/CHECK_PB_DEFINITIONS`.

On the selection screen, choose **▶Program ▶Execute in Background ▶** to execute the report in batch mode. The system executes all checks of the categories, which you selected.

Related Information

Packing Instruction

This unified package building (UPB) engine processes a set of master data that you have defined in your packing instruction. For more information, see [Packing with Packing Instructions](#).

In the UPB profile, you can define additional, UPB-specific parameters. For more information, see [Unified Package Building Profile](#).

UPB applies the following stackability logic. A top-level package item in the resulting packaging hierarchy is marked as stackable if the following prerequisites are met:

- The underlying rule contains a stacking factor of zero (or initial value) or greater than one.
- The package is considered being complete.

If these prerequisites are not met, the top-level package item is considered as being non-stackable.

Packaging Specification

This unified package building (UPB) engine processes a set of master data that you have defined in your packaging specification. For more information, see [Packaging Specification](#).

In the UPB profile, you can define additional, UPB-specific parameters. For more information, see [Unified Package Building Profile](#).

Test Cockpits for Unified Package Building

The test cockpits for unified package building allow you to simulate the building of packages and the determination of unified package building rules. You can then analyze the result. You can also display and analyze the master data for products and display products or reference products in the product hierarchy. The result shows the outcome of unified package building without any additional processing in Transportation Management.

Two variants are available:

- Document-based test cockpit

For more information, see [Test Cockpit for Unified Package Building \(Document-Based\)](#).

- Product-based test cockpit

For more information, see [Test Cockpit for Unified Package Building](#).

Activities

To start the document-based test cockpit from the SAP Easy Access menu, choose ► **Transportation Management** ► **Administration** ► **Unified Package Building** ► **Unified Package Building - Test Cockpit (Document-Based)** ►.

You can also use the transaction code /SCMTMS/TEST_PLN_CPB.

To start the product-based test cockpit from the SAP Easy Access menu, choose ► **Transportation Management** ► **Administration** ► **Unified Package Building** ► **Unified Package Building - Test Cockpit** ►.

You can also use the transaction code /SCMB/TEST_UPB.

Test Cockpit for Unified Package Building (Document-Based)

With this report, you can simulate the building of packages and analyze the result. You can also display and analyze the master data for products and display products or reference products in the product hierarchy. The result shows the outcome of unified package building (UPB) without any additional TM-specific processing.

With this report you can do the following:

- You can simulate the building of packages out of business documents (for example freight orders).
- Optionally, select a UPB profile. If you don't select a UPB profile, the system checks if a planning profile is defined, for example in the corresponding freight order type, and if this planning profile contains a UPB profile.

The document-based test cockpit for UPB offers the same functionality as the **Build Packages** mode of the product-based test cockpit for UPB. For more information, see [Test Cockpit for Unified Package Building](#).

To start the document-based simulation of UPB from the SAP Easy Access menu, choose **►Transportation Management ►Administration ►Unified Package Building ►Unified Package Building - Test Cockpit (Document-Based).** ►

You can also use the transaction code /SCMTMS/TEST_PLN_CPB.

Related Information

[Unified Package Building in TM](#)

[Package Building](#)

[Unified Package Building Profile](#)

Test Cockpit for Unified Package Building

With this report, you can simulate the building of packages and the determination of unified package building rules (UPB rules). You can then analyze the result. You can also display and analyze the master data for products and display products or reference products in the product hierarchy.

The test cockpit for unified package building (UPB) offers several execution modes. You can do the following:

- **Build Packages:** Simulate the building of packages out of products and analyze the result
- **Determine UPB Rules:** Simulate the determination of UPB rules and analyze the result
- **Read UPB Rules:** Display the details of a UPB rule.
- **(Ref.) Prod. Hier: Top->Down/Bottom->Up:** Display a hierarchy for products or reference products
- **Determine Pkg. Type Assignment:** Display the package types assigned to a product or reference product

Build Packages

You can simulate the building of packages out of products and analyze the result. You have the following options to start the simulation:

- Select products by various product-based selection criteria

- Select a UPB profile. The system executes UPB using the engine that you select in the unified package building profile.
- Activate the 3D scene to display a 3D visualization of the packages that were built during UPB
- Start the report with an empty input section and select products later

You can navigate to the product master data and change data, if needed.

Messages from the UPB, like error messages, warnings or success messages, are displayed to help you understand and analyze the result of UPB.

In the result, you can see, for example, which UPB engine and UPB rule (that is, packing instruction or packaging specification) the system used. For more information, see [Unified Package Building](#).

The simulation of UPB includes the following functions:

Function	Where to Find	Further Information
Add products	Toolbar of input section	Insert rows to add (further) products to the input list, then execute the simulation and analyze the result.
Get Products per Reference Product	Toolbar of input section	Add all products in one go to the input list that are located below the entered reference product.
Export/Import data	Toolbar of input section	Download currently displayed input list. If you plan to upload data, you should first download data, to get the data structure.
Open Product Master	Result section	Double-click the icon to navigate to the product master data. You can change the data, if needed.
Display Product Hierarchy (Bottom->Up)	Result section	Double-click the icon to display all products of the hierarchy that are above the entered product.

Determine UPB Rules

This execution mode determines UPB-engine specific UPB rules based on your settings in the UPB profile, such as the relevant packaging level. For the UPB engines "packing instruction" and "packaging specification", the system determines the defined sets of packaging master data and indicates their sequence (main UPB rule and alternative UPB rules). For the UPB engine "package building", however, the system returns a dummy entry only.

From the result screen, you can select one of the determined UPB rules and start the [Build Packages](#) mode.

i Note

The UPB rule is an abstraction of the used packing instruction or packaging specification, which both work based on rules. If you use package building as your UPB engine, the system does not determine UPB rules because package building does not work based on rules.

Read UPB Rules

This execution mode provides the details of UPB rules and their levels. You can use it to read information regarding the content of a packing instruction or packaging specification.

Display Hierarchy for Product or Reference Product

With the product hierarchy, you can display products (or reference products) and their location in the product hierarchy. Data like product master data, package type assignment or the alternative unit of measure is displayed in the hierarchy. You can navigate to the transactions and change the data, if needed.

You have the following options to start the product hierarchy:

- **Hierarchy Bottom->Up:** Show all (reference) products in a hierarchy tree that are located **above** the entered product.
- **Hierarchy Top->Down:** Show all (reference) products that are located **below** the entered product. The entered product serves as top node.

The product hierarchy includes the following functions:

Function	Where to Find	Further Information
Add products	Toolbar of input section	Insert rows to add (further) (reference) products to the input list and then display the hierarchy trees of the added (reference) products.
Remove products	Toolbar of input section	Remove rows to delete (reference) products from the input list and then display the hierarchy trees of the remaining (reference) products.
Open Product Master	Result section (tree)	Double-click the icon to navigate to the product master data. You can change data, if needed.
UoM	Result section (tree)	Double-click the icon to display the alternative unit of measure (UoM) of the corresponding (reference) product.
PTA	Result section (tree)	Double-click the icon to display the package type assignment of the corresponding (reference) product.
Show Assigned Package Types	Toolbar of result section	Mark one or several nodes to display the package type assignment. The symbol in column PTA shows if a package type is assigned to a (reference) product.
Show Alternative UoM	Toolbar of result section	Mark one or several nodes to display the alternative unit of measure (UoM) of a product.
Open Package Type Assignment Transaction	Toolbar of result section	Navigate to the transaction where you can define package type assignments for products (/SCMB/PB_PKG_ASS).
Show Alternative UoM	Toolbar of package type assignment section	Mark one or several rows to display the alternative unit of measure (UoM) of a packaging material.

Display Package Type Assignments

With this execution mode you can display the package types assigned to a product. Select your products by various product-based selection criteria.

Activities

To start the test cockpit for UPB (program /SCMB/TEST_PB), enter transaction code /SCMB/TEST_UPB in the command field.

Note that when you select **Execute** on the selection screen of the **Build Packages** mode and the **Determine UPB Rules** mode, a second screen is displayed that allows you to change your selection. You start UPB by selecting **Build Packages** on this second screen. Similarly, you start the determination of UPB rules by selecting **Determine UPB Rules**.

Related Information

[Defining Product-Specific Master Data](#)

[Defining Package Type Assignments](#)

[Unified Package Building Profile](#)

[Reference Products for Package Building](#)

Profiles and Settings

Use

A number of profiles and settings are available with which you can perform transportation planning. You can use them to group planning-relevant settings, for example, capacities and planning costs. The use of profiles and settings makes it easier for you to start manual and automatic planning and is a prerequisite for VSR optimization in the background (see [VSR Optimization](#)).

Features

The following is a list of the profiles and settings that are available to you:

- Profiles

Two profiles are available to which you can assign attributes and settings. These attributes and settings represent a type of subprofile with its own ID that you define once and can then assign to multiple profiles.

- Selection profile (see [Selection Profile](#))
 - Time-related selection attributes
 - Geographical selection attributes
 - Additional selection attributes
- Planning profile (see [Planning Profile](#))
 - Capacity selection settings
 - Capacity availability settings
 - Settings for capacity document creation
 - Settings for manual planning

- Scheduling settings
 - Optimizer settings
 - Constraints and costs settings
 - Incompatibility settings
 - Settings for carrier selection
 - Load planning settings
 - Settings for unified package building
 - Settings for the transportation cockpit
- Decreasing capacities (see [Capacity Decrease](#))
 - Freight unit building rules (see [Freight Unit Building Rule](#))
 - Incompatibilities and settings for incompatibilities (see [Incompatibilities](#))

i Note

- Note that the profiles and settings involve Customizing that you can transport into the production system.
- You transport your profiles and settings by choosing the appropriate pushbutton and entering a transport request. If your administrator has activated automatic change recording (transaction SCC4), a dialog box will automatically appear if you make changes to your profiles and settings. Here you must select a transport request. The automatic change recording ensures that all changes to profiles and settings are assigned to a transport request. We recommend that you activate automatic change recording, otherwise it will be difficult for you to ensure consistency between the different profiles and settings.

If profiles and settings are only to be changeable in the Customizing system and not in the production system, you must control this via the corresponding roles.

The following authorization objects are relevant in this case:

- T_PRF_CR for the creation of profiles and settings
- T_PRFOPPID for the planning profile
- T_PRFSELID for the selection profile
- T_PRFINCID for incompatibility settings
- T_FUBRID for freight unit creation rules
- T_COND for conditions

Selection Profile

Definition

A user-specific grouping of business documents (that is, forwarding orders, freight units, and freight orders) that is taken into account during transportation planning.

Use

The system takes into account the settings that you make in the selection profile during manual planning, during VSR optimization, during planning in the background, and during carrier selection.

i Note

You can define an authorization check for the change and usage requirements of each selection profile. You can configure the relevant settings using authorization object T_PRFSELID.

Structure

In the selection profile and in the assigned selection attributes, you define which business documents the system is to take into account as well as the maximum number of documents. You can assign the following selection attributes to a selection profile:

- Time-related selection attributes

Here you define the demand horizon, for example (see [Demand Horizon](#)).

- Geographical selection attributes

Here you define source and destination locations, for example.

- Additional selection attributes

Here you define additional attributes for database queries (see [Additional Selection Attributes](#)).

You can use your selection profile as follows:

- As a selection profile for freight units

The system uses the profile to select freight units.

- As a selection for freight orders and freight bookings

The system uses the profile to select existing freight orders and freight bookings that are to be taken into account in planning.

This means that the selection of business documents made by the system depends on where and how you use your selection profile. For example, if, when calling the transportation cockpit, you specify your selection profile as a selection profile for freight units in your profile and layout set (see [Use of Profile and Layout Sets](#)), the system selects freight units. However, if you specify your selection profile as a selection profile for freight orders and freight bookings, the system selects freight orders and freight bookings.

i Note

To define selection profiles, open the [Create Selection Profile](#) app on the launchpad.

Note that the selection criteria in the geographical and additional selection attributes have an effect on performance.

Demand Horizon

Definition

Period containing all demands that are to be taken into account in planning.

Use

You define the demand horizon in the time-related selection attributes that you assign to your selection profile (see [Selection Profile](#)). You can define a demand horizon for pickup and a demand horizon for delivery. The system chooses all freight units and freight orders whose pick-up date/time or delivery date/time lies within the relevant demand horizon.

Structure

You can define the demand horizon as absolute or relative:

- Absolute

You define the demand horizon precisely by defining a start and end date and a start and end time.

- Relative

You do not define the demand horizon with precise dates and times, but specify instead a duration starting from the current date. The system then automatically determines the start and end date and the start and end time. If the demand horizon is not to start on the current date, but instead at a later time, you can define an offset. According to this, the relative demand horizon is defined as follows:

- Start of demand horizon = current date + defined offset

Here, the offset is made up of the offset in days and the additional offset (in hours and minutes).

- End of demand horizon = start of demand horizon + defined duration of demand horizon

Here, the duration of the demand horizon is made up of the duration in days and the additional duration (in hours and minutes).

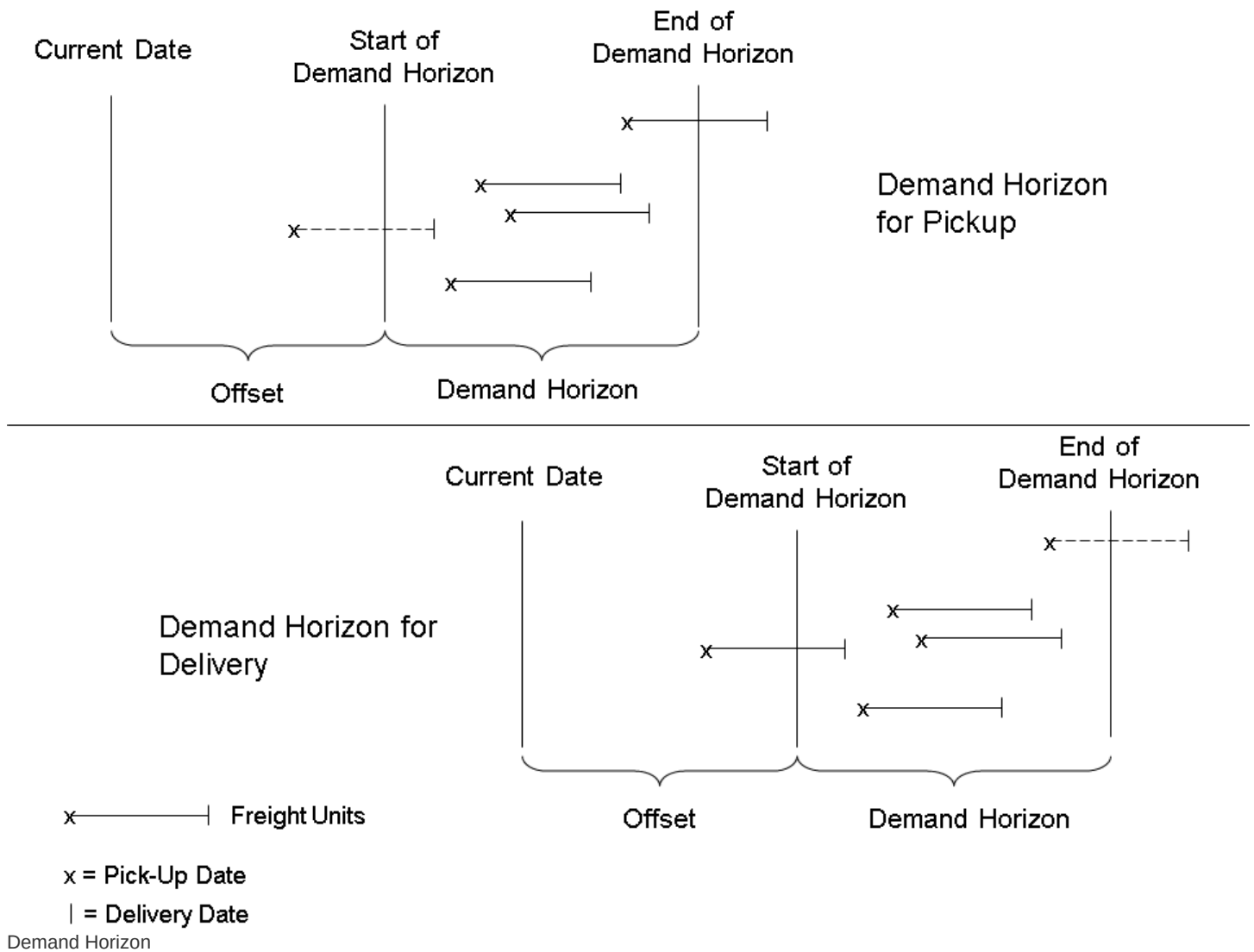
If you specify a factory calendar, the system takes into account non-working days when calculating the start of the horizon. In this case, the planning horizon always begins on a working day.

You can simulate the relative demand horizon. You enter a reference date that the system uses to determine and display the start and end date and the start and end time for the pick-up or delivery.

- Relative horizon with a fixed start time or a fixed end time

You define a relative demand horizon as described above. You also enter a fixed start time as of which you want the horizon to start or a fixed end time by which you want the horizon to end. You can also simulate this demand horizon.

The following figure shows how the demand horizon is defined:



Additional Selection Attributes

Use

To restrict the selection of requirements data even more specifically, you can define additional attributes for database queries.

Features

Definition of Additional Attributes for Database Queries

Here you define an additional filter for certain business documents:

- Business documents in forwarding order management (forwarding orders, forwarding quotations, order-based transportation requirements and delivery-based transportation requirements)
- Business documents in freight order management (freight orders, freight bookings, consignment orders, transportation units, freight units)

Activities

To define additional selection attributes, open the [Create Additional Selection Attributes](#) app on the launchpad.

To assign your additional selection attributes to a selection profile, open the [Edit Selection Profile](#) app on the launchpad and then open the required selection profile.

Planning Profile

Definition

A user-specific grouping of parameters that are taken into account during transportation planning.

Use

The system takes into account the settings that you configure in the planning profile during manual planning, load planning, and VSR optimization (interactive and in the background; see [VSR Optimization](#)).

i Note

- You can define an authorization check for the change and usage requirements of each planning profile. You can configure the relevant settings using authorization object T_PRFOPPID.
- A planning profile is selected as a standard profile. If the system does not find a planning profile, it uses the values from this standard profile.

Structure

The planning profile comprises a range of settings (subprofiles) for which own apps are also available. The planning profile and the settings are described below.

Planning Profile

In the planning profile, you define the following, for example:

- Planning horizon (see [Planning Horizon](#))
- Subprofile assignments

Here you assign existing settings and selection profiles to your planning profile. You can also create new subprofiles from scratch or by copying an existing subprofile. For each assigned subprofile (except selection profiles and carrier selection settings), you can specify whether profile data can be changed during planning sessions.

- Means of Transport and Vehicle Types

Here you can activate determination of means of transport and vehicle types based on conditions. For more information, see [Means of Transport and Vehicle Type Determination](#).

- Parallel processing profiles

Here you can enter parallel processing profiles for different functions, such as distance and duration determination.

- Planning strategies (see [Process Controller Configuration](#))

You can either specify the standard strategy or use your own strategy. You can add additional methods to the standard strategy.

- Manual planning (standard strategy VSRI_SCH)

You define this strategy in the manual planning settings.

- Scheduling (standard strategy VSS_EMBED)

You define this strategy in the scheduling settings.

- Checks (standard strategy VSR_CHECK)

The system checks capacities, incompatibilities, and multiresources, and it performs a dangerous goods check.

- VSR optimization (standard strategies VSR_DEF and VSR_1STEP)

You define this strategy in the optimizer settings.

- Load planning (standard strategy ALP_DEF)

You define this strategy in the settings for load planning.

- Generating transportation proposals (standard strategies VSR_DEF and VSR_1STEP)

You define this strategy in the optimizer settings.

- Integrating real-time execution data (see [Integration of Real-Time Execution Data](#))
- Handling empty stages

Here you specify how the system is to handle empty transportation stages when solving warnings about missing transportation stages in the Gantt chart (see also "Settings for manual planning" below).

Capacity Selection Settings

Here you choose the required capacities, such as resources. If you use resources for which you have defined an ADR limit, VSR optimization takes into account the number of ADR points for this resource during the optimization run (see [ADR 1.1.3.6 Points Check Calculation](#)). Moreover, you can specify that resources for which you have set a planning block are displayed in the resource lists of the transportation cockpit.

If you want to carry out your planning based on equipment types, enter your vehicle and container types here under **Vehicle Types** and **Container Types**. For more information, see [Planning Based on Equipment Types](#).

Capacity Availability Settings

Here you can restrict the availability of resources and equipment types. Using rules that you define here, the system determines when certain resources or equipment types are available based on shift sequences and factory calendars. For more information about planning based on equipment types, see [Planning Based on Equipment Types](#).

You can also assign driving and working constraint profiles to your vehicles (truck resources or vehicle types) and drivers. For more information, see [Considering Driving Time and Working Hours Constraints](#).

Capacity Document Creation Settings

Here you can specify a standard business document type or a condition with which the business document type is to be determined.

Moreover, you can enter a requirement grouping profile that is used to create capacity documents based on requirement groups. For more information, see [Creation of Capacity Documents Based on Requirement Groups](#).

Optimizer Settings

For example, you can define the following here:

- Optimizer runtime
- Maximum number of transshipment locations and processes
- Freight order building rule (see [Freight Order Building Rule](#))
- Required process controller strategy
- Settings for generating transportation proposals (see [Generation of Transportation Proposals](#))
- Settings for incremental planning (see [Incremental Planning](#))
- Whether you require rough or detailed information for your planning activities (see [Rough-Cut Planning](#))
- Whether the system is to consider a subsequent planned transportation stage as a hard time constraint
- Whether freight orders that are ready for execution or in execution can be changed
- Whether setup warnings are displayed or hidden
- How default routes are to be used when freight units, container units, railcar units, and package units are processed in the VSR optimization run

In the advanced settings, you can also specify whether the main aim in your transportation proposals is to ensure the lowest transportation costs possible or the shortest transportation duration possible based on the planned delivery date/time. You can also define preferences in relation to your relative weighting of the variance of carriers, routes, and departure dates, or activate or deactivate capacity constraints at transportation mode level. You can also specify whether the system is to ignore certain settings such as the capacity or ADR limit of a resource during VSR optimization.

Unified Package Building Settings

- Unified package building

Here you enter a unified package building profile and a profile for parallel processing. You can also specify whether the items are to be filtered (for example, outbound delivery only or pick-up only). In this way, you can filter out returns and leave them unpackaged, for example. For more information, see [Unified Package Building in TM](#).

- Creating package units based on unified package building for freight units

Here you enter a default package unit type (mandatory) and a default rule for the creation of package units (optional). In addition, you can enter a package unit type for package units with only one destination location and a corresponding rule. These settings are taken into account if the system creates first packages and then package units for freight units for which you have selected **Exclusive Package for Customer**. This means that the package units only receive products for a specific customer. To use the function, you must also enter a unified package building profile. For more information, see [Use of Package Units in Planning](#).

Load Planning Settings

Here you can define the optimizer runtime and activate automatic runtime regulation. You can also specify the engine you want to use (row-stack approach or free positioning) and various rules for load planning and load consolidation. For example, you can define the maximum height difference between stacks in a row.

Constraints and Costs Settings

Here you define costs related to freight units and means of transport. In most cases, these costs are not actual costs. They simply offer a means of controlling the result of the optimization run (for example, earliness costs and lateness costs). For more information, see [Constraints and Costs Settings](#).

Incompatibility Settings

Here you define settings for your incompatibilities (see [Incompatibilities](#)).

You can also specify whether the constraints for loading (loading direction and attached equipment) are to be taken into account in VSR optimization, load planning and load consolidation, and for the creation of transportation proposals. The system considers these as incompatibilities between location and resource. For manual planning, you can specify whether the system is allowed to violate the incompatibilities or not or whether it should issue a warning if they are violated.

As soon as you define these constraints, we recommend that you add the profiles for loading direction and attached equipment to the locations and vehicle resources that are to take these constraints into account. For more information about defining profiles for loading direction and attached equipment, see [Loading Requirements for Locations](#) and [Loading Requirements for Vehicle Resources](#).

Carrier Selection Settings

Here you specify whether the system is to use transportation allocations or business shares, for example.

Manual Planning Settings

Here you specify how you want the system to behave when it assigns documents and resources:

- Assignment of documents

You use these settings to control how the system is to assign one or more requirement documents to a capacity document.

For more information, see [Assignment of Requirement Documents to Existing Capacity Documents](#).

- Removal of the assignment of documents

You use this setting to control how the system is to remove the assignment of requirement documents to capacity documents.

- Assignment of resources and creation of documents

You use these settings to control how the system is to assign resources to documents and how it is to create and assign reference documents.

- Assignment of drivers

You use this setting to control whether a default truck is to be assigned together with the default driver.

- Solving of warnings

You use this setting to control how the system is to handle empty transportation stages when solving warnings about missing transportation stages in the Gantt chart.

For more information, see the corresponding field help.

Scheduling Settings

Here you can control how the system schedules documents. For example, you can define loading and unloading durations, coupling and uncoupling durations, and the durations required to prepare and finalize a stop based on locations and resources.

You can specify whether driving time and working hours constraints are to be considered. For more information, see [Considering Driving Time and Working Hours Constraints](#). You can also override operating times to be used for a location based on the resource. For more information, see [Consideration of Scheduling Constraints](#).

Transportation Cockpit Settings

Here you define settings that control the behavior of the transportation cockpit. For example, you can activate context determination for the selection. For more information, see [Context Determination](#). You can also specify that your selection is to be deleted after an action has been executed successfully. For more information, see SAP Note [3181915](#) .

Activities

To define a planning profile, open the [Create Planning Profile](#) app on the launchpad.

Planning Horizon

Definition

Planning period in which transportation activities are to be executed.

Use

You define the planning horizon in the planning profile (see [Planning Profile](#)). The planning horizon defines the possible period of time in which the system can schedule transportation activities (for example, loading and unloading, transport).

Structure

You can define the planning horizon as absolute or relative:

Absolute Planning Horizon

Enter a start and end date and a start and end time.

Relative Planning Horizon

You do not define the planning horizon with precise dates and times, but enter instead a duration starting from the current date. The system then automatically determines the start and end date and the start and end time. If the planning horizon is not to start on the current date, but instead at a later time, you can define an offset. Here you can specify whether the system is to round the planning horizon to complete days. For the planning horizon, you can specify the time zone in which the system is to calculate the horizon.

According to this, the planning horizon is defined as follows:

- Start of planning horizon = current date + defined offset

Here, the offset is made up of the offset in days and the additional offset (in hours and minutes).

- End of planning horizon = start of planning horizon + defined duration of planning horizon

Here, the duration of the planning horizon is made up of the duration in days and the additional duration (in hours and minutes).

If you specify a factory calendar, the system takes into account non-working days when calculating the start of the horizon. In this case, the planning horizon always begins on a working day.

You can simulate the relative planning horizon. You enter a reference date that the system uses to determine and display the start and end date and the start and end time.

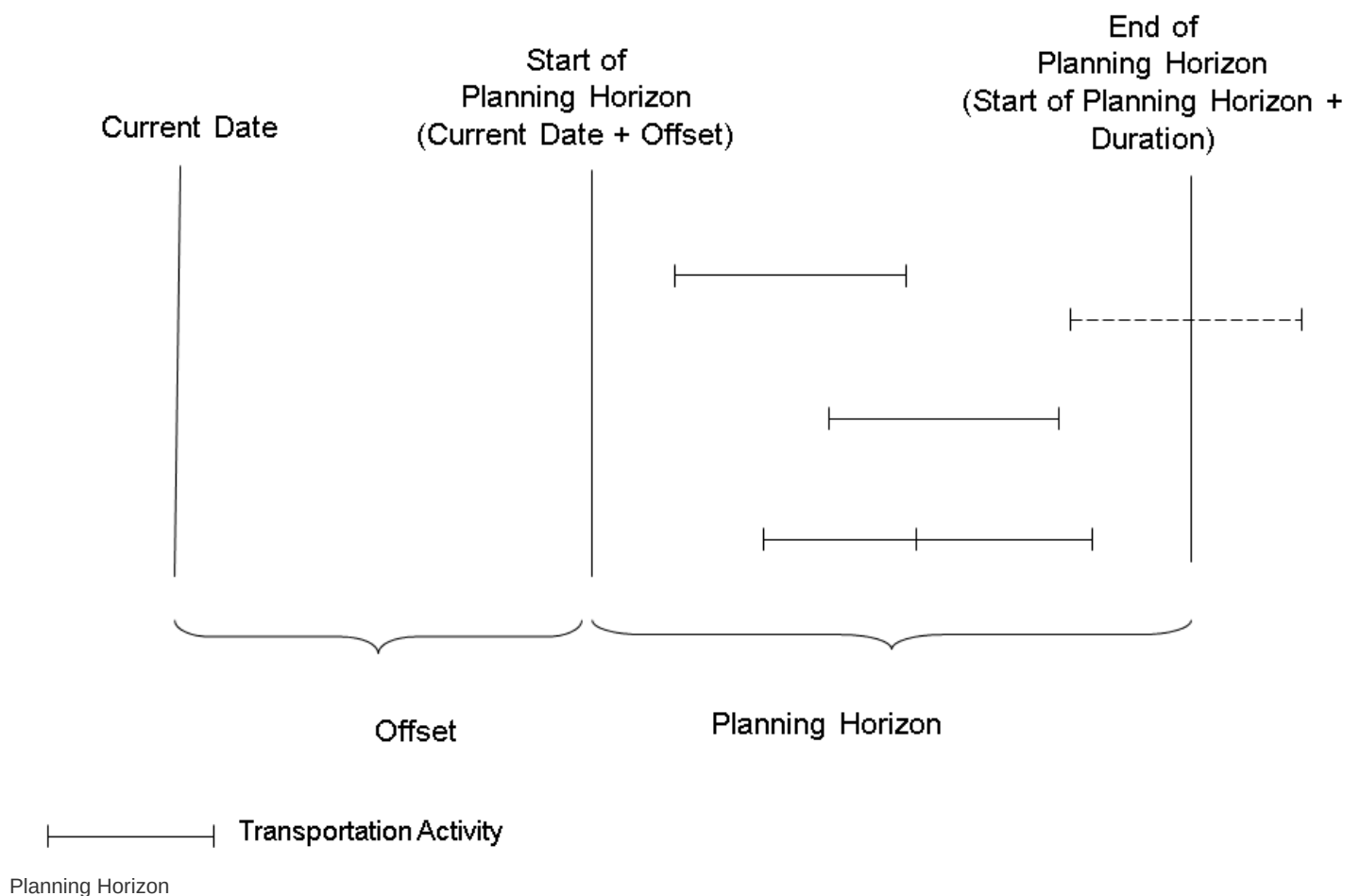
Relative Planning Horizon with a Fixed Start Time or a Fixed End Time

You define a relative planning horizon as described above. You also enter a fixed start time as of which you want the horizon to start or a fixed end time by which you want the horizon to end. You can also simulate this planning horizon.

→ Recommendation

You always define the planning horizon in the future. Otherwise, the system takes into account documents that lie in the past, which can lead to performance problems. You can define a planning horizon in the past to correct documents that lie in the past, for example. However, we recommend in this case that you define a separate planning profile with its own planning horizon that takes the past and the future into account.

The following figure shows how the planning horizon is defined:



Constraints and Costs Settings

Use

You can define freight-unit-dependent and equipment-dependent costs and constraints. These costs do not refer to real costs. They can be used to influence the optimization run so that certain solutions are prioritized. By defining higher lateness costs than earliness costs, for example, you can indicate that delivering earlier than requested is the preferred solution (see [VSR Optimization](#) and [Load Consolidation](#)).

Features

Requirement Document-Dependent Costs

These are non-delivery costs as well as earliness costs and lateness costs that the system calculates for each requirement document used in the VSR optimization run.

i Note

VSR optimization understands requirement documents to be freight units, container units, railcar units, and consignment orders.

You can enter non-delivery costs, earliness costs, and lateness costs directly or by specifying a condition (see [Condition](#)) with the condition type /SCMTMS/FU_PNLT_COST.

Equipment-Dependent Costs

Fixed Costs

These are costs that the system calculates once for each capacity (for example, vehicle type or schedule) used in the VSR optimization run.

i Note

For schedules, the fixed costs are incurred per departure.

→ Recommendation

- You can use **fixed costs for each capacity document** and optionally combine them with variable costs to estimate the effort required when using different equipment types.

Example: You want to transport one freight unit that fits into two vehicle types. If you define fixed costs for each capacity document, you can influence the optimization result in such a way that the smaller vehicle type is selected.

- If you want to influence the optimization result so that as few vehicle resources are used as possible, we recommend that you define **fixed costs for each vehicle resource**.

In VSR optimization, the costs are applied for each instance of a vehicle resource:

- The VSR optimizer adds the costs once for each single resource to which at least one freight order is assigned within the planning horizon. This means that these costs are not added again when another freight order is assigned.
- In the case of multiresources and equipment types, VSR optimization ignores these fixed costs because freight documents on multiresources and equipment types are independent of each other and the number of instances is usually unlimited.

You can find an example in the input help.

Dimension-Relevant Costs

These are variable transportation costs per unit for the following dimensions:

- Time (duration)
- Distance
- Transported quantity per distance
- Intermediate stop

These costs refer to the use of the dimensions of a capacity (including empty runs) during the VSR optimization run.

These costs and constraints are described in detail in the following sections:

- Costs per duration

The costs per duration define the time-dependent costs that are calculated per unit of measure for a specific means of transport during VSR optimization. You can also define a maximum duration of an order.

You can also define the maximum duration between the first and last pick-up stop and the maximum duration between the first and last delivery stop that the VSR optimizer must consider when creating a freight order or a trailer unit using this means of transport. In this way, you can control that all pick-up or delivery stops are served within a certain time period, such as on the same day.

- Costs per distance

The costs per distance define the distance-related costs that are calculated per unit of measure for a specific means of transport during VSR optimization. You can choose from various cost bases. You specify whether the system is to use the costs per distance from the planning costs in the planning profile (that is, from the [Distance Costs per Unit of Measure](#) field) or from the transportation lane. Alternatively, you can specify that the system takes into account the total of both values.

If the system is to take into account the distance costs from the planning profile or from the planning profile **and** the transportation lane, you can define a maximum value and unit. The unit refers to both the costs and the maximum value. If you do not specify a unit, the system uses the unit [Kilometer](#).

❖ Example

You want to model a case where means of transport MT1 is to be used everywhere and with constant costs per distance. Means of transport MT2 is to be used primarily in region A, and it is to be used in region B for bottleneck situations.

Therefore, choose **Yes** in the [Consider Distance Costs from Constraints and Costs Settings](#) field (calculation basis: planning costs) for MT1, and define distance-dependent costs per unit of measure. You choose **No** in the [Dist. Costs from Transportation Lane](#) field.

For MT2, you choose **No** in the [Consider Distance Costs from Constraints and Costs Settings](#) (calculation basis: planning costs), and **Yes** in the [Distance Costs from Transportation Lane](#) field.

In this case, the system adds the costs from the planning costs and the costs from the transportation lane.

If the system is to take into account the distance costs from the transportation lane only, you can choose between two types of distance costs, each of which applies a different geographical basis in its calculation logic:

- Route-based distance costs

Route-based distance costs are determined using a calculation logic that adds up the costs for the transportation of the goods from the source location to the destination location via a number of transshipment locations based on the costs for each stage.

- Destination-based distance costs

Destination-based distance costs are determined using a calculation logic that adds up the distances between the source and target locations, including all distances between transshipment locations, and multiplies the result by a cost factor. This cost factor is part of the direct transportation lane between the source location and target location and is based on the geographical location of the destination. The probability that the destination location of a transportation is the source location of the next transportation determines the value of the cost factor.

i Note

You must have defined a direct transportation lane between the source location and the target location before the system can take this calculation logic for distance costs into account in the VSR optimization run.

❖ Example

You want to transport goods from New York City to Minneapolis via Chicago. The distance between New York and Chicago is 800 and the distance between Chicago and Minneapolis is 400. The total distance that the system is to take into account in the VSR optimization run, for both the route-based distance costs and the destination-based distance costs, is therefore 1200 (800 + 400).

In the case of the route-based distance costs, the cost factor for the transportation of the goods from New York to Chicago is 3. The cost factor for the transportation of the goods from Chicago to Minneapolis is 4.

In the case of the destination-based distance costs, the cost factor for the transportation of the goods from New York to Minneapolis is 2.

If you select route-based distance costs for your VSR optimization run, the system calculates the transportation charges as follows: $800 \times 3 + 400 \times 4 = 4000$.

If you select destination-based distance costs for your VSR optimization run, the system calculates the transportation charges as follows: $(800 + 400) \times 2 = 2400$.

You can also define the maximum distance between the first and last pick-up stop and the maximum distance between the first and last delivery stop that the VSR optimizer must consider when creating a freight order or a trailer unit using this means of transport. In this way, you can control that all pick-up or delivery stops are within a specific radius. Here, too, the system uses the unit **Kilometer** if you do not specify a unit.

- Costs per quantity

You define the costs per quantity for each VSR optimization run. You must also define a unit. You cannot define a maximum value for the costs per quantity. However, you can specify the cost basis, in other words, whether the system is to use the costs per quantity from the transportation lane or from the schedule, or from the planning costs of the planning profile (that is, from the **Costs per Quantity** field).

In the case of costs per quantity from the transportation lane or the schedule, you can specify whether the system is to calculate the costs in a distance-independent way or whether it is to multiply them by the distance.

For nonscheduled means of transport, you take the costs per quantity from the transportation lane. For a means of transport of a schedule, you take the costs per quantity from the schedule.

In the case of costs per quantity from the planning costs, you can specify whether the system is to calculate the costs per freight document stage (in a distance-independent way) or whether it is to multiply them by the distance.

❖ Example

The following example shows how the system calculates the costs in the case of costs per quantity from the planning costs per freight document stage (in a distance-independent way):

You want to transport a freight unit (FU) with a weight of 10 kg from A to B. You have defined costs of 5 per KG. In this case, the system calculates costs per quantity to the amount of 50.

If you transport the FU from A via C to B, the system calculates costs per quantity to the amount of 100, in other words, 50 per stage.

- Costs per intermediate stop

You can define the following:

- Maximum number of intermediate stops

- Costs per intermediate stop
- Maximum number of pickup stops
- Maximum number of delivery stops

Utilization

You can enter in percent the minimum target utilization of a means of transport that is to be taken into account for the generation of road freight orders during the VSR optimization run. Moreover, you can specify whether this utilization should be a hard or soft constraint.

Cost Factors for Penalty Costs

You can use these cost factors to weight the earliness and lateness costs that you have defined for each freight unit on the basis of the means of transport. The VSR optimization multiplies these cost factors by the earliness costs or lateness costs of each freight unit. The following cost factors are available:

- Cost factor for premature pickup
- Cost factor for delayed pickup
- Cost factor for premature delivery
- Cost factor for delayed delivery

i Note

You define earliness costs and lateness costs by either entering them directly or by specifying a condition with the condition type /SCMTMS/FU_PNLT_COST. You specify this condition when you define time windows for the pickup and delivery. You must then assign these time windows to a planning profile.

For more information, see [Time Windows](#).

Costs for Greenhouse Gas Emissions

You can define costs for greenhouse gas emissions (measured in CO₂ equivalents) to set the priority of such emissions during VSR optimization.

For more information, see [Greenhouse Gas Emissions](#).

Capacity Adjustment

You can define adjustment factors for mass, volume, and normalized quantity. These factors determine to what extent the actual vehicle capacity available for planning can be increased or reduced. These factors are considered in VSR optimization and when generating transportation proposals. They are not considered for the following:

- Manual planning
- Schedules
- Load consolidation
- Capacity decrease
- Compartments
- Combinations of vehicle resources

For the adjustment factor for the normalized quantity, you have to define a unit of measure for the normalized quantity in the settings for the transportation cockpit.

Activities

To define constraints and costs settings, open the [Create Constraints and Costs Settings](#) app on the launchpad.

To assign your constraints and costs settings to a planning profile, open the [Edit Planning Profile](#) app on the launchpad, and then select the required planning profile.

Freight Order Building Rule

Use

The system takes into account freight order building rules when creating and changing more than one freight order on one resource during VSR optimization (VSR = Vehicle Scheduling and Routing).

You use freight order building rules to control how the system assigns activities on one resource to different freight orders.

Features

VSR optimization can build freight orders according to the following rules:

- [New Freight Order when Resource Is Empty](#)

VSR optimization creates a new freight order if the resource is empty, that is, no freight units and no trailers are assigned to the resource.

Note that if one or several outbound freight units are assigned to a freight order and there is one consecutive inbound freight unit that is delivered to the source location of the freight order, the freight order also contains the transportation stage of the resource to which the inbound freight unit is assigned.

- [New FO when Resource Is Empty and Depot Location Reached](#)

VSR optimization creates a new freight order when the following prerequisites are met:

- The resource is empty, that is, no freight units and no trailers are assigned to the resource.
- The resource has reached the depot location.

i Note

- An exception to these rules are schedules. The system creates a freight order for every departure of a schedule. Empty runs within the route of the schedule are part of the freight order as well.
- If you have selected [Empty Stage Should Be Last Stage of Capacity Document](#) in the [Preferred Position of Empty Stage](#) field in your planning profile, the empty stage is added to a preceding freight order even if this means that the new freight order doesn't start directly after the resource is empty but after the empty stage.

Activities

You can define freight order building rules in the [Optimizer Settings](#) screen area of a planning profile (see [Planning Profile](#)).

Example

For examples of how the system builds freight orders depending on the freight order building rule, see [Examples of Freight Order Building Rules in Use](#).

More Information

[VSR Optimization](#)

Examples of Freight Order Building Rules in Use

The following examples demonstrate how the system builds freight orders for a given set of freight units based on the selected freight order building rule.

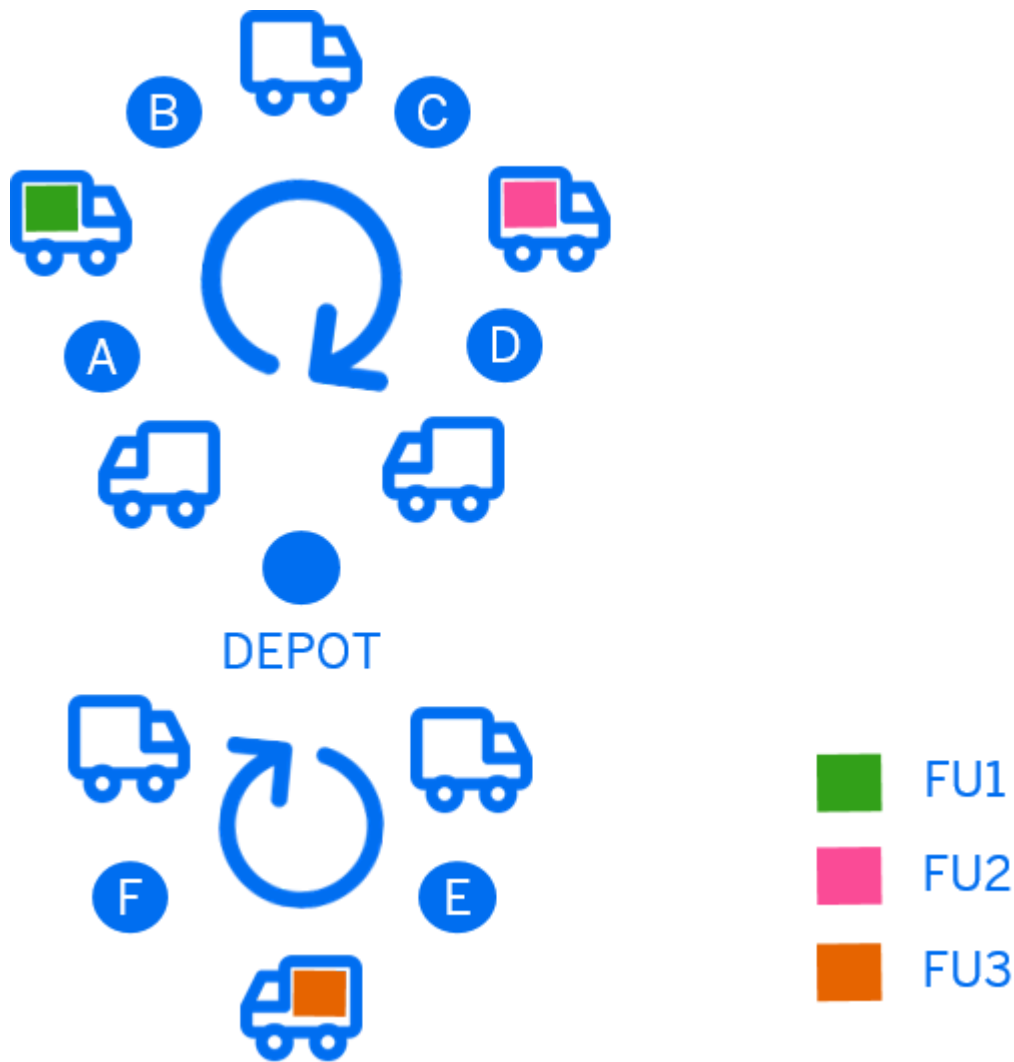
Example 1

You have the following freight units:

Freight Unit	From	To
1	A	B
2	C	D
3	E	F

You have a resource with enough capacity and a depot location.

The figure below illustrates the transportation plan that a truck uses to transport the freight units, as well as the empty runs, and is followed by an explanation:



The truck uses the following transportation plan to transport the freight units:

It starts at the depot location and drives to location A without any load (empty run). At location A, freight unit 1 is loaded onto the truck. The truck then drives to location B, where freight unit 1 is unloaded. The truck continues to location C without any load (empty run). At location C, freight unit 2 is loaded onto the truck. The truck then drives to location D, where freight unit 2 is unloaded. After that, the truck returns to the depot location without any load (empty run).

The truck then drives to location E without any load (empty run). At location E, freight unit 3 is loaded onto the truck. The truck then drives to location F, where freight unit 3 is unloaded. Finally, the truck returns to the depot location without any load (empty run).

The system builds the following freight orders depending on which freight order building rule you have selected:

- **New Freight Order when Resource Is Empty**

The system builds the following freight orders:

- Freight order 1 that contains freight unit 1.
- Freight order 2 that contains freight unit 2.
- Freight order 3 that contains freight unit 3.

- **New FO when Resource Is Empty and Depot Location Reached**

The system builds the following freight orders:

- Freight order 1 that contains freight unit 1 and freight unit 2.

- o Freight order 2 that contains freight unit 3.

Example 2

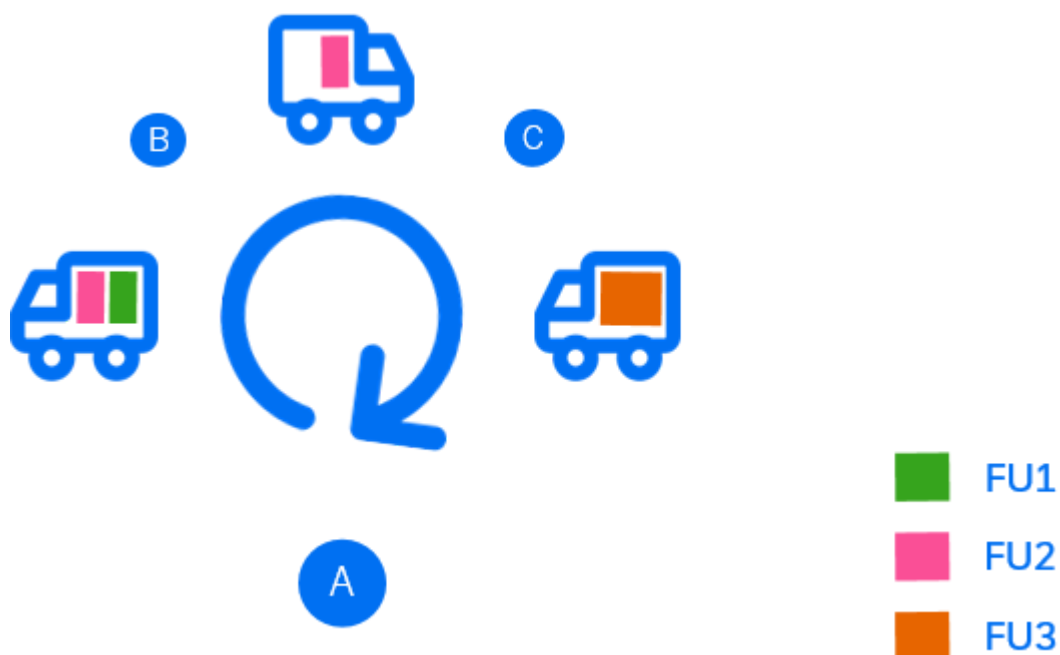
You have the following freight units:

Freight Unit	From	To	Comment
1	A	B	This freight unit is an outbound freight unit.
2	A	C	This freight unit is an outbound freight unit.
3	C	A	This freight unit is an inbound freight unit. It contains an empty crate that needs to be returned to location A.

You have a resource with enough capacity for transporting the freight units. Further, you have selected the following freight order building rule:

- **New Freight Order when Resource Is Empty**

The figure below illustrates the transportation plan that a truck uses to transport the freight units and is followed by an explanation:



The truck uses the following transportation plan to transport the freight units:

At location A, freight unit 1 and 2 are loaded onto the truck. The truck drives to location B, where freight unit 1 is unloaded. The truck then drives to location C where freight unit 2 is unloaded and freight unit 3 is loaded onto the truck. The truck then returns to location A, where freight unit 3 is unloaded.

The system builds one freight order for the roundtrip that contains freight unit 1, 2, and 3.

Capacity Decrease

Use

When you transport goods for multiple customers on a vehicle resource (including passive vehicle resources), in other words, the vehicle resource stops at multiple intermediate stops on the way from the source location to the destination location, you can use partitions to separate the goods. However, this results in a reduction of the total capacity of the vehicle resource since each partition consumes capacity itself. You can use this function to define how the capacity is to decrease.

i Note

- This function refers only to the capacity of the vehicle resource. It is independent of compartments (see [Compartment](#)) that you have defined for the vehicle resource.
- You define decreasing capacities once for your system. You do not assign them to a profile.
- You cannot define decreasing capacities for schedule vehicles.

Features

Fixed Value

You can define a fixed value or a percentage by which the capacity is to decrease for each loading-relevant intermediate stop where loading, unloading, or transshipment activities take place. The system continues to reduce the capacity for the VSR optimization (see [VSR Optimization](#)) until one of the following conditions is met:

- The total capacity is consumed, in other words, it is zero.
- The maximum number of intermediate stops defined for the means of transport has been reached (user interface for the definition of planning costs).
- The maximum number of intermediate stops defined for the capacity decrease (end of intermediate stop area) has been reached.

Single Values

You can define a value by which the capacity is to decrease for a specific number of loading-relevant intermediate stops. If you do not define a value for a certain number of loading-relevant intermediate stops, the capacity remains the same.

i Note

Note that in both cases the capacity decrease refers to the capacity that the vehicle resource has **before** the transportation start, in other words, the capacity with which you can load the vehicle resource at the source location.

Activities

To define capacity decreases, open the [Create Decreasing Capacity Settings](#) app on the launchpad.

More Information

[Definition of Decreasing Capacities](#)

Definition of Decreasing Capacities

You have defined the following data:

- Means of transport 0001
- Load unit KG
- Vehicle resource RES1 with means of transport 0001 and a capacity of 10,000 kg

Fixed Value or Fixed Percentage

If you want to use a fixed percentage, specify the following values for the capacity decrease:

TM	Load Unit	Stop Range: Start	Stop Range: End	Value Type	Capacity Decrease	Percentage
0001	KG	2	11	Fixed Value	10	Capacity Decrease Is a Percentage

If you want to use a fixed value, specify the following values for the capacity decrease:

TM	Load Unit	Stop Range: Start	Stop Range: End	Value Type	Capacity Decrease	Percentage
0001	KG	2	11	Fixed Value	1000	Capacity Decrease Refers to Load Unit

i Note

- In the case of the value type **Fixed Value**, only one entry (one line) is possible for each means of transport.
- Only when using the value type **Fixed Value** does it make sense to enter a value in the **Stop Range: End** field.

In both cases, the reduced capacity is as follows:

Number of Loading-Relevant Intermediate Stops	Reduced Capacity
1	10.000
2	9.000
3	8.000
4	7.000
5	6.000
6	5.000
7	4.000
8	3.000

Number of Loading-Relevant Intermediate Stops	Reduced Capacity
9	2.000
10	1.000
11	0

If, on the way from the source location to the destination location, the means of transport makes two intermediate stops where loading, unloading, or transshipment activities take place, you can load it with a capacity of 9,000 kg at the destination location. However, if the means of transport makes 10 intermediate stops where loading, unloading, or transshipment activities take place, the available capacity is only 1,000 kg. If the number of loading-relevant intermediate stops is greater than 11, the capacity remains at zero.

Single Values

If you want to define single values, specify the following values for the capacity decrease:

Stop Range: Start	Value Type	Capacity Decrease	Percentage
2	Single Value	1000	Capacity Decrease Refers to Load Unit
4	Single Value	500	Capacity Decrease Refers to Load Unit
5	Single Value	500	Capacity Decrease Refers to Load Unit
6	Single Value	700	Capacity Decrease Refers to Load Unit
8	Single Value	2000	Capacity Decrease Refers to Load Unit
9	Single Value	2300	Capacity Decrease Refers to Load Unit
10	Single Value	1000	Capacity Decrease Refers to Load Unit

i Note

In the case of single values, it does not make sense to use the option **Capacity Decrease Is a Percentage** in the **Percentage** field.

In this case, the reduced capacity is as follows:

Number of Loading-Relevant Intermediate Stops	Reduced Capacity
1	10.000
2	9.000
3	9.000
4	8.500

Number of Loading-Relevant Intermediate Stops	Reduced Capacity
5	8.000
6	7.300
7	7.300
8	5.300
9	3.000
10	2.000
11	2.000

If, on the way from the source location to the destination location, the means of transport makes two intermediate stops where loading, unloading, or transshipment activities take place, you can load it with a capacity of 9,000 kg at the destination location. However, if the means of transport makes 10 intermediate stops where loading, unloading, or transshipment activities take place, the available capacity is only 2,000 kg. Note that the value by which the capacity is reduced each time always refers to the value in the previous line and not to the original capacity. Only the first entry refers to the original capacity.

If the number of loading-relevant intermediate stops is greater than 11, the capacity remains at 2,000 kg.

Freight Unit Building Rule

Use

The system takes into account freight unit building rules (FUB rules) when creating and editing freight units (see [Creation and Editing of Freight Units](#)).

Features

Freight Unit Building Strategy

In the FUB rule, you define how the system is to consolidate freight units. To do this, you can use various freight unit building strategies (FUB strategies):

- Consolidate as much as possible

In this case, the items come from different business documents, for example, forwarding orders. The system groups the items in freight units.

i Note

In addition to forwarding orders, other possible business documents are forwarding quotations, order-based transportation requirements, and delivery-based transportation requirements.

- Consolidate per item

In this case, the system generates one freight unit per business document item.

- Consolidate per request (compatible parts)

In this case, the system groups all items of a business document to form a freight unit.

For all three FUB strategies, the system takes into account hard constraints and incompatibilities (see [Incompatibilities](#)).

i Note

The FUB strategy also applies to the creation of transportation units and freight orders (see below under [Document Type](#) and [Automatic Creation of Freight Orders](#)).

Planning and Split Quantities

In the FUB rule, you enter planning quantities (for example, gross weight and gross volume) with the corresponding units of measure. The system copies the units of measure to the freight units that it creates. The units of measure form the basis for checking the capacity of resources. You define the maximum capacity of a freight unit using the split quantity.

i Note

- If you create equipment-based freight units or transportation units, the system automatically fills the planning and split quantities from Customizing for the equipment. For more information, see below under [Equipment-Based Freight Units and Transportation Units](#).
- You can also create freight unit building rules without indicating a unit of measure. For more information, see [UoM Setting for Quantity and Alternative Quantity](#) below.

❖ Example

You want to transport 5 tons of cement, but you can only transport it in batches of 500 kg. In this case, you define a split quantity of 500 kg. The system generates 10 freight units.

If the system cannot find the planning unit in the business document, it converts the unit using the values that you have defined in the product master.

❖ Example

You want to transport 5 tons of cement, but you can only transport it in batches of 0.5 m³. In the product master, you have defined that 0.8 m³ of cement is equivalent to 1 ton of cement. The system then converts 5 tons of cement into 4 m³ and generates 8 freight units.

In addition to the split quantity, you can also specify a rounding value. In this case, the system calculates the freight unit quantity as a multiple of the rounding value.

❖ Example

You define the following values:

Product	1 computer = 1 piece = 0.06 m ³
Split quantity	1 m ³
Rounding value	1 piece

Since 1 m³ divided by 0.06 m³ makes 16.67 m³, with this split quantity, the freight unit can contain a maximum of 16 computers.

If you enter 3 pieces as the rounding value, for example, because you can only pack three computers in a box, the freight unit can contain a maximum of 15 computers, since 15 is a multiple of 3 and is less than 16.67.

In the FUB rule, you can use the item split checkbox to control whether items are allowed to be split if the split quantity is exceeded.

❖ Example

You have defined the following values:

Freight unit building strategy	Consolidate as much as possible
Critical quantity	Gross weight
Item split allowed	No
Split quantity	20 tons

You want to schedule four items with a total weight of 21 tons. Items 1 to 3 each contain 5 tons and item 4 contains 6 tons. Taking into account your settings, the system creates two freight units:

- Freight unit 1 with items 1 to 3
- Freight unit 2 with item 4

Freight Unit Update Strategy

You can specify whether it is possible to move items from one freight unit to another when freight units are updated. You do so by defining the freight unit update strategy in the freight unit building rule.

❖ Example

There are two cargo items in a forwarding order. The system creates one freight unit for each item. One freight unit is planned (FU1) and the other freight unit is not planned (FU2). Item 1 belongs to FU1 and item 2 belongs to FU2.

If item 1 is deleted from the forwarding order, the following occurs:

- In the standard system (that is, **Item Swap Allowed (Standard Behavior)** is selected in the freight unit building rule), item 2 is moved to FU1 and FU2 is canceled.
- If you have selected **Item Swap Not Allowed** in the freight unit building rule, item 2 stays in FU2 and FU1 is canceled.

UoM Setting for Quantity and Alternative Quantity

You can indicate which units of measure are allowed for the quantity and alternative quantity that are to be used as planning quantity. You can either allow dimensionless units of measure only, such as EACH, Pieces, and PAL, or units of measure of all dimensions. You can also indicate that no unit of measure is required. The system then uses the units of measure defined in the predecessor document items. In this case, splitting of freight units is not supported.

Other planning quantities are not affected by this setting.

i Note

Quantities or alternative quantities that have a unit of measure with a dimension are ignored by vehicle scheduling and routing (VSR) optimization, transportation proposal, and load consolidation in the respective requirement documents, such as freight units.

Document Type

You can control which document type is to be created by the freight unit builder, that is, freight unit type, transportation unit type, or freight order type. You can either enter a document type directly or enter a condition for determining the document type in the advanced settings. You define freight unit types, transportation unit types, and freight order types in Customizing. For more information, see Customizing for [Transportation Management](#) under the following:

- [►Planning►Freight Unit►Define Freight Unit Types►](#)
- [►Planning►Transportation Unit►Define Transportation Unit Types►](#)
- [►Freight Order Management►Freight Order►Define Freight Order Types►](#)

If the system determines a freight order type as the document type, it automatically generates freight orders without generating freight units or transportation units (see below under [Automatic Creation of Freight Orders](#)).

Default Business Document Categories

During freight unit building, the system can create different categories of transportation units. If you do not specify a freight document type in the freight unit building rule (either directly or using a condition), the system creates the following business document categories:

Item Category	Business Document Category
Product / package	Freight unit
Package	Package unit as an instance of the transportation unit with TU category "Package"
Container	Container unit as an instance of the transportation unit with TU category "Container"
Railcar	Railcar unit as an instance of the transportation unit with TU category "Railcar"
Trailer	Trailer unit as an instance of the transportation unit with TU category "Trailer"

In this case, the system reads the default document type from Customizing. You must therefore ensure that a default type has been defined in the Customizing activities [Define Freight Unit Types](#) and [Define Transportation Unit Types](#).

If you want the system to create "Freight unit" instead of "Container unit" as the business document category for container items, you must specify the document type of the freight unit in the freight unit building rule (either directly or via a condition).

Equipment-Based Freight Units and Transportation Units

If you enter an equipment type and group in the freight unit building rule, the freight unit builder creates equipment-based freight units or transportation units. Depending on the equipment type, the freight unit builder creates transportation units of the category container, railcar, or trailer.

The system automatically derives the planning quantities, units of measure, and split quantities from Customizing for the equipment (Customizing activity [Define Equipment Groups and Equipment Types](#)). You can delete or change these quantities in the freight unit building rule. This enables you to create transportation units or freight orders (shortcut scenario) with multiple items (see [Example: Freight Unit Building for Railcar Units](#)).

If you want the system to create "Freight unit" instead of "Container unit" as the business document category for container items, you must specify the document type of the freight unit in the freight unit building rule (either directly or via a condition). We recommend that you enter it directly, rather than using a condition.

For more information, see [Consideration of Equipment Data During FU or TU Creation](#).

Automatic Creation of Freight Orders

If you want a freight order to be created automatically instead of a freight unit (shortcut scenario), enter a freight order type as the document type or a condition for determining this freight order type. This function is also supported for transportation units (see [Transportation Unit](#)).

❖ Example

You specify in a condition that if the quantity exceeds 20 tons, freight order type 1 is to be used; otherwise freight unit type 1 is to be used.

This function is suitable, for example, for the following cases:

- In general you want to send small orders that are below one of the upper limits you have defined with an express delivery company.
- You always have to transport orders as a full load due to their size.

Process Controller Strategies

You can use strategies to define the way in which freight units are created in accordance with your own requirements. SAP delivers the standard strategy FUB_AUTO for freight unit building. For more information, see [Process Controller Configuration](#).

Unified Package Building

On the tab page **Advanced Settings** you can specify if during freight unit building the system creates freight units, freight orders, or package units with a complete hierarchy of package and product items based on the information from the order documents. Moreover, the system determines the exact position of each product in the package and displays these positions in the load plan.

Alternatively, you can define that the system calculates only a rough estimation of the number of packages that can be built. This estimation is displayed in the header data of a freight unit on the **General Data** tab page. It can also be used by the VSR optimizer during planning.

As a prerequisite, you must have entered a unified package building profile. For the determination and display of the exact product positions, you must have selected **Optimizer-Based Mixed Package Building** as the mixed package building mode in your unified package building profile.

Whether the system creates a freight unit, freight order, or package unit during freight unit building depends on which document type you enter on the **General Data** tab page of the freight unit building rule. If you want the system to create a complete package hierarchy, you can choose a document type of the categories **FU**, **FO** (Freight Order), or **TU** (Transportation Unit). If you use a TU type, the category must be **Package** (that means package unit). We recommend to use a TU type with category **Package** instead of an FU type.

You can also specify the maximum number of top-level packages that a freight unit, freight order, or package unit may contain when the system creates them during freight unit building. The system considers this number when it creates a complete package hierarchy.

Further, you can specify that a separate freight unit, freight order, or package unit is to be created for unpackaged items, that means, one common business document is created for all unpackaged items that contains no other items. The system considers this checkbox when it creates a complete package hierarchy. If you use the package builder as your unified package building engine, you must select the **Only Full Packages in PB Result** checkbox in your unified package building profile as a prerequisite. In this case, the system builds full packages for a product and the remaining quantities of this product stay unpackaged.

For more information, see [Unified Package Building in TM](#).

Activities

You can define FUB rules by opening the app **Create Freight Unit Building Rule** from the launchpad.

More Information

[Determination of Freight Unit Building Rules](#)

[Definition of Dimensions and Units of Measurement](#)

Determination of Freight Unit Building Rules

Use

When creating freight units, the system determines which freight unit building rule (FUB rule) it is to take into account for a business document item. To find out which business documents are supported, see [Creation and Editing of Freight Units](#).

Prerequisites

- Optionally, you have defined FUB rules. For more information, see [Freight Unit Building Rule](#).
- You have defined conditions with the condition type /SCMTMS/FUBR. For more information, see [Condition](#).

Features

To determine FUB rules, the system does the following:

1. The system checks whether you have specified a condition in the business document type, for example, forwarding order type. There can be only one condition per business document type.
2. If the system does not find a condition or if this condition does not return a result, the system checks whether you have specified a standard FUB rule in the business document type.
3. If you have not specified one, the system uses default settings.

i Note

You can see which FUB rule the system has used in the freight unit.

Definition of Dimensions and Units of Measurement

Use

You use dimensions to group units of measurement. You need units of measurement to display product quantities, for example, for freight unit building rule (FUB rules), and to describe the capacity of vehicle resources (see [Freight Unit Building Rule](#) and [Vehicle Resource](#)).

i Note

The system might display quantities with 14 decimal digits. If you want the system to round quantities to a specific number of decimal digits, follow the instructions described in SAP Note [1132219](#) .

Prerequisites

- You have checked in Customizing under **General Settings** > **Check Units of Measurement** that the dimensions delivered as a standard and the associated units of measurement are complete.
- When using FUB rules, the system checks if the forwarding order contains the relevant quantities based on the FUB rules. If the forwarding order does not contain the relevant quantities, the system uses the product data. For this purpose, you have defined the following in the product master data:

- Base unit of measure (BUoM)
- Conversion factors

You need conversion factors to convert the base units of measure into the required units of measurement. You define these in the product master on the **Units of Meas.** tab page. For more information, see [Definition of Products](#).

If you have not defined a conversion factor, the system cannot take split quantities or rounding values into account when building freight units from business documents (forwarding orders, for example). In this case, it sets the quantities to zero.

Example

Examples for Dimensions and Units of Measurement

The following table provides you with examples of dimensions and of some units of measurement you can assign to them.

Dimension	Units of Measurement
Mass	Kilogram; milligram
Volume	Cubic centimeter; cubic decimeter

Example for the Conversion of Units of Measurement

The following units of measurement are predefined as a default:

- **Kilogram** (dimension **mass**)
- **Cubic decimeter** (dimension **volume**)

You also define the following unit of measurement:

- **Pallet** (no dimension)

Your forwarding order contains product A and product B each of which have the quantity "1 piece". You have defined **piece** as BUoM for both products.

You have defined the following conversion factors for product A:

- 1 piece = 2 kg

- 1 piece = 3 dm³

You have defined the following conversion factor for product B:

- 1 piece = 0.5 pal

In this case, product A is displayed with quantity 2 kg or 3 dm³. In the unit of measurement **pallet** (pal), the quantity 0 is displayed because you have not defined a conversion factor.

Product B is displayed with the quantity 0.5 pal. Since you have not defined conversion factors for **kilogram** or **cubic decimeter**, no conversion occurs. The system displays the value 0.

Incompatibilities

Use

You use incompatibilities to define the compatibility of planning data that the system is to take into account during transportation planning, for example:

- Freight units with different Incoterms must not be transported together.
- A container of a certain equipment type cannot be transported on a particular resource because it is too small.
- Refrigerated goods must only be transported in appropriate means of transport (refrigerated trucks).
- A certain means of transport cannot be unloaded at a location since the location does not have a suitable loading ramp.
- A truck that needs to be unloaded from the side cannot be unloaded at a location since the location does not offer the prerequisites for unloading from the side.
- A driver must not transport dangerous goods or must not drive a truck because he does not have the corresponding license.

In this way, you can control how freight units or container units are assigned to vehicle resources or vehicle types in manual planning, for example.

i Note

For consistency reasons, you can define incompatibilities only. To define a compatibility, you have to define an incompatibility with all other attribute values.

Features

You can define incompatibilities in different ways.

On the one hand, you can define incompatibilities between any two attributes of two business object nodes. To do so, you define two conditions and specify the relevant results. Two business object instances are then incompatible if the result of the condition matches these relevant results.

On the other hand, you can enter only one condition and select the **Ident. Values Only** (Identical Values Only) checkbox. In this way, you can define an incompatibility between two instances of the same business object (BO), for example, two freight units. The two BO instances are then only incompatible if their values differ. If a planning run comprises many business objects and the attribute can have many values, performance might be reduced. In this case, we do not recommend that you use this option.

Alternatively, you can use customer-specific logic to determine the pairs of incompatible objects. To do so, choose the option **External Determination Strategy** in the **Determination Method** field and specify the required strategy in the **External Determination Strategy** field. For more information, see [External Strategies](#). If you define incompatibilities with conditions and a freight unit has several products, the system only determines the first product, and it does this randomly. You can avoid this restriction by using example strategy **TM_INCOMP**. In this case, you specify two products that are incompatible in your incompatibility. The example strategy then checks whether the freight units contain these products. **TM_INCOMP** serves as an example strategy for incompatibilities, the class assigned describes the interface and serves as reference implementation for customer-specific classes. The unit tests serve documentation purposes.

The example strategy shows that condition-based incompatibilities are not always suitable. However, if a freight unit has only a single product item (package items or container items can, however, also exist), a filter on the item type **Product** in the condition is sufficient.

You use the **incompatibility area** to define the area in which the system is to take into account an incompatibility:

- Automatic planning (that is, VSR optimization, load planning and load consolidation, and generating transportation proposals) and manual planning
- Creating freight units
- Carrier selection (including continuous move)
- Creating delivery proposals

VSR optimization considers incompatibilities as hard constraints.

i Note

VSR optimization does not support planning with drivers. Therefore, VSR optimization supports driver incompatibilities with default drivers only.

You use the **incompatibility type** to define the BOs between which an incompatibility is to be valid. For example, you can use the incompatibility type **FU – Compartment** to ensure that a freight unit is not transported in a certain compartment. You can also use the incompatibility type to define the incompatibility level.

i Note

- You can define incompatibilities between both freight units and container units on the following incompatibility levels:
 - Vehicle level (this means that the two freight units must not be transported in the same vehicle or in the same compartment)
 - Compartment level (this means that the two freight units must not be transported in the same compartment)
 - Means-of-transport combination level (this means that the two freight units must not be transported in the same vehicle, in the same compartment, or in the same means-of-transport combination)
 - Consignment order level (this means that the two freight units must not be transported with the same consignment order)
- Conditions for the compartment level must use data access definitions with the following data access classes:
 - **/SCMTMS/CL_COND_DAD_COMP_DATA**: Data access class for compartment data, for example, compartment type, sequence
 - **/SCMTMS/CL_COND_DAD_COMP_ATTR**: Data access class for compartment attributes to extract the attribute values as defined for the compartment type

- Conditions for the means-of-transport combination level must use data access definitions with the following classes:
 - /SCMTMS/CL_COND_DAD_MTR_COMB: Data access class for means-of-transport combination data
 - /SCMTMS/CL_COND_DAD_MTR_CO_ATT: Data access class for means-of-transport combination attributes
- A condition type exists for each BO node that the system uses in an incompatibility type.

Furthermore, when defining your incompatibility, you can specify whether the system is allowed to violate incompatibilities during manual planning, VSR optimization, or background processing. You can override this setting in the incompatibility settings. You can also specify if the incompatibility applies to vehicle resources or vehicle types or both. Note however that this is only possible for incompatibility types that involve vehicle resources or vehicle types.

In the incompatibility settings, you specify which incompatibilities the system is to consider in a certain incompatibility area. You can assign them to your planning profile, your carrier selection settings, and, for freight units, your freight unit building rule. Moreover, you can define default incompatibility settings, for example, for a certain incompatibility area. This system considers these if you have not assigned any incompatibility settings to the specified profiles.

i Note

When you access the transportation cockpit, the system checks for incompatibility settings in the following order: in the profile and layout sets, in the planning profile, and in the default incompatibility settings.

If you then create a freight order or a freight booking directly in the transportation cockpit, the incompatibility settings that were identified when you accessed the cockpit are assigned to the business document. When you open the freight order or freight booking document directly, these settings are then used for the incompatibility check.

After you have defined an incompatibility, you can perform a simulation.

If you want to simulate an incompatibility with the incompatibility type **Freight Unit – Freight Unit (Vehicle Level)**, for example, you specify two freight units. In this case, the system checks whether an incompatibility exists between the two freight units.

Activities

You can define incompatibilities by opening the app **Create Incompatibility** from the Launchpad.

You can define incompatibility settings by opening the app **Create Incompatibility Settings** from the Launchpad.

Related Information

[Definition of Incompatibilities](#)

[Planning Profile](#)

[Condition](#)

Definition of Incompatibilities

Incompatibility Between Two Freight Units with Different Transportation Groups

You want to prevent freight units with the transportation group **Liquid** and **Solid** from being delivered together. In this case, you define a condition (see [Condition](#)) with the condition type for freight unit incompatibilities. Here the output value is the transportation group for the item.

You then define an incompatibility with the incompatibility type **FU-FU (Vehicle Level)**, which specifies that freight units with these two transportation groups must not be delivered at the same time on one truck. You choose the incompatibility area for manual planning, VSR optimization, and for generating transportation proposals.

You then specify the condition that you defined before twice. You specify **Liquid** as the relevant result of the first condition. You specify **Solid** as the relevant result of the second condition.

Incompatibility Between Certain Freight Units and a Certain Means of Transport

All freight units with the transportation group **Non-Liquid** are to be incompatible with the means of transport **Tanker**. In this case, you define two conditions: one for the freight unit (with the condition type for freight unit incompatibilities) and one for the capacity, in this case, the vehicle resource, for example (with the condition type for vehicle resource incompatibilities).

You then define an incompatibility with the incompatibility type **FU-Vehicle Resource**. You specify **Non-Liquid** as the relevant result for the condition for the freight unit. You specify **Tanker** as the relevant result for the condition for the vehicle resource.

i Note

Alternatively, you can specify **Liquid** as the relevant result for the condition for the freight unit and **Not a Tanker** as the relevant result for the condition for the vehicle resource.

Incompatibility Between Freight Units with Different Transportation Groups

You want to prevent freight units with different transportation groups from being delivered together.

In this case, you define an incompatibility and specify only one condition in which the system determines the transportation group for the freight unit. You also select the **Ident. Values Only** (Identical Values Only) checkbox. Freight units with different transportation groups are then incompatible.

i Note

You can define the incompatibilities listed here in the same way for container units.

Context Determination

Use

This function allows you to expand your selection of resources, drivers, and documents to be displayed in the transportation cockpit. The expanded selection includes resources, drivers, and documents that, within the planning horizon, are assigned to the resources, drivers, and documents you selected on the selection screen of the transportation cockpit.

You can use context determination to expand your selection to all documents assigned to the following resources:

- Truck
- Trailer
- Driver
- Container
- Locomotives

- Railcars

You can use context determination for the following documents:

- Road freight orders

Your selection includes all trucks and drivers that transport the selected road freight orders as well as the container and trailer units that are assigned to the selected road freight orders.

- Trailer units

Your selection includes all trailers that transport the selected trailer units as well as the road freight orders and container units that are assigned to the selected trailer units.

- Container units

Your selection includes all containers that are assigned to the selected container units as well as the freight orders, trailer units, and railcar units to which the selected container units are assigned.

- Rail freight orders

Your selection includes all locomotives that transport the selected rail freight orders as well as the container and railcar units that are assigned to the selected rail freight orders.

- Railcar units

Your selection includes all railcars as well as the rail freight orders and container units that are assigned to the selected railcar units.

- Package units

Your selection includes all trucks, trailers, and containers as well as the freight orders, trailer units, and container units that are assigned to the selected package units.

Activities

In your planning profile you set the **Context Determination** checkbox in order to activate context determination. To edit your planning profile, open the **Edit Planning Profile** app on the launchpad. For more information, see [Planning Profile](#).

Example

You have three road freight orders in your planning horizon that you want to transport with the same truck. You have activated context determination in your planning profile.

You select the truck and enter the Transportation Cockpit. All three road freight orders are added to the selection. Alternatively, you can select one of the road freight orders and enter the Transportation Cockpit. The truck and the other two road freight orders are added to the selection.

Context Determination in VSR Optimization

Use

This function ensures that the system processes all dependent objects, in other words, freight units and road freight orders, together during VSR optimization. It cannot be switched off.

By taking the context into account, the system creates a transportation plan that is consistent with regard to all possible dependencies. Possible dependencies include:

- Incompatibilities between freight units
- Windows of the freight units
- Capacity decrease

Features

The system performs context determination automatically when you start VSR optimization in the transportation cockpit. In addition to the objects that you have selected for VSR optimization in the transportation cockpit (that is, either individual objects or all objects), it also considers indirectly linked objects. The system determines the following objects as context and hands them over to VSR optimization:

- The system determines indirectly linked freight orders from freight units.

❖ Example

A freight unit is to be delivered from A via B and C to D. You have selected the freight unit for the stage from B to C in the transportation cockpit. A freight order exists for each of the stages A to B and C to D. The system determines these two freight orders as context.

- The system determines indirectly linked freight units and transportation units from freight units. Here, the system transfers the entire transportation chain to VSR optimization, even if you have selected only one freight unit for a stage.

❖ Example

A freight unit is to be delivered from A via B and C to D. You have selected the freight unit for the stage from B to C in the transportation cockpit. Additional freight units exist for the stages A to B and C to D. The system determines these freight units as context.

- The system determines indirectly linked freight orders from resources. However, this applies to finite resources only (for multiresources, you must have specified the number of individual resources).

❖ Example

A vehicle resource is to travel from A via B and C to D. A freight order is planned for each of the stages A to B and C to D. The system determines these two freight orders as context.

i Note

- The system determines only those objects as context that you have not explicitly selected.
- VSR optimization uses the objects that the system has determined as context for information purposes only. VSR optimization is not permitted to change these objects.
- The system determines the context for multiresources if, in Customizing for the related means of transport, you have selected the checkbox for blocking multiresources and have entered a number of single resources that is greater than zero.

The system determines the following objects as additional changeable input and hands them over to VSR optimization:

- VSR optimization determines indirectly linked freight units from road freight orders that have already been planned. That means it adds freight units that are already planned on a selected road freight order to the selection.

- **Example**

You have selected a road freight order. Two freight units are already planned on this road freight order. The system adds these two freight units to the selection.

Related Information

[Context Determination](#)

Manual Planning

Use

Manual planning allows you to manually create or change a transportation plan. To create a transportation plan, you assign requirement documents, for example freight units, to capacities, for example vehicle resources. The system creates capacity documents, for example freight orders, which you can process further. For example, you can assign drivers to your freight orders or perform load planning. If you assign freight units to a container, for example, a container unit is created that you can directly assign to a freight booking.

Planning strategies determine the steps that the system is to carry out during the planning process and the order in which it does so. In the planning profile, you can specify which planning strategy you want to use. We provide the following standard planning strategies, for example:

- VSRI_DEF for manual planning
- VSRI_SCH for manual planning with subsequent scheduling

Manual planning is done in the transportation cockpit. Profile and layout sets as well as page layouts let you configure the data used for planning and the layout of the transportation cockpit (see [Starting the Transportation Cockpit](#)). You can display areas of the transportation cockpit in more than one window (see [Planning with Multiple Windows](#)).

i Note

Note that the tools and checks provided for manual planning are intended to provide support only. You, as the planner, remain responsible for ensuring that the planning results are consistent.

Prerequisites

- You have specified a planning strategy in the planning profile. For more information, see [Planning Profile](#).
- You have configured the settings for freight orders in the Customizing activity [Define Freight Order Types](#) under [Transportation Management > Freight Order Management > Freight Order](#).
- You have configured the settings for freight bookings in the Customizing activity [Define Freight Booking Types](#) under [Transportation Management > Freight Order Management > Freight Booking](#).
- You have configured the settings for transportation units in the Customizing activity [Define Transportation Unit Types](#) under [Transportation Management > Planning > Transportation Unit](#).
- If you work with schedules, you have configured the settings in the Customizing activity [Define Schedule Types](#) under [Transportation Management > Master Data > Transportation Network > Schedule](#).
- If you work with vehicle resources, you have created active and passive vehicle resources. To do so, on the [SAP Easy Access Screen](#), choose [Logistics > Transportation Management > Master Data > Resource > Resource](#).

- If you want to undo and redo the manual planning actions you perform in the transportation cockpit and in carrier selection, you have configured the settings in the Customizing activity [Define Maximum Number of Consecutive Undo and Redo Actions](#) under [Transportation Management > Basic Functions > General Settings](#).

Features

You have the following options, for example:

- You can create capacity documents directly from freight units (see [Creation of Capacity Documents from Freight Units](#)).
- You can assign freight units to existing capacity documents (see [Assignment of Requirement Documents to Existing Capacity Documents](#)) or remove freight units from capacity documents.
- You can create capacity documents from vehicle resources, trailers, railcars, containers, or schedules. You can also create them as blank documents, that is, with no freight units assigned to them (see [Creation of Capacity Documents from Resources](#)).
- You can create capacity documents based on requirement groups (see [Creation of Capacity Documents Based on Requirement Groups](#)).
- You can assign a resource (for example, a trailer) to an existing capacity document (for example, a freight order) to create a requirement document (for example, a trailer unit). Alternatively, you can first create the requirement document (for example, the trailer unit) and assign it to the capacity document (for example, the freight order) or the resource (for example, the truck) in which case a capacity document (for example, a freight order) is created.
- You can assign freight units to an equipment type, such as a vehicle type, to create a capacity document, such as a road freight order. For more information, see [Planning Based on Equipment Types](#).
- You can have the system create local transportation unit items. For more information, see [Manual Planning with Local Transportation Unit Items](#).
- You can perform various planning and reassignment activities using drag and drop (see [Reassignment](#)).
- You can manually overwrite some of the results of your manual planning or an automatic planning run. For example, you can change the vehicle resource, the carrier, or departure date in a freight order.
- You can generate multiple alternative transportation proposals, that is, transportation plans, for each freight unit and then choose one of these proposals (see [Generation of Transportation Proposals](#)).
- You can create delivery proposals, which you can send to Logistics Execution (LE) where deliveries are then created (see [Creation of Deliveries from TM](#)).
- You can calculate charges for your freight orders and freight bookings (see [Charge Calculation](#)).
- You can combine a transportation unit for empty provisioning and a transportation unit for empty return in a single transportation unit (triangulation transportation unit). This feature is only available for container units and railcar units. For more information, see [Triangulation](#).
- During manual planning, the system makes sure that two freight unit stages or two freight units with different paying company codes are not planned on the same freight document. For more information, see [Transportation Management in Advanced Intercompany Business Processes](#).
- The system can calculate greenhouse gas (GHG) emissions during manual planning. For more information, see [Greenhouse Gas Emissions](#).
- You can carry out various planning operations using natural language. For more information, see [Conversational Planning](#).
- You can display and evaluate key performance indicators (KPIs). For more information, see [Key Performance Indicators in the Transportation Cockpit](#).

- You can define and change appointments for freight orders. For more information, see [Appointment](#).

More Information

[Utilization](#)

[Outer Volume](#)

[Considering Dangerous Goods](#)

[Change Controller](#)

[Company Codes](#)

Starting the Transportation Cockpit

Context

The transportation cockpit is the main user interface for transportation planning.

Prerequisites

- If you want to start the transportation cockpit with a predefined set of fields, you can define field catalogs according to your needs and assign them to your page layouts. For more information, see [Field Configuration for the Transportation Cockpit](#).
- If you want to use profile and layout sets, you have defined profiles according to your needs, for example a planning profile and several selection profiles. For more information about profiles and settings, see [Profiles and Settings](#).
- Requirements and capacities exist.

Procedure

1. Open the **Transportation Cockpit** app from the launchpad. Note that you can control the behavior of the transportation cockpit by defining transportation cockpit settings in the planning profile. For more information, see [Planning Profile](#).
2. The **Profile and Layout Sets** screen is displayed. Select an existing profile and layout set or create a new one and then choose **Continue**. For more information, see [Use of Profile and Layout Sets](#). Alternatively, you can switch to the **Transportation Cockpit: Selection Criteria** screen using **Switch Start Screen**. This screen allows you a quick start of the transportation cockpit without having to define several selection profiles. Enter your selection criteria and choose **Continue**. For more information, see [Selection Criteria for Transportation Cockpit](#).
 - If you want to bypass the **Profile and Layout Sets** screen the next time you call the transportation cockpit, mark one profile and layout set as the default set by selecting the checkbox in the **Default Set** column. Then select the **Hide Screen** checkbox below the table. When you next call the transportation cockpit, the system loads the planning data using the criteria that you specified as the default set. You can return to the **Profile and Layout Sets** screen by choosing **►Change Profile Selection►Profile Selection Screen►**.
 - You can assign page layouts to your profile and layout sets. They allow you to define which buttons and functions, for example, are available in the transportation cockpit and on the result screen of vehicle scheduling and routing (VSR) optimization. For more information, see [Page Layouts](#).
3. In the transportation cockpit, you have the following options:
 - Planning with lists and hierarchies

For more information, see [Manual Planning](#).

- o Planning on the geographical map

For more information, see [Interactive Planning on the Map](#).

- o Planning in the Gantt chart

For more information, see [Gantt Chart for Transportation Planning](#).

Selection Criteria for Transportation Cockpit

Use

Selection criteria enable you to carry out your planning activities without having to configure selection profiles.

Prerequisites

- Transportation demands (freight units, for example) and capacities exist.
- We recommend that you have defined a planning profile. If you do not define one, the system uses default values that might not meet your planning needs.

Features

On the [Transportation Cockpit: Selection Criteria](#) screen, you have the following options:

Function	Description
Continue	Starts the planning function
Switch Start Screen	Toggles the view between the Selection Criteria and the Profile and Layout Sets screen
Layout Personalization	Enables you to customize the layout of the transportation cockpit
Display Settings	Enables you to change the settings for the time zone, unit for distances, and alternative identification types
Saved Searches	Enables you to load or delete a search variant that you previously saved
Clear Entries	Clears all of the search criteria that you entered
Save Search As	Saves the criteria that you entered as a search variant
Default Start Screen	Specifies the screen that is displayed by default when you start the transportation cockpit. Note that if you change the default start screen, you have to restart the app for the change to be applied.

Activities

To access the [Transportation Cockpit: Selection Criteria](#) screen, open the [Transportation Cockpit](#) app from the launchpad.

Use of Profile and Layout Sets

Use

When starting the transportation cockpit, carrier selection and delivery creation apps, you can select a profile and layout set (PLS) to define the layout of the apps and which objects are available in the app for you to work on.

Features

Central PLS Creation and Management

The PLS management applications enable an administrator to centrally create and manage PLSs for a team or organization and assign users to them. This approach simplifies the process of working with PLSs by removing the need for individual users to create and maintain their own.

When an administrator creates a PLS and assigns a user to it, the PLS is available for the user to select on the relevant application start screen.

The administrator can create and manage PLSs using the following apps:

- PLS Management for Transportation Cockpit
- PLS Management for Carrier Selection
- PLS Management for Delivery Creation

Tip: To help the administrator, Quick Help is available on the PLS management apps. To display Quick Help, on the UI, right-click and select [Display Quick Help](#).

When creating a PLS, the following recommendations should be considered:

- Transportation cockpit

You are recommended to enter a selection profile for freight units as well as a planning profile to which you have already assigned a selection profile for freight orders and freight bookings.

Optionally, you can also specify other settings, such as incompatibility settings or settings for capacity selection. If you have already assigned incompatibility settings or settings for capacity selection to the specified planning profile, for example, the system overwrites these settings with the settings that you specified here.

The system considers the profiles that you specify in your PLS for all planning options available in the transportation cockpit.

i Note

In the transportation cockpit, the user can use the [Change Planning Settings](#) function to temporarily change the planning settings and load planning settings in the selection profiles. The system discards these changes when you leave the transportation cockpit or change the profile selection.

- Carrier selection

You are recommended to enter a selection profile for freight orders and freight bookings as well as carrier selection settings.

For more information, see [Carrier Selection](#).

- Creating delivery proposals

You are recommended to enter a selection profile for freight units, freight orders and freight bookings as well as a delivery profile.

PLS Selection on Application Start Screens

On the Transportation Cockpit, Carrier Selection and Delivery Creation app start screens the user can select the profile and layout sets that the administrator has assigned them to.

If authorized, on the app start screen the user can create a new PLS or edit a PLS that they are assigned to. If the user has already assigned settings to the specified planning profile, the system overwrites these settings with the settings that the user specified here. In addition, the user can edit the page layout used in a PLS by clicking [Layout Personalization](#).

Tip: To help the user, Quick Help is available on the app start screens where the user selects the PLS they want to use. To display Quick Help, on the UI, right-click and select [Display Quick Help](#).

Activities

The administrator can use the authorization object T_PL_SETS to define if the user has authorization to create, edit and assign PLSs. The authorization applies to the app start screens and the PLS management apps.

More Information

[Page Layouts](#)

Field Configuration for the Transportation Cockpit

As a power user or administrator, you can predefine the fields that are offered in the lists and hierarchies of the [Transportation Cockpit](#) app as well as their sequence and names.

Use

SAP delivers maximum user interface (UI) structures for the lists and hierarchies of the [Transportation Cockpit](#) app that contain all possible fields for these lists and hierarchies. Based on the respective maximum UI structure, you create a field catalog for each list and hierarchy. Each field catalog defines which fields are to be displayed; whether they are displayed by default, using personalization, or not at all; and the sequence in which they are to be displayed. You can also specify the names (labels) of the fields and tooltips. Configuring the fields can improve the response time of the transportation cockpit, which mainly depends on fields displayed by default but also on fields that you can display using personalization. Fields that are not displayed at all, that is, neither by default nor using personalization, do not have an impact on the response time.

You assign these field catalogs to your page layouts. When you start the [Transportation Cockpit](#) app with a certain page layout, the assigned field catalogs are taken into account and the corresponding fields are displayed for the lists and hierarchies. You can create several field catalogs per list and hierarchy if you want to support several scenarios for this list or hierarchy. You can also copy and delete field catalogs. Note that the maximum user interface structures also contain the fields that you could add using the enhancement concept for the [Transportation Cockpit](#) app in the past.

SAP delivers field catalogs for lists and hierarchies of the [Transportation Cockpit](#) app. These are the default field catalogs. You can't change or delete them. You can copy them and then change them. If you create new page layouts, the system automatically assigns the default field catalog defined for the corresponding list or hierarchy. Note that for your existing page layouts, you have to assign the field catalog manually. For more information about the available default field catalogs, see SAP Note [3362076](#).

If you extend one of the nodes ROOT, STOP, STOP_SUCCESSOR, or ITEM_TR of business object TOR with your own fields, these fields are automatically inserted into the corresponding field catalogs. You can then edit them in the field catalog as any other field. For example, you can add or change a label or indicate that they are to be displayed by default on the user interface.

You can transport your field catalogs to other systems.

Prerequisites

To define field catalogs, use the authorization object T_LAYOUT, activity [Administer](#).

Activities

1. You define field catalogs for your lists and hierarchies in the [Define Field Catalogs for Transportation Cockpit](#) app based on the maximum UI structures. You specify, which fields are to be displayed by default in the [Transportation Cockpit](#) app and which are to be displayed using personalization. Additionally, you specify the sequence of the fields as well as their names and tooltips.

You can mark one field catalog as the default field catalog per content object, for example road freight order list or hierarchy.

2. You assign your field catalogs to your page layouts in the [Define Page Layouts for Transportation Cockpit](#) app.

If you have marked a field catalog as the default field catalog, this field catalog is automatically assigned to the page layout when you add the corresponding content object to the page layout.

3. When you start the [Transportation Cockpit app](#) with a certain page layout, the assigned field catalogs are taken into account and the corresponding fields are displayed for the lists and hierarchies.

Page Layouts

A page layout defines the buttons, functions, levels and hierarchies that are displayed in the Transportation Cockpit, Carrier Selection and Delivery Creation app UIs.

To create and manage page layouts, use the following apps:

- [Define Page Layouts for Carrier Selection](#)
- [Define Page Layouts for Delivery Creation](#)
- [Define Page Layouts for Transportation Cockpit](#)
- [Define Page Layouts for Transportation Proposal](#)

On the start screen of these apps, you can mark a page layout as your default page layout. In the [Define Page Layouts for Transportation Cockpit](#) app and the [Define Page Layouts for Carrier Selection](#) app, you can also specify that quick views are not to be displayed for a specific page layout. Note that you need to add the [Hide Quick View](#) field from the [Settings](#) first. If you work with multiple windows, you can activate this function for each of the corresponding page layouts.

After creating a page layout, you can include it in a profile and layout set. The app for which you have defined a page layout (for example, the transportation cockpit), is then always opened with this page layout. Note that you can also switch to another page layout from within the app. In the page layout, you can define the page layouts that should be available for switching.

You can transport your page layouts to other systems.

For an overview of the page layouts delivered by SAP for the transportation cockpit, see SAP Note [2358887](#)  and SAP Note [3155675](#) .

Activities

- To define page layouts for all users, specific user roles, or an individual user, use the authorization object T_LAYOUT, activity [Manage, Administer](#).
- To define a selection of available functions for your page layouts, use the settings in Customizing under [Transportation Management](#) [» Planning](#) [» General Settings](#) [» Define Settings for Page Layouts](#).
- To synchronize page layouts with the Customizing settings, use the app [Synchronize Page Layouts with Customizing](#). This will synchronize all page layouts.

Note that you can also synchronize an individual page layout. To do so, select [Show Configuration Data](#) for the local toolbar within the page layout. Then select [Restore Pushbutton Data](#). For the global toolbar, select [Restore Pushbutton Data](#) within the page layout.

- To improve the overview of your rescheduling activities and the coupling and uncoupling of trailers, you can define a layout for which you select the [Dual View](#) checkbox on the [Orders Area](#) or [Transportation Units Area](#) tab page. You define this layout in the layout settings for the transportation cockpit. In the transportation cockpit, you can display the list with the documents to be rescheduled in two views. Alternatively, you can select the [Dual View](#) checkbox on the [Hierarchies Area](#) tab page. For more information about reassignment and coupling and uncoupling trailers, see [Reassignment](#) and [Use of Trailers](#).

Related Information

[Use of Profile and Layout Sets](#)

[Profiles and Settings](#)

Planning with Multiple Windows

You can display areas of the transportation cockpit in up to three separate and flexible windows that you can position across multiple monitors and can resize independently. This means you can display areas such as the geographical map and the Gantt chart in separate windows, and drag and drop objects between windows.

Context

Complete the following steps to enable multiple windows. For more information about restrictions, see SAP Note [2551221](#) .

Procedure

1. Create a page layout for an additional window.

Standard SAP page layouts that are compatible with multiple windows use the following naming convention:

- The suffix "MW" indicates the page layout is for the main window.
- The suffix "A1" and "A2" indicate the page layout is for additional window 1 and additional window 2.

2. In the (main) page layout for the transportation cockpit, add the additional-window page-layout in the [Additional Windows](#) area.
3. To display the additional window, in the transportation cockpit, on the application toolbar, click [Open Additional Window](#).

The additional window is displayed and "Additional Window 1" is displayed on the title bar; similarly, "Main window" is displayed on the title bar of the original window that has become the main window.

Key Performance Indicators in the Transportation Cockpit

In the transportation cockpit, you can display and evaluate key performance indicators (KPIs).

As a transportation planner, KPIs help you evaluate whether your transportation plan is efficient and optimized. For example, if you create road freight orders that are subcontracted, these road freight orders should be optimized regarding utilization and costs. If you plan your own fleet, minimizing the distances that are traveled without load might be another KPI you are interested in. In both cases, you want to see the impact of your planning decisions immediately for the relevant KPIs in the transportation cockpit.

Prerequisites

- You have defined a transportation KPIs profile and selected at least one KPI from a list of predefined KPIs. The KPIs profile controls which KPIs are displayed in the transportation cockpit and how they are displayed. The predefined KPIs cover areas such as distance and duration, utilization, weight, volume, quantity, normalized quantity, charges, and CO₂ emissions. Optionally, you have defined filter criteria, threshold values, and semantic colors for the KPIs. For example, for a utilization KPI, you could define that a utilization of 90 % and higher is fine (positive). The corresponding value is then displayed in green. You can also indicate whether the previous value and the delta value are to be displayed for a KPI.

For each KPI, you can show details and documentation.

Nested KPIs, that is KPIs that consist of more than one KPI, such as CO₂ emissions per distance, are displayed on the **Nested KPIs** tab.

For each KPI (including nested KPIs), you need to specify the unit of measure in which you want the KPI values to be displayed.

You can change the heading and subheading of the predefined KPIs according to your needs or even define your own KPIs. For more information, see SAP Note [3612870](#) .

To create a transportation KPIs profile, open the **Create Transportation KPIs Profile** app from the launchpad.

- You have specified the transportation KPIs profile on the **Profile and Layout Sets** screen or the **Transportation Cockpit: Selection Criteria** screen. By doing so, the transportation cockpit will contain a KPIs screen area and the KPIs will be determined automatically after each planning operation.
- In the field catalogs for road freight orders and freight unit stages, you have selected fields that allow you to see whether a road freight order or freight unit stage is critical for a specific KPI. Further, you have indicated whether these fields are to be displayed by default or using personalization. For example, you can display the calculation status, the duration per weight, or the number of pick-up stops per distance for road freight orders. For more information about field catalogs, see [Field Configuration for the Transportation Cockpit](#).

Features

The KPIs screen area is collapsible and expandable. It is visible for a planning session all the time, even if you maximize the transportation cockpit or work in dual view mode.

The KPIs displayed within the KPIs screen area are updated immediately after each planning operation, undo, or redo operation. Further, you can see how a KPI changes when you perform a planning operation. The delta value for the KPI, such as +1.5% or -2.0%, lets you easily see whether the KPI has improved or deteriorated or not changed at all. An arrow indicates the trend based on the delta value. It's either red or green depending on the KPI objective. The previous value of a KPI allows you to compare a KPI before and after a planning operation. Using the context menu, you can display documentation and a log for each KPI.

In the lists and hierarchies of the transportation cockpit, you can see whether a road freight order, freight unit stage, or truck resource is critical for a specific KPI. You can also display the calculation status, which indicates whether a certain document was taken into account by the KPI calculation.

i Note

- KPIs are supported for road freight orders, truck resources, and freight unit stages only.
- The display of KPIs allows you to analyze the data of selected freight unit stages, road freight orders, and truck resources. This feature is not intended to be used as a pure analytical tool that allows analyzing data such as all freight orders of the last half year. Therefore, technical limitations such as the KP calculation horizon and warning messages are implemented to prevent the use of the transportation cockpit as a pure analytical tool.
- The system takes into account selected documents only if they lie within the KPI calculation horizon (partly or completely). This horizon is determined automatically based on the planning horizon. It starts seven days before the end date of the planning horizon and ends at the same date as the planning horizon. If the planning horizon is less than seven days, the calculation horizon starts with the planning horizon. If the planning horizon lies entirely in the future, the start date of the KPI calculation horizon is moved to the current date.
- KPIs that are not displayed in the transportation cockpit don't have a negative impact on the performance.
- If you work with multiple windows, the KPIs screen area is displayed on the main window only.

Related Information

[Calculation of Distances Traveled With or Without Load](#)

Use of Hierarchical Views

Use

You organize the view in the transportation cockpit by creating hierarchical views for freight orders, freight bookings, and transportation units (trailer units, railcar units, container units, and package units).

If you organize these business documents into hierarchies, you can filter and sort various areas in the transportation cockpit according to specific criteria, such as the vehicle resource.

Defining hierarchical views simplifies your rescheduling activities and the coupling and uncoupling of trailers. For more information, see [Reassignment](#) and [Use of Trailers](#).

Prerequisites

- In Customizing, you have made the relevant settings so that you can configure your standard layout for rescheduling in the layout settings for the transportation cockpit.

The following hierarchy types are predefined by default:

- FBSEA for planning activities in sea transportation
- FB AIR for planning activities in air transportation
- FOLD for planning activities in overland transportation
- FORAI for planning activities in rail transportation
- FUSTG for planning freight unit stages
- PUSTG for planning package unit stages
- FPSTG for planning freight unit stages and package unit stages

- TUTRL for location-based planning of trailer units with the assigned freight units
- TUTRS for stage-based planning of trailer units with the assigned freight units
- TURAI for location-based planning of railcar units with the assigned freight units
- TURAS for stage-based planning of railcar units with the assigned freight units
- TUCNT for location-based planning of container units with the assigned freight units
- TUCLD for location-based planning of container units with the assigned freight documents
- TUCNS for stage-based planning of container units with the assigned freight units
- TUCSD for stage-based planning of container units with the assigned freight documents
- TUPCK for location-based planning of package units with the assigned freight units
- TUPLD for location-based planning of package units with the assigned freight documents
- TUPUS for stage-based planning of package units with the assigned freight units
- TUPSD for stage-based planning of package units with the assigned freight documents
- TUCO for planning consignment orders
- TUCOC for capacity document-based planning of consignment orders

You can also define hierarchies with separate hierarchy types and groupings and a view switch. You can specify the function (consumer) that each hierarchy type can use. For more information, see Customizing for [Transportation Management](#) under [►Planning ►General Settings ►Define Hierarchical Views for Business Documents ►](#).

- You have defined a layout in the layout settings for the transportation cockpit in which you have assigned a hierarchy type to the corresponding hierarchies on the [Transportation Units Area](#), [Orders Area](#), [Requirements Area](#) or [Hierarchies Area](#) tab pages.

If you do not make this assignment, the system uses the applicable default hierarchy type. For more information, see [Use of Profile and Layout Sets](#).

Features

In the transportation cockpit, you can show hierarchies as alternative order lists. You can define grouping levels for each hierarchy level and specify whether the grouping level in question is expanded by default when displayed in the transportation cockpit.

If you organize freight orders, freight bookings, or transportation units into hierarchies, you can filter and sort various areas in the transportation cockpit according to specific criteria, such as the vehicle resource.

Creation of Capacity Documents from Freight Units

Use

When planning manually, use this process to assign freight units to any of the following documents:

- Freight documents:
 - Road freight order
 - Rail freight order

- Air freight booking
- Ocean freight booking
- Transportation units:
 - Container unit
 - Trailer unit
 - Railcar unit
 - Package unit

Prerequisites

- For more information, see [Manual Planning](#).

Process

1. In the transportation cockpit, select the freight units you want to assign to a capacity document, that is a freight document or a transportation unit. You can select the freight units either in the freight unit hierarchy or on the map, if displayed. Alternatively, you can also select freight unit stages.
2. On the respective toolbar, click the **Create Capacity Document** or **Create Multiple Capacity Documents** pushbutton to display the list of documents you can create.

i Note

- The number of multiple documents created depends on how the setting **Rule for Creating Capacity Documents** is configured (**► Edit Planning Profile ► Manual Planning Settings tab ►**). The system creates either a document per freight unit (default) or a document for each freight unit stage.
- If you have multiple types of a capacity documents, you can specify which type is used by default in the planning profile. Alternatively, you can use the **Change Planning Settings** to temporarily overrule the default setting for the duration of the session.

3. In the menu, select the capacity document you want to create.

The capacity documents are created but you can still edit and add to them as you continue planning. For more information, see [Manual Planning](#).

i Note

- If you create a freight document for one or more freight unit stages generated from a delivery-based or order-based transportation requirement, the system determines the freight document type from the information about the respective default business document type that you entered in the planning profile. However, if you have assigned a schedule to one of the stages, the system determines the freight document type from the Customizing activity **Define Schedule Types**. For more information, see the **Creation of a Freight Document for Several Stages** section of [Direct Creation of Freight Documents and Selection of Schedules](#).
- If execution information already exists for a business document such as a freight order, the system cannot change the corresponding transportation stop during manual planning or vehicle scheduling and routing (VSR) optimization. Examples of execution information include a freight unit that has already been loaded or a freight order that has already been delivered to a location.

Creation of Capacity Documents from Resources

Use

You use this process to create capacity documents in the transportation cockpit on the basis of resources (such as trucks, trailers, railcars, containers, and schedules). If required, you can also create blank capacity documents (that is, capacity documents that do not contain a freight unit), which you can then process at a later time.

Prerequisites

See [Manual Planning](#).

Process

1. You open the **Transportation Cockpit** app on the launchpad. For more information, see [Starting the Transportation Cockpit](#).
2. On the relevant tab page, you select one or more resources and choose the corresponding pushbutton to create a capacity document for your resource type. Alternatively, you assign freight units to the required resources using drag and drop or [Assign Selected Items](#).
 - o If you create capacity documents from vehicle resources, railcars, trailers, or containers, the system displays a dialog box in which you must enter the appropriate capacity document type (the system proposes the document type specified in the planning profile or, if no document type is specified here, it proposes the default document type from the corresponding Customizing activity ([Define Freight Order Types](#) or [Define Freight Booking Types](#) for freight orders and freight bookings and [Define Transportation Unit Types](#) for transportation units)). Note that you do not have to enter the location data at this point. You can create capacity documents without location data.
 - o If you create blank capacity documents from schedules or departures, the system displays a dialog box in which you must enter the appropriate capacity document type (the system proposes the document type specified in the Customizing activity [Define Schedule Types](#)). Note that if you create capacity documents on the basis of a schedule, the system creates one capacity document for each departure belonging to the selected schedule. Alternatively, you can create freight documents for one or more schedule departures.

When you create a capacity document by assigning a freight unit to a schedule or schedule departure, the system adds the locations specified in the freight unit to the capacity document.

i Note

You can also create blank freight orders in the [Freight Orders/Freight Bookings](#) area of the transportation cockpit by choosing the [New](#) pushbutton and then the [Freight Order](#) option.

You can create blank road freight orders in the [Road Freight Order Hierarchy](#) and in the [Road Freight Orders](#) area of the transportation cockpit by choosing the [New Road Freight Order](#) pushbutton.

You can create blank road freight orders in the [Road Freight Order Hierarchy](#) and in the [Road Freight Orders](#) area of the transportation cockpit by choosing the [New Road Freight Order](#) pushbutton.

3. You carry out any subsequent planning activities.

For more information, see [Manual Planning](#).

4. You check your planning.

The system performs a consistency check during which it checks incompatibilities, for example.

5. You start scheduling.

The system performs backward or forward scheduling. Note that if you do not complete this step, dates and times may be incorrect.

6. You save the planning results.

Creation of Capacity Documents Based on Requirement Groups

You can create capacity documents based on requirement documents that are grouped according to specific attributes.

Prerequisites

- You have defined requirement grouping profiles in the [Create Requirement Grouping Profile](#) app on the launchpad and specified grouping rules and capacity limits. SAP delivers several grouping rules but you can also define your own grouping rules in Customizing (see below).

For example, you can indicate in your requirement grouping profile that requirement documents are to be grouped by source or destination location when you want to create road freight orders.

- Optional: In your requirement grouping profile, you have specified that the system is to determine capacity limits using load consolidation instead of considering the capacity limits you have defined in the profile.

In your load planning settings, you have indicated the consolidation mode you want to use (capacity-based or load-planning-based load consolidation).

- You have defined capacity document creation settings in the [Create Capacity Document Creation Settings](#) app and specified a requirement grouping profile per capacity document category.

Note that if you do not specify a requirement grouping profile, all requirement documents are placed into the same group.

- Optionally, you have defined your own grouping rules. For more information, see Customizing for Transportation Management under [►Planning ►General Settings ►Define Grouping Rules for Requirement Grouping ►](#).

Use

Requirement grouping

The generic grouping framework first groups requirement document stages, such as freight unit stages, based on a list of attributes, for example, source location or carrier. The attributes are represented by grouping rules.

The system applies the maximum capacity limit to these groups of requirement document stages. That means, groups above the maximum capacity limit are split into smaller groups so that the resulting freight document that is created for each group fits into a vehicle resource or equipment type. A set of grouping rules and capacity limits is bundled in a requirement grouping profile.

In the requirement grouping profile, you can also specify incompatibility settings as a further criterion for the grouping. Note, however, that you must have selected the [Identical Values Only](#) checkbox in your incompatibilities for them to be considered.

Creation of capacity documents

For all groups of requirement document stages that are above the minimum capacity limit, the system creates a capacity document. All groups of requirement document stages that are below the minimum capacity limit remain unplanned that means, no freight document is created for them. Single requirement document stages that are bigger than the maximum capacity limit also remain unplanned.

Creation of capacity documents with load consolidation

For each group of requirement document stages, the system uses load consolidation to find the optimal set of equipment. It does this by determining capacity limits based on the capacities of the selected equipment types. The system considers constraints like orientation, stacking factor, incompatibilities, and the costs defined for your equipment types.

If several equipment types that are relevant for load consolidation are available in the planning session, a dialog box is displayed that allows you to select the equipment types you want to use. In the transportation cockpit settings, you can specify that this dialog box is skipped. In this case, all equipment types that are available in the planning session are used.

i Note

- Since load planning only supports one source and one destination, the source location and destination location are used as grouping criteria.
- This option supports equipment types only.

Activities

In the transportation cockpit, select requirement document stages. For these stages select **Create Capacity Documents Based on Groups** and then the requested capacity document category, for example **Create Road Freight Orders Based on Groups**.

For more information about the allowed combinations of requirement documents and capacity documents, see [Supported Document Combinations](#).

To run the creation of capacity documents based on requirement groups in the background, on the SAP Easy Access screen, choose **Logistics > Transportation Management > Administration > Background Processing > Create Capacity Documents Based on Groups**. For more information, see [Creation of Capacity Documents Based on Requirement Groups](#).

Related Information

[Planning Profile](#)
[Incompatibilities](#)

Supported Document Combinations

The type of requirement document selected affects the type of capacity document allowed.

The following combinations of requirement documents and capacity documents are allowed during capacity document creation based on requirement groups:

- Freight units (FUs):
 - All capacity document types are allowed.
 - Road freight orders, trailer units, and container units support load planning-based and capacity-based consolidation (LP & CB), rail freight orders and railcar units support capacity-based consolidation (CB).
- Package units (PUs):
 - Road freight orders (LP & CB)
 - Consignment orders
 - Trailer units (LP & CB)

- Container units (LP & CB)
- Trailer units (TUs):
 - Only road freight orders are allowed.
- Railcar units (RUs):
 - Only rail freight orders are allowed.
- Container units (CUs):
 - Road freight orders
 - Rail freight orders
 - Ocean freight bookings
 - Air freight bookings
 - Trailer units
 - Railcar units
- Consignment orders (COs):
 - Road freight orders (LP & CB)
 - Rail freight orders (CB)
 - Ocean freight bookings
 - Air freight bookings
 - Trailer units (LP & CB)
 - Railcar units (CB)
 - Container units (LP & CB)

The table below displays the supported document combinations: green (Yes) indicates a supported combination, while yellow indicates an unsupported combination.

Additionally, the supported consolidation mode is indicated as follows:

- LP = load planning-based consolidation
- CB = capacity-based consolidation

Note that TU stands for trailer units.

Supported combinations

Requirement Docs/Capacity Docs	Road FOs	Rail FOs	Ocean FBs	Air FBs	COs	TUs	RUs	CUs	PUs
FUs	Yes (LP & CB)	Yes (CB)	Yes	Yes	Yes	Yes (LP & CB)	Yes (CB)	Yes (LP & CB)	Yes
PUs	Yes (LP & CB)				Yes	Yes (LP & CB)		Yes (LP & CB)	

Requirement Docs/Capacity Docs	Road FOs	Rail FOs	Ocean FBs	Air FBs	COs	TUs	RUs	CUs	PUs
TUs	Yes								
RUs		Yes							
CUs	Yes	Yes	Yes	Yes		Yes	Yes		
COs	Yes (LP & CB)	Yes (CB)	Yes	Yes		Yes (LP & CB)	Yes (CB)	Yes (LP & CB)	

Manual Planning with Local Transportation Unit Items

In the transportation cockpit, you can create transportation units as freight document items instead of separate business documents.

This is useful in scenarios in which you don't need a separate business document for your trailer, railcar, or container. The freight document contains all relevant information.

Prerequisites

- In the manual planning settings of your planning profile, you have selected **Create TU as Freight Document Item** in the **Create TU Item or TU Document** field.
- If you work with containers, you have also specified the target document category of the freight document, such as **Air Freight Booking**, in the **Target Document Category for Container** field of the manual planning settings.

Use

If you perform manual planning by assigning freight units to railcar resources, trailer resources, or container resources or railcar types, trailer types, or container types using drag and drop or **Assign Selected Items** for example, the system creates a transportation unit (TU) item in the corresponding freight document locally (a local trailer item, for example) instead of a separate TU document, such as a trailer unit.

Examples:

- You assign a freight unit to a trailer type by selecting **Assign Selected Items**. The system creates a road freight order with a local item for the trailer.
- You assign a freight unit to a container resource using drag and drop. The system automatically creates an ocean freight booking with a local item for the container if you have selected **Ocean Freight Booking** as the target document category in the manual planning settings.
- You assign a railcar type to a rail freight order using drag and drop. The system automatically creates a railcar item in the rail freight order locally.

In the rail freight order hierarchy, you can manually insert a railcar with or without a railcar type:

- **Insert Railcar with Railcar Type**

You insert a railcar with a railcar type into the rail freight order and indicate the number of railcar items that the system is to create. The system creates the indicated number of railcar items in the rail freight order locally.

- [Insert Railcar](#)

You insert a railcar without a railcar type into the rail freight order. The system creates a railcar item in the rail freight order locally. Then you can assign a railcar type to this railcar item using drag and drop, for example.

In the same way, you can insert a trailer with or without a trailer type in the road freight order hierarchy or insert a container with or without a container type in the air freight booking hierarchy or ocean freight booking hierarchy.

Once the local TU item is created, you can do the following, for example:

- You can assign a container type to a local container item using drag and drop.
- You can reassign a container item from one air freight booking to another using drag and drop.
- You can change the railcar type, trailer type, or container type in the freight order or freight booking hierarchy using the [Equipment Type](#) and [Equipment Group](#) fields. You can also use these fields to add railcar types, trailer types, or container types or remove them.

Unified package building considers product items and package items in local trailer items and local container items. You can perform unified package building either for the entire road freight order or for selected local trailer items or local container items. You can also remove packages from selected local trailer items and local container items.

i Note

- Load planning doesn't support local TU items.
- Load consolidation doesn't support local TU items. That means, the system always creates a separate TU document even if you have specified in the manual planning settings that a local TU item is to be created.
- Vehicle scheduling and routing and the generation of transportation proposals don't support local TU items.

Related Information

[Planning Profile](#)

Assignment of Requirement Documents to Existing Capacity Documents

Use

You use this process to assign requirement documents, such as freight units, to existing capacity documents, such as freight orders.

Prerequisites

See [Manual Planning](#).

Process

1. You open the [Transportation Cockpit](#) app.

For more information, see [Starting the Transportation Cockpit](#).

2. You can select how the requirement documents are to be assigned by making settings in [Manual Planning Settings](#), under [Document Assignment](#).

For example, you can control how the location sequence of the capacity document is to be extended. For more information, see the corresponding field help.

3. You choose one or more requirement documents and assign them to an existing capacity document by using drag and drop. Alternatively, you can enter a command in the command field or select one or more requirement documents and a capacity document and choose [Assign Selected Items](#).

4. You carry out any subsequent planning activities.

For more information, see [Manual Planning](#).

5. You check your planning.

The system performs a consistency check during which it checks incompatibilities, for example.

6. You start scheduling.

The system performs backward or forward scheduling. Note that if you do not complete this step, dates and times may be incorrect.

7. You save the planning results.

Related Information

[Document Assignment When Execution Has Already Started](#)

Document Assignment When Execution Has Already Started

You may need to assign a requirement document, such as a freight unit, to a capacity document, such as a road freight order, even after the execution of the capacity document has already started or has already been completed.

i Note

These features also apply to warehouse loading and unloading stops in Advanced Shipping and Receiving.

Correcting the Capacity Document

One reason this situation occurs is when there is a mismatch between the information in the capacity document and what is actually happening, or has already happened, at the time of loading or unloading. Sometimes during loading or unloading, additional cargo items that are not assigned to the capacity document are loaded or unloaded along with the items that are already assigned. To correct this, you can update the capacity document by assigning the additional items to the same stop.

If the processing of the items has not yet started, the statuses of the new items match the statuses of the other items in the capacity document.

However, if you add a requirement document after the vehicle has loaded or unloaded the goods at the stop, has checked out, or has departed, then the status of the items added later doesn't match the status of the stop and of the other items in the capacity document. You can correct these errors using document error handling.

For further information, see [Late Changes to Capacity Documents](#).

Creating New Stops

A different reason this situation occurs is if cargo items have not been loaded or unloaded where and when they should have been, and you want the vehicle to return to load or unload these items.

If you were to follow the same procedure as above, and correct the capacity document, the status of the items would not reflect reality. Therefore, instead of reusing a stop that has already been executed, you create a new stop for the same location. The system automatically adapts the routing of the capacity document so the vehicle can return and load or unload a second time at the location.

To do this, under **Manual Planning Settings**, choose **Consider Handling Execution Status**. This means that when you assign a requirement document to a capacity document during manual planning, the system disregards those stops where the status is **Checked Out** or **Departed**. If necessary, it creates a new stop for the same location. This means that the handling execution status of stops and the requirement items is consistent.

i Note

The setting only takes effect when you assign the requirement document to the capacity document header. If, for example, you assign a freight unit directly to a freight order location, the system uses the freight order location you choose.

You can make further settings as to where exactly the stop should be inserted under **Extension Strategy for Location Sequence of Capacity Documents**; for example, you can choose to insert the new stop where the distance added to the total is the least. For more information, see the corresponding field help.

Advanced Shipping and Receiving

If you are using this feature in Advanced Shipping and Receiving, the system also considers the handling execution status of warehouse loading and unloading stops. For example, if loading is completed for a warehouse loading stop, but you want to add another freight unit, the system creates a new warehouse loading stop for the same location, if the truck has not yet checked out from the shipping or receiving point.

However, in the Advanced Shipping and Receiving inbound scenario, the system ignores the **Consider Handling Execution Status** setting if it can only determine stops that have already been executed with the correct location in the stop sequence. In this case, the system assigns the freight unit to a previously executed stop. The procedure is then as described in [Late Changes to Capacity Documents](#). However, if the system determines a stop that has not yet been executed at the correct location, then it assigns the freight unit to this stop.

Reassignment

Use

In the transportation cockpit, you can use this function to reschedule freight documents by using drag and drop. Alternatively, you can reassign freight documents by selecting the relevant rows and choosing the **Reassignment** pushbutton. For example, you can assign a freight unit to a new freight order if the scheduled delivery date is moved.

Prerequisites

- In the layout settings for the transportation cockpit, you have defined a layout for which you have selected the **Reassignment** checkbox on the **Orders Area** or **Transportation Units Area** tab page. Alternatively, you can select the **Reassignment** checkbox on the **Hierarchies Area** tab page. For more information, see [Use of Profile and Layout Sets](#).
- In Customizing, you have made the relevant settings so that you can configure your standard layout for rescheduling in the layout settings for the transportation cockpit.

The standard system contains a range of predefined hierarchy types. In addition, you can define your own hierarchy types. For more information, see Customizing for Transportation Management under [►Planning►General Settings►Define Hierarchical Views for Freight Documents ►](#).

Features

Once you have configured the relevant settings in Customizing and in the layout settings for the transportation cockpit, you can also use various hierarchy levels as drag and drop sources and targets in the hierarchy in question. For more information, see [Reassignment Activities](#).

Activities

You select the layout that you have defined in the layout settings for the transportation cockpit. In the **Freight Orders/Freight Bookings** screen area, you choose the **Dual View** pushbutton. You can now execute your reassignment activities.

Related Information

[Use of Hierarchical Views](#)

Reassignment Activities

Once you have configured the relevant settings in Customizing and in the layout settings for the transportation cockpit (see [Reassignment](#)), you can use the following hierarchy levels as drag and drop sources and targets within the various hierarchies:

i Note

The drag and drop source and the drag and drop target is always interchangeable.

Freight Order Hierarchies (Road Freight Order Hierarchy and Rail Freight Order Hierarchy)

Drag and Drop Source	Drag and Drop Target	System Activity
Freight order	Transportation unit	You couple the part of the transportation unit that is not yet coupled to the freight order. i Note The coupling locations must already be available in both documents.
Location in freight order	Another location in the freight order	You change the positions of the locations in question in the location sequence. This also transfers the requirements assigned to the location to the location in its new position in the sequence. This type of planning enables you to stop at the same location twice in succession in the same freight order. i Note If you add a location at which a freight unit is unloaded to the location sequence of another freight order, the system checks whether the location at

Drag and Drop Source	Drag and Drop Target	System Activity
		which the relevant freight unit is loaded also exists in the freight order. If this location is already included in the freight order, the system enhances only those requirements for this location. If the location is not included in the freight order, the system adds it to the freight order with the corresponding requirements. The same applies if you add a location at which a freight unit is loaded to the location sequence of another freight order. In this case, the system checks whether the location at which the freight unit is unloaded already exists in the freight order and carries out the relevant planning operations.
Location in freight order	Same location in freight order	You transfer the requirements assigned to the location acting as the drag & drop source to the location acting as the drag & drop target. If the location does not exist in the freight order, this action adds the location to the location sequence of the freight order. At the same time, you delete the location acting as the drag & drop source from the location sequence of the freight order to which the location and its requirements was originally assigned. i Note If you add a location at which a freight unit is unloaded to another freight order, the system checks whether the location at which the relevant freight unit is loaded also exists in the freight order. If this location is already included in the freight order, the system enhances only those requirements for this location. If the location is not included in the freight order, the system adds it to the freight order with the corresponding requirements. The same applies if you add a location at which a freight unit is loaded to another freight order. In this case, the system checks whether the location at which the freight unit is unloaded already exists in the freight order and carries out the relevant planning operations.

Drag and Drop Source	Drag and Drop Target	System Activity
Location in freight order	Freight order	You transfer the requirements assigned to the location acting as the drag & drop source to the same location in the freight order acting as the drag & drop target. At the same time, you delete the location acting as the drag & drop source from its original position in the location sequence. i Note If you add a location at which a freight unit is unloaded to the location sequence of another freight order, the system checks whether the location at which the relevant freight unit is loaded also exists in the freight order. If this location is already included in the freight order, the system enhances only those requirements for this location. If the location is not included in the freight order, the system adds it to the freight order with the corresponding requirements. The same applies if you add a location at which a freight unit is loaded to the location sequence of another freight order. In this case, the system checks whether the location at which the freight unit is unloaded already exists in the freight order and carries out the relevant planning operations.
Freight order stage	Transportation unit	You assign the freight order stage to the transportation unit and add missing locations to the location sequence for the transportation unit.
Freight unit stage	Freight order	You assign the freight unit stage to the freight order and add missing locations to the freight order.
Freight unit stage	Compartment of a freight order	You assign the freight unit stage to the compartment of a freight order and add missing locations to the freight order.
Freight unit stage	Location in freight order	You assign the freight unit stage to the freight order and add missing locations to the freight order. At the same time, you integrate the newly assigned location into the location sequence at the position predefined by the location.
Freight unit stage	Transportation unit	You assign the freight unit stage to the transportation unit and add missing locations to the transportation unit.

Transportation Unit Hierarchies (Trailer Unit Hierarchy and Railcar Unit Hierarchy)

Drag and Drop Source	Drag and Drop Target	System Activity
Transportation unit	Freight order	You couple the part of the transportation unit that is not yet coupled to the freight order. i Note The coupling and uncoupling locations must already be assigned to both documents.
Location in the transportation unit	Other location in the transportation unit	You change the positions of the locations in question in the location sequence. At the same time, you transfer the requirements assigned to the location to the location in its new position in the sequence. i Note If you add a location at which a freight unit is unloaded to the location sequence of another transportation unit, the system checks whether the location at which the relevant freight unit is loaded also exists in the transportation unit. If this location is already included in the transportation unit, the system enhances only those requirements for this location. If the location is not included in the transportation unit, the system adds it to the transportation unit with the corresponding requirements. The same applies if you add a location at which a freight unit is loaded to the location sequence of another transportation unit. In this case, the system checks whether the location at which the freight unit is unloaded already exists in the transportation unit and carries out the relevant planning operations.
Location in the transportation unit	Same location in the transportation unit	You assign the freight units that were assigned to the drag & drop source to the drag & drop target. At the same time, you delete the drag & drop source from the location sequence. i Note If you add a location at which a freight unit is unloaded to the location sequence of another transportation unit,

Drag and Drop Source	Drag and Drop Target	System Activity
		the system checks whether the location at which the relevant freight unit is loaded also exists in the transportation unit. If this location is already included in the transportation unit, the system enhances only those requirements for this location. If the location is not included in the transportation unit, the system adds it to the transportation unit with the corresponding requirements. The same applies if you add a location at which a freight unit is loaded to the location sequence of another transportation unit. In this case, the system checks whether the location at which the freight unit is unloaded already exists in the transportation unit and carries out the relevant planning operations.
Location in the transportation unit	Transportation unit	You transfer the requirements assigned to the location acting as the drag & drop source to the same location in the transportation unit acting as the drag & drop target. At the same time, you delete the location acting as the drag & drop source from its original position in the location sequence. i Note If you add a location at which a freight unit is unloaded to the location sequence of another transportation unit, the system checks whether the location at which the relevant freight unit is loaded also exists in the transportation unit. If this location is already included in the transportation unit, the system enhances only those requirements for this location. If the location is not included in the transportation unit, the system adds it to the transportation unit with the corresponding requirements. The same applies if you add a location at which a freight unit is loaded to the location sequence of another transportation unit. In this case, the system checks whether the location at

Drag and Drop Source	Drag and Drop Target	System Activity
		which the freight unit is unloaded already exists in the transportation unit and carries out the relevant planning operations.
Location in the transportation unit	Freight unit stage	You assign the freight unit stage to the transportation unit and add missing locations to the transportation unit. At the same time, you integrate the newly assigned location into the location sequence at the position predefined by the location.
Freight order	Transportation unit	You couple the part of the transportation unit that is not yet coupled to the freight order. i Note The coupling and uncoupling locations already exist in both documents.
Transportation unit stage	Freight order	You couple the transportation unit stage to the freight order and add missing locations to the freight order.
Freight unit stage	Compartment in the transportation unit	You assign the freight unit stage to the compartment of a transportation unit and add missing locations to the transportation unit.
Freight unit stage	Transportation unit	You assign the freight unit stage to the transportation unit and add missing locations to the transportation unit.
Freight unit stage	Location in the transportation unit	You assign the freight unit stage to the transportation unit and add missing locations to the transportation unit. At the same time, you integrate the newly assigned location into the location sequence at the position predefined by the location.
Freight unit stage	Freight order	You assign the freight unit stage to the freight order and add missing locations to the freight order.

i Note

The freight order is included in the transportation unit hierarchy as information only. All of the planning operations listed in the table above that have a freight order as the drag & drop source or target are exclusively cross-hierarchy planning operations between the freight order hierarchy and the transportation unit hierarchy. These planning operations cannot be carried out in the transportation unit hierarchy using drag & drop.

Drag and Drop Source	Drag and Drop Target	System Activity
Freight unit stage	Freight booking	You assign the freight unit stage to a freight booking.
Freight unit stage	Capacity reservation	You assign a full-container-load freight unit stage to the capacity reservation. At the same time, you replace the container from the capacity reservation with the full-container-load freight unit if the container type matches.
Freight unit stage	Container	You assign a less-than-container load freight unit stage to a container. i Note The container must be part of the freight booking.

Freight Unit Hierarchy

Drag and Drop Source	Drag and Drop Target	System Activity
Freight unit stage	Freight order	You assign the freight unit stage to the freight order and add missing locations to the freight order.
Freight unit stage	Location in freight order	You assign the freight unit stage to the freight order and add missing locations to the freight order. At the same time, you integrate the newly assigned location into the location sequence at the position predefined by the location.
Freight unit stage	Transportation unit	You assign the freight order stage to the transportation unit and add missing locations to the location sequence for the transportation unit.
Freight unit stage	Compartment of a freight order	You assign the freight unit stage to the compartment of a freight order and add missing locations to the freight order.
Freight unit stage	Location in the transportation unit	You assign the freight unit stage to the transportation unit and add missing locations to the transportation unit. At the same time, you integrate the newly assigned location into the location sequence at the position predefined by the location.
Freight unit stage	Compartment in the transportation unit	You assign the freight unit stage to the compartment of a transportation unit and add missing locations to the transportation unit.
Freight unit stage	Freight booking	You assign the freight unit stage to a freight booking.

Drag and Drop Source	Drag and Drop Target	System Activity
Freight unit stage	Capacity reservation	You assign a full-container-load freight unit stage to the capacity reservation. At the same time, you replace the container from the capacity reservation with the full-container-load freight unit if the container type matches.
Freight unit stage	Container	You assign a less-than-container load freight unit stage to a container. i Note The container must be part of the freight booking.

i Note

All of the planning operations listed here are possibly only if either the drag & drop source or the drag & drop target do not originate from the freight unit hierarchy.

Container Unit Hierarchy

Drag and Drop Source	Drag and Drop Target	System Activity
Container unit stage	Freight order	You assign the container unit stage to the freight order and add missing locations to the freight order.
Container unit stage	Location in freight order	You assign the container unit stage to the freight order and add missing locations to the freight order. At the same time, you integrate the newly assigned location into the location sequence at the position predefined by the location.
Container unit stage	Compartment in freight order	You assign the container unit stage to the compartment of a freight order and add missing locations to the freight order.
Container unit stage	Railcar unit	You assign the container unit stage to the railcar unit and add missing locations to the location sequence for the railcar unit.
Container unit stage	Location in the railcar unit	You assign the container unit stage to the railcar unit and add missing locations to the railcar unit. At the same time, you integrate the newly assigned location into the location sequence at the position predefined by the location.
Container unit stage	Trailer unit	You assign the container unit stage to the trailer unit and add missing locations to the location sequence for the trailer unit.
Container unit stage	Location in the trailer unit	You assign the container unit stage to the trailer unit and add missing locations to the trailer unit. At the same time, you integrate

Drag and Drop Source	Drag and Drop Target	System Activity
		the newly assigned location into the location sequence at the position predefined by the location.
Container unit stage	Freight booking	You assign the container unit stage to a freight booking and add missing locations to the container unit.

Interactive Planning on the Map

Use

This function allows you to switch to a display of a map in the transportation cockpit (visual transportation cockpit) and perform planning there.

Prerequisites

- In Customizing for Transportation Management under [Basic Functions > Geographical Map > Define Settings for Geographical Map](#), you have activated the map display.
- You have defined a georouting strategy (optional). You can define georouting strategies in Customizing for Transportation Management under [Basic Functions > Geographical Map > Define Settings for Geographical Map](#). Note that the system ignores these strategies if you have defined more specific georouting strategies under [Master Data > Transportation Network > Transportation Lane > Distance and Duration Determination > Define Strategy for Distance and Duration Determination and Georouting](#). For more information, see [Settings for Distance and Duration Determination](#).
- If you want to use the map in the transportation cockpit, you have defined a page layout that includes the map. You do this by opening the [Define Page Layouts for Transportation Cockpit](#) app on the launchpad. For more information about page layouts for the transportation cockpit, see [Use of Profile and Layout Sets](#).
- If you want to use your own layout for the map display in the transportation cockpit, transportation network cockpit, or in the document overviews, you must have created your own layout for the map display in Customizing for Transportation Management under [Basic Functions > Geographical Map > Define Layouts for Geographical Map](#). In your layout, for example, you can do the following:
 - You can activate the display of the [Aggregated Information](#) window and specify whether only requirement documents or requirement documents and freight documents are to be considered.
 - You can specify which colors are to be used on the map.
 - You can specify that certain information is to be displayed as a tooltip or label.
 - You can specify that the first or last stage of a freight order, or both stages, are to be hidden.
 - You can specify that a specific map content is used for this layout by entering a country in the [Context "Country"](#) field.

Features

Adding Objects to the Map

All objects listed in the transportation cockpit can be displayed on the map for reference and interactive planning.

You can use the following methods for adding objects to the map individually or in combination with each other:

- Using the **Update Map** dropdown list.

The dropdown list on the application toolbar includes multiple functions for adding and removing objects to and from the map.

- Toggling an object's action button.

In the transportation cockpit tables, toggling an object's action button adds or removes the object to or from the map. (To display the actions buttons, in the transportation cockpit page layout, click the details button in the area's **Actions** column and then select the options **Add to Map** and **Remove from Map**.)

- Selecting the object.

In Customizing, you can specify that when you select an object in the transportation cockpit, it is immediately displayed on the map and remains displayed until deselected. You can activate this function under **Transportation Management > Basic Functions > Geographical Map > Define Layouts for Geographical Map** and select the option **Quick Add**.

If you select the Quick Add option, you can specify the selection behavior:

- Selection on the map overrules the selection on the transportation cockpit by document or resource type only.
- Selection on the map overrules the selection on the transportation cockpit completely.

To specify that the objects you add to the map continue to be displayed even after you refresh the transportation cockpit, select the **KeepObject** setting in Customizing under **Transportation Management > Basic Functions > Geographical Map > Define Layouts for Geographical Map > Layouts**.

To ensure the map fulfills your requirements, you can use the **Display Profile** dialog (on the map toolbar, click **Show Display Profile**) to specify how objects are represented on the map and what object information is displayed. Additionally, you can specify if the objects you add to the map are displayed or hidden depending on their status.

Also, when adding objects to the map, the **Aggregated Information** window in the map area updates automatically to display the current total weight and volume of the displayed objects (requirement documents). If you have selected **Consider All Documents in Aggregated Information Window** in Customizing, freight documents are considered as well. If you select a freight document, all stops and stages of this freight document are selected. The weight and volume of the most critical stage of the freight document are considered. If you only select a stop, the weight and volume of the assigned requirement documents (that are loaded or unloaded at this stop) are considered.

Planning Functions

You can carry out your planning on the map. If you select unplanned freight units and start planning from the context menu, for example, the system shows the possible assignments of the freight units to the resources. Select the assignments that you require.

You can also assign unplanned freight units or multirelations that contain unplanned freight units to resources by using drag and drop. When you do so, you can either assign the freight units to the resource or the resource to the freight units. You can use the proximity search in the context menu to find unplanned freight units and transportation units that are within a specific radius of a resource or a location.

If you want to split a stage that is displayed on the map, you can assign an unplanned freight unit or multirelation to a location using drag and drop. Once you have selected the required planning option, the system adds the selected location to the freight unit stage and updates the map accordingly. This process also applies to freight orders.

You can also display transshipment locations that are not part of your currently selected planning activities. You display these locations by choosing **Show Assigned Transshipment Locations** from the context menu.

In the context menu for a freight unit, you can add locations.

If a freight order does not have a vehicle assigned to it, you can also assign the vehicle by using drag and drop. To do so, you assign the freight order to a single vehicle resource or a multispot that contains at least one vehicle resource or passive vehicle resource and select the relevant planning option from the menu. You can use the proximity search in the context menu to find available resources that are within a specific radius. You can also specify the exact position of a resource at any time in the context menu of the map.

You can also assign a road freight order to another road freight order using drag and drop. Its stages are added to the target road freight order and then it is deleted.

In the case of resources on the map, you can find the assigned business documents by selecting the corresponding option from the context menu for the resource. The system searches for already planned freight orders and adds these to the map.

You can use the context menu of the map to do the following for example:

- Display different colors for multiple business documents selected simultaneously. This enables you to better differentiate between the documents. The color of each respective document is displayed in the **Color** column in the document list.
- Hide the first or last stage of a freight order or both stages.

The system displays all of your transportation planning activities on the map. When you leave the display of the map, your planning activities are also displayed in the table view of the transportation cockpit.

More Information

[Map Display](#)

[Map Functions](#)

Gantt Chart for Transportation Planning

Use

The Gantt chart provides a visual platform for performing transportation planning. With an intuitive and graphical user interface, the Gantt chart enables you to perform transportation planning visually, interactively, and more efficiently.

Prerequisites

You have defined the following for the Gantt chart:

- Color schemes and patterns
- Label schemes and tooltips
- Fields and field lists
- Hierarchies, views, and layouts

For more information, see Customizing for [Transportation Management](#) under [Basic Functions](#) > [Gantt Chart](#) .

Features

The Gantt chart has the following features:

- Transparent availability of resources and drivers

The chart panel presents you with an overview of the availability of vehicle resources, handling resources, and drivers over time. For example, different colors indicate whether a truck has a document assigned to it, or whether it is traveling, in downtime, or in non-working time. You can also check whether a handling resource is available or in downtime at a location and if a driver is available or if he is absent. Not just handling resources, you can use the Operating Time Calendar hierarchy to visualize the operating time configured for a specific location. The Operating Time Calendar displays the non-working times and activities occurring in that location. These activities are: loading, unloading, prepare, and finalize. The hierarchy also displays the overlaps of loading activities.

For more detailed information on how to enable operating times in Gantt Chart, see SAP Note [2671572](#) 

- Visibility of utilization of resources

You can display the level of consumed capacity of a vehicle resource at a specific point in time by choosing . The system displays the utilization curve of the selected vehicle resource in the chart panel. The tooltip along the utilization curve displays the loading utilization of a vehicle resource in terms of volume, weight, alternative quantity, or normalized quantity, depending on your settings in Customizing. The tooltip along the utilization line of a handling resource presents you with information on the available capacity, required capacity, and over-capacity of this handling resource at a specific point in time. By choosing , you can display loading or unloading activities that are planned for a handling resource at different times.

- Option to display or hide document separators in the Gantt chart

The Gantt chart presents you with document separators indicating the precise document start and document end. Document separators are shown as square brackets ("[]") with the left square bracket ("["") representing the document start and the right square bracket ("]") representing the document end.

You can choose to display or hide document separators in the Gantt chart by selecting the corresponding checkbox in the legend area.

Document separators apply to the following types of resources:

- truck
- trailer
- driver
- locomotive
- railcar
- container

For more detailed information on how to enable document separators, see SAP Note [2502946](#) 

- Option to display or hide time windows in the Gantt chart

The Gantt chart presents you with time windows indicating when goods are to be picked up and delivered. In the Gantt chart, time windows are shown as dotted lines and solid lines. You can choose to display the time windows by using the corresponding checkboxes in the legend area.

For more detailed information on how to enable document separators, see SAP Note [2502910](#) 

- Option to display and solve warnings in the Gantt chart

The Gantt chart presents you with warnings such as the **Missing Stage** warning or the **Incorrect Driver Assignment** warning that are indicated by a warning icon. You can solve a warning using the context menu. You can choose to display or hide this type of warning in the Gantt chart by selecting the respective checkboxes in the legend. The legend area also

displays the number of warnings for each warning type. Moreover, the number is dynamically updated once a warning of a certain type is solved or generated. Besides of the legend area, the **Number of Warnings** column in the table area presents you the total number of warnings on each row in a real-time manner. You can also deselect a certain warning type in the legend area so that warnings of that type are not considered when calculating the total number.

- Insights into planning and execution status of documents and transportation activities

If you transfer event-related data from Event Management to Transportation Management, you can gain real-time insights into the planning status and execution status of the documents and transportation activities in the chart panel. The statuses of documents and activities are presented in predefined colors and patterns, as described in the legend. Note that the Gantt chart automatically hides a legend item in the legend area if no corresponding documents or activities are shown in the chart area.

- Flexible and dynamic switch between different views

You can switch between a single view and a dual view (with its horizontal and vertical versions). In the chart panel of each view, you have the option to switch between a document view and an activity view. The dropdown lists and the related pushbuttons in the toolbar enable you to switch dynamically between views and hierarchies that are pre-defined in Customizing. For example, in a horizontal dual view, you can choose to display freight orders in the upper view and trucks in the lower view, so that you can assign unplanned freight orders to trucks using drag and drop.

- View all objects with one click

Bird Eye provides you with the capability to have a panoramic view of all objects with one simple click. With this feature turned on, the system intelligently adjusts the zoom rate so that all items are displayed on the chart without a need to move the horizontal scroll bar.

This feature can be implemented on all rows or just on visible rows, which can be configured in Gantt Chart Customizing. For more information, see SAP Note [2671506](#) .

You can also use the Bird Eye feature on a specific row using the **Bird Eye on Row** context menu. In this case, the zoom level will be adjusted to fit all objects on that row.

- Personalize table area appearance

You can personalize the appearance of the table area with the following options:

- Display or hide a column
- Modify the column sequence
- Adjust the column width

Moreover, you can save all personalization settings as a variant so that the table area is presented with the saved settings for the next time you load the Gantt chart.

- Configurable visualization of notifications

You can define the color and height of a certain type of notifications in the Gantt chart. Moreover, you can configure when to display the following types of notifications by specifying a utilization range value of vehicle resources:

- Empty run
- Low utilization
- Overcapacity

- Interactive planning in the Gantt chart

- Drag and drop

You can use drag and drop to perform planning in the Gantt chart. For example, you can assign documents to resources, change the location sequence of a document, and perform reassignments by using drag and drop in the chart panel.

To use resources more efficiently, you can simultaneously drag multiple capacity documents and then drop them to another one. For example, if you drag two freight orders and drop them to another one, the freight units previously assigned to the orders being dragged will be assigned to the order where you drop them. Additionally, you can perform cross-type drag and drop. For example, you can drag two ocean freight bookings and drop them to an air freight booking if needed.

You can perform drag and drop operations in the following modes:

- Assign and schedule based on drop time (default)

In this mode, the start time and end time of the object you drag and drop is determined by the position where you drop it. If you drag a document and drop it very close to another one, the system will automatically move the document you dropped right after or right before the other document.

- Assign and schedule based on original time

In this mode, you explicitly keep the original start time and end time of a freight unit or freight order when you drag and drop it to a resource. This mode makes sure that the schedule of the object you drag and drop remains intact while the default mode may cause a discrepancy in the timeline even if you try to avoid it.

This mode is typically used in a scenario where you want to assign a scheduled order to a vehicle resource, such as a driver or a truck. In this scenario, you usually want to keep the schedule of the order.

For more information on drag and drop options, see Activities.

o Attribute-based planning

In attribute-based planning, loading and unloading activities are color-coded to differentiate one freight unit from another. For example, the loading activities for freight units carrying chemical products and those for freight units carrying foods are in different colors. This helps you efficiently group or split freight units based on their visual appearances and thus generate optimized transportation plans from a holistic view.

For more information about how to enable attribute-based planning, see SAP Note [2671540](#) .

o Hyperlink and context menu

In the selection panel, the hyperlink allows you to navigate directly to the detailed screen of the selected item. In the chart panel, you can double-click a document or an activity of a document to go to the detailed screen of the document. Alternatively, you can use the **Show Document** or **Show Resource** context menu in the chart panel to display the details of a document or a resource. The context menu in the chart panel offers you options that provide easy navigation in a row, such as **Jump to First** and **Jump to Last**. There are also context menus that allow you to remove an assignment, split or merge freight unit stages, insert or remove a location, assign or unassign a driver, and solve warnings. Moreover, you can use the **Assign Schedule** context menu to plan a freight unit or a container unit. This feature enables you to assign a freight unit or container unit to a scheduled flight, voyage, road trip, or rail trip that you have found by search. This feature also plans loading and unloading activities for the freight unit or container unit. For example, in air freight bookings and ocean freight bookings, the loading time and the unloading time are set to the cutoff time and the available time in the schedule.

o Quick view

Gantt Chart presents a quick view page for both objects in the table area and in the chart area if you hover the mouse over an object for about one second or more. The quick view page displays basic information on the object. Take the railcar unit as an example - the quick view page displays the following information:

- Source Location
- Destination Location

- Departure Date
- Arrival Date
- Resource Capacity
- Remaining Capacity
- Equipment Type

Hyperlinks on the quick view page enable you to perform various operations depending on its type. For example:

- You can click the Location hyperlink to navigate to the details page of a location.
- You can click the Show on Map hyperlink to display the object on the Map.
- You can also click the Email hyperlink to launch an email program.

For more information about how to configure quick view, see SAP Note [2671529](#) .

- o Selection of items in the Gantt chart

In the chart panel, you can select two or more items using the CTRL key. You can also select all items in a row by choosing the **Select All** context menu. If you select one document or activity that appears more than once in multiple rows of the Gantt chart, the system automatically selects the same document or activity in all rows. In a dual view, selection in one view does not influence selection in the other view, and you can select items in both views at the same time.

- o Editable fields for capacity documents

You can edit certain fields for a capacity document directly in the table area of a Gantt chart without needing to open the capacity document. Further , depending on how you update the field, the Gantt chart dynamically updates the corresponding row in the chart area to reflect the change. For certain fields, you can also specify their value using search results.

The following table lists the fields that can be edited directly in Gantt charts.

Capacity Document	Editable Field
Road freight order	Departure date
	Departure time
	Arrival date
	Arrival time
	Carrier
	Means of Transport
	Vehicle
Freight booking	Departure date
	Departure time
	Carrier
	Means of Transport
	Airline code (air booking only)

Capacity Document	Editable Field
	Flight number (air booking only)
	Voyage ID (ocean booking only)
	Train number (rail freight booking only)
	Vessel (ocean booking only)
	Availability date
	Availability time
	Cargo Cut off date
	Cargo Cut off time
Trailer Unit / Container Unit / Railcar Unit	Departure date
	Departure time
	Carrier
	Trailer/Railcar/Container
	Train number (railcar unit only)
	Equipment Type (container unit only)
	Equipment Group (container unit only)
	Means of Transport

For more detailed information on how to enable editable fields for capacity documents, see SAP Note [2502783](#) 

- Integration with the transportation cockpit

The Gantt chart, integrated into the transportation cockpit, interacts well with other existing functions. For example, you can select one or more freight units in the freight unit list, and a resource in the Gantt chart, and then choose **Plan Selected Items** to assign the selected freight units to the resource. You can also update maps, generate transportation proposals, and run optimizer planning for the selected items in the Gantt chart and in the lists and hierarchies of the transportation cockpit.

If you enable the Smart Add function, any capacity documents or requirement documents that have been selected in other UIBBs, for example the freight units list, in the Transportation Cockpit are automatically visualized in the Gantt chart. Furthermore, the Gantt chart also displays the objects related to those selected documents. For example, if a freight unit is selected in the freight units list, in addition to the freight unit itself, the Gantt chart also displays the freight order related to the freight unit and the vehicle resource assigned to the freight order. If the vehicle resource has other freight orders assigned, those freight orders are also displayed. This function also works in a multi-window environment.

i Note

- If you enable the Smart Add function, the Gantt chart does not show any shapes until you select one or more items in other UIBBs. This makes it very likely that the Gantt chart shows nothing when the cockpit is initially loaded as no items are selected in any UIBBs at that time.
- If you enable the Smart Add function, you can specify the selection behavior:
 - Selection on the Gantt chart overrules the selection on the transportation cockpit by document or resource type only.

- Selection on the Gantt chart overrules the selection on the transportation cockpit completely.

For more information about the Smart Add function, see SAP Note [2671518](#) .

- Configurable visual appearance and layout

You can define color schemes and patterns, field lists, hierarchies, views, and layouts for the Gantt chart. For more information, see Customizing for [Transportation Management](#) under [Basic Functions > Gantt Chart](#) .

Activities

You can perform the following planning operations in the activity view of the chart panel in the Gantt chart using drag and drop:

- You can assign an unplanned trailer unit stage to a road freight order by dragging the travel activity of the unplanned trailer unit stage and dropping it on the travel activity of the road freight order.
- You can change the location sequence including source or destination location of a document by dragging one travel activity and dropping it on another travel activity. Alternatively, you can change the location sequence of a document by dragging a loading or unloading activity and dropping it on a travel activity.
- You can reschedule a document by dragging any activity of the document and dropping it on the desired time in the chart panel.
- You can couple a truck and a trailer by dragging the travel activity of the trailer unit and dropping it on the travel activity of the road freight order.
- In a multi-pick scenario, you can change the sequence of loading locations in a document by dragging the loading activity at one location and dropping it on a travel activity.
- In a multi-drop scenario, you can change the sequence of unloading locations in a document by dragging the unloading activity at one location and dropping it on a travel activity.
- You can reassign freight units from one road freight order to another by dragging the loading or unloading activity of one road freight order and dropping it on a travel activity, loading activity, or unloading activity of another freight order.

i Note

By dragging the loading or unloading activity of one road freight order, you remove all the loaded or unloaded freight units.

- You can move a location from one road freight order to another, including any freight units that are planned to be transported from or to this location. You do this by dragging the loading or unloading activity of one road freight order and dropping it on a travel activity, loading activity, or unloading activity of another freight order. The location and the freight units are then deleted from the original road freight order.
- You can reassign the resource of one or more documents by dragging one travel activity of the document and dropping it on the chart panel of another resource.

i Note

In this case, you can drop the travel activity on the chart panel of a resource in the document view.

- You can create a trailer unit and couple it to an existing road freight order by dragging a travel activity of the road freight order and dropping it on the chart panel of the trailer.

i Note

In this case, you can drop the travel activity on the chart panel of a resource in the document view.

- You can create a trailer unit and assign a freight unit by dragging a travel activity of the freight unit and dropping it on the chart panel of the trailer.
- You can create a freight order and assign a freight unit by dragging a travel activity of the freight unit and dropping it on the chart panel of the truck.
- You can reassign one or more freight units by dragging the travel activities of the freight units and dropping them on the chart panel of another document.
- You can assign one or more transportation stages of a road freight order to a driver by dragging the travel activity of the transportation stages and dropping them on the chart panel of a driver.

i Note

This function is only available for drivers who are to be assigned to road freight orders per transportation stage.

You can perform the following planning operations in the document view of the chart panel in the Gantt chart using drag and drop:

- You can assign an unplanned trailer unit stage to a road freight order by dragging the unplanned trailer unit stage and dropping it on the road freight order.
- You can reschedule a document by dragging that document and dropping it on the desired time in the chart panel.
- You can reassign the resource of one or more documents by dragging a document and dropping it on the chart panel of another resource.
- You can reschedule a document by dragging it and dropping it on another position in the chart panel of the same resource or driver.
- You can create a trailer unit and couple it to an existing road freight order by dragging a road freight order and dropping it on a trailer.
- You can create a trailer unit and assign a freight unit by dragging a freight unit and dropping it on the chart panel of the trailer.
- You can create a freight order and assign a freight unit by dragging a freight unit and dropping it on the chart panel of the truck.
- You can reassign one or more freight units by dragging the freight units and dropping them on the chart panel of another document.
- You can reassign the driver of one or more road freight orders by dragging the freight orders and dropping them on the chart panel of the new driver to be assigned to these freight orders.

i Note

This function is only available for single drivers who are assigned to entire freight orders.

- You can create a road freight order with a driver already assigned to it by dragging freight units or trailer units and dropping them on the chart panel of the driver.

i Note

This function is only available for drivers who are assigned to a particular truck by default.

Additionally, in the context menu, you can insert locations to and remove locations from trailer units, freight orders, and freight units. You can also use default routes for the transport of trailer units and freight units. For more information on default routes, see [Use of Default Routes in VSR Optimization](#).

More Information

[Overview of the Gantt Chart Functions](#)

Overview of the Gantt Chart Functions

Use

The Gantt chart screen provides different functions that allow you to work effectively.

Features

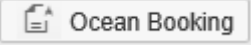
The following table lists and explains some of the available functions:

Icon on the User Interface	Function	Description
n/a	Display Single View	A view that comprises the following elements: • Selection panel that refers to the left pane of the Gantt chart and is constructed as a hierarchy • Chart panel that shows the planning status over time for each item on the left • Time axis above the chart panel
n/a	Display Multiple Views	A view that consists of multiple single views that are aligned either vertically or horizontally. You can choose to display different items in the views, for example, freight orders in the first view and trucks in the second view. You can drag and drop items in the chart panel of one view to the chart panel of another view. You can switch between display of single view and multiple views using the Add Gantt Chart and Remove Gantt Chart functions.
n/a	Synchronize Scrollbars	If you have multiple views displayed, you can choose whether the displayed views are scrolled simultaneously when you move the horizontal scrollbar of either of the views or not. This function enables you to display the same time frame of the views when moving the scrollbar of either of them or to display different time frames in different views.
	Document View	The chart panel displays documents such as freight orders and trailer units. You can switch to an activity view at any time.
	Activity View	The chart panel displays transportation activities such as loading, coupling, and traveling. You can switch to a document view at any time.

Icon on the User Interface	Function	Description
	Rescheduling Mode	In the rescheduling mode, you can drag one activity and drop it to another position in the same row. The system reschedules the document by scheduling the dragged activity to the desired time, without changing the stop sequence of the document.
	Resequencing Mode	In the resequencing mode, you can drag one activity and drop it to another activity in the same row. The system changes the stop sequence of the document.
	Settings	You can switch on and off specific functions for your current session, for example, Display Current Time , Display Cursor Time , Show Planning Horizon Boundaries .
	Legend	The legend shows the color schemes and patterns for the documents, transportation activities, planning and execution statuses, notifications, nonavailable times, vehicle load utilization, handling resource utilization, and warnings. Also, you can choose to display or hide the following types of legend items: • Notifications • Warnings • Document separators • Time windows
	Zoom In	This function enables you to gain a closer view of the items in the chart panel across shorter time intervals.
	Zoom Out	This function enables you to gain a broader view of the items in the chart panel across longer time intervals.
	Bird Eye	You can use this pushbutton to make the system intelligently adjust the zoom rate so that all items are displayed on the chart without a need to move the horizontal scroll bar.
n/a	Sort Sequence	In the selection panel, you can sort the sequence of entries in the table/hierarchy by one individual column (ascending or descending) for each view.
n/a	Filter	You use this function to find entries that meet certain filter conditions. You can apply one or more filter conditions to a list in the Gantt chart. If a list consists of two or more levels, you can only filter at the root level. For example, the system displays a “truck with driver” list. The truck is the root level of the list. The drivers that are assigned to the truck are a sub level. You can only filter the list by truck.

Icon on the User Interface	Function	Description
		To delete your filter, choose the Filter pushbutton. In the Change Filter dialog box, choose Delete Filter. Note that you can filter by the following date formats: • DD.MM.YYYY • MM/DD/YYYY • MM-DD-YYYY • YYYY.MM.DD • YYYY/MM/DD • YYYY-MM-DD You can also filter by the 24-hour time format “HH:MM:SS” (for example 14:05:10). We recommend you enter the date and time in the filter in the same format that you use in the date and time column of the selection panel in the Gantt chart. When you enter both a date and a time, you must leave a space between the date and the time.
	Show Utilization	The system displays the utilization curve of the selected vehicle resource in the chart panel. The tooltip along the utilization curve shows the percentages of consumed weight, volume, alternative quantity, or normalized quantity, depending on your settings in Customizing. You are notified of low load utilization and over-capacity by pre-defined colors, as shown in the legend.
	Hide Utilization	The system hides the utilization curve of a vehicle resource.
	Show Details	You can use this pushbutton to display the documents or transportation activities of a vehicle resource that overlap at a particular period of time. You are notified of an overlap by a pre-defined color, as shown in the legend. Alternatively, you can double-click the overlap bar to display overlap details. For handling resources, you can use this pushbutton to display the loading and unloading activities that are planned for a handling resource at different times.
	Hide Details	You can use this pushbutton to hide the documents or transportation activities that overlap for a vehicle resource. You can only see the overlap notification in a pre-defined color. Alternatively, you can double-click the overlap bar to hide overlap details. You can also use this pushbutton to hide loading and unloading activities that are planned for a

Icon on the User Interface	Function	Description
		handling resource at different times.
n/a	Fix	This pushbutton fixes the documents and transportation activities to prevent them from being changed by automatic and manual planning.
n/a	Unfix	This pushbutton removes the fixed status of the documents and transportation activities.
n/a	Scheduling	Scheduling determines start and end times for transportation activities, considering multiple constraints such as time windows for the activities. Scheduling is triggered for freight orders and is based on your settings in the planning profile. If you run scheduling for a resource, all freight orders assigned to this resource are scheduled. For more information, see Scheduling .
	Cancel	This pushbutton deletes freight orders and trailer units. It also removes assignment of freight units, stops, and trailer units from documents. You can delete a freight order or a trailer unit either by deleting the whole document or by deleting all of its travel activities. You can unassign a freight unit from a document by deleting the freight unit's load, unload, or travel activity. You can unassign a trailer unit from a freight order by deleting the coupling or uncoupling activity. If you delete a travel activity from a freight order or a trailer unit, all loaded and unloaded freight units are removed from both stops of this stage. The stops are also removed from the freight order or the trailer unit. Note that if you delete the travel activity of the first stage, only the first stop is removed; if you delete the travel activity of the last stage, only the last stop is removed. If you remove load or unload activity from a freight order or a trailer unit, you unassign freight units from a stop, but the stop is not removed.
 Road Freight Order	Create Road Freight Order	You use this pushbutton to create a new road freight order from scratch. If you select a truck and use this pushbutton, the system creates a new road freight order and assigns the truck to the new road freight order.
 Trailer Unit	Create Trailer Unit	You use this pushbutton to create a new trailer unit from scratch. If you select a trailer and use this pushbutton, the system creates a new trailer unit and assigns the trailer to the new trailer unit.
Create Freight Document	Create Freight Unit	You use this pushbutton to create a new road freight order for an unplanned freight unit stage.
 Air Booking	Create Air Booking	You use this pushbutton to create an air freight booking from scratch. You can then plan the loading and unloading activities for the air freight booking. To do this, save the newly created freight booking, click the corresponding hyperlink in the selection

Icon on the User Interface	Function	Description
		panel to navigate to the detailed screen, and then edit the loading and unloading activities in the Locations section. Alternatively, you can assign the air freight booking to a scheduled flight. The system then plans the loading and unloading activities automatically according to the data specified in the schedule.
	Create Ocean Booking	You use this pushbutton to create an ocean freight booking from scratch. You can then plan the loading and unloading activities for the ocean freight booking. To do this, save the newly created ocean freight booking, click the corresponding hyperlink in the selection panel to navigate to the detailed screen, and then edit the loading and unloading activities in the Locations section. Alternatively, you can assign the ocean freight booking to a scheduled voyage. The system then plans the loading and unloading activities automatically according to the data specified in the schedule.
	Copy	You use this pushbutton to copy one or more objects, such as a road freight order and a trailer unit.
n/a	Display Planning Horizon Boundaries	You can display vertical lines in the chart panel that indicate the start and end date and time of your planning horizon.
n/a	Display Current Time	You can display a vertical line in the chart panel that indicates the current time.
n/a	Display Cursor Time	You can display the time to which the cursor points in the chart panel. The time is indicated by a vertical line that moves with the cursor.
n/a	Resize Panels	You can adjust the size of the selection panel and the chart panel by moving the divider between them for each view.
n/a	Resize Views	You can resize the views by moving the divider between them.

More Information

[Gantt Chart for Transportation Planning](#)

Freight Document Overview

Use

You can use this function to display a location-based view of your resources and requirements (for example, freight units, transportation units (passive vehicle resource), and compartments):

- All transportation stops of a freight unit as well as their sequence
- All activities performed at these transportation stops, for example, loading and unloading of freight units and containers or coupling and uncoupling of railcars and trailers
- Information, for example, the start of the loading and unloading, as well as of the transport

In addition, you can use various editing options.

Features

You have the following editing options:

- Change the sequence of the transportation stops (not possible for freight orders that are based on a schedule)
- Add transportation stops (not possible for freight orders that are based on a schedule)
- Delete transportation stops (not possible for freight orders that are based on a schedule)
- Add freight units (that is, stages) to a freight order
- Delete freight units and transportation units (that is, stages) from a freight order
- Move freight units and container units from one compartment to another (within a freight order)
- Change assignment of freight units and transportation units to transportation stops

For all of these options, the system performs a series of checks. This allows the system to check, for example, dependencies between freight units as well as incompatibilities (see [Incompatibilities](#)).

Activities

The freight document overview is available on the user interfaces for the freight order and the freight booking as well as in the transportation cockpit. To display the freight document overview in the transportation cockpit, open the **Transportation Cockpit** app on the launchpad. Note that the freight document overview is only displayed for **one** freight order.

Generation of Transportation Proposals

Use

The VSR optimizer (see [VSR Optimization](#)) uses this function to generate alternative transportation proposals for the selected freight unit of which you can then use one. Unlike in the case of an optimization run, here the VSR optimizer searches for alternative transportation proposals.

Prerequisites

- You have defined optimizer settings and have assigned them to your planning profile (see [Planning Profile](#)). For example, you have made the settings for generating transportation proposals and have specified which planning strategy you want to use. If you want to specify your transportation proposal preferences in more detail, you have selected the **Do Not Create Transp. Prop. Immediately** checkbox. You do this by opening the **Create Transportation Cockpit Settings** app on the launchpad.

i Note

You can use the settings from your planning profile and adjust them for each transportation proposal generation. To do this, select the [Change Proposal Settings](#) checkbox.

- If you want to personalize the page layout to display transportation proposals, open the [Define Page Layouts for Transportation Proposal](#) app on the launchpad.

You can then include the personalized page layout in the page layouts of the transportation cockpit.

i Note

If you want to specify and change specific transportation proposal preferences for planning a freight unit in the transportation cockpit, you have selected the [Calculate](#) and [Reset Preferences](#) checkboxes in the page layouts for the transportation proposals. You previously made the [Transportation Proposal Preferences](#) screen area visible in your page layouts.

Features

A transportation proposal contains the following information:

- Route (sequence of locations of a freight unit)
- Freight order
- Assignment of freight unit to freight order

When it generates transportation proposals, the system considers the selected freight unit as well as indirectly linked freight orders, transportation units, and freight units (see [Context Determination](#)). The system does **not** change these indirectly linked freight orders and freight units. However, it does consider them as a constraint for the selected freight unit, which means that they can result in restrictions for the selected freight unit. For example, no vehicle resources are available for the selected freight unit if they are used by indirectly linked freight units.

The system generates transportation proposals only for the unplanned transportation stages of a freight unit that you have selected in the transportation cockpit. In doing so, the system generates either new freight orders or, if you have selected existing freight orders in the transportation cockpit, it upsizes these existing freight orders for the transportation proposal and adjusts them to the times specified in the transportation proposal.

In the case of the transportation stages of a freight unit that have already been planned, the system only generates transportation proposals if these transportation stages are not fixed.

i Note

- Planned transportation stages that the system determines using context determination are always fixed in the VSR optimizer.
- The generation of transportation proposals for multiple transportation stages is only supported if these transportation stages belong either to the same requirement document or to multiple requirement documents that were created during freight unit building for the same preceding document.

SAP delivers the standard planning strategies VSR_DEF and VSR_1STEP for generating transportation proposals. You use the standard planning strategy VSR_1STEP to specify that carrier selection is to be performed directly after transportation proposals have been generated. For more information, see [One-Step Planning](#) and [Planning Strategies](#).

When transportation proposals are generated, a dangerous goods check is performed if you have activated it. For more information, see [Considering Dangerous Goods](#).

You can use the route planner to display the connections between the locations in the transportation proposal on a geographical map. For more information, see [Path Finder](#).

To plan a freight unit with one or more freight unit stages in the transportation cockpit, you can specify in detail your preferences for the transportation proposal in question. For more information, see [Planning Profile](#).

If you want the system to generate transportation proposals for a package unit, container unit, or railcar unit to which freight units are assigned, the VSR optimizer only takes the package unit, container unit, or railcar unit into account. However, if you have not defined time windows for the package unit, container unit, or railcar unit, the VSR optimizer takes the time windows of the assigned freight units into account.

Activities

You can generate transportation proposals as follows:

- From interactive planning
- From the user interface for creating forwarding orders
- From the user interface for creating forwarding quotations

Command Line Planning

Use

Within the transportation cockpit, you can carry out various planning operations without having to use the mouse.

Prerequisites

In the page layouts for the transportation cockpit, you have activated the command line. You do this by opening the [Define Page Layouts for Transportation Cockpit](#) app on the launchpad.

Features

You define the commands for carrying out the required planning operations according to the following rule:

<Input Element ID> – <Target Element ID>

The element ID is defined in the [Index](#) column, which is available for all tables in the transportation cockpit. The element ID is a number that is unique for every element on the user interface. An element ID that is used for a freight unit, for example, is not also used for the resource. This means that the element ID identifies the source and the target uniquely.

To decrease the time required for entry, you can combine multiple commands into a single entry. To combine multiple commands, you enter them in the command line and separate each command with a plus (+). These commands are then processed sequentially after you press ENTER.

Activities

Enter the required command in the command line.

Example

Example 1: Simple Planning of Freight Units on a Vehicle Resource

The freight units to be planned have the indexes 5, 6, and 7. The vehicle resource that you want to use to transport the freight units has the index 2. To assign the freight units to the vehicle resource, you enter the following command in the command line:

```
5 6 7 – 2
```

For more examples, see the system documentation.

Conversational Planning

Within the transportation cockpit, you can carry out various planning operations using natural language.

i Note

- Use of this AI-assisted feature may require additional entitlement and authorization. Please consult your SAP account executive for more information.
- The features described here are only available in SAP S/4HANA Cloud Private Edition.

Prerequisites

- You have configured the settings related to Intelligent Scenario Lifecycle Management (ISLM) that are required to use generative AI in Transportation Management (TM).

First, you have requested an SAP Business Technology Platform (SAP BTP) service key for the intelligent scenario GENAI_TM_COPL_GPT40. Make sure you have noted down the name of the intelligent scenario, as you will need it later in the setup process. For more information, see [Requesting Access to SAP AI Services on SAP Business Technology Platform \(SAP BTP\)](#). Then you have performed the subsequent steps. For more information, see Customizing for **Transportation Management** under **Basic Functions** > **Generative Artificial Intelligence**.

For more information about ISLM setup, see [SAP Delivered Generative AI Scenarios](#).

- You have defined a generative AI profile of type **Conversational Planning** (conversational planning profile). In this profile, you have indicated one or several capabilities, that is, planning operations, that you would like to perform in the transportation cockpit using generative AI.

For more information, see Customizing for **Transportation Management** under **Basic Functions** > **Generative Artificial Intelligence** > **Define Generative AI Profile**.

- You have assigned your conversational planning profile to your page layout. By doing so, a new area will be available at the top of the transportation cockpit including an input field that allows you to enter your prompts. To assign the profile to the page layout, open the **Define Page Layouts for Transportation Cockpit** app from the launchpad.
- If you want to enrich certain lists and hierarchies of the transportation cockpit with information provided by generative AI, you need to indicate in the corresponding field catalogs whether the field **Conversational Planning Response** is to be displayed by default or using personalization.

For more information, see [Field Configuration for the Transportation Cockpit](#).

For more information about setting up conversational planning, see SAP Note [3573759](#) (AI use case Conversational Planning in S/4HANA Cloud Private Edition).

Features

A new screen area at the top of the transportation cockpit allows you to enter your prompts. You can do the following:

- You can select one or several of the following objects using natural language:
 - Freight unit stage
 - Freight order
 - Truck type
 - Truck resource

❖ Example

- Select all freight orders going to Spain
 - Select the truck with the highest weight capacity
- You can perform the following planning operations using natural language:
 - Assign freight units to freight orders
 - Assign freight units to trucks
 - Assign trucks to freight orders

❖ Example

- Assign all freight units from New York to the small truck
 - Assign all freight unit stages from Paris to freight order 123
- You can fill an extra column **Conversational Planning Response** in some lists and hierarchies with additional information generated by generative AI. The benefit of this extra column consists in providing information, which is not visible otherwise saving you from the time-consuming task of manually checking all objects for the required information.

The following lists and hierarchies are supported:

- Road freight order list and hierarchy
- Freight unit stage list and hierarchy
- Truck resource list

❖ Example

- Which freight units contain milk?
- Which freight units go to Bavaria?

Several Business Add-Ins (BAIs) are available to enhance this function according to your needs. For example, you can use these BAIs to adjust data, such as model parameters, associated with your ISLM scenario or data that is associated with the generative AI calls, such as messages and responses. Or you can enhance the context that is taken into account for a given capability for example by adding additional object information. For more information, see Customizing for **Transportation Management** under:

- [Business Add-Ins \(BAIs\) for Transportation Management](#) > [Basic Functions](#) > [Generative Artificial Intelligence](#) >
- [Business Add-Ins \(BAIs\) for Transportation Management](#) > [Planning](#) > [Conversational Planning](#) >

Activities

Enter your prompt in the input field and select **Submit**. The system performs the requested capability and displays the result in the corresponding lists and hierarchies.

i Note

- The system only considers selected entries as context.
- The system considers entries in the order that is given by the object index.

Additional Information

This chapter contains additional information that is relevant for using conversational planning, such as supported context information.

The following context information can be made available to the large language model (LLM):

Supported Context Information

Freight Unit Stage	Road Freight Order	Truck Resource/Truck Type
Index	Index	Index
Selected	Selected	Selected
Destination City	Destination City	Registration Number *
Destination Country	Destination Country	Temperature Control
Destination Location	Destination Location	Home Location *
Source City	Source City	Weight Capacity
Source Country	Source Country	Volume Capacity
Source Location	Source Location	Quantity Capacity
Weight	Weight	Status
Products	Weight Capacity	Equipment Group
Status	Volume Capacity	Equipment Type
Earliest Pick-Up Date	Quantity Capacity	-
Latest Pick-Up Date	Location Sequence	-
Earliest Delivery Date	Registration Number	-
Latest Delivery Date	Products	-
-	Status	-
-	Departure Date	-
-	Arrival Date	-

Automatic Planning

Automatic planning offers the following functions:

- Vehicle scheduling and routing (VSR) optimization

For more information, see [VSR Optimization](#).

- Load consolidation

For more information, see [Load Consolidation](#).

- Load planning

For more information, see [Load Planning](#).

VSR Optimization

Use

The aim of VSR optimization (VSR = vehicle scheduling and routing) is to assign freight units to capacities (for example, vehicle resources) in a cost-effective way while adhering to constraints (see [Constraints for VSR Optimization](#)), and to determine the sequence of deliveries and transportation dates/times.

Prerequisites

- You have defined the master data for VSR optimization, for example, the network connections to the optimization server and the maximum number of users. You do this on the SAP Easy Access screen under **► Cross-Application Components ► Remote Control and Communication Framework ► Administration ► Settings ► Edit Destinations ►**.

- You have created a planning profile (see [Planning Profile](#)) and have assigned the following settings to it:

- Optimizer settings

For example, you have specified the planning strategy (see [Planning Strategies](#)) that you want to use. Open the **Create Optimizer Settings** app on the launchpad.

- Constraints and costs settings

To do this, open the **Create Constraints and Costs Settings** app on the launchpad.

- Capacity selection settings

Open the **Create Capacity Selection Settings** app on the launchpad.

You can also assign a selection profile for freight orders and freight bookings as well as incompatibility settings. If you want to perform one-step planning, you must also assign carrier selection settings (see [One-Step Planning](#)).

- You have defined transportation lanes. Open the **Define Transportation Lane** app on the launchpad.
- In the case of multimodal transportations or transshipments, you have defined transshipment locations (see [Transshipment Location](#)) or default routes (see [Use of Default Routes in VSR Optimization](#)). Open the **Assign Transshipment Location** or the **Create Default Route** app on the launchpad.

- If you want to control the input or the results of the VSR optimization, you have made the settings for the Business Add-Ins (BAdIs) [BAdI: VSR Optimizer Preprocessing](#) and [BAdI: VSR Optimizer Postprocessing](#).

For more information, see Customizing for Transportation Management under ► [Transportation Management](#) ► [Business Add-Ins \(BAdIs\) for Transportation Management](#) ► [Planning](#) ► [VSR Optimizer](#) ► [VSR Optimizer Preprocessing and Postprocessing](#) ►.

Features

VSR optimization generates a transportation plan consisting of planned freight orders from the optimization data (freight units and capacities, see [Optimization Data](#)). When doing this, it processes freight orders already available from a previous VSR optimization run or manual planning. VSR optimization inserts the freight units step-by-step into the transportation plan. Then VSR optimization modifies these initial solutions step-by-step, for example, by loading a freight unit on to another capacity or by changing the delivery sequence of a capacity. When doing this, the optimizer tries to minimize the total costs (see [Total Costs](#)) while adhering to the constraints. At the end of the planning run, VSR optimization returns the best solution found. For example, if the costs defined for non-delivery of a freight unit are lower than the costs for transporting the freight unit, the freight unit is not delivered.

→ Recommendation

To achieve the best possible planning result, we recommend that you find a happy medium between creating extremely large freight units and extremely small ones. The larger the freight units are, the simpler the planning activities in VSR optimization. If the freight units are smaller, there may well be a higher theoretical potential for optimization, but in practice, planning in VSR optimization is made more difficult due to the higher number of freight units.

❖ Example

You have a truck with a capacity of 20 tons. You are planning to transport 20 tons of flour in 1 kg packages. Neither extreme is advisable here for the reasons given above. You should not create a single freight unit with a weight of 20 tons, nor should you create 20,000 freight units with a weight of 1 kg each.

Furthermore, you can generate multiple alternative transportation proposals, in other words, transportation plans, for each freight unit and can then choose to use one of them. For more information, see [Generation of Transportation Proposals](#).

SAP delivers the planning strategies VSR_DEF and VSR_1STEP as standard for the optimization. You can use the planning strategy VSR_1STEP to control whether the system is to perform carrier selection immediately after VSR optimization. For more information, see [One-Step Planning](#) and [Planning Strategies](#).

The results of the VSR optimization run are explained in the explanation tool. For more information, see [Explanation Tool](#).

Activities

You start VSR optimization in the transportation cockpit with [Optimizer Planning](#). You have the following options:

- You can perform VSR optimization for all entries.
- You can perform VSR optimization for selected entries only. In this case, if you do not choose any capacities, the system takes into account all capacities. If the system is to consider freight orders and freight bookings, you must also specify this.
- You can perform VSR optimization for selected entries as well as for all resources and equipment types. If the system is to consider transportation units, freight orders, and freight bookings, you must also specify this.

To perform VSR optimization in the background, call report /SCMTMS/PLN_OPT. Choose a planning profile and a requirements selection profile (and a freight order selection profile, if required), and execute the report in the background. The system displays

the generated freight orders as the result. For more information, see [Planning Run](#).

To end a VSR optimization run prematurely, on the SAP Easy Access screen choose **Cross-Application Components** > **Remote Control and Communication Framework** > **Administration** > **Display Active Session**. Then choose the pushbutton to stop the process.

Related Information

[Consideration of Scheduling Constraints](#)

[Validity of Objects](#)

Optimization Data

Definition

Data for which the system optimizes planning during VSR optimization (see [VSR Optimization](#)).

Structure

Optimization data includes the following data:

- All selected freight units
- All freight units (planned and unplanned) that lie within the planning horizon
- All data that the system determines using context determination

The VSR optimizer does not change this data.

- All capacities that match your settings for capacity selection

In addition, transportation lanes between all the relevant locations of the VSR optimization run (source and destination locations of the chosen freight units, transshipment locations, and depot locations) are part of the optimization data.

The following example describes the effects of transportation lanes on vehicle resources (that is, vehicle resources for which you have **not** selected the **Passive** checkbox for the relevant means of transport) without a schedule. These restrictions do not apply to schedule vehicles.

If a vehicle with means of transport 'truck' transports a freight unit from A to B, then you must define a direct transportation lane from A to B for the defined validity period for the means of transport 'truck'. It is not sufficient to have defined transportation lanes from A to C and from C to B, for example. The VSR optimizer does not combine these to create the transportation lane A to B.

This is also the case if you want to deliver one freight unit from A to C and one freight unit from A to B. It is also not sufficient in this case to have defined transportation lanes from A to C and from C to B. The freight unit can only be delivered from A to B if you have defined a direct transportation lane from A to B.

We also recommend that you define a direct transportation lane between A and B in the following case: You want to deliver a freight unit from A to B. There is no direct transportation lane defined between A and B but there is a transfer location. The freight unit can only be delivered if it is transferred at the transshipment location. To avoid unnecessary loading and unloading at a transshipment location, if a freight unit is shipped in a single vehicle on both transportation stages, we recommend that you define a direct transportation lane.

Total Costs

Definition

Costs that the system minimizes during VSR optimization (see [VSR Optimization](#)).

Structure

The total costs comprise the following costs:

- Freight unit-dependent costs
 - Earliness costs
 - Lateness costs
 - Costs for non-delivery
- Capacity-related costs
 - Fixed costs per freight order
 - Fixed costs per vehicle resource or equipment type
 - Load costs
 - Variable transportation costs per unit for the following dimensions:
 - Time
 - Distance
 - Intermediate stops
 - Transported quantity per distance
 - CO₂ emissions (see [Greenhouse Gas Emissions](#))

You define these costs in the constraints and costs settings that you assign to a planning profile (see [Planning Profile](#)).

Initial Solution

Definition

Solution that is generated, evaluated, and then optimized in VSR optimization (see [VSR Optimization](#)). There are two types of initial solution:

- The existing transportation plans from a previous VSR optimization run or manual planning form an initial solution.
- VSR optimization generates additional initial solutions by distributing freight units to capacities (for example, vehicle resources) using heuristic criteria.

Use

VSR optimization can only process, and, where necessary, improve existing transportation plans from a previous VSR optimization run or manual planning as an initial solution, if they meet the following prerequisites:

- They must use connections with transportation lanes.
- They must lie in selected capacities.
- They must not violate any incompatibilities (unless you have permitted this in the **Incompatibilities** field in the optimizer settings).
- If they are part of a multimodal transportation, this transportation must be valid for VSR optimization (for example, the transshipment locations must correspond to the transshipment locations defined in the hierarchy).

If the initial solution is invalid, VSR optimization tries to rebuild it as much as possible.

Constraints for VSR Optimization

Use

VSR optimization (see [VSR Optimization](#)) takes into account constraints when it assigns freight units to capacities. You can differentiate between hard and soft constraints:

- VSR optimization always adheres to hard constraints.
- You model soft constraints using penalty costs, for example, lateness costs that are part of the total costs.

Features

VSR optimization minimizes the total costs in the VSR optimization run while adhering to the hard constraints. By appropriately weighting the penalty costs in the total costs, you can influence the priority of adhering to the soft constraints. If, for example, you give a greater weighting to the lateness costs, this might lead to longer distances to fulfill the requested dates/times.

❖ Example

Hard constraints: Loading capacities of a vehicle resource. VSR optimization can only schedule freight units within the existing transportation capacities.

Soft constraints: Requested dates/times of freight units. VSR optimization can also schedule a freight unit after the requested delivery date. In this case, delay costs are incurred.

The following table shows the planning data that VSR optimization considers as constraints:

Planning Data	Constraint	Remark
Handling resources for loading and unloading as well as operating times	Hard	In addition to the operating times, VSR optimization also takes into account loading capacities for the handling resources.
Incompatibilities (see Incompatibilities)	Hard	N/A

Planning Data	Constraint	Remark
Pickup	Soft	VSR optimization can also schedule the pickup of the freight unit before or after the requested pick-up date/time. In this case, earliness or delay costs are incurred. You can define tolerances for the premature or delayed pickup without penalty costs (see Time Windows).
Pickup	Hard	VSR optimization can only schedule the pickup for the acceptable pick-up date/time. You can define tolerances within which a premature or delayed pickup is allowed (see Time Windows).
Pickup	Soft and hard	A combination of the options described above
Delivery	Soft	VSR optimization can also schedule the delivery of the freight unit before or after the requested delivery date/time. In this case, earliness or delay costs are incurred. You can define tolerances for the premature or delayed delivery without penalty costs (see Time Windows).
Delivery	Hard	VSR optimization can only schedule the delivery for the acceptable delivery date/time. You can define tolerances within which a premature or delayed delivery is allowed (see Time Windows).
Delivery	Soft and hard	A combination of the options described above
Journey to depot for own means of transport	Hard	N/A
Loading and unloading durations	Hard	In the case of schedule-based freight orders and freight bookings, the system does not take into account loading and unloading durations.
Driving time and working hours constraints	Hard	You must specify in your scheduling settings that the system is to consider these constraints (see Considering Driving Time and Working Hours Constraints).
Schedules of means of transport (trains, ships, aircraft, and trucks)	Hard	N/A
Minimum goods wait time at the transshipment location	Hard	If there is a freight unit stage before or after the schedule and if additional cut-off times have been defined in the schedule or freight booking, VSR optimization takes these cut-off times into account instead of the minimum goods wait time at the transshipment location.

Planning Data	Constraint	Remark
Maximum goods wait time at the transshipment location	Hard	If there is a freight unit stage before or after the schedule and if additional cut-off times have been defined in the schedule or freight booking, VSR optimization takes these cut-off times into account instead of the maximum goods wait time at the transshipment location.
Distance traveled	Soft	VSR optimization attempts to minimize the distance.
Distance traveled (for each freight order)	Hard	The distance must not be greater than the value defined in the constraints and costs settings.
Distance traveled (for each freight order)	Soft and hard	A combination of the options described above
Duration	Soft	VSR optimization attempts to minimize the duration of the road freight orders.
Duration (for each freight order)	Hard	The duration must not be greater than the value defined in the constraints and costs settings.
Duration (for each freight order)	Soft and hard	A combination of the options described above
Number of intermediate stops	Soft	VSR optimization attempts to minimize the number of intermediate stops.
Number of intermediate stops (for each freight order)	Hard	The number of intermediate stops must not be greater than the value defined in the constraints and costs settings.
Number of intermediate stops (for each freight order)	Soft and hard	A combination of the options described above
Decreasing capacities (see Capacity Decrease)	Hard	N/A
Compartment capacities	Hard	N/A
Capacities of means-of-transport combinations	Hard	N/A
ADR limit	Hard	The number of ADR points for a resource must not be exceeded (see ADR 1.1.3.6 Points Check Calculation).

Explanation Tool

Use

You can obtain a detailed overview of the data that flowed into the VSR optimization run and the results that were output. In this way, you can quickly and easily recognize how the VSR optimization has optimized the costs.

In the same way, you can obtain a detailed overview of the data that flowed into the load optimization run and the results that were output. In this way, you can quickly and easily recognize how the load optimization has optimally used the loading space while taking into account various restrictions such as the maximum axle load.

Prerequisites

- You have set user parameter /SCMTMS/EXP to X to allow for the logging of the optimizer data for optimization runs that you start interactively.
- You have the authorization to work with the explanation tool (authorization object T_EXP_USR).

Features

The explanation tool for **VSR optimization** displays the following information:

- Input for the VSR optimization run, for example, information about the following objects:
 - Incoming business documents (freight documents or requirement documents)
 - Network
 - Resources
- Results of the VSR optimization run, for example:
 - Status information about business documents (freight documents or requirement documents, or newly created business documents)
 - Analysis data, for example, reasons why business documents could not be delivered
 - Solution overview

All planning-relevant costs per transportation plan are listed here. If you created transportation proposals, the costs for each transportation plan are displayed. You can see a breakdown of the costs.

In addition, you receive information about the utilization of the solution and the freight documents.

From the explanation tool you can, in part, go to the associated business documents directly. You can use the pushbutton for calling customer functions to call the Business Add-In (BAI) /SCMTMS/PLN_EXP, to execute your own functions for particular selected objects.

The explanation tool for **load optimization and load consolidation** displays the following information:

- Input for the load optimization or load consolidation run (incoming business documents and resources)
- Results of the load optimization or load consolidation run
 - Here, the exact position of each package in the loading space is displayed. In addition, the stack, level, and row of each package is displayed.
 - Furthermore, the current weight and the maximum permitted weight of the vehicle are displayed along with the current and maximum axle load for each axle group.

You can use the pushbutton for calling customer functions to call the Business Add-In (BAI) /SCMTMS/PLN_EXP, to execute your own functions for particular selected objects.

Activities

To call the explanation tool for VSR optimization and load optimization, choose **Explanation** in the transportation cockpit. To call the transportation cockpit, open the **Transportation Cockpit** app on the launchpad.

You can call the application log from the SAP Easy Access screen by choosing ► **Cross-Application Components** ► **Processes and Tools for Enterprise Applications** ► **Remote Control and Communication Framework** ► **Log Display** ►.

For more information about the available application IDs, see SAP Note [3703775](#) .

For more information about the planning run in the background, see [Planning Run](#).

We recommend scheduling the deletion program /SCMTMS/PLN_EXP_DELETE regularly in the background.

Incremental Planning

Use

You can use this function to plan freight orders and freight bookings in automatic planning incrementally and to assign freight units to existing freight documents. This function is available for VSR optimization and the transportation proposal.

Prerequisites

In the optimizer settings of your planning profile, you have specified which type of incremental planning you want to use. To do this, open the **Edit Optimizer Settings** app on the launchpad.

Features

You can use incremental planning for the following automatic planning functions:

- Transportation proposal

You use this function to consider existing freight documents in the generation of transportation proposals for new freight units. The system adds the capacities and locations of the freight units that you want to assign to the existing freight documents. If the new locations do not fit in the location sequence of the existing freight document or if there is not enough capacity to assign the new freight units to the existing freight documents, the system creates a new freight document with the new freight units and a new location sequence.

- VSR optimization

You use this function to specify whether the VSR optimizer can delete existing freight documents or only enhance them. You have the following options:

- VSR optimization without incremental planning

The VSR optimizer does not plan incrementally. Instead, it deletes existing freight documents and automatically creates new ones.

- VSR optimization with incremental planning

The VSR optimizer cannot delete existing freight documents. Instead, it adds the capacities and locations of the freight units that you want to assign to the existing freight documents. If the new locations do not fit in the location sequence of the existing freight document or if there is not enough capacity to assign the new freight units to the existing freight documents, the system creates new freight documents.

- Incremental planning without inserting new locations

The system assigns freight units to existing freight documents only if there is enough capacity. Furthermore, the system assigns freight units to existing freight documents if the locations of the freight units fit into the existing location sequence. For example, a freight unit from A → C can be added to a freight order from A → B → C but not to a freight order from A → D or from B → A. You are not permitted to insert new locations into existing freight documents.

- o Incremental planning without new capacity documents

The system assigns freight units to existing capacity documents until the existing capacity has been reached. New capacity documents are not created. As a result, some freight units may remain unplanned.

- o Incremental planning with changes to location sequence

During incremental planning the system adjusts the existing location sequence in the capacity documents. Only the first and last locations cannot be changed.

For example, you have a location sequence with four locations: location A in the west, location B in the southwest, location C in the northeast, and location D in the east. If you have planned the location sequence manually the locations could have the sequence A → C → B → D.



Incremental Planning Without Changes to Location Sequence

The VSR optimizer can now swap the locations B and C so that the sequence is A → B → C → D.



Incremental Planning with Changes to Location Sequence

Note, however, that the VSR optimizer must not change the location sequence in such a way that B or C is at the beginning or end of the sequence. This means, for example, that the following location sequences are not allowed:

- $B \rightarrow A \rightarrow C \rightarrow D$
- $A \rightarrow C \rightarrow D \rightarrow B$

However, the VSR optimizer is allowed to insert a new location X at any point, such as $X \rightarrow A \rightarrow C \rightarrow B \rightarrow D$ or $A \rightarrow C \rightarrow B \rightarrow X \rightarrow D$. The prerequisite for this is that you have **not** chosen incremental planning without inserting new locations.

The VSR optimizer is not allowed to move the locations A and D.

Activities

Select the freight units and freight documents to be planned in the transportation cockpit and choose **Optimizer Planning**. For more information about optimizer planning, see [VSR Optimization](#).

Select the freight units and freight documents to be planned in the transportation cockpit and choose **Transportation Proposals**. For more information, see [Generation of Transportation Proposals](#).

Rough-Cut Planning

Use

You can use rough-cut planning to schedule the pre-carriage and on-carriage of a transportation without having to define specific transportation lanes.

Prerequisites

- You have assigned a transshipment location to your transportation zone or a location in the master data and have defined a fixed duration for the rough-cut planning. To assign transshipment locations, open the [Assign Transshipment Location](#) app on the launchpad.
- You have configured the relevant settings and assigned the optimizer settings to your planning profile in the optimizer settings under [Consider Constraints](#). For more information, see [Planning Profile](#).

Features

Rough-cut planning is based on a fixed transportation duration within which you can reach the transshipment location assigned to specific location from this location.

i Note

When rough-cut planning is used, it must be used for both the pre-carriage and on-carriage. You cannot use rough-cut planning for the pre-carriage and detailed planning for the on-carriage, or vice versa.

If you have made assignments at various levels such as transportation zone or location level, the system uses the value for the most precise geographical location.

Example

You want to transport goods from Walldorf to Minneapolis. The source location `Walldorf` is in the transportation zone `Southern Germany`. Frankfurt airport is the transshipment location assigned to the transportation zone `Southern Germany`. The destination location `Minneapolis` is in the transportation zone `Midwest`. Chicago airport is the transshipment location assigned to the transportation zone `Midwest`.

You have a master flight schedule that connects Frankfurt and Chicago airports but do not have a transportation lane for transporting the goods between the source location in Walldorf to Frankfurt airport or between the destination location in Minneapolis and Chicago airport.

In the master data for the transportation zone `Southern Germany`, you have specified a rough-cut planning duration of six hours, and for the location `Walldorf` two hours.

If you select the [Use Rough-Cut Planning Where Defined](#) option in the [Rough-Cut Planning](#) selection list in the [Optimizer Settings](#) screen area in the planning profile, the system takes the rough-cut planning duration of two hours into account during planning for the transportation of goods to Frankfurt airport.

Validity of Objects

You can define a validity for transportation lanes and default routes. However, this cannot be displayed in VSR optimization.

When you start VSR optimization and VSR optimization comes across a transportation lane or default route, it assumes that this is valid across the whole planning horizon.

In optimizer preprocessing, the system uses the start of the planning horizon as a reference and transfers to VSR optimization all objects that are valid on this date and are relevant for the scenario. This results in the following behavior:

- VSR optimization does **not** consider objects that only become valid during the planning horizon.
- VSR optimization may still use objects that become invalid during the planning horizon for the whole planning horizon.

i Note

The validity of vehicle resources can be displayed in VSR optimization.

Related Information

[Definition of Transportation Lanes](#)

[Default Route](#)

[Use of Default Routes in VSR Optimization](#)

Load Consolidation

Use

You can use this function to optimize the distribution of freight units, package units, and consignment orders to the following resources: trucks, containers, trailers, and railcars or the corresponding equipment types. The system then generates the corresponding business documents: freight orders, container units, trailer units, and railcar units.

Prerequisites

- If you want to work with load planning-based consolidation, you have to fulfil the prerequisites for load planning. For more information, see [Load Planning](#).
- You have made the settings for load consolidation, for example, you have defined the engine that you want to use (row-stack approach or free positioning). You do this by opening the **Create Load Planning Settings** app on the launchpad.
- If you want to take the orientation of your products into account during load consolidation, you have to define product orientation profiles for your products and assign these to your products (**WM Packaging** view in the material master data). For more information, see Customizing for **Transportation Management** under **Master Data > Product > Define Product Orientation Profile**. If you want to use all orientations, you need to use the **Free Positioning** engine. The row-stack approach only supports the orientations **Bottom Straight** and **Bottom Turned**. If you have not defined a product orientation profile for your product or for the corresponding reference product, the system uses the two standard orientations **Bottom Straight** and **Bottom Turned**.
- If you want to take the stackability of products into account during load consolidation, you have to make the required settings. For more information, see [Stackability](#).

Features

The following modes are available for load consolidation:

- Capacity-based consolidation

With this form of load consolidation, the system creates freight orders or transportation units taking the dimensions of the affected resources or equipment types, such as weight or volume, into consideration.

- Load planning-based consolidation

With this form of load consolidation, the system creates freight orders or transportation units and then executes load planning.

For more information, see [Load Planning](#).

The system takes incompatibilities into consideration in both modes (see [Incompatibilities](#)).

Activities

You start load consolidation in the transportation cockpit. You have the following options:

- The system is to consolidate only the selected freight units and resources or equipment types.
- The system is to consolidate the selected freight units and all resources or equipment types.
- The system is to consolidate all freight units and resources or equipment types.

You can also run load consolidation in the background. For more information, see [Load Consolidation Run](#).

Load Planning

Use

You can plan the loading space of box trucks, trailers, semi-trailers, and containers in the transportation cockpit or directly in one of the following business documents:

- Road freight order
- Trailer unit
- Container unit

Planning is carried out at the level of business document items.

i Note

Load planning returns an approximate calculation of how the available loading space can be optimally used, taking into account the maximum axle loads and the maximum weight of a vehicle resource. The calculation is based on the master data that you enter for the resource. It is essential that you check the results calculated during load planning.

Prerequisites

- You have entered master data for the resources such as the height, width, and length of the loading space. For more information about the master data for load planning, see [Master Data in Load Planning](#).

Note that load planning doesn't support every attribute available within the vehicle resource master data. For more information, see [3696255](#) .

- You have made the settings for load planning, for example, you specified the engine you want to use (row-stack approach or free positioning). You do this by opening the [Create Load Planning Settings](#) app on the launchpad.
- You have specified in the business document the height, width, and length of the items to be planned.

To define the dimensions of a freight order item in the **forwarding order**, open the [Edit Forwarding Order](#) app on the launchpad. In the item hierarchy on the [Details](#) tab page, you can enter various package details such as the height, width or length for each package.

To define the dimensions of a freight order item in the **freight order**, open the [Edit Road Freight Order](#) app on the launchpad. In the item hierarchy on the [Details](#) tab page, you can enter various package details such as the height, width or length for each package.

To define the dimensions of an item in the **container unit**, open the [Edit Container Unit](#) app on the launchpad. In the item hierarchy on the **Details** tab page, you can enter various package details such as the height, width or length for each package.

To define the dimensions of an item in the **trailer unit**, open the [Edit Trailer Unit](#) app on the launchpad. In the item hierarchy on the **Details** tab page, you can enter various package details such as the height, width or length for each package.

To define the dimensions of an item in the **package unit**, open the [Edit Package Unit](#) app on the launchpad. In the item hierarchy on the **Details** tab page, you can enter various package details such as the height, width or length for each package.

Alternatively, you can define the dimensions of a business document item in the transportation cockpit by using drag and drop to assign a freight unit to a business document. The system automatically copies the dimensions of the freight unit item to your business document.

- In the page layouts for the transportation cockpit, you have activated the following functions:
 - In the **Orders Area**, you have activated load planning for freight orders and freight bookings
 - In the **Area for Transportation Units**, you have activated load planning for container units and trailer units
 - In the **Order Details Area**, you have activated the display of the load plan

To edit page layouts for the transportation cockpit, open the [Define Page Layouts for Transportation Cockpit](#) app on the launchpad. For more information, see [Use of Profile and Layout Sets](#).

- Optionally, you have defined your own layouts for the 3D load plan in Customizing for **Transportation Management** under [Basic Functions](#) > [Load Planning](#) > [Define Layouts for 3D Load Plan](#) and assigned them to your page layout. This is necessary, for example, to adjust the display and to activate manual load planning or drag and drop. For more information about the different layout settings, see [Load Plan](#).
- If you want to display layers as a separate hierarchy level in the load plan hierarchy, you have selected the **Display Layers** checkbox in Customizing for your hierarchy type for the load plan (**Consumer of the Hierarchy: Freight Document: Load Plan**). For more information, see Customizing for Transportation Management under [Planning](#) > [General Settings](#) > [Define Hierarchical Views for Business Documents](#).

You have also specified this hierarchy type for the load plan in your page layout (under [Pushbuttons for Content Toolbar](#) > [Embedded Configurations of Load Plan](#)).

- If you want to take the orientation of your products into account during load planning, you have to define product orientation profiles for your products and assign these to your products (**WM Packaging** view in the material master data). For more information, see Customizing for **Transportation Management** under [Master Data](#) > [Product](#) > [Define Product Orientation Profile](#). If you want to use all orientations, you need to use the **Free Positioning** engine. The row-stack approach only supports the orientations **Bottom Straight** and **Bottom Turned**. If you have not defined a product orientation profile for your product or for the corresponding reference product, the system uses the two standard orientations **Bottom Straight** and **Bottom Turned**.
- If you want to take the stackability of products into account during load planning, you have to make the required settings. For more information, see [Stackability](#).
- You are using a pure inbound or outbound scenario. Load planning supports the following scenarios:
 - Single pickup and multi-drop (outbound)
 - Multi-pickup and single drop (inbound)

However, “multi-pickup and multi-drop” is not supported. If you have an outbound scenario with returns or empties, “multi-pickup and multi-drop” can be supported. In this case, load planning ignores returns and empties (see SAP Note [3382660](#) .

Features

When you plan a road freight order on a box truck, trailer, or semi-trailer, the system takes into account capacity restrictions in terms of the height, width, length, and weight of the loading space. The same capacity restrictions also apply to the planning of a trailer unit on a trailer or semi-trailer. It also takes into account the maximum axle load of your box truck.

Furthermore, in the system you can add a split deck to your trailer or semi-trailer and consequently plan two decks.

When you plan a container unit on a container, the system also takes into account capacity restrictions in terms of the height, width, and length of the loading space as well as the total weight and the empty weight of the container.

In both the transportation cockpit and in the freight order, you can check the results of load planning as a 3D load plan or a hierarchical table (load plan hierarchy) containing the business document items. In the 3D load plan, you can show or hide individual objects as well as entire rows, columns, or stacks. The system also provides you with information about the current load distribution and utilization of the loading space. You can also create a load chart using the 3D load plan that shows the exact weight restrictions for each area of the resource, provided you have entered a value for the maximum axle load in the **Max. Weight on Axle Group** field. To display the load chart, right-click the load plan and choose **Switch Load Chart**.

More Information

[Master Data in Load Planning](#)

[Automatic Load Planning](#)

[Load Plan](#)

[Rules in Automatic Load Planning](#)

[Double-Deck and Split Deck Resources](#)

Double-Deck and Split Deck Resources

Transportation Management supports planning with both fixed and flexible split deck (double-deck) resources.

Prerequisites

- In the master data of your vehicle resource, you have specified the split deck position under **►►Load Optimization► Basic Data** .
- In the **Resource Viewer** app under **Sections**, you have specified information about the split deck, including the type (platform or beam), size, weight, and movement range. To assist you, there is help available for each input field.
- If there are packages that cannot be stacked, you have selected the setting **Non-Stackable Package** in the requirement document.

i Note

- Only flexible beam split-decks are supported. (The vertical position of a platform split-deck is fixed.)
- Flexible split decks are only supported in load planning and not in load consolidation.
- All packages must have the same footprint (length & width).
- Only one split deck per resource is supported, which must cover the length of the cargo body.

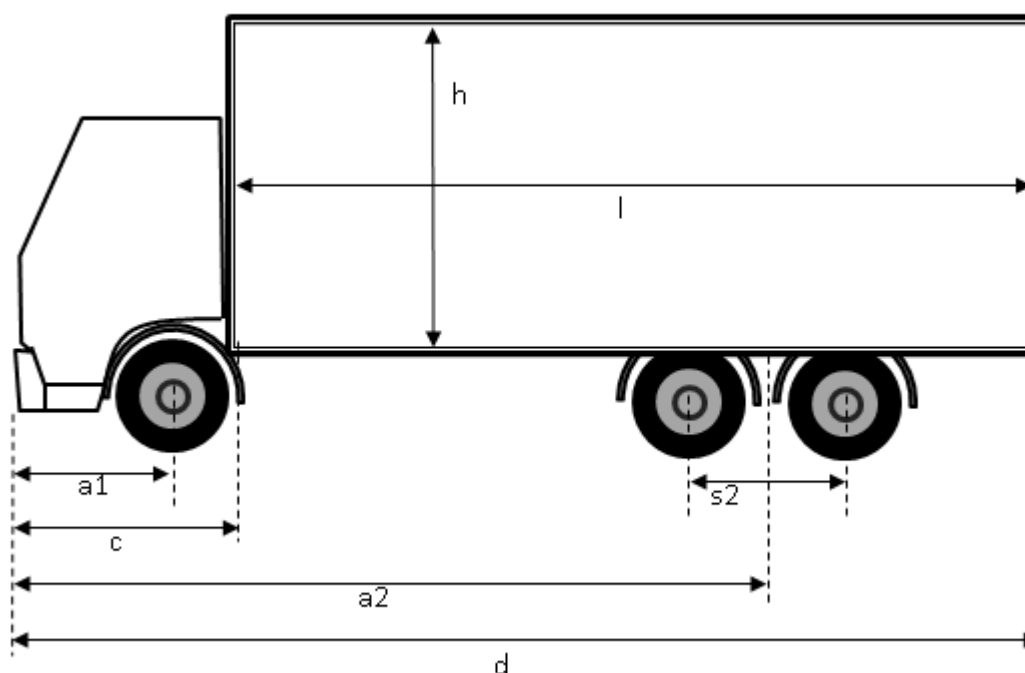
- All packages are planned with a straight orientation unless the package width is the longest dimension, in which case the package is planned with a turned orientation to ensure the package fits safely upon the beams of the split deck.

For example, a standard Europallet (length: 1.20m, width: 0.80m) is placed with a straight orientation to fit on the beam. If the pallet has a length of 1.0m and a width of 1.20m, it is placed with a turned orientation.

Master Data in Load Planning

Definition

You can enter various information about your vehicle resources in the master data for load planning. The figures below show a box truck and a trailer with some sample measurements:



Box Truck

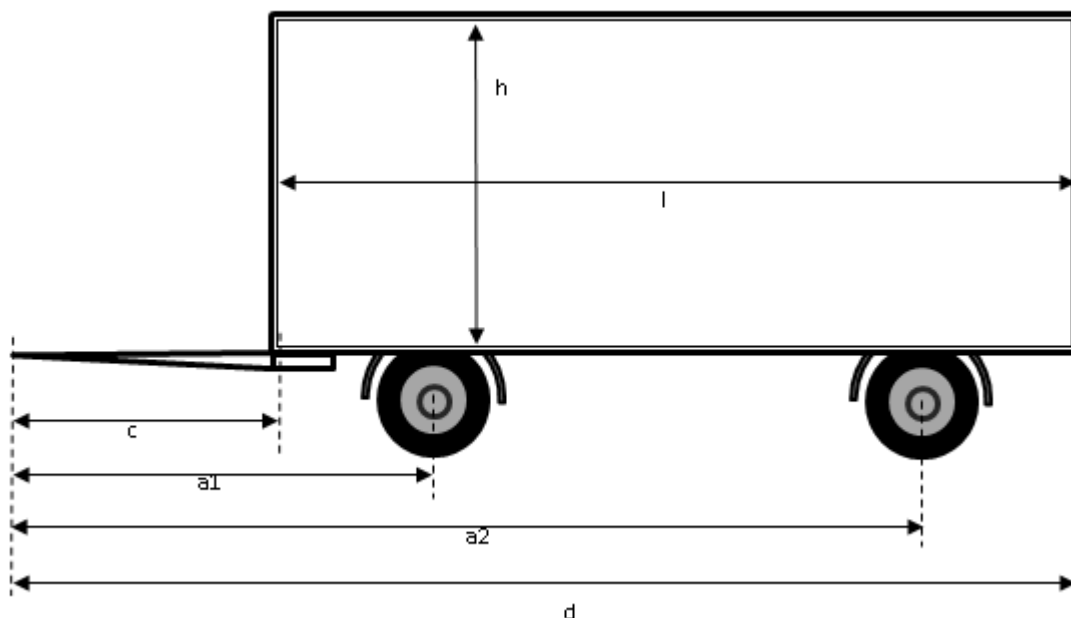
The following table explains the measurements from the figure above:

Measurement in Figure	Name in Master Data	Description
a1	Axle Group Distance 1	Distance between axle group 1 and the front of the box truck
c	Cargo Body Distance	Distance between the start of the loading space and the front of the box truck
a2	Axle Group Distance 2	Distance between axle group 2 and the front of the box truck
d	Connector Distance	Distance between the trailer coupling of the box truck and the front of the vehicle
s2	Distance Between Axles	Distance within axle group 2
h	Interior Height	Height of the loading space

Measurement in Figure	Name in Master Data	Description
l	Interior Length	Length of the loading space

i Note

You must also specify the width of the box truck so that the system can include the loading space of the box truck in automatic load planning.



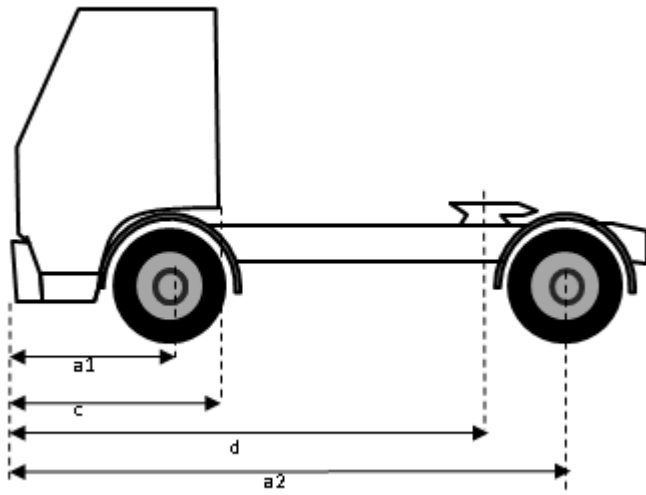
Trailer

The following table explains the measurements from the figure above:

Measurement in Figure	Name in Master Data	Description
c	Cargo Body Distance	Distance between the start of the loading space and the front of the trailer
a1	Axle Group Distance 1	Distance between axle group 1 and the front of the trailer
a2	Axle Group Distance 2	Distance between axle group 2 and the front of the trailer
d	Connector Distance	Distance between the trailer coupling of the trailer and the end of the trailer
h	Interior Height	Height of the loading space
l	Interior Length	Length of the loading space

i Note

You must also specify the width of the trailer so that the system can include the loading space of the trailer in automatic load planning.



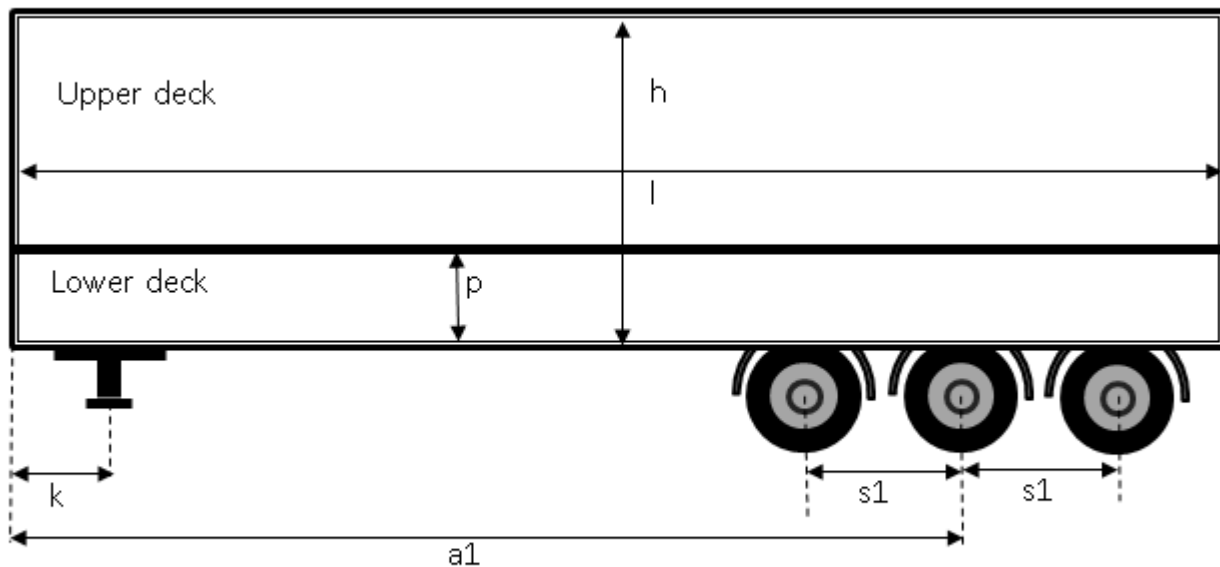
Tractor

The following table explains the measurements from the figure above:

Measurement in Figure	Name in Master Data	Description
c	Cargo Body Distance	Distance between the start of the loading space and the front of the tractor i Note Since a tractor does not have any loading space, Cargo Body Distance designates the length of the driver's cab as the start of the loading space.
a1	Axle Group Distance 1	Distance between axle group 1 and the front of the tractor
a2	Axle Group Distance 2	Distance between axle group 2 and the front of the tractor
d	Connector Distance	Distance between the tractor coupling and the front of the vehicle

i Note

You must also specify the width of the tractor so that the system can display the tractors in the 3D load plan.



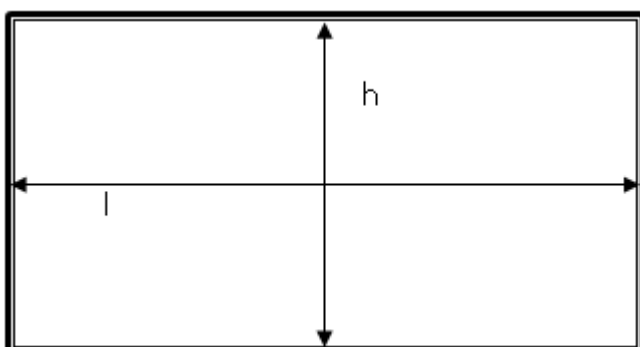
Semi-Trailer

The following table explains the measurements from the figure above:

Measurement in Figure	Name in Master Data	Description
a1	Axle Group Distance 1	Distance between axle group 1 and the front of the trailer
s1	Distance Between Axles	Distance within axle group 1
k	King Pin Distance	Distance between the king pin and the front of the trailer
h	Interior Height	Height of the loading space
l	Interior Length	Length of the loading space
p	Split Deck Position	Position in the trailer at which you have added the split deck

i Note

You must also specify the width of the semi-trailer so that the system can include the loading space of the semi-trailer in automatic load planning.



Container

The following table explains the measurements from the figure above:

Measurement in Figure	Name in Master Data	Description
h	Interior Height	Height of the loading space
l	Interior Length	Length of the loading space

i Note

You must also specify the width of the container so that the system can include the loading space of the container in automatic load planning.

Activities

To define this and other master data, open the [Define Resource](#) app on the launchpad.

Alternatively, on the SAP Easy Access screen choose [►Transportation Management►Master Data►Resources►Define Resources►](#).

Automatic Load Planning

Use

In automatic load planning, also known as load optimization, the optimizer for load planning and load consolidation creates a load plan for your business document. When it does so, it takes into account capacity restrictions in terms of the height, width, and length of the loading space as well as the weight. For vehicle resources with two axle groups, it also takes into account the maximum axle load.

You can perform automatic load planning based on different optimization approaches:

- Row-stack approach

This approach is grid-based. It is recommended for homogeneous loads, that means, packages with the same or similar footprint, such as Euro pallets.

- Free positioning

This approach allows positioning of the objects independent from rows and stacks. The goal is to both reduce the used loading space and improve the overall utilization of the truck by avoiding gaps between packages. This approach is recommended for heterogeneous loads, that means, packages with different footprints, such as boxes, cartons, and individually packaged large products.

For more information, see SAP Note [3349183](#) .

i Note

- Free positioning does not support all rules and functions of automatic load planning.

Prerequisites

- You have created master data for the resources for which you want to optimize the load. For more information about the master data for load planning, see [Master Data in Load Planning](#).
- You have defined the master data for load optimization, for example, the network connections to the optimization server and the maximum number of users. You can do this on the SAP Easy Access screen under ► **Cross-Application Components** ► **Processes and Tools for Enterprise Applications** ► **Remote Control and Communication Framework** ► **Administration** ► **Settings** ► **Edit Destinations** ►.
- You have created a planning profile (see [Planning Profile](#)) and have assigned the following settings to it:

- Settings for load optimization

You have specified the optimizer engine that you want to use for load planning. You do this by opening the **Load Planning Settings** app on the launchpad. Here you can also define various rules for load optimization. For example, you can specify the maximum difference in height between two adjacent stacks or define a loading pattern (see [Rules in Automatic Load Planning](#)). You can also specify the optimizer runtime here.

- Capacity selection settings

You do this by opening the **Create Capacity Selection Settings** app on the launchpad.

- Two Business Add-Ins (BADIs) are available that you can use to change the results of the optimizer for load planning and load consolidation. For more information, see Customizing for **Transportation Management** under ► **Business Add-Ins (BADIs) for Transportation Management** ► **Planning** ► **Optimizer for Load Planning and Load Consolidation** ►:

- **BADI: Optimizer for Load Planning and Load Consolidation - Preprocessing**
- **BADI: Optimizer for Load Planning and Load Consolidation - Postprocessing**

For more information, see SAP Note [3349184](#) .

- During an upgrade from an older release in which the old interface for the load planning optimizer was the default, the checkbox **Use Old Interface for Load Planning Optimizer** is automatically selected. If you want to use the optimizer engine for free positioning, you need to deselect this checkbox. For more information, see Customizing for **Transportation Management** under ► **Basic Functions** ► **General Settings** ► **Define General Settings for TM** ►.

If you still want to use load planning with the old interface, see SAP Note [3349183](#)  for more information.

Features

You can start automatic load planning from the transportation cockpit or directly in a business document. You can check the results of automatic load planning in a 3D load plan or in a hierarchical table (load plan hierarchy) containing all the planned items. For more information, see [Load Plan](#).

If automatic load planning was successful, the system automatically sets the load plan item status in the business document to **Planned**. If some items cannot be planned due to reasons of capacity, the system automatically sets the load plan status to **Invalidated**. For more information, see [Explanation of Statuses of Business Documents](#).

You can use the explanation tool for automatic load planning to compare your entries for automatic load planning with the results of the optimizer run. You can also use the explanation tool to view information on the current status of a load optimization run. For example, you can view the actual axle load of your box truck and trailer or your tractor and semi-trailer based on the business document items that were previously planned. For more information, see [Explanation Tool](#).

Activities

To start automatic load planning in the transportation cockpit, choose **Load Planning**.

To start automatic load planning from a business document, choose ► [Follow-Up Actions](#) ► [Perform Load Planning](#) ►.

To start automatic load planning from your worklist, select the business document to be planned and choose [Perform Load Planning](#).

You can call the explanation tool for automatic load planning using transaction RCC_LOG.

Rules in Automatic Load Planning

Use

In automatic load planning, you can define rules to be considered by the system during a load optimization run. You can choose between the following types of rules:

- Rules that apply to automatic load planning for all resource types. For more information, see [Load Planning Rules Independent of the Resource Type](#).
- Rules that apply only to the automatic load planning of the upper decks of double-deck trailers. For more information, see [Rules for the Load Planning of Upper Decks](#).

You can define different rules for the lower and upper deck of double-deck trailers.

However, you cannot combine one rule with another rule if the two rules contradict each other logically. For example, you cannot define two rules that simultaneously specify that you want the system to load all packages turned and all packages straight in automatic load planning.

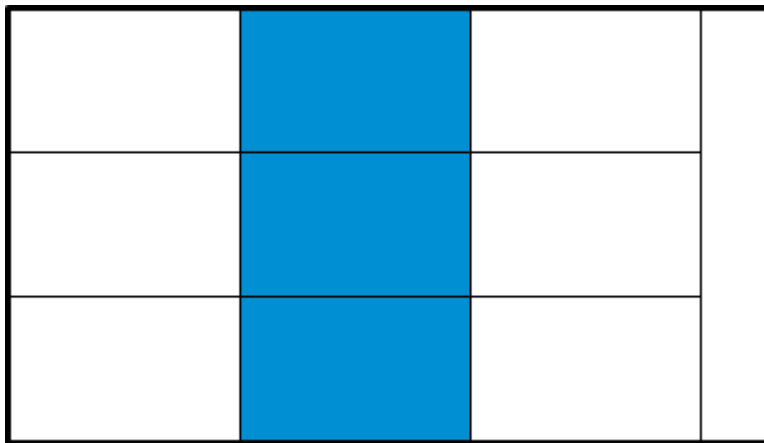
Structure

The combination of stack, row, and line indicates the exact position of each package loaded onto the resource.

The figures below show the meaning of the terms "stack", "row", and "line" in automatic load planning when viewed from above.

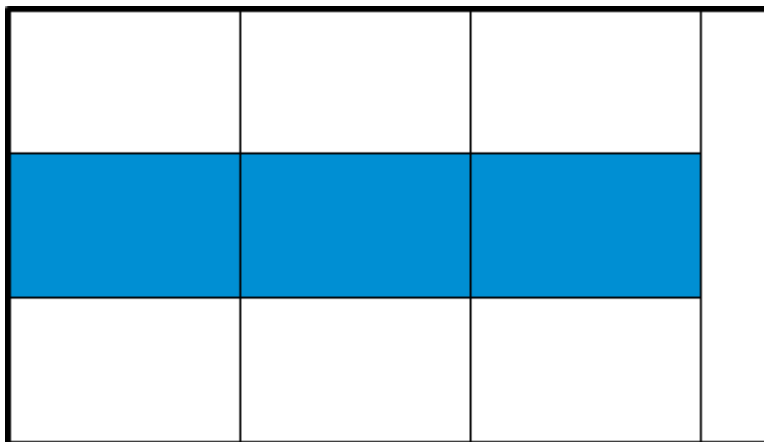
Stack

This figure shows several packages that are loaded onto the resource **one above the other**. In automatic load planning, these packages form a stack.



Row

This figure shows several packages that are loaded onto the resource **next to each other** from left to right. In automatic load planning, these packages form a row. The first row of a deck is always towards the start of the loading space.



Line

This figure shows several packages that are loaded onto the resource **one behind the other** from front to back. In automatic load planning, these packages form a line. The first line of a deck is always on the left side of the deck in the direction of travel.

Features

In your rules you can specify a combination of loading pattern and orientation for each row of your vehicle resource. The system considers this combination during a load optimization run.

Loading Pattern

You can define different loading patterns for each vehicle resource that apply to the entire resource. You can define different loading patterns for the upper and lower deck of a double-deck trailer that both apply to the entire deck. Alternatively, you can define rules for each resource that apply only to one adjacent row or to several adjacent rows.

When you define the rules for automatic load planning, you can select the following loading patterns:

- S (straight)

This loading pattern comprises only packages that are loaded onto the resource straight.

- T (turned)

This loading pattern comprises only packages that are loaded onto the resource turned.

- E

This loading pattern comprises the combination of several same-sized packages that are loaded straight and several same-sized packages that are loaded turned. The straight and turned packages sit flush at the end of the loading pattern. In loading pattern E, the packages that are loaded straight are on the **left** of the packages that are loaded turned.

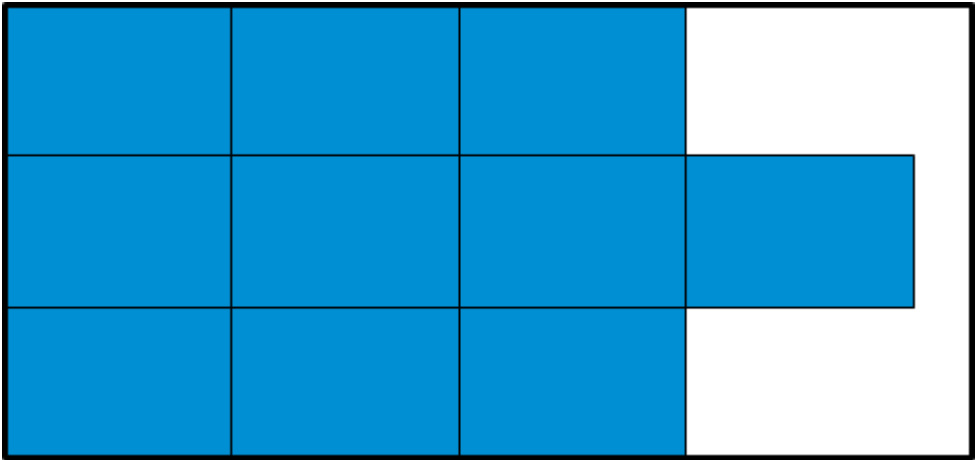
- F

This loading pattern comprises the combination of several same-sized packages that are loaded straight and several same-sized packages that are loaded turned. The straight and turned packages sit flush at the end of the loading pattern. In loading pattern F, the packages that are loaded straight are on the **right** of the packages that are loaded turned.

- P (pinwheel)

This loading pattern comprises a total of four same-sized packages that are loaded straight and turned in alternation so that they form a square with a small area that is free in the middle. As such, the shape of this loading pattern is similar to a chimney or a pinwheel.

The following figures show examples of the different loading patterns. In each of the selected examples, the loading pattern applies to the entire resource or the entire deck. The orientation of the packages is always centered.



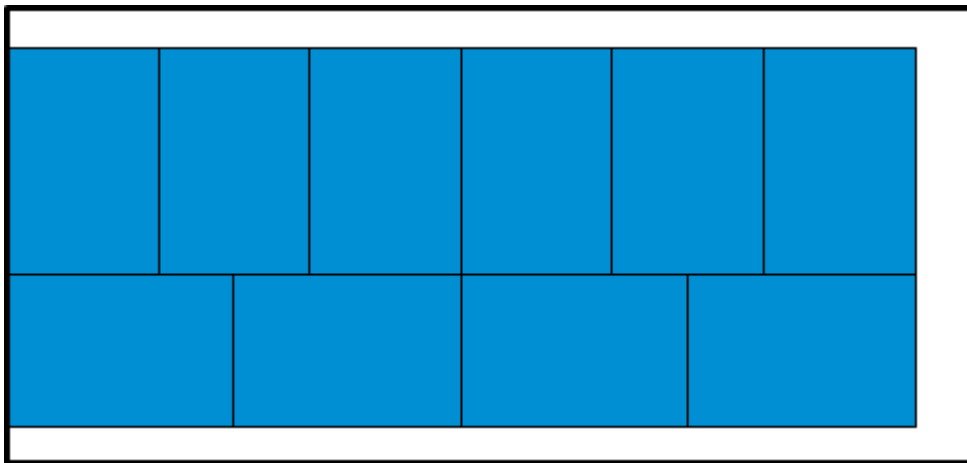
Loading Pattern S

This figure shows how all packages are loaded onto a resource in each row straight.



Loading Pattern T

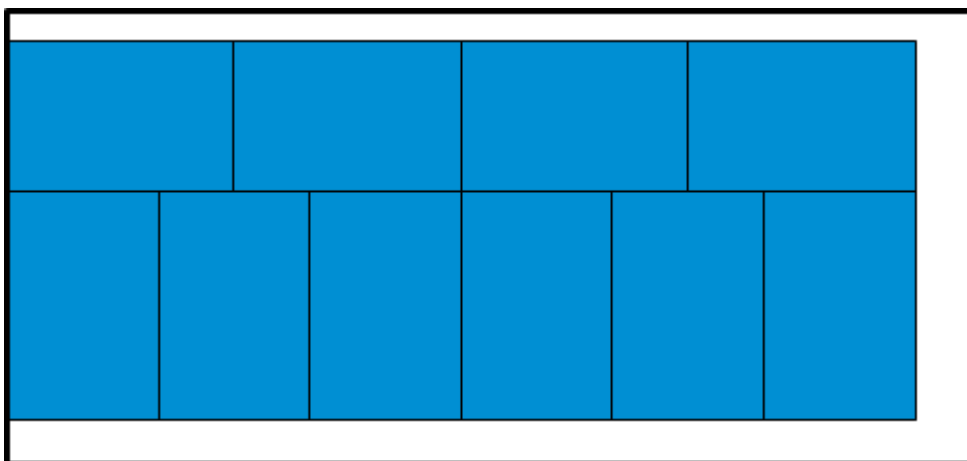
This figure shows how all packages are loaded onto a resource in each row turned.



Loading Pattern E

This figure shows how all packages are loaded onto a resource in line 1 straight and in line 2 turned. Loading pattern E always comprises several same-sized packages in line 1 that are loaded straight and several same-sized packages in line 2 that are loaded turned. The straight and turned packages sit flush at the end of a loading pattern. The loading pattern E is shown twice in this figure.

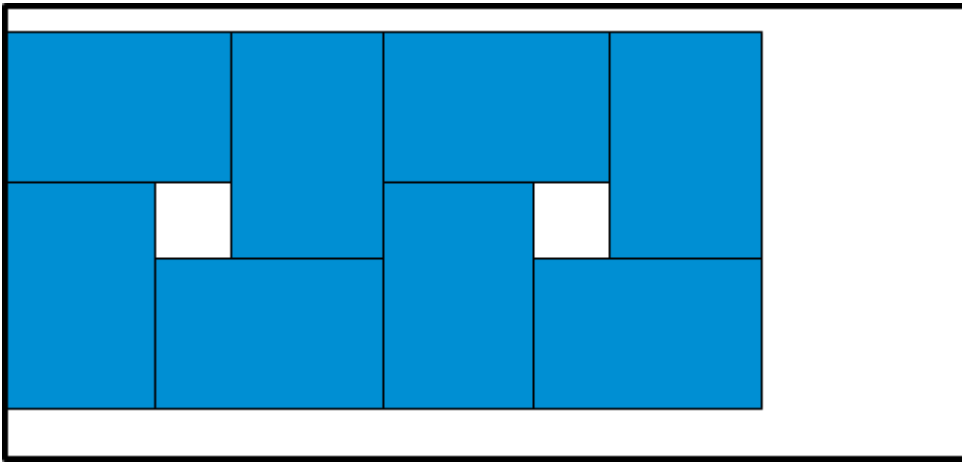
Depending on the resource type, you have to enter the following formula in the corresponding rule to get the loading pattern shown in this figure: E ; E



Loading Pattern F

This figure shows how all packages are loaded onto a resource in line 1 turned and in line 2 straight. Loading pattern F always comprises several same-sized packages in line 1 that are loaded turned and several same-sized packages in line 2 that are loaded straight. The straight and turned packages sit flush at the end of a loading pattern. The loading pattern F is shown twice in this figure.

Depending on the resource type, you have to enter the following formula in the corresponding rule to get the loading pattern shown in this figure: F ; F



Loading Pattern P

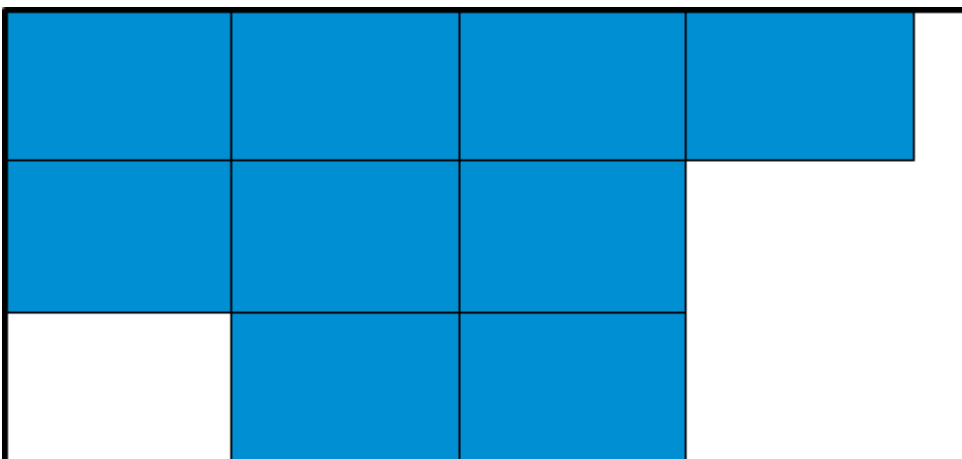
This figure shows how four same-sized packages are loaded onto a resource straight and turned in alternation so that they form a square with a small area that is free in the middle. As such, the shape of this loading pattern is similar to a chimney or a pinwheel.

Orientation

You can define each of these loading patterns with a specific orientation for all packages that are loaded in a row onto the resource. You can choose the following orientations:

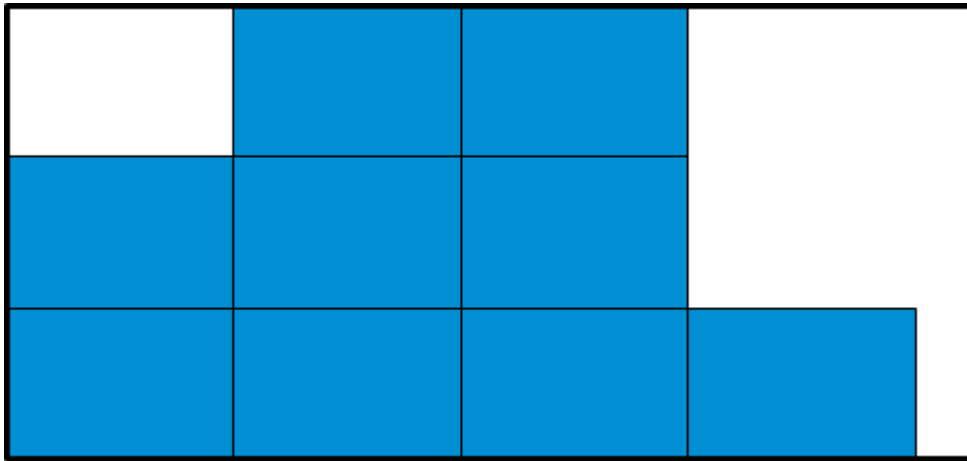
- R - right
- L - left
- C - centered
- D - distributed

The following figures show examples of the different orientations. In each of the selected examples, the orientation always applies to the entire resource or the entire deck.



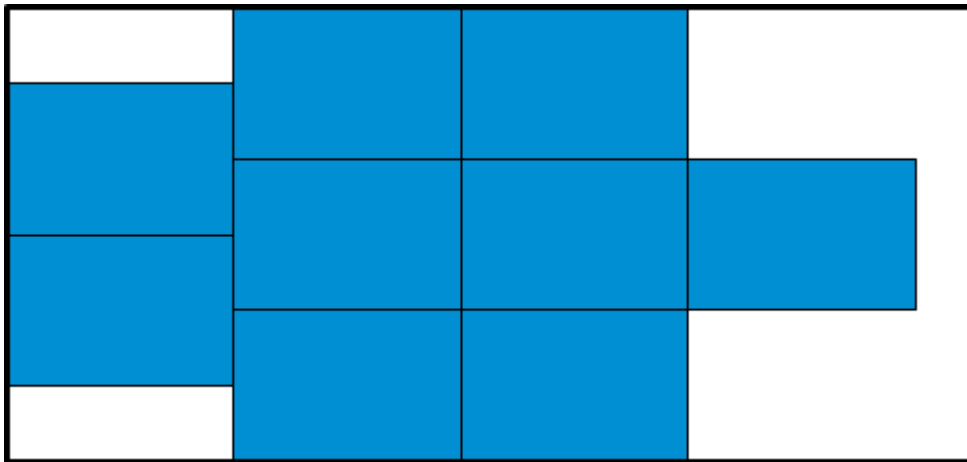
Orientation R

This figure shows how all packages are oriented on a resource in each row on the right in the direction of travel.



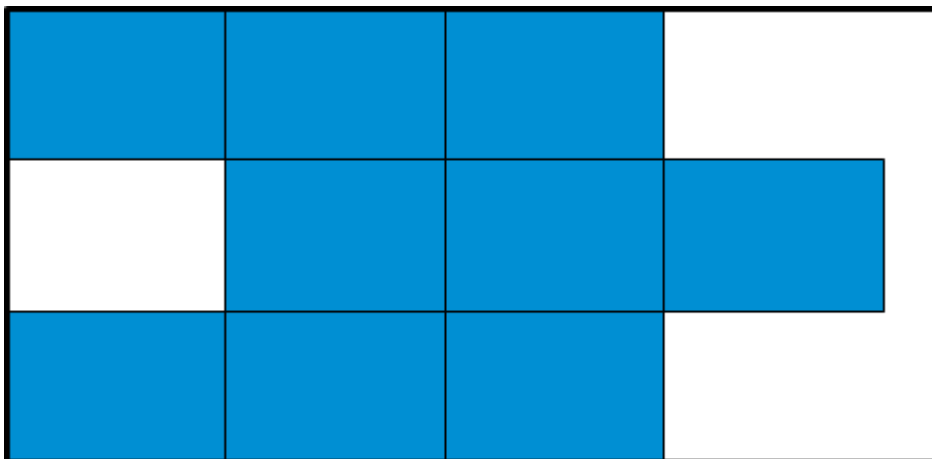
Orientation L

This figure shows how all packages are oriented on a resource in each row on the left in the direction of travel.



Orientation C

This figure shows how all packages are centered on a resource. If you do not enter any other orientation, during a load optimization run the system plans all packages on the resource with orientation C.



Orientation D

This figure shows how all packages are distributed across the row on a resource.

Activities

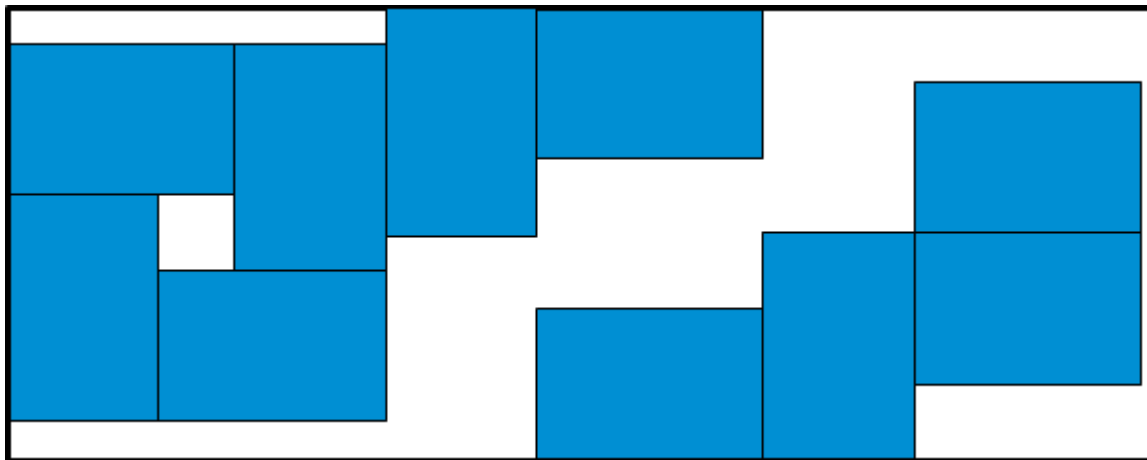
To define rules for your load optimization run, open the [Create Load Planning Settings](#) app on the launchpad.

To define a specific combination of loading pattern and orientation for the individual rows of a deck, insert a semicolon in the corresponding rules after each combination of loading pattern and orientation. This means that the first combination always

relates to the first row of the deck.

Example

This figure shows the result of a load optimization run using a rule that comprises different combinations of loading pattern and orientation for a single deck:



Example

You have to define the following rule to get the loading pattern shown in this figure: PC; TR; SSD; TL; SSC.

Load Planning Rules Independent of the Resource Type

Use

You can define the following rules for the optimizer runs for load planning and load consolidation for a specific combination of equipment group, such as truck or trailer, and equipment type, such as 12-ton box truck or 40-ton trailer.

Features

Load Planning Rules Independent of the Resource Type

Number	Explanation	Comment
1000	Stack height increases in the direction of travel	Violation of this rule is liable to penalty costs. You cannot combine this rule with rule 1001.
1001	Stack height decreases in the direction of travel	Violation of this rule is liable to penalty costs. You cannot combine this rule with rule 1000.
1100	Maximum height difference of adjacent stacks	Any deviations that are greater than the value defined in this rule are liable to penalty costs. You cannot combine this rule with rules 1101 and 1102.
1101	Maximum height difference of stacks within a row	Any deviations that are greater than the value defined in this rule are liable to penalty costs. You cannot combine this rule with rule 1100.
1102	Maximum height difference of stacks within a line	Any deviations that are greater than the value defined in this rule are liable to penalty costs. You cannot combine this rule with rule 1100.

Number	Explanation	Comment
2000	Prioritize weight over volume, and load packages and products from heaviest to lightest	You can use this rule to load the heaviest packages and products first instead of loading the biggest packages and products first. You cannot combine this rule with rule 2001. i Note Packages and products that cannot be loaded are penalized based on their weight. The default is to penalize based on volume. If you do not define this rule, the system automatically calculates penalty costs based on the volume of the packages and products that were not loaded. i Note You cannot define this rule for a specific combination of equipment group and equipment type; instead, you can define this rule only for all resources in an optimizer run.
2001	Prioritize weight and volume instead of volume only, and load packages and products that are both big and heavy first	You can use this rule to load the packages and products that are both the heaviest and the biggest first instead of loading the biggest packages and products first. You cannot combine this rule with rule 2000. i Note Packages and products that cannot be loaded are penalized based on their weight and their volume. The default is to penalize based on volume only. If you do not define this rule, the system automatically calculates penalty costs based on the volume of the packages and products that were not loaded. i Note You cannot define this rule for a specific combination of equipment group and equipment type; instead, you can define this rule only for all resources in an optimizer run.
2100	Packages and products with high density should be at bottom of the stack	You cannot combine this rule with rule 2101. i Note If you do not define a rule for the stacking of packages and products, the system considers only the weight of each package and product when it builds stacks. If the maximum weight that can be loaded in a specific package or product falls below the results of an optimizer run using these rules, the system does not consider this rule for the package or product in question. i Note You cannot define this rule for a specific combination of equipment group and equipment type; instead, you can define this rule only for all resources in an optimizer run.
2101	Packages and products can be positioned anywhere in stack regardless of weight and density	You cannot combine this rule with rule 2100. i Note If you do not define a rule for the stacking of packages and products, the system considers only the weight of each package and product when it builds stacks.

Number	Explanation	Comment
		If the maximum weight that can be loaded in a specific package or product falls below the results of an optimizer run using these rules, the system does not consider this rule for the package or product in question. i Note You cannot define this rule for a specific combination of equipment group and equipment type; instead, you can define this rule only for all resources in an optimizer run.
2102	Heavy packages and products should be positioned at the bottom	The system tries to position heavy packages and products on the floor of the vehicle as far as possible. You cannot combine this rule with rules 2100 and 2101.
2130	Maximum deviation of dimensions of smaller packages and products on top of larger packages and products within a stack	You can use this rule to allow stacking of a smaller package or product on top of a larger package or product in the same stack if the deviation of the package or product dimensions (length and width) is within the given tolerance. However, if you do not use this rule during automatic load planning or if you enter the value 0 as the maximum deviation, the optimizer plans only identically-sized packages and products on top of each other within a stack. i Note You cannot use this rule for packages and products in stacks within loading patterns E, F and P. For more information about loading patterns, see Rules in Automatic Load Planning.
2140	Packages and products that are unloaded at different stops can be loaded in a stack	You can use this rule to ensure that the optimizer also builds stacks with packages and products for different destination locations. i Note You cannot combine this rule with rule 2150.
2150	Ignore last in first out (LIFO) principle	You can use this rule to bypass the application of the LIFO principle during automatic load planning, otherwise the optimizer always applies this rule when you plan packages and products on a resource. i Note You cannot define this rule for a specific combination of equipment group and equipment type; instead, you can define this rule only for all resources in an optimizer run.
2151	Load packages for a specific customer at the back of the truck	This rule considers only packages that you have created as exclusive packages for customers. For more information, see the section "Automatic Creation of Package Units" under Use of Package Units in Planning.
2152	Minimum width for last in first out (LIFO) corridor	The system tries to provide a corridor between each package or product and the vehicle's back door that doesn't contain any packages or products that will be unloaded at a later stop
2160	Load resource as compactly as possible	The system tries to position all packages and products as close as possible to the driver's cab and the roof of the vehicle. This can result in a lower number of stacks in order to minimize the used length of the resource. i Note You cannot combine this rule with rule 2165.

Number	Explanation	Comment
2161	Minimize number of stacks	The system tries to keep the number of stacks as low as possible.
2162	Reduce gaps	The system tries to keep the gaps between the packages and products as small as possible.
2165	Load resource as broadly as possible	The system tries to position all packages and products as close as possible to the floor of the vehicle. This can result in a higher number of stacks in order to maximize the used length of the resource. i Note You cannot combine this rule with rule 2160.
2180	Disable stacking	If this rule is activated, no packages and products are stacked in the resource.
2200	Maximum weight per stack	In the case of a double-deck trailer, the maximum weight per stack that you define here applies to both decks. You can define an alternative maximum weight for the upper deck of your double-deck trailer in rule 2201.
3000	Maximum deviation of weight from center of cargo body to left and right in percent	You can use this rule to control the left-right weight balance of the load. You define the maximum deviation of the center of gravity of the load from the center of the cargo body that is allowed. Any deviation that is greater than the value defined in this rule is liable to penalty costs.
3050	Maximum deviation of axle utilization in percent	The automatic load planning minimizes the deviation of axle utilization between a vehicle's axle groups. Penalty costs are incurred for deviations that are higher than the defined maximum limit (parameter value). The utilization of the king pin is taken into account in the case of semi-trailers. For a combination of tractor and semi-trailer, only the second and third axle group are taken into consideration.
3100	Minimum weight on driving axle relative to total weight of vehicle resource in percent	If you have defined an axle as a driving axle in the master data, the axle load must not fall below the percentage of the current total weight of the loaded resource that you have specified here.
3101	Minimum weight on steering axle relative to total weight of vehicle resource in percent	If you have defined an axle as a steering axle in the master data, the axle load must not fall below the percentage of the current total weight of the loaded resource that you have specified here.
4000	U.S. Federal Bridge Formula	In the Federal Bridge Formula, the U.S. government defines specific rules that have to be considered for road transportation in the United States. For more information, see http://ops.fhwa.dot.gov/freight/publications/brdg_frm_wghts/index.htm . i Note You cannot define this rule for a specific combination of equipment group and equipment type; instead, you can define this rule only for all resources in an optimizer run.
8001	Load all packages and products with same loading pattern	The combination of loading pattern and orientation that you define in this rule applies to your entire resource. You cannot combine this rule with the following rules: 8011, 8021, 8022, 8023, 8024, 8025, 8026, 8052, and 8056.

Number	Explanation	Comment
8011	Load all packages and products with a loading pattern specific to each row	In this rule you can define a specific combination of loading pattern and orientation for each row of your resource. You cannot combine this rule with the following rules: 8001, 8021, 8022, 8023, 8024, 8025, 8026, 8052, and 8056.
8021	Load all packages and products turned	You cannot combine this rule with the following rules: 8001, 8011, 8022, 8023, 8024, 8025, 8026, 8052, and 8056.
8022	Load all packages and products in each row straight; if not enough space turn packages and products in first rows	If you use this rule for your automatic load planning, the system plans all packages and products in your resource in straight rows. If there is not enough space for other straight rows, the system turns the remaining packages and products in the first rows of the resource. You cannot combine this rule with the following rules: 8001, 8011, 8021, 8023, 8024, 8025, 8026, 8052, and 8056.
8023	Load all packages and products in each row straight; if not enough space turn packages and products in last rows	If you use this rule for your automatic load planning, the system plans all packages and products in your resource in straight rows. If there is not enough space for other straight rows, the system turns the remaining packages and products in the last rows of the resource. You cannot combine this rule with the following rules: 8001, 8011, 8021, 8022, 8024, 8025, 8026, 8052, and 8056.
8024	Load all packages and products straight	You cannot combine this rule with the following rules: 8001, 8011, 8021, 8022, 8023, 8025, 8026, 8052, and 8056.
8025	Load all packages and products in each row turned; if not enough space load all packages and products in first rows straight	If you use this rule for your automatic load planning, the system turns all packages and products in your resource. If there is not enough space for other turned rows, the system plans the remaining packages and products in the first rows of the resource in straight rows. You cannot combine this rule with the following rules: 8001, 8011, 8021, 8022, 8023, 8024, 8026, 8052, and 8056.
8026	Load all packages and products in each row turned; if not enough space load all packages and products in last rows straight	If you use this rule for your automatic load planning, the system turns all packages and products in your resource. If there is not enough space for other turned rows, the system plans the remaining packages and products in the last rows of the resource in straight rows. You cannot combine this rule with the following rules: 8001, 8011, 8021, 8022, 8023, 8024, 8025, 8052, and 8056.
8052	Load all packages and products in first row turned	If you use this rule for your automatic load planning, the system turns all packages and products in the first row in your resource regardless of your remaining loading patterns. You cannot combine this rule with the following rules: 8001, 8011, 8021, 8022, 8023, 8024, 8025, 8026, and 8056.
8056	Load all packages and products in last row turned	If you use this rule for your automatic load planning, the system turns all packages and products in the last row in your resource regardless of your remaining loading patterns. You cannot combine this rule with the following rules: 8001, 8011, 8021, 8022, 8023, 8024, 8025, 8026, and 8052.
8060	Stabilize load by turning packages and products in last rows if necessary	You can use this rule to reduce the risk of packages and products shifting during transportation. However, you cannot use this rule for the last rows of loading patterns E, F, and P.

Rules for the Load Planning of Upper Decks

You can define the following rules for the load optimization runs for the upper decks of double-deck trucks and double-deck trailers.

Rules for the Load Planning of Upper Decks

Number	Explanation	Comment
2201	Maximum weight per stack on the upper deck	You can use this rule to define a maximum weight per stack for the upper deck of your resource that differs from the maximum weight per stack for the lower deck of your resource that you defined in rule 2200.
2211	Maximum weight per row on the upper deck	The maximum weight per row for the upper deck does not apply to rows that are part of the loading patterns P, E, and F.
2221	Maximum weight per stack in the middle of a row on the upper deck	This rule applies only to rows with at least three stacks. The rule does not apply to rows that are part of the loading patterns P, E, and F. If you combine this rule with rule 2201, the system considers the lower of the two specified maximum weights.
8002	Upper deck: Load all packages and products with same loading pattern	The combination of loading pattern and orientation that you define in this rule applies to the entire upper deck of your resource. You cannot combine this rule with the following rules: 8012, 8031, 8032, 8033, 8034, 8035, 8036, 8053, and 8057.
8012	Upper Deck: Load all packages and products with a loading pattern specific to each row	In this rule you can define a specific combination of loading pattern and orientation for each row of the upper deck of your resource. You cannot combine this rule with the following rules: 8002, 8031, 8032, 8033, 8034, 8035, 8036, 8053, and 8057.
8031	Upper deck: Load all packages and products in each row turned	You cannot combine this rule with the following rules: 8002, 8012, 8032, 8033, 8034, 8035, 8036, 8053, and 8057.
8032	Upper deck: Load all packages and products in each row straight; if not enough space turn packages and products in first rows	If you use this rule for your automatic load planning, the system plans all packages and products on the upper deck of your resource in straight rows. If there is not enough space for other straight rows, the system turns the remaining packages and products in the first rows of the upper deck. You cannot combine this rule with the following rules: 8002, 8012, 8031, 8033, 8034, 8035, 8036, 8053, and 8057.

Number	Explanation	Comment
8033	Upper deck: Load all packages and products in each row straight; if not enough space turn packages and products in last rows	If you use this rule for your automatic load planning, the system plans all packages and products on the upper deck of your resource in straight rows. If there is not enough space for other straight rows, the system turns the remaining packages and products in the last rows of the upper deck. You cannot combine this rule with the following rules: 8002, 8012, 8031, 8032, 8034, 8035, 8036, 8053, and 8057.
8034	Upper deck: Load all packages and products in each row straight	You cannot combine this rule with the following rules: 8002, 8012, 8031, 8032, 8033, 8035, 8036, 8053, and 8057.
8035	Upper deck: Load all packages and products in each row turned; if not enough space load all packages and products in first rows straight	If you use this rule for your automatic load planning, the system turns all packages and products on the upper deck of your resource. If there is not enough space for other turned rows, the system plans the remaining packages and products in the first rows of the upper deck in straight rows. You cannot combine this rule with the following rules: 8002, 8012, 8031, 8032, 8033, 8034, 8036, 8053, and 8057.
8036	Upper deck: Load all packages and products in each row turned; if not enough space load all packages and products in last rows straight	If you use this rule for your automatic load planning, the system turns all packages and products on the upper deck of your resource. If there is not enough space for other turned rows, the system plans the remaining packages and products in the last rows of the upper deck in straight rows. You cannot combine this rule with the following rules: 8002, 8012, 8031, 8032, 8033, 8034, 8035, 8053, and 8057.
8053	Upper deck: Load all packages and products in first row turned	If you use this rule for your automatic load planning, the system turns all packages and products in the first row on the upper deck of your resource regardless of your remaining loading patterns. You cannot combine this rule with the following rules: 8002, 8012, 8031, 8032, 8033, 8034, 8035, 8036, and 8057.

Number	Explanation	Comment
8057	Upper deck: Load all packages and products in last row turned	If you use this rule for your automatic load planning, the system turns all packages and products in the last row on the upper deck of your resource regardless of your remaining loading patterns. You cannot combine this rule with the following rules: 8002, 8012, 8031, 8032, 8033, 8034, 8035, 8036, and 8053.

Stackability

During load planning and load consolidation, the system takes stackability of products into account.

You can define the following settings:

- Stackability between two products

You define stacking settings for your products or reference products in a product relationship profile, which you then indicate in your load planning settings.

i Note

- Only the stacking settings defined in the product relationship profile are taken into account, not the consolidation settings.
- These stacking settings only indicate whether a product can be directly stacked on another product. The system doesn't check whether this product can be indirectly stacked on another product, that means, with a third product inbetween. If you want the system to also check indirect stacking, you can enable this with a Business Add-In (BAI) for the row-stack approach.

To define a product relationship profile, open the [Change Product Relationship Profile](#) app from the launchpad. To indicate the profile in your load planning settings, open the [Edit Load Planning Settings](#) app from the launchpad.

- Stackability of products on themselves

You indicate the maximum stacking factor for your products and packaging materials in your master data.

To create a product, on the SAP Easy Access screen, choose [Logistics](#) [Transportation Management](#) [Master Data](#) [Material](#) [MM01 - Create](#). You can define the maximum stacking factor in the following places:

- On product header level
- On the [Additional Data](#) [Units of measure](#) view

i Note

Due to compatibility reasons, load planning and load consolidation ignore the maximum stacking factor defined on the [Units of measure](#) view if a maximum stacking factor greater than zero is defined on product header.

For example, you can specify a maximum stacking factor of two for pallets (PAL). This means, that up to two pallets can be stacked on top of each other. If you don't indicate a maximum stacking factor or indicate a maximum stacking factor of zero, this means that the product or package can be stacked an infinite number of times on itself. A maximum stacking factor of one means that the product or package can't be stacked on itself.

You can define different maximum stacking factors per unit of measure.

- Weight put on top of a package

The system checks the maximum top load that you have defined in the master data of your products. The maximum top load describes the maximum weight that you can place on top of a product's full package.

- Load planning rules

You can define load planning rules that control stackability, such as rule 2180. For more information, see [Rules in Automatic Load Planning](#).

The system considers these settings in the following way:

- Stacking settings

Stacking settings are considered for products and packages. For packages, the handling is different depending on whether a closed packaging material or open packaging material is used.

The stacking settings are considered for **closed packaging materials**, such as pallet cages. However, stacking settings for products inside closed packages are ignored.

The stacking settings are also considered for top-level packages with **open packaging material** and all products inside. The stacking settings for reference products are considered, too.

Package A can be stacked on package B if the following prerequisites are met:

- All products in package A can be stacked on all products in package B.
- The packaging material of package A can be stacked on the package material of package B.
- Package B is stackable (defined by unified package building), that is, UPB allows that something can be stacked on it.

However, stacking settings for products contained in cartons put on pallets are ignored.

- Maximum stacking factor

The maximum stacking factor is considered for products and packages. However, stacking is only allowed for top-level packages that have been defined as stackable by unified package building (or by a customer enhancement).

For **closed packaging materials**, only the maximum stacking factor defined for the packaging material and the unit of measure of the package item is considered. Thus, the content of the package does not matter.

For an **open packaging material**, its maximum stacking factor is only considered if empty packages are stacked on each other. If the package contains at least one product or another package, the maximum stacking factor of the packaging material is ignored by load planning and load consolidation, and the maximum stacking factor of the product or package inside is considered.

The maximum stacking factor on reference product level is considered, too. However, only the reference product level directly above the real products is considered. Maximum stacking factors defined for reference products of reference products are ignored.

Load Plan

Definition

The load plan is the result of load planning. It provides you with an overview of how the loading space available for transporting a business document is currently used. A load plan is also created for optimizer-based mixed package building. It provides you with

an overview of the loading of a package or a pallet, for example.

In both the transportation cockpit and in the business document, you can display the complete load plan as a 3D load plan or as a hierarchical table (load plan hierarchy).

You have the following options in the 3D load plan:

- You can show and hide gridlines in your decks.
- You can show and hide individual objects as well as entire rows, columns, or stacks and fill with different colors. You can define colors for your 3D load plan in Customizing for [Transportation Management](#) under [Basic Functions](#) [Colors](#) [Define Colors for Gantt Chart, Load Planning, and Geographical Map](#).

You can also specify colors for the x, y, and z axis. These colors are then used when you drag objects along these axes. For more information, see Customizing for [Transportation Management](#) under [Basic Functions](#) [Load Planning](#) [Define Layouts for 3D Load Plan](#) (Colors view).

- You can display labels for higher-level packages (outer packaging). The labels can contain up to three package attributes. You can also display a ruler for resources (such as trucks) and equipment types. You must have selected the [Show Item Labels](#) or [Show Ruler for Resources/Equipment Types](#) checkbox in Customizing for [Transportation Management](#) under [Basic Functions](#) [Load Planning](#) [Define Layouts for 3D Load Plan](#). You have also defined the required attributes for item labels under [Additional Texts](#). You can define these separately for loaded and not loaded items.
- You can adjust a load plan manually using pushbuttons in the toolbar, keyboard shortcuts, or drag and drop to move products and packages on a truck. The status and the axle load are adjusted accordingly. You must have selected the [Enable Manual Load Planning](#) checkbox in Customizing for [Transportation Management](#) under [Basic Functions](#) [Load Planning](#) [Define Layouts for 3D Load Plan](#) and have also selected [Enable Drag and Drop](#) for drag and drop. You can also define in this Customizing activity which combination of keys execute which command and how far an object should be moved (length of the movement increment).

Drag and drop also allows you to create a load plan, that is, to move one or more unplanned objects into the truck's loading space or to remove them from the truck again.

You can also define your own toolbar in this Customizing activity. First, you create an ID for the toolbar, which you can optionally enter in the [Layouts](#) view under [Display Settings](#). Then you select the items for your toolbar, such as buttons and button choices, as well as the sequence in which they are to be displayed. You can select these items from a list of predefined actions and change their description. For a button choice, you must also specify the subitems.

For example, you can arrange items in the 3D load plan flat or in stacks, lift them up, and drop them down again.

Actions such as positioning items in the 3D load plan as well as selecting and deselecting items can be performed for several items at the same time. In addition, you can select all items with the same source or destination location, product, or items in the same stack or row. You can select individual items in two ways: by using a single left mouse click or by using rectangle selection.

- You can change the orientation of your top-level products in the load plan using the context menu, the [Orientation](#) pushbutton, or keyboard shortcuts. You must have selected the [Enable Manual Load Planning](#) checkbox in Customizing for [Transportation Management](#) under [Basic Functions](#) [Load Planning](#) [Define Layouts for 3D Load Plan](#). You can also define in this Customizing activity which combination of keys execute which change of orientation. Note that you can only change the orientation according to the product orientation profile that you have defined for the product (see "Prerequisites" under [Load Planning](#)).
- If you have created your load plan with layer-based mixed package building based on the package building optimizer (that is, if you have selected optimizer-based mixed package building as the mixed package building mode), each layer is represented as a block instead of the individual products on the layer.
- You can use a Business Add-In (BAI) to display products as COLLADA models. For more information, see SAP Note [3478427](#).

You have the following options in the load plan hierarchy:

- You can create a load plan manually by entering a value greater than zero for a product in the **Loading Sequence** column. You can also specify the position and orientation. If you work with split deck resources, you must also specify whether the product is to be loaded onto the upper or lower deck. The system recalculates the axle loads.
- You can adjust the position and orientation of products and packages by manually changing the coordinates.
- For combination resources, you can move the items from one trailer to the other using drag and drop.
- You can display which profile, which engine (that is, which package builder, packing instruction, or packaging specification), and which rule (that is, which packing instruction or packaging specification) was used for unified package building. If you make changes to the package item itself or to the packed items that result in the load plan status of the package item being automatically set to **Not Planned** or **Invalidated**, the information in these fields is deleted. For more information, see [Unified Package Building](#).
- You can add items for auxiliary packaging materials manually. Top-level auxiliary packaging material items are considered in load planning.

You can set the load plan item status (cargo) for auxiliary packaging materials manually. Alternatively, the system can set it automatically during unified package building or load planning.

The load plan status (capacity) is not relevant for auxiliary packaging materials because auxiliary packaging materials do not have their own capacity.

For more information about packaging materials, see [Use of Auxiliary Packaging Materials](#).

- You can change the orientation of your products in the **Orientation** field. For more information about the prerequisites, see the corresponding paragraph for the 3D load plan.
- Note that there are dependencies between some actions in the load plan and the load plan status value **Finalized**. For example, you cannot delete the load plan if the load plan status of a resource has the value **Finalized**. For more information, see the section "Fixing of Packages" under [Creation or Update of Consignment or Freight Orders on Inbound Messages](#).
- You can display layers as separate hierarchy levels.

Structure

The load plan hierarchy contains the following information about the business document item:

- Item
- Loading sequence
- Row, stack, and level in the loading space
- Gross weight
- Height, width, and length
- Stackability

You also get information about the load distribution as well as additional statistical data about the current load on separate tab pages. This includes the following:

- Number of loaded business document items
- Used area on the lower deck of your double-deck trailer
- Utilization of the loading volume in percent

- Used area on the upper deck of your double-deck trailer
- Maximum weight for each axle group compared to the current weight for each axle group
- Maximum trailing load compared to the current trailing load

A load plan that was created for optimizer-based mixed package building also contains the following information:

- Products and product items in the package (product hierarchy)
- Exact position of the products in the package

You can overwrite this exact position.

- Orientation of the products in the package
- Parent location, in other words the parent location of the location hierarchy that you have defined in your unified package building profile

You can also display packages graphically in the 3D load plan. You can use [Switch Package Details](#) to change the display so that the exact contents of the package are displayed (the individual products and product items).

You must have selected the [Show Package Details](#) checkbox in Customizing for Transportation Management under [Basic Functions](#) > [Load Planning](#) > [Define Layouts for 3D Load Plan](#).

You can print the load plan with or without the 3D load plan. For more information, see [Load Plan Printing](#).

Related Information

[Optimizer-Based Mixed Package Building](#)

[Unified Package Building Profile](#)

[Utilization](#)

Load Plan Printing

You can print the hierarchical table of the load plan (load plan hierarchy) with or without 3D images (3D load plan).

First, a PDF file of the load plan is created for each business document. You can then print this PDF file on paper, for example, or send it as an e-mail attachment. This printout can be used by the warehouse staff or the truck driver, for example, while loading or unloading at the warehouse door.

Prerequisites

- To print the load plan with one or more 3D images, you need a subscription to the image creation service provided by SAP Integrated Product Development (IPD).
- You have made the following settings in Customizing for Transportation Management under [Basic Functions](#) > [Load Planning](#) > [Printing](#):
 - You have defined a load plan printing profile in the Customizing activity [Define Load Plan Printing Profile](#).

In the load plan printing profile, you specify a load plan printing layout and a layout for the 3D load plan as well as whether the 3D load plan is to be printed as one or more 3D images that represent the individual loading steps.

- You have assigned your load plan printing profile to the desired business document type, such as the freight order type, in the Customizing activity [Assign Load Plan Printing Profile](#).

- You have defined a load plan printing layout in the Customizing activity [Define Load Plan Printing Layout](#).
- You have specified the sections that are to be contained in the load plan printing layout in the Customizing activity [Configure Load Plan Printing Layout](#).
- You have made the settings for output management in your freight order type and container unit type in Customizing ([Output Options](#) area); for example, you have entered the additional output profile /SCMTMS/TOR_PRINT_ROAD. For more information, see [Output Management](#).

i Note

You can check whether your setup for the image creation service is correct and complete using report /SCMTMS/PLN_LP_IMG_CHECK. For more information, see the report documentation.

To access the report, on the SAP Easy Access screen, choose ► [Logistics](#) ► [Transportation Management](#) ► [Administration](#) ► [Load Plan Printing](#) ► [Check Image Creation Service](#) ►.

Features

You print the load plan from the business document or from the transportation cockpit. A background report is also available. The system creates **one** PDF file for each business document.

Depending on how you have configured the load plan printing layout and on which type of business document you want to print, the PDF file can contain the following sections:

- Header, for example, document number, page number, and logo
- General information, for example, carrier name and departure date
- Stop sequence, in other words, location and address data as well as arrival and departure times
- Capacity and cargo overview, for example, information about available capacity and utilization
- Packaging material overview, in other words, the aggregated quantities of the top-level packaging materials and the number of unpackaged products
- Item list, in other words, top-level packages and unpackaged products in their loading sequence
- Detailed item list, in other words, an expanded view of the items including the complete packaging hierarchy
- One or more 3D images

The following business documents are supported:

- Road freight order
- Container unit

Load plan printing is based on the standard printing framework (output management).

Activities

- You start load plan printing via the output management of the relevant business document ([Output Management](#) tab page, [Loading/Unloading Instruction \(Loading Plan\)](#) action).
- Personal object worklist for freight orders

You start load plan printing using the **Mass Output** button ([Loading/Unloading Instruction \(Loading Plan\)\(Archived Issue\)](#) action).

- Transportation cockpit

You start load plan printing using the **Print** button.

- Background report

On the SAP Easy Access screen, choose **Logistics** > **Transportation Management** > **Administration** > **Background Processing** > **Print Load Plan**.

After the report has been executed, you can open the PDF files with the respective load plan in the following places:

- Business document

Choose **Open Archive Link** on the **Output Management** tab page under **Communication Data**.

- Application log

Copy the link in the **Archive Link** field or choose **Open Archive Link**.

Related Information

[Load Plan](#)

[Load Planning](#)

[Output Management](#)

One-Step Planning

Use

In the case of one-step planning, the system performs a complete planning run that comprises any combination of VSR optimization, carrier selection, and load optimization.

Features

For one-step planning, SAP delivers the standard planning strategy VSR_1STEP. This combines the two standard planning strategies VSR_DEF for VSR optimization and TSPS_DEF for carrier selection.

You can use VSR_1STEP for VSR optimization and for the generation of transportation proposals. When transportation proposals are generated, carrier selection is performed for each transportation proposal generated. If you then want to perform a load optimization, you can add method VSR_ALP to VSR_1STEP. Alternatively, you can define your own planning strategy that comprises any combination of the methods of VSR optimization, carrier selection, and load optimization. The following methods can be used, for example:

- VSR_PRE for preprocessing the VSR optimization run
- VSR_OPT for the VSR optimization run
- VSR_POST for postprocessing the VSR optimization run
- VSR_TSPS for carrier selection
- VSR_ALP for load planning

You can define planning strategies in Customizing under ► [Transportation Management](#) ► [Basic Functions](#) ► [Process Controller](#) ► [Define Strategy](#) ►.

i Note

SAP also delivers the standard planning strategy VSRI_1STEP for one-step planning in manual planning. If you enter this strategy as a strategy for manual planning in your planning profile, the system automatically performs carrier selection at the end of manual planning (that is, if you assign a freight unit to a resource). Here you can also add load optimization to the standard planning strategy. Alternatively, you can also define here your own planning strategy that comprises any combination of VSR optimization, carrier selection, and load optimization.

Activities

If you want to use the standard planning strategy VSR_1STEP, specify it in the optimizer settings. To do this, open the [Create Optimizer Settings](#) app on the launchpad.

More Information

[Planning Strategies](#)

[VSR Optimization](#)

[Carrier Selection](#)

[Load Planning](#)

Scheduling

Use

Scheduling determines start and end times for a set of dates and times such as departure and arrival of a freight order as well as the pick-up and delivery of the assigned freight units or the coupling and uncoupling of trailers. In scheduling, the system considers multiple constraints like a predefined relative ordering among the activities of the freight order as well as the pick-up windows and the delivery windows of the assigned freight units or the availability of the driver that is to transport a particular road freight order. Scheduling can be triggered for one or more selected freight orders and is based on your settings in the planning profile (see [Planning Profile](#)).

i Note

For schedule-based freight orders and freight bookings, scheduling does not consider loading and unloading durations.

Prerequisites

You have made the following settings in the planning profile:

- You have specified a planning horizon. If you do not do so, the system uses the values defined in the default profile.
- You have specified loading and unloading durations for your freight units. If you do not do so, the system sets the loading and unloading durations for your freight units to 0. For more information, see [Loading and Unloading Durations](#).
- You have specified coupling and uncoupling durations. If you do not do so, the system sets the coupling and uncoupling durations to 0.

- You have specified if the system is to consider requested or acceptable pick-up and delivery dates for your freight units.
- You have specified if you want to use scheduling in a forward or in a backward direction. By default, the system uses forward scheduling.
- You have specified a scheduling strategy.

Features

Scheduling Strategy

In the scheduling settings of your planning profile, you can specify which scheduling strategy you want to use. We provide the following standard scheduling strategies:

- Scheduling (Embedded) (VSS_EMBED)

This is the default scheduling strategy. It doesn't require a remote communication and control framework (RCCF) call. You can indicate in the scheduling settings, that the system is to take into account cost-based requested dates/times. However, this strategy doesn't take into account scheduling constraints that you have defined in Customizing.

- Scheduling (Optimizer Server) (VSS_DEF)

Forward and Backward Scheduling

You can use scheduling in a forward or in a backward direction:

- In forward scheduling, the system first defines the start and end time of the **first** activity to be scheduled. Afterwards, the system assigns start and end times for all succeeding activities.
- In backward scheduling, the system first defines the start and end time of the **last** activity to be scheduled. Afterwards, the system assigns start and end times for all preceding activities.

For both types of scheduling, the system tries to schedule the activities with as little idle time between the individual activities as possible. Equally, the system considers the coupling and uncoupling of trucks and trailers as well as the availability of drivers and vehicle resources. In this case, the system considers the coupling of the trailer to a vehicle and the loading of a freight unit into the trailer as separate activities.

Constraints

Scheduling considers the following constraints:

- Time windows for loading and unloading activities for your freight units
- Loading and unloading durations of your freight units
- Appointments for the freight order your freight units are assigned to
- Location sequence of your freight order
- Existing freight orders on resources

i Note

This does not apply to multiresources.

- Fixed departure time of a freight order stage
- Coupling and uncoupling durations

- Absences and non-working times of the involved drivers
- Downtimes and non-working times of the involved vehicle resources
- Calendar resources as well as the calendars and capacities for the loading and unloading activities of handling resources
- Calendars of the involved vehicle resources and drivers for loading, unloading, transportation, coupling, and uncoupling activities
- Minimum and maximum goods wait times defined for transshipment locations

Additionally, you can specify specific scheduling constraints such as working times. For more information, see [Customizing for Transportation Management](#) under [►Planning ►General Settings ►Define Scheduling Constraints ►](#).

→ Recommendation

We recommend that you define driving and working constraint profiles instead of scheduling constraints. While the scheduling constraints can lead to longer travel or waiting times, which are hard to understand, the advantage of driving and working constraint profiles is that rests of a certain duration are scheduled without interruption and that these rests can be displayed on the user interface. Further, scheduling constraints are no longer supported by embedded scheduling. Weitere Informationen finden Sie unter [Considering Driving Time and Working Hours Constraints](#).

Scheduling in the Freight Order User Interface

If you start scheduling from the user interfaces for freight orders, the system calculates the planned dates for your freight orders. You can select the scheduling direction either in the freight order or in the planning profile.

For the scheduling of freight orders, you have the following options:

- Scheduling based on the planning profile

With this option, the system determines the scheduling direction based on the planning profile. If you have not defined a planning profile, the system uses forward scheduling by default.

- Forward scheduling

With this option, the system uses forward scheduling irrespective of the planning profile. If you have specified a departure date/time, the system then fixes this and carries out forward scheduling starting from this date. In doing so, the system always refers to the current transportation stop.

- Backward scheduling

With this option, the system uses backward scheduling irrespective of the planning profile. If you have specified an arrival date/time, the system then fixes this and carries out backward scheduling starting from this date. In doing so, it always relates to the last transportation stop, meaning the destination location.

Execution-Based Scheduling

If a business document is already in execution and an event is reported for an activity, the system fixes this activity at the reported position. During scheduling, the system ignores all activities before the last executed activity.

Note that if at least one activity at a stop has status **In Execution** or **Executed**, the system ignores all activities of this stop.

Explanation Tool for Scheduling

You can analyze your scheduling results. For more information, see [Explanation Tool for Scheduling](#).

Activities

To start scheduling, you have the following options:

- You can start scheduling from the transportation cockpit by choosing **Scheduling** in the transportation cockpit areas for freight orders and transportation units or in the Gantt chart of the **Transportation Cockpit** app. Additionally, in the Gantt chart, you can schedule road freight orders and trailer units per drag and drop.
- You can start scheduling from the user interfaces for freight orders by choosing **Scheduling** and then selecting one of the options listed above. For more information, see [Scheduling of Freight Orders](#).
- If you have activated the user parameter /SCMTMS/SCH_FORCE, you can start scheduling automatically by entering the departure date of a freight order and pressing ENTER. Here, you can choose between the following options:
 - **X** activates this function for freight orders on all user interfaces where you can use scheduling.
 - **T** activates this function for freight orders in the transportation cockpit only.

Example

Forward Scheduling

Customer A orders 50 pallets of product X. Customer A asks for a delivery date in three weeks. Currently, the capacity of your depot is 100 m³. Product X takes up 95 m³ of your depot. Therefore, you decide to move the goods out of your depot as soon as possible in order to minimize your inventory. That is why you decide to use forward scheduling.

Backward Scheduling

Customer B orders 50 pallets of product Y. Customer B asks for a delivery date in three weeks. Currently, the capacity of your depot is 100 m³. Product Y takes up 5 m³ of your depot. Therefore, you decide to stick to the delivery time window as closely as possible. That is why you decide to use backward scheduling.

More Information

SAP Note [1908165](#) 

Schedule Unplanned Requirement Documents

To improve your planning decisions, you can quickly generate the approximate transportation time for requirement documents without needing to plan the documents. In addition, the Gantt chart can display the stage sequence to provide a useful visual aid.

Prerequisite

In Customizing, under **►Transportation Management ► Planning ► Freight Unit ► Define Freight Unit Types ►**, you have selected **Enable Automatic Determination** from the **Distance/Duration** dropdown list.

Use

There are multiple ways to schedule an unplanned requirement document:

- Applying a default route to the document
- Clicking **Schedule**

- Starting the freight unit building process

You can view the scheduling results on the Gantt chart and in the requirement document (see Activities below).

Features

- After scheduling, the requirement document contains the transportation times for each stage (planned arrival and planned departure times) but the document is still unplanned.
- Stop activities are used to calculate the transportation times of the stages but the stop data is not contained in the requirement documents (and therefore not visible on the Gantt chart.)

Stop durations consider the following activities:

- Preparing for the stop
 - Loading and unloading
 - Finalizing the stop
 - In addition, operating times of the location and the handling resource are considered
- Scheduling ignores the handling resource capacity.
- Scheduling can run forwards or backwards. The direction is defined in [Scheduling Settings](#).
- Scheduling uses the requested and accepted loading or unloading time-windows of the requirement document and uses the settings defined in [Scheduling Settings](#).
- Scheduling considers minimum and maximum wait-time constraints defined either in the location master or in the default route – if a default route is applied.
- Stages that are already planned in the requirement document are unaffected by this scheduling process. Additionally, newly scheduled stages do not overlap planned stages.

Activities

To view the scheduling results in the requirement document, complete the following steps:

1. Display a requirement document by opening a requirement document worklist app from the lauchpad, for example [Freight Units](#) and then selecting a freight unit.
2. In the [Stages](#) tab, select a stage.
The stage information is displayed.
3. Select the [Planning/Execution Data](#) tab.
4. Optionally, you can save the schedule to the requirement document.

Consideration of Scheduling Constraints

Use

You can use this function to consider additional constraints during scheduling. This allows the system to extend the trip durations in transportation plans so that a driver can take their required breaks.

i Note

The system considers scheduling constraints both during scheduling and during vehicle scheduling and routing (VSR) optimization. However, it does not consider scheduling constraints during scheduling if you are using the scheduling strategy [Scheduling \(Embedded\)](#).

❖ Example

A truck would require two days for a trip from the east coast to the west coast of the United States. However, since the driver must take breaks along the way, a longer trip duration must be scheduled.

i Note

- No legal regulations are delivered with this function.
- The aim of this function is **not** to map legal regulations. It only allows the system to extend the trip durations in transportation plans that give the driver a sufficient buffer and allow them to take the required breaks.
- This function only applies for vehicle resources. Passive vehicle resources and schedules are not supported.

→ Recommendation

We recommend that you define driving and working constraint profiles instead of scheduling constraints. While the scheduling constraints can lead to longer travel or waiting times, which are hard to understand, the advantage of driving and working constraint profiles is that rests of a certain duration are scheduled without interruption and that these rests can be displayed on the user interface. Further, scheduling constraints are no longer supported by embedded scheduling. For more information, see [Considering Driving Time and Working Hours Constraints](#).

Prerequisites

- You have defined constraints for scheduling in Customizing. These constraints represent your rules. You combine several constraints in one set. You can assign this set to one or more means of transport. For more information, see Customizing for [Transportation Management](#) under [►►Planning►General Settings►Define Scheduling Constraints](#) ►.
- In the scheduling settings of your planning profile, you have indicated in the [Consider Time Constraints](#) field that scheduling constraints are to be considered. In your transportation cockpit session, you can overrule this setting in the [Change Planning Settings](#) dialog box.
- You have defined the duration of the activities:
 - Transportation
You define this duration in the transportation lane. To do so, on the [SAP Easy Access](#) screen, choose [►►Transportation Management►Master Data►Transportation Network►Define Transportation Lane](#) ►.
 - Loading/Unloading
You define this duration in the scheduling settings of your planning profile. Open the [Edit Scheduling Settings](#) app on the launchpad. For more information, see [Loading and Unloading Durations](#).
 - Coupling/Uncoupling
You define this duration in Customizing. For more information, see Customizing for [Transportation Management](#) under [►►Master Data►Resources►Define Coupling/Uncoupling Duration](#) ►.

Features

Constraints

- Fixed constraint

For fixed constraints, you define the exact start time of the rule.

♣ Example

Example 1: You want to specify that a driver is not to work more than 10 hours per day. In Customizing, you first define an activity group "Work". You define work as driving, coupling, and uncoupling. You therefore assign the activity types "transport" and "coupling/uncoupling" to your activity group. If a transportation plan consists of more than 11 hours per day, the system extends it to the next day by adding an additional duration to an activity or pushing it between activities.

Example 2: You want to specify that a driver may only spend a maximum of 8 hours behind the wheel. In this case, you define an activity group "steering" in Customizing and assign the activity type "transportation". If the transportation plan takes longer than 8 hours per day, the system extends it to the next day by adding an additional duration to an activity or pushing it between activities.

- Rolling constraint

For rolling constraints, you do not define a start time of the rule. This means that the rule must be valid at any given time on the time axis.

♣ Example

You specify that a truck may drive a maximum of 4 hours within 5 hours. However, you do not define when these 5 hours begin. This rule must therefore be valid for any 5-hour segment on the time axis. If an activity takes longer, the system also extends the transportation plan here.

i Note

The fulfillment of scheduling constraints does not necessarily lead to a connected break per rule.

For example, you have planned two freight units with two delivery locations. You have also defined a scheduling constraint that allows a maximum travel time of eight hours within 24 hours. If there are opening hours that lead to a wait time of one hour before delivery of the first goods, this hour can be reused as a break. Therefore, the trip from the start location to the first delivery location is extended by one hour. In addition, a break of 15 hours is added on the return trip from the second delivery location. This is sufficient to fulfill the constraint that requires a non-travel time of 16 hours within 24 hours.

You can define multiple fixed and rolling constraints. The scheduled additional duration can be used to fulfill several constraints. If there is idle time between activities, this time is also used to adhere to rest periods.

♣ Example

- Example 1: You have defined two scheduling constraints:
 - Constraint 1: A maximum of four hours and 30 minutes of working time is permitted within five hours.
 - Constraint 2: A maximum of eight hours of working time is permitted within 24 hours.

This results in a transportation plan in which 30 more minutes are scheduled after four and a half hours of travel. After another three and a half hours, an additional 15 hours and 30 minutes are scheduled.

- Example 2: You have specified in Customizing that a truck may drive a maximum of eight hours per day. Now the system is to schedule an activity with 10 hours. In this case, the system extends the transportation plan. This means it schedules eight hours on the first day, shifts an additional activity duration for the rest of the day, and schedules the remaining two hours on the following day.

- Example 3: In Customizing, you have specified that only eight hours may be planned for activities per day. The system is now to schedule two activities of a freight order, each of five hours. In this case, the system schedules the first activity on the first day. It schedules the second activity with three hours on the first day, and it moves the remaining two hours to the next day.

You cannot tell in the system when exactly the additional duration was inserted. You can only see that the activities took longer than the durations that you defined.

Consideration of Context

The system considers the constraints for the scheduling per resource, not per freight order. Freight orders that are determined as context (and are not explicitly included in the scheduling or VSR optimization run) as well as fixed freight orders are ignored in any case during the determination of the new transportation plan.

❖ Example

The system schedules two freight orders for one truck, each with six hours, one after the other. You have specified in Customizing that the truck may drive a maximum of eight hours per day. In this case, the system extends the second freight order, that is, it adds an additional duration. If the first freight order was determined as context, however, the system does not extend the second freight order.

Considering Driving Time and Working Hours Constraints

Based on different sets of constraints (rules), rests are scheduled during planning.

To prevent accidents caused by driver fatigue, many countries have implemented official driving time and working hours regulations for drivers. When using planning functions such as scheduling and vehicle scheduling and routing (VSR) optimization, driving time regulations can have a significant impact on the planning process.

Driving time and working hours constraints are a set of planning constraints that allow modelling and consideration of a variety of requirements related to driving time and working hours. These constraints enable the system to produce meaningful scheduling and planning results that are beyond pure approximation.

For more information on modeling of regulations see [Examples of Constraints](#).

Prerequisites

- You have defined profiles for driving time and working hours constraints (driving and working constraint profiles) in Customizing.

For more information, see Customizing for [Transportation Management](#) under [►Planning►General Settings►Define Driving and Working Constraint Profile►](#).

For more information about defining driving time and working hours constraints, see [Definition of Driving Time and Working Hours Constraints](#).

- In the capacity availability settings of your planning profile, you have assigned your driving and working constraint profiles to your vehicles (truck resources or vehicle types) and drivers.

To do so, you need to create entries for your vehicles on the [Vehicle](#) tab, or entries for your drivers on the [Driver](#) tab, or both. You can indicate vehicles by specifying a means of transport, a vehicle group and type, or a resource. You can indicate drivers by specifying a driver ID.

On the **Activities (Local Staff)** tab, you can optionally indicate locations at which certain activity groups are handled by the local staff and are thus not considered by the driving time and working hours constraints.

The system evaluates and applies the entries in the capacity availability settings from top to bottom. Wildcards are allowed.

- In the scheduling settings of your planning profile, you have indicated in the **Consider Time Constraints** field that driving and working constraint profiles are to be considered. In your transportation cockpit session, you can overrule this setting in the **Change Planning Settings** dialog box.

Note that you need to select scheduling strategy **Scheduling (Embedded)** (VSS_EMBED) in the scheduling settings for the system to consider driving time and working hours constraints during scheduling.

- In Customizing, you have added the following fields if you want to display rests in the Gantt chart:
 - **Total Driving Time**
 - **Total Working Time**
 - **Total Rest Time**
 - **Total Idle Time**
 - **Total Unavailability**

They are available for the truck list, the driver list and the road freight order list.

For more information, see Customizing for **Transportation Management** under **Basic Functions > Gantt Chart > Define Field Lists and Label Schemes for Gantt Chart**.

- You have optionally defined color settings for rests in the Gantt Chart.

For more information, see Customizing for **Transportation Management** under **Basic Functions > Gantt Chart > Define Color Schemes and Patterns for Gantt Chart**.

- You have optionally defined label schemes for rests in the Gantt Chart. You can also define tooltips as labels. The standard tooltip for rests displays the duration only. However, you can configure it so that the rest type and the start and end time are displayed as well.

For more information, see Customizing for **Transportation Management** under **Basic Functions > Gantt Chart > Define Field Lists and Label Schemes for Gantt Chart**.

- You can use user parameter /SCMTMS/SCH_FORCE to force automatic forward scheduling after data has been changed.

Use

Driving time and working hours constraints are the rules that define when drivers need to take a rest, such as a lunch break or a night's rest. You define them in Customizing as driving and working constraint profiles. In the capacity availability settings and the scheduling settings of your planning profile, you enable planning to take these constraints into account. Scheduling and vehicle scheduling and routing (VSR) optimization can then schedule rests in road freight orders based on these constraints. You can see the duration of the scheduled rests in the corresponding lists and hierarchies of the transportation cockpit. They are also graphically visualized in the Gantt chart.

You can define several sets of constraints, that is, driving and working constraint profiles, to consider different requirements, such as requirements from different countries.

The system considers driving and working constraint profiles in the following way:

- If you have assigned a multiresource or vehicle type to your road freight order, the system considers the driving and working constraint profile assigned to this multiresource or vehicle type. It ignores all activities from other road freight

orders assigned to the same resource but assumes that the driver (vehicle) is well rested and the constraint calculation can start afresh before the road freight order is executed.

- If you have assigned a single resource to your road freight order, the system considers the driving and working constraint profile assigned to this single resource. It also considers activities from other road freight orders assigned to the same resource (within the planning horizon).
- If you have assigned a driver to your road freight order, the system considers the driving and working constraint profile that belongs to this driver. The system ignores driving and working constraint profiles of an additional truck resource assigned to the road freight order. It also considers activities from other road freight orders assigned to the same driver (within the planning horizon).

A rest can only interrupt a travel activity. It can't interrupt other activities, such as loading, unloading, or preparing but can occur between these activities at a stop. For example, a rest can occur between loading and coupling.

The system assumes that a driver (vehicle) is well rested at the beginning of the first road freight order in the planning horizon. No explicit rest is planned before the first scheduled road freight order.

Whenever a road freight order overlaps with unavailability, the unavailability is filled with the largest rest available in the current driving and working constraint profile. An unavailability can also be used to enforce a rest within a certain time slot.

Scheduling and VSR optimization consider other road freight orders that are planned on the same resources before and within the planning horizon when determining rests.

If you want the system to consider other road freight orders that are planned on the same single resource **before** the planning horizon when determining rests, you need to select the **Consider Road Freight Orders Before Planning Horizon** checkbox in your scheduling settings.

If you want subsequent road freight orders assigned to the same truck resource within the planning horizon to be re-scheduled too when you re-schedule a road freight order, you need to select the **Reschedule Subsequent Documents** checkbox in your scheduling settings.

i Note

- Considering driving time and working hours constraints leads to a longer duration of the road freight order due to the included scheduled rests.
- Driving and working constraint profiles are only supported for embedded scheduling. They are not taken into account during scheduling if you use scheduling strategy **Scheduling (Optimizer Server)** (see Prerequisites).
- VSR optimization doesn't support the following functions:
 - Rolling constraints
 - Accumulation constraints at the start of the week
 - Drivers

VSR optimization only considers driving and working constraint profiles assigned to vehicle types or resources. For vehicles with a default driver, the profile assigned to the default driver is used instead of the profile assigned to the vehicle.

If a truck resource with an assigned default driver has a road freight order assigned which is not assigned to the default driver, the truck resource is blocked for further use in optimization.

- Driver teams aren't supported.

- When you use driving and working constraint profiles in scheduling, there is no guarantee, that scheduling finds a solution if one exists.

Display of Rests

The following times related to driving time and working hours are displayed in the road freight order list and hierarchy, in the truck list, and in the driver list of the transportation cockpit:

- **Total Driving Time**
- **Total Working Time**
- **Total Rest Time** (the quick view offers a list of individual rests)
- **Total Idle Time**
- **Total Unavailability**

i Note

For multiresources and equipment types, the total unavailability is displayed only.

Rests are displayed in the Gantt chart in the following way:

- Resource- and driver-based lists
 - In the **Document** view, only rests between documents are shown.
 - In the **Activity** view, all rests are shown.
- Document-based lists
 - In the **Document** view, no rests are shown.
 - In the **Activity** view, all rests inside the document are shown.

Rests are graphically visualized as rectangles in the Gantt chart.

i Note

If service activities, such as loading or coupling, are allowed during a rest, the "true" rest is visualized only. That means, no rest is shown in parallel to the service activities because the driver is actually working and not resting. For more information, see [Definition of Driving Time and Working Hours Constraints](#).

Definition of Driving Time and Working Hours Constraints

You define driving time and working hours constraints based on activity groups and rest types.

Activity Groups

Activity groups are sets of activity types. Available activity types are traveling, loading, unloading, etc.

❖ Example

- Activity group **Driving** with activity type **Travel**

- Activity group **Work** with activity types **Travel, Pickup, Delivery, Couple, Uncouple, Prepare, Finalize, Prepare First Stop, Finalize Last Stop**
- Activity group **Service** with activity types **Pickup, Delivery, Couple, Uncouple, Prepare, Finalize, Prepare First Stop, Finalize Last Stop**

Rest Types

Rest types describe a time period in which certain activities can't be executed and are used to reset the limits of a constraint. By default, a rest type doesn't allow any activities but you can overrule that by specifying an activity group to allow certain activities during a rest.

❖ Example

- In the EU, a daily rest lasts at least 11 hours.
- In the EU, a break lasts at least 45 minutes, but can be split into at least 15 minutes followed by at least 30 minutes break.
- In the US, a driving break needs to last at least 30 minutes, but service activities, such as loading or coupling, are allowed during the rest.

i Note

In your driving and working constraint profile, you can indicate that it's possible to split a rest into two shorter rests, for example, due to legal regulations. However, split rests are supported for constraint type **Accumulation** only.

Constraint Types

A constraint can enforce rests during scheduling and therefore defines the rules for driving times and working hours. This will increase the duration of a road freight order.

Constraint types determine how constraints are calculated. The following constraint types are available:

- Accumulation constraints

Accumulation constraints are used to model rules, where the sum of certain activities must not exceed a specific limit. The constraint only sums up specific activity types (that are defined by an activity group) and hence the end point of the limit is not a fixed point in time but depends on the scheduled activities. Formally, an accumulation constraint is defined by an accumulation limit, a restricted activity group, an accumulated activity group, and a rest type. Given a specified limit (capacity), the total duration of accumulated activity types (that are part of the activity group) must not exceed the limit. If the limit is exceeded, a rest of the defined rest type is scheduled before continuing scheduling restricted activities.

The accumulation constraint can sum up a group of activities since a defined starting point, which is either after the last rest or at the start of the week.

Vehicle scheduling and routing (VSR) optimization only supports constraints that start after the last rest.

❖ Example

- After accumulating nine hours of driving since the last rest, all work is restricted until a night's rest of 11 hours is taken.
- After accumulating four and a half hours of driving since the last rest, all work is restricted until a 45-minute rest is taken.

- Horizon constraints

Horizon constraints are used to model rules, where activities are only allowed within an interval of a fixed length (time horizon). The constraint requires a rest after the time interval (limit) has passed. The horizon constraint always counts the time that has elapsed since the last rest. Formally, a horizon constraint is defined by an interval limit, a restricted activity group, and a rest type. Given a specified limit (time horizon), and a starting point of the interval, activities can only be scheduled within the interval. If the interval boundaries are exceeded, a rest of the defined rest type is scheduled before continuing scheduling restricted activities.

❖ Example

14 hours after the last rest, a night's rest of 10 hours is required.

- Rolling constraints

Rolling constraints are used to model rules, where the sum of certain activities must not exceed a specified limit within any arbitrary interval of a defined duration. The constraint ensures that in any possible interval within the planning horizon, the accumulated (summed up) duration of a group of activities doesn't exceed the defined limit. Optionally, a rest type can be specified. If specified, a rest of this rest type can reset the constraint. Formally, a rolling constraint is defined by an accumulation limit, a restricted activity group, an accumulated activity group, a rest type, and a rolling interval length.

This constraint type isn't supported by VSR optimization.

❖ Example

Within any interval of seven consecutive days, a maximum accumulated driving time of 60 hours is allowed. A rest of 34 hours or more can be planned to restart the consecutive day period.

i Note

If a certain constraint enforces a rest of a certain duration, the system uses this rest to fulfill all other constraints with rests of a shorter duration.

For example, you have two accumulation constraints:

- After accumulating nine hours, a night's rest of 11 hours is needed.
- After accumulating four and a half hours, a 45-minute rest needed.

When the system plans the night's rest of 11 hours, it also considers the 45-minute rest as planned without planning an additional rest for it.

Several constraints are bundled in a driving and working constraint profile. For more information, see [Customizing for Transportation Management](#) under [►Planning ►General Settings ►Define Driving and Working Constraint Profile ►](#).

Examples of Constraints

You can use driving time and working hours constraints to model requirements related to driving time and working hours.

Examples: EU

In the following, examples that are relevant for the EU are described. Note that the solution provided in the system is recommended for short planning horizons. Further, it contains continuous rest periods, which implies that the scheduled activities and freight orders are valid with respect to the examples listed below.

- **EU 1 (Daily Rest):** Within each period of 24 hours after the end of the previous daily rest period, a driver shall have taken a new daily rest period of at least 11 hours. A rest is any uninterrupted period during which a driver may freely dispose of his or her time.
- **EU 2 (9h Driving Time):** The accumulated driving time between two rests of 11 hours shall not exceed 9 hours.
- **EU 3 (4.5h Driving Time):** After a driving period of 4.5 hours, a driver shall take an uninterrupted break of minimum 45 minutes unless he takes a rest period. A break is any period during which a driver may not carry out any driving or any other work and which is used exclusively for recuperation.

Split breaks: The break may be replaced by a first break of at least 15 minutes followed by a second break of at least 30 minutes each distributed over the period in such a way as to comply with the provisions of the first paragraph.

- **EU 4 (6h Working Time):** The driver needs to take a break of at least 30 minutes after maximum 6 hours working time (not only driving time).

Note that waiting times that occur inside a freight order can be used for rest periods.

Examples: US

In the following, examples that are relevant for the US are described. Note that the solution provided in the system is recommended for short planning horizons. Further, it contains continuous rest periods, which implies that the scheduled activities and freight orders are valid with respect to the examples listed below.

- **US 1 (Daily On-Duty Limit):** A truck driver may not drive beyond the 14th consecutive hour after coming on duty, following 10 consecutive hours off duty. Off-duty time doesn't extend the 14-hour period.
- **US 2 (11h Driving Time):** The accumulated driving time between two rests of 10 hours shall not exceed 11 hours.
- **US 3 (8h Driving Time):** Drivers must take a 30-minute break when they have driven for a period of 8 cumulative hours without at least a 30-minute interruption. The break may be satisfied by any non-driving period of 30 consecutive minutes that is, on-duty not driving, off-duty, or any combination taken consecutively.

i Note

Breaks in this chapter are not related to the break calendar but to the rests that you can define in the driving and working constraint profile.

Frequently Asked Questions

The following is a list of frequently asked questions (FAQs) regarding driving time and working hours constraints.

- **Can I combine multiple accumulation, horizon, and rolling constraints?**

Yes, any reasonable number of different constraint types can be combined. Note that a larger number of constraints increases the complexity and may, hence, have an impact on the runtime.

- **Can I use driving time and working hours constraints to model EU and US regulations?**

Yes, you can use driving time and working hours constraints to model official regulations. All constraints are considered as hard constraints during scheduling (scheduling strategy **Scheduling (Embedded)** (VSS_EMBED)) and vehicle scheduling and routing (VSR) optimization and therefore always lead to results that comply with the defined constraints. Since rests are scheduled with minimal durations as defined in the rest types, results should be executable in reality.

- **What is the difference between driving time and working hours constraints and scheduling constraints?**

Driving time and working hours constraints have two major advantages. They enable you to model more general constraints, and the scheduled activities and freight orders require the rests to be continuous, which implies that they fulfill the official regulations. Note that the costs of a solution with driving time and working hours constraints and costs of a solution with similar scheduling constraints can't be directly compared. In particular, driving time and working hours constraints are more restrictive, which may lead to a solution with increased costs.

- **What happens if I combine scheduling constraints and driving time and working hours constraints?**

In this case, only driving time and working hours constraints are considered during scheduling (scheduling strategy [Scheduling \(Embedded\)](#) (VSS_EMBED)) and VSR. Scheduling with this scheduling strategy doesn't support scheduling constraints.

- **Can I create different sets of rules for different types of vehicles?**

Yes, this is possible by using different driving and working constraint profiles. In the capacity availability settings of your planning profile, you can assign the profiles to your vehicle types and resources.

- **Are driving time and working hours constraints also supported by scheduling strategy [Scheduling \(Optimizer Server\)](#) (VSS_DEF)?**

Please use scheduling strategy [Scheduling \(Embedded\)](#) (VSS_EMBED). This scheduling strategy fully supports driving time and working hours constraints. Scheduling strategy [Scheduling \(Optimizer Server\)](#) (VSS_DEF) doesn't support driving time and working hours constraints.

- **Are rests and breaks in already planned road freight orders considered by driving time and working hours constraints when subsequent tours, that is road freight orders, are planned?**

Rests and breaks in already planned road freight orders are considered by driving time and working hours constraints if you have activated context determination in the transportation cockpit settings.

- **Is it possible to exclude specific activities at certain stops such as loading and unloading from the working hours calculation because they are done by local staff such as warehouse workers?**

Yes, it is possible to define activities that are done by local staff per location. You can define them in the capacity availability settings of your planning profile. These activity types aren't considered during driving time and working hours calculation.

- **There are no rests in the solution obtained by scheduling strategy [Scheduling \(Embedded\)](#) (VSS_EMBED). What could be a possible cause?**

First of all, check whether the obtained activity sequence is long enough to require rests. If this is the case, check in the scheduling settings and the capacity availability settings whether driving time and working hours constraints have been activated and relevant for your vehicle or driver. Further, you could check your driving and working constraint profile in Customizing.

- **There are no rests in the solution obtained by VSR optimization. What could be a possible cause?**

First of all, check whether the obtained activity sequence is long enough to require rests. If this is the case, check in the explanation tool, whether driving time and working hours constraints have been activated. If this is the case, compare the output of the optimizer displayed in the explanation tool with the solution displayed in the freight order overview. If they are different, scheduling strategy [Scheduling \(Optimizer Server\)](#) (VSS_DEF) might have been called after the VSR optimization run. This scheduling strategy doesn't support driving time and working hours constraints. We recommend to use scheduling strategy [Scheduling \(Embedded\)](#) (VSS_EMBED) instead.

- **How can I analyze the impact of driving time and working hours constraints on scheduling?**

You can use the Gantt chart to display the activities and rests of your road freight orders. If you want to analyze scheduling (scheduling strategy [Scheduling \(Embedded\)](#) (VSS_EMBED)), you can use the [Explain Scheduling Results](#) app.

Effects of Date/Time Deviations During Execution

Use

You can check the effects that deviations in dates/times during the execution of freight orders have on the rest of your planning. Deviations in dates/times can occur if a delay is reported.

Features

If a freight order cannot be executed as planned, for example, because the truck left late, you can adjust your planning. You do this by performing forward scheduling. The system reschedules the freight order.

Scheduling only takes place for the resource that is assigned to the freight order. If there are dependencies to other resources or freight orders, this can lead to conflicts, for example, for multimodal transports. You must solve this conflict manually or using VSR optimization. We recommend solving conflicts manually. If you use VSR optimization, you must make sure that all objects are in the planning horizon.

Activities

You can start scheduling on the following user interfaces:

- Transportation cockpit

We recommend scheduling from the transportation cockpit if the system is to consider not only the late freight order but also all of the other freight orders at the resource during scheduling. To start scheduling, open the [Transportation Cockpit](#) app on the launchpad.

- User interface for changing freight orders

We recommend scheduling from this user interface if you only want to schedule late freight orders. To start scheduling, open the [Edit Road Freight Order](#) or [Edit Rail Freight Order](#) app on the launchpad.

- Freight order overview

To start scheduling, open the [Freight Orders](#) app on the launchpad.

Loading and Unloading Durations

Use

Loading and unloading (stop) times are calculated by the system during manual planning and during VSR optimization based on the rule-set specified in the [Create Scheduling Settings](#) app. The app also gives you the ability to export the rule-set to a spreadsheet so that you can share it with someone who does not use the app. This feature lets you use a spreadsheet application to analyze or modify the rule-set. You can then import the data modified in the spreadsheet application back to the app. This can be used as an alternative to modifying the data directly in the app.

Features

The rule-set lets you specify the following:

- A fixed duration for loading and unloading activities at a location

- A variable duration (time per unit) for loading and unloading activities at a location
- The duration required for preparing and finalizing a location. This duration can be used to account for activities such as completing paperwork and driving through the depot.
- Operating times and handling resources for loading and unloading

Note that these values overwrite operating times and handling resources that you have defined in master data.

- Operating times for loading and unloading based on requirement document attributes, such as dangerous goods

The fields for entering the details are categorized by the direction of travel (inbound and outbound):

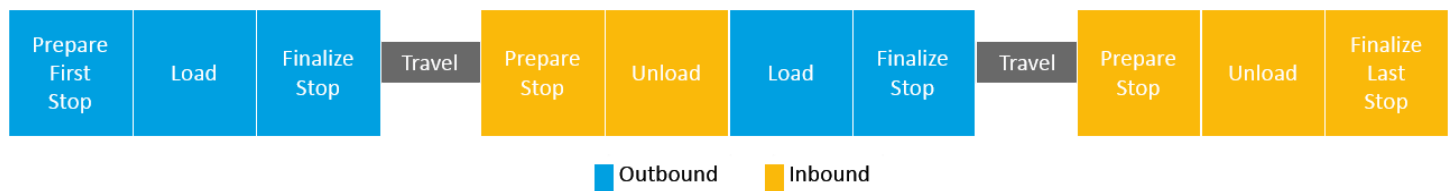
- Outbound tab

Use the tab to enter details for loading durations, the duration for finalizing stops, and preparing the first stop.

- Inbound tab

Use the tab to enter details for unloading durations, the duration for preparing stops, and finalizing the last stop.

The figure below shows how stops and activities are categorized.



i Note

- For schedule-based freight orders and freight bookings, scheduling does not consider loading and unloading durations.
- If you specify multiple rules, the system uses the first rule that matches the location and resource.
- The loading and unloading durations are calculated by a fixed duration that is applied once per capacity document stop, plus the variable durations per requirement document stop. The overall duration is applied to both the capacity document and the requirement document stops.

Activities

To specify a rule-set, on the Launchpad, open the [Create Scheduling Settings](#) app.

To export the rule-set to a spreadsheet, click [Download](#). Then you can analyze and modify the rule-set in the spreadsheet application and import the file back to the app by clicking [Upload](#).

Time Windows

Use

You use time windows to control when goods are to be picked up and delivered. VSR optimization (see [VSR Optimization](#)) then schedules the pickup or delivery exactly in the time window that you have defined.

You can also control how the system is to deal with cases in which these time windows are not taken into account during VSR optimization, that is, whether VSR optimization is to calculate penalty costs. You can model the following cases:

- The goods are picked up too early or too late.
- The goods are delivered too early or too late.

Features

You create time windows by defining tolerances within which a premature or delayed pickup or delivery is to be allowed. For each tolerance, you specify whether VSR optimization is to consider the tolerance as a hard, soft, or hard and soft constraint (see [Constraints for VSR Optimization](#)). In this way, you control whether VSR optimization calculates penalty costs when the constraints are not adhered to.

You can define the following tolerances:

- Maximum earliness
- Maximum delay
- Delay without penalty costs
- Earliness without penalty costs

The dates and times that you have defined for the pickup and delivery in the business document, for example, the forwarding order, form the basis for calculating the tolerances. For more information, see [Definition of Delivery Windows](#).

You can also specify that the system is to take into account the exact time for the pick-up date or delivery date. The system always uses the current point in time and rounds up or down to full days.

For more information about the link between tolerances and constraints, see [Delivery Window](#).

i Note

- The system also takes into account windows for pickup and delivery in manual planning (see [Manual Planning](#)). However, this applies to hard constraints only. If it cannot adhere to the tolerances, it simply issues a warning message.
- The system already takes time windows into account when creating freight units and determines the dates of the freight units. It does not evaluate the time windows further when you access planning.
- The system takes into account a factory calendar of a calendar resource assigned to a location when creating freight units.

Activities

You define your time windows by entering a rule for windows for pickup and delivery in the freight unit type or a condition (condition type /SCMTMS/TOR_TIMEWIND). For more information, see Customizing for Transportation Management under [Planning > Freight Unit > Define Freight Unit Type](#).

For more information about defining conditions for time windows, see [Definition of a Condition for Time Windows](#).

Delivery Window

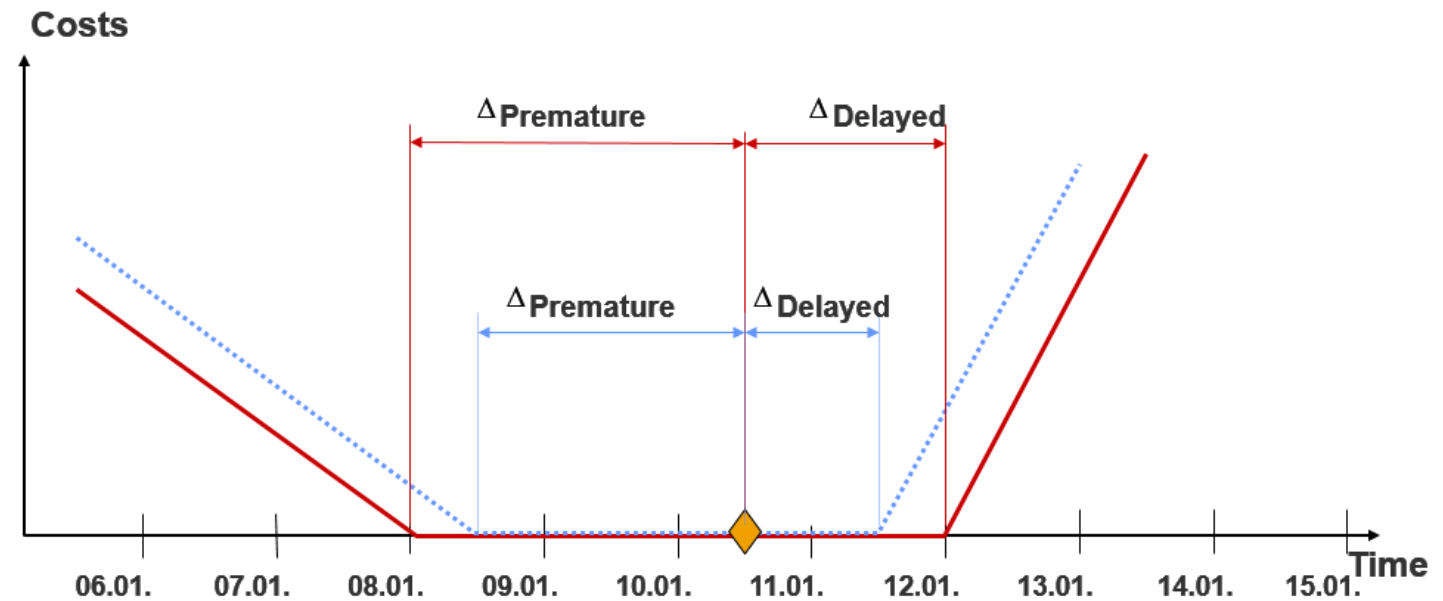
You have defined the following tolerances:

- Delta premature = 2 days (tolerance for premature delivery without penalty costs)

- Delta earliest = 3 days (tolerance for maximum earliness)
- Delta delayed = 1 day (tolerance for delayed delivery without penalty costs)
- Delta latest = 2 days (tolerance for maximum delay)

The requested delivery date/time (from date/time) is 10.01., 12:00. The solid red curve in the following figures specifies that you do not want to take into account the exact time of the delivery date. The dotted blue curve specifies that you want to take into account the exact time.

The following figures show how these tolerances affect the delivery date, depending on whether you define them as hard, soft, or hard and soft constraints.



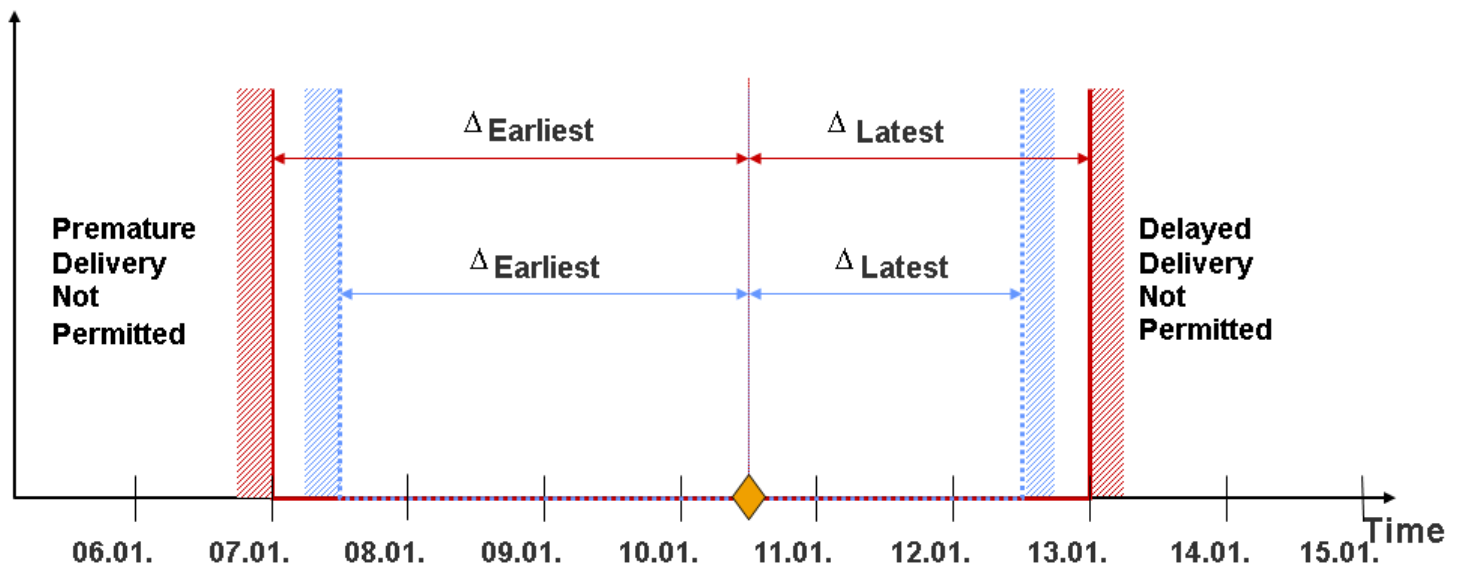
Soft Constraints for Premature and Delayed Delivery

In this case, a delayed delivery is allowed until 12.01. (solid red curve). Since the system is not to take into account the exact time for the delivery date, it rounds up to the full day, in other words, from 11.01., 12:00 to 12.01., midnight. No penalty costs are calculated. A delivery after 12.01. is allowed, but penalty costs are incurred.

A premature delivery can occur at the earliest on 08.01. without penalty costs being incurred. If the delivery occurs before 08.01., penalty costs are incurred. Since the system is not to take into account the exact time for the delivery date, it rounds down to the full day, in other words, from 08.01., 12:00 to 08.01., midnight.

If you have specified that the system is to take into account the exact time of the delivery date (dotted blue curve), penalty costs are incurred for a delayed delivery after 11.01., 12:00 and for a premature delivery before 08.01., 12:00.

Costs

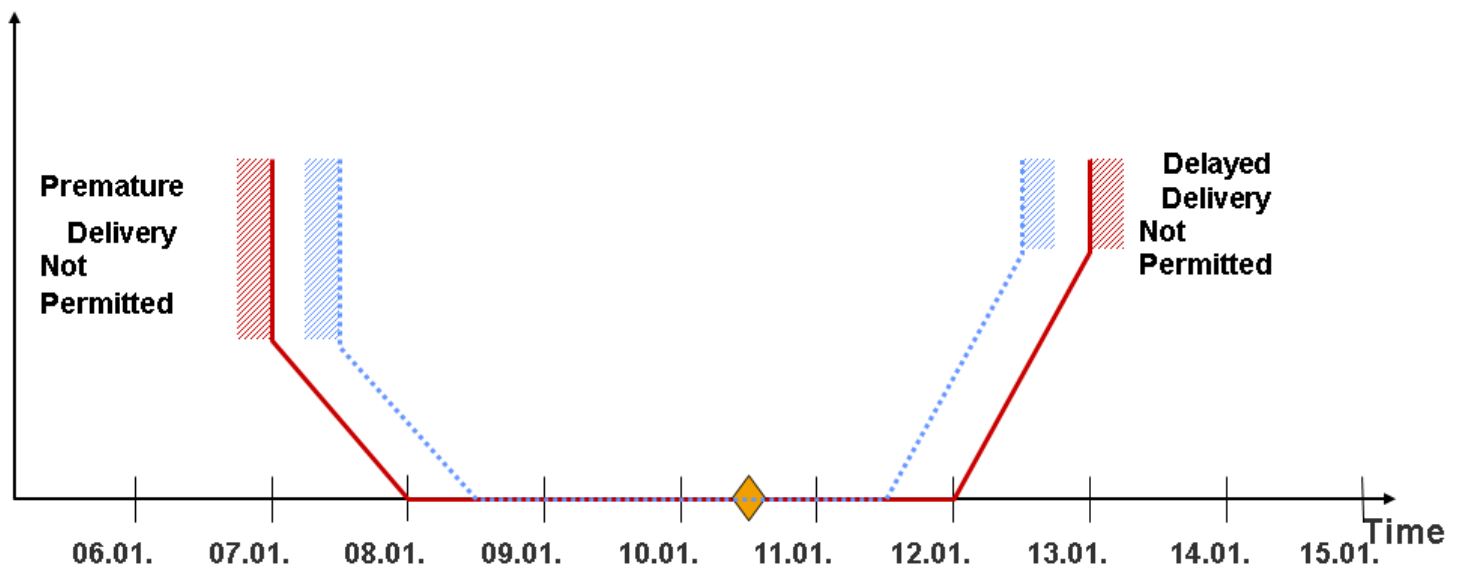


Hard Constraints for Premature and Delayed Delivery

In this case, a delayed delivery is not allowed after 13.01. (solid red curve). The delivery can occur at the earliest on 07.01. Since the system is not to take into account the exact time for the delivery date, it rounds up or down to the full day, in other words, from 12.01., 12:00 to 13.01., midnight and from 07.01., 12:00 to 07.01., midnight.

If you have specified that the system is to take into account the exact time of the delivery date (dotted blue curve), a delivery is not allowed after 12.01., 12:00 and before 07.01., 12:00.

Costs



Hard and Soft Constraints for Premature and Delayed Delivery

In this case, a delayed delivery is allowed after 10.01. (solid red curve). Since the system is not to take into account the exact time for the delivery date, it rounds up to the full day, in other words, from 11.01., 12:00 to 12.01., midnight. However, penalty costs are incurred if the delivery occurs on 12.01. A delivery after 13.01. is not allowed.

A premature delivery is allowed from 07.01. Since the system is not to take into account the exact time for the delivery date, it rounds down to the full day, in other words, from 08.01., 12:00 to 08.01., midnight. However, penalty costs are incurred if the delivery occurs on 07.01. A delivery before 07.01. is not allowed.

If you have specified that the system is to take into account the exact time of the delivery date (dotted blue curve), a delayed delivery is not allowed after 12.01., 12:00. Penalty costs are incurred for a delivery between 11.01., 12:00 and 12.01., 12:00.

A premature delivery before 07.01., 12:00 is not allowed. Penalty costs are incurred for a delivery between 07.01., 12:00 and 08.01., 12:00.

Definition of Delivery Windows

The following examples show the options that you have when defining time windows. First of all, the relevant settings that you make are shown. A figure is then used to show how the system interprets your entries. The examples all refer to the delivery. However, they also apply in the same way to the pickup.

Example 1

In the forwarding order, you have defined the following date/time for the delivery at the destination location:

Earliest delivery (acceptable dates/times)	N/A
Delivery date/time (from) (requested dates/times)	17.09.07 00:00:00
Delivery date/time (to) (requested dates/times)	N/A
Latest delivery (acceptable dates/times)	N/A

i Note

You define dates/times for delivery on the user interface for defining the forwarding order ([Locations and Dates](#) tab).

You have also defined the following tolerances:

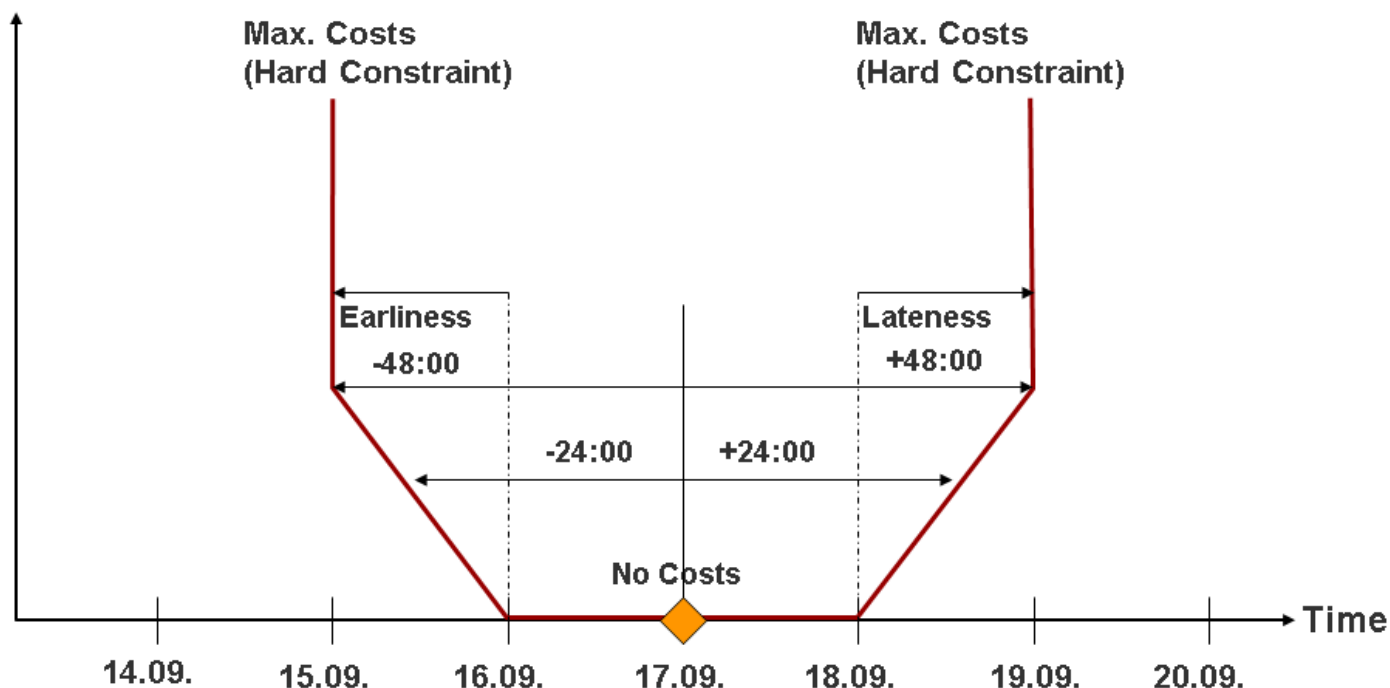
Tolerance for early delivery	24:00 hrs
Tolerance for earliest delivery	48:00 hrs
Tolerance for late delivery	24:00 hrs
Tolerance for latest delivery	48:00 hrs

i Note

You define penalty costs for earliness and lateness in the costs settings for the planning profile either directly or by entering a condition (condition type /SCMTMS/FU_PNLT_COST).

The following figure shows how the system takes these values into account and what the time window looks like:

Costs



Time Window (Example 1)

In this case, the system takes into account all tolerances. A delayed delivery up until 18.09 does not incur any costs. Penalty costs are incurred between 18.09 and 19.09. A delivery is not permitted as of 19.09. A premature delivery can occur at the earliest on 16.09. without penalty costs being incurred. If the delivery occurs between 15.09 and 16.09, penalty costs are incurred. A delivery is not permitted before 15.09.

Example 2

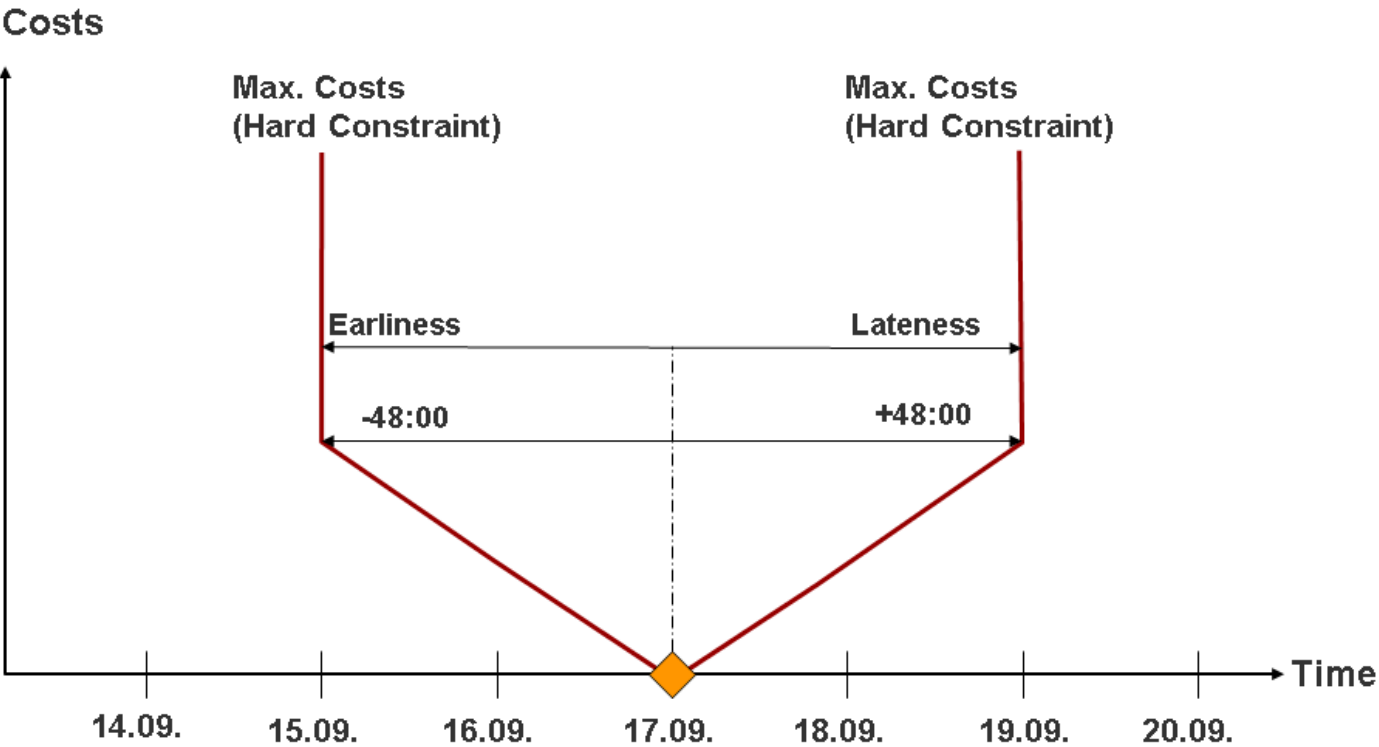
In the forwarding order, you have defined the following dates/times for the delivery:

Earliest delivery (acceptable dates/times)	N/A
Delivery date/time (from) (requested dates/times)	17.09.07 00:00:00
Delivery date/time (to) (requested dates/times)	17.09.07 00:00:00
Latest delivery (acceptable dates/times)	N/A

You have also defined the following tolerances:

Tolerance for early delivery	24:00 hrs
Tolerance for earliest delivery	48:00 hrs
Tolerance for late delivery	24:00 hrs
Tolerance for latest delivery	48:00 hrs

The following figure shows how the system takes these values into account and what the time window looks like:



Time Window (Example 2)

In this case, the system does not take into account the tolerances for early and late delivery, since it takes into account the dates/times that you have defined in the forwarding order. A delayed delivery is possible until 19.09, but penalty costs are incurred. A delivery is not permitted as of 19.09. A premature delivery can occur at the earliest on 15.09. If the delivery occurs between 15.09 and 17.09, penalty costs are incurred. A delivery is not permitted before 15.09.

Example 3

In the forwarding order, you have defined the following dates/times for the delivery:

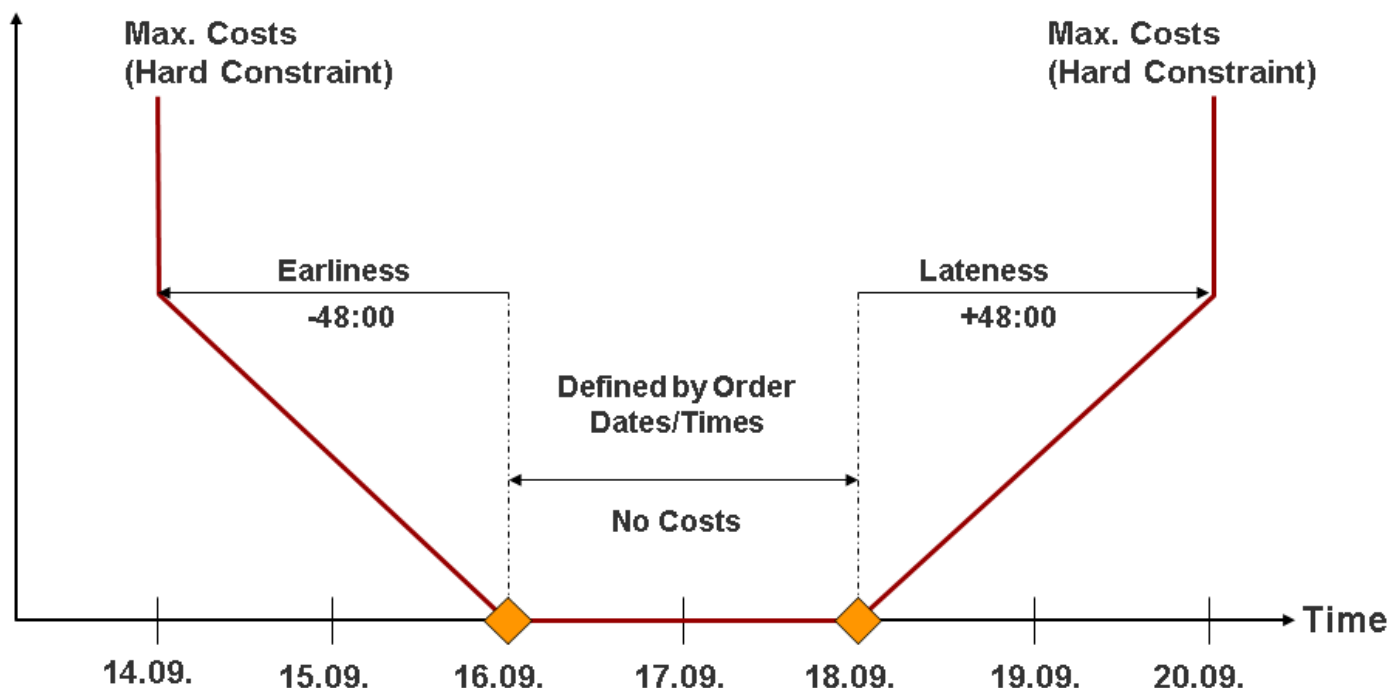
Earliest delivery (acceptable dates/times)	N/A
Delivery date/time (from) (requested dates/times)	16.09.07 00:00:00
Delivery date/time (to) (requested dates/times)	18.09.07 00:00:00
Latest delivery (acceptable dates/times)	N/A

You have also defined the following tolerances:

Tolerance for early delivery	24:00 hrs
Tolerance for earliest delivery	48:00 hrs
Tolerance for late delivery	24:00 hrs
Tolerance for latest delivery	48:00 hrs

The following figure shows how the system takes these values into account and what the time window looks like:

Costs



Time Window (Example 3)

In this case, the system only takes into account the tolerances for early and late delivery. A delayed delivery up until 18.09 does not incur any costs. Penalty costs are incurred between 18.09 and 20.09. A delivery is not permitted as of 20.09. A premature delivery can occur at the earliest on 16.09. without penalty costs being incurred. If the delivery occurs between 14.09 and 16.09, penalty costs are incurred. A delivery is not permitted before 14.09.

Example 4

In the forwarding order, you have defined the following dates/times for the delivery:

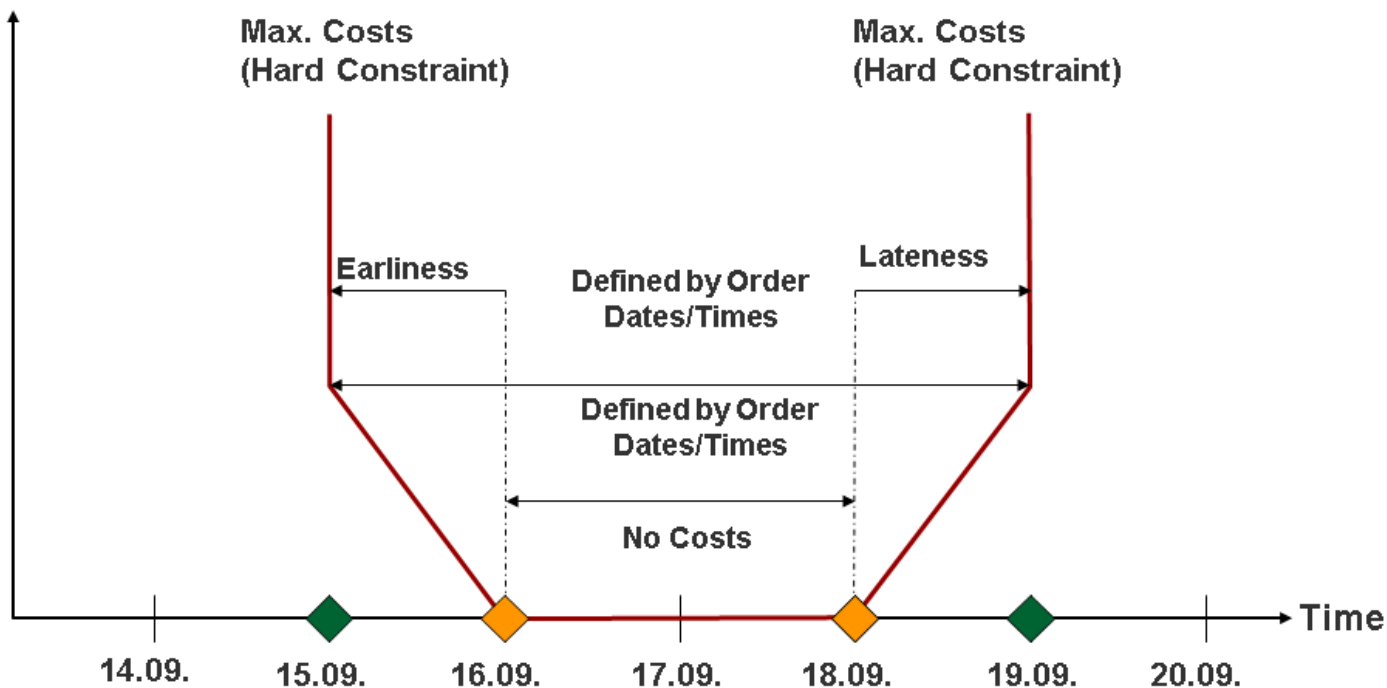
Earliest delivery (acceptable dates/times)	15.09.07 00:00:00
Delivery date/time (from) (requested dates/times)	16.09.07 00:00:00
Delivery date/time (to) (requested dates/times)	18.09.07 00:00:00
Latest delivery (acceptable dates/times)	19.09.07 00:00:00

You have also defined the following tolerances:

Tolerance for early delivery	24:00 hrs
Tolerance for earliest delivery	48:00 hrs
Tolerance for late delivery	24:00 hrs
Tolerance for latest delivery	48:00 hrs

The following figure shows how the system takes these values into account and what the time window looks like:

Costs



Time Window (Example 4)

In this case, the system does not take any of the tolerances into account, since it takes into account the dates/times that you have defined in the forwarding order. A delayed delivery up until 18.09 does not incur any costs. Penalty costs are incurred between 18.09 and 19.09. A delivery is not permitted as of 19.09. A premature delivery can occur at the earliest on 16.09. without penalty costs being incurred. If the delivery occurs between 15.09 and 16.09, penalty costs are incurred. A delivery is not permitted before 15.09.

Example 5

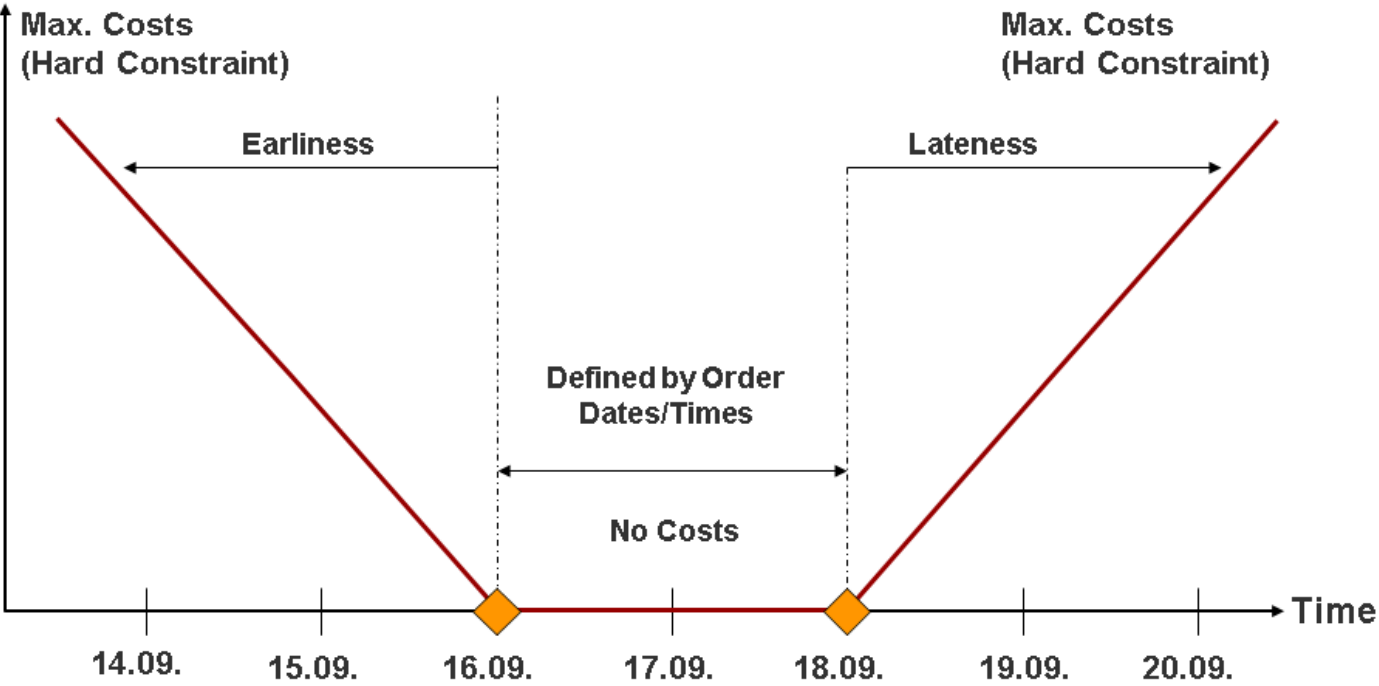
In the forwarding order, you have defined the following dates/times for the delivery:

Earliest delivery (acceptable dates/times)	N/A
Delivery date/time (from) (requested dates/times)	16.09.07 00:00:00
Delivery date/time (to) (requested dates/times)	18.09.07 00:00:00
Latest delivery (acceptable dates/times)	N/A

In this case, you have not defined any tolerances.

The following figure shows how the system takes these values into account and what the time window looks like:

Costs



Time Window (Example 5)

Penalty costs are incurred for a delayed delivery as of 18.09. Penalty costs are not incurred for a delayed delivery between 17.09 and 18.09. Penalty costs are incurred for a premature delivery before 16.09. Penalty costs are not incurred for a premature delivery between 16.09 and 17.09.

Example 6

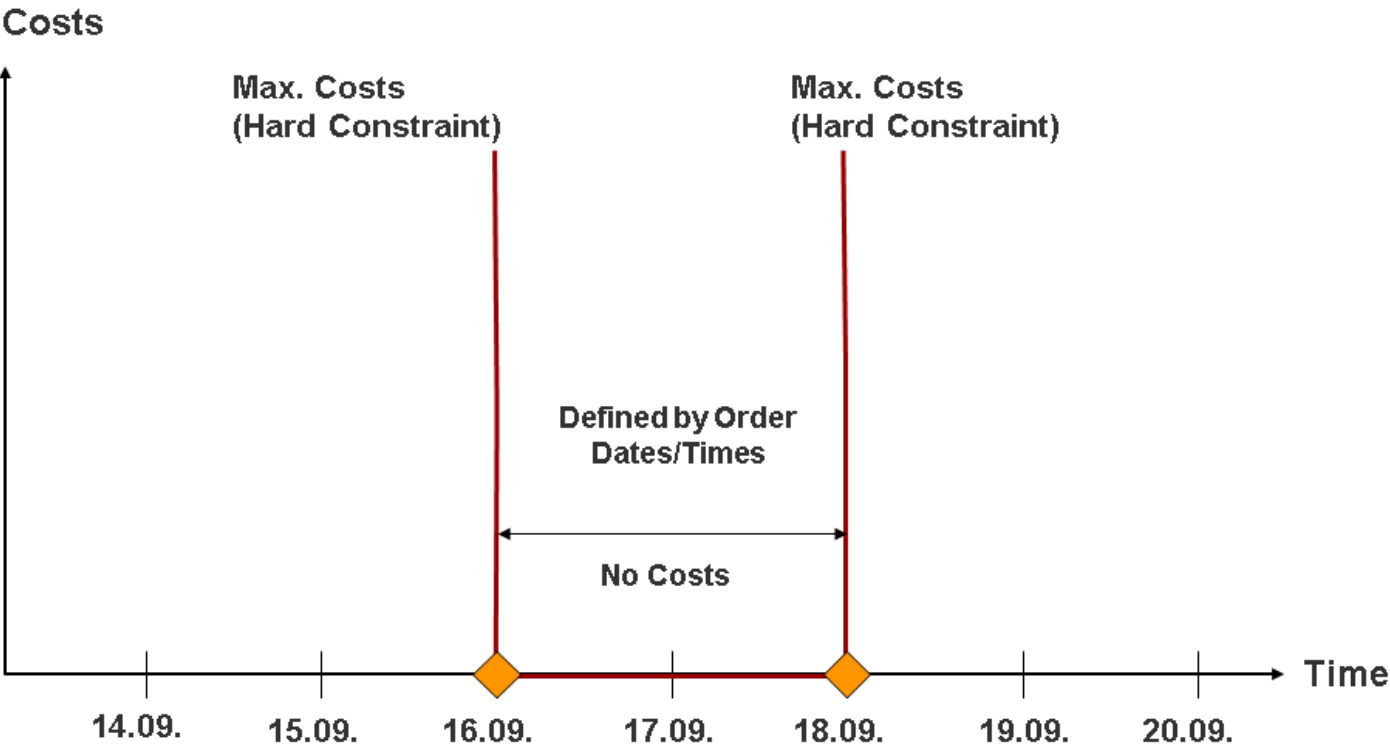
In the forwarding order, you have defined the following dates/times for the delivery:

Earliest delivery (acceptable dates/times)	16.09.07 00:00:00
Delivery date/time (from) (requested dates/times)	N/A
Delivery date/time (to) (requested dates/times)	N/A
Latest delivery (acceptable dates/times)	18.09.07 00:00:00

You have also defined the following tolerances:

Tolerance for early delivery	24:00 hrs
Tolerance for earliest delivery	48:00 hrs
Tolerance for late delivery	24:00 hrs
Tolerance for latest delivery	48:00 hrs

The following figure shows how the system takes these values into account and what the time window looks like:



Time Window (Example 6)

In this case, the system does not take any of the tolerances into account. Penalty costs are not incurred for a delayed delivery between 17.09 and 18.09. A delivery is not permitted after that. Penalty costs are not incurred for a premature delivery between 16.09 and 17.09. A delivery is not permitted before that.

Example 7

In this example, the system interprets the dates/times from the forwarding order as hard constraints without you having to define hard constraints. In the forwarding order, you have defined the following dates/times for the delivery:

Earliest delivery (acceptable dates/times)	N/A
Delivery date/time (from) (requested dates/times)	16.09.07 00:00:00
Delivery date/time (to) (requested dates/times)	18.09.07 00:00:00
Latest delivery (acceptable dates/times)	N/A

You have also defined the following tolerances:

Tolerance for early delivery	N/A
Tolerance for earliest delivery	00:00 hrs
Tolerance for late delivery	N/A
Tolerance for latest delivery	00:00 hrs

The following figure shows how the system takes these values into account and what the time window looks like:

Costs



Time Window (Example 7)

In this case, the system only takes into account the tolerances for earliest and latest delivery. Penalty costs are not incurred for a delayed delivery between 17.09 and 18.09. A delivery is not permitted after that. Penalty costs are not incurred for a premature delivery between 16.09 and 17.09. A delivery is not permitted before that.

Definition of a Condition for Time Windows

This example shows how you define a condition for a time window depending on the freight unit type.

In this case, you define a condition with the condition type /SCMTMS/TOR_TIMEWIND based on a decision table.

As the data access definition, enter /SCMTMS/TOR_TYPE (business document type of freight order management).

Depending on the freight unit type, you specify in the decision table which durations and rounding rules the system is to use when calculating the times for the time window.

Earliness and Lateness

In the **Earliness** and **Lateness** fields, you specify whether the system is to treat an earliness or lateness as a constraint and, if yes, as which sort of constraint:

- As a soft constraint
- As a hard constraint
- As a soft and hard constraint

If you have entered **Soft Constraint** or **Soft and Hard Constraint**, in the **Prem. Stay** (premature stay) field and the **Del. Stay** (delayed stay) field, enter the required time period in the format HH:MM.

If you entered **hard constraint** or **soft and hard constraint**, in the **EarlstStay** (earliest stay) field and the **Lat. Stay** (latest stay) field, enter the required offset in the format HH:MM.

Round

In addition, you can specify whether and how you want the system to round (to complete days, for example).

Operating Times

You can also specify whether you want the system to consider the operating times of a location. As a prerequisite, you must have defined a calendar resource with operating times in the master data and assigned this to the location.

Note that VSR optimization (VSR = vehicle scheduling and routing) only takes the operating times into account if you have defined loading and unloading durations that are greater than zero.

Delivery and Pickup

You can enter different values for the delivery and the pickup. To do so, enter a second data access definition /SCMTMS/TOR_STOP_CAT.

In the decision table, specify the following values for pickup:

Stop Type	Earliness	Lateness	EarlStStay (earliest stay)	Round
0 (pickup)	2 (hard constraint)	0 (no consideration as constraint)	0	True

This means that the pickup may not take place earlier than agreed. A late pickup is acceptable. The system is to round the pickup date to complete days.

You specify the following values for the delivery:

Stop Type	Earliness	Lateness	Lat. Stay (latest stay)	Round
I (delivery)	0 (no consideration as constraint)	2 (hard constraint)	0	False

This means that the delivery may not take place later than agreed. An early delivery is acceptable.

Since the time of the delivery date defined by the customer is to be adhered to, specify that the system is not to round it to full days.

Calculation Base

You can specify how loading and unloading activities are to be scheduled in terms of the pick-up/delivery windows of the freight unit or transportation unit, that is, whether the activities must start or end in the time window.

The following values are possible:

- **Start of (Un-)Loading Activity**
- **End of (Un-)Loading Activity**
- **Default** (this is the start of the loading activity for outbound scenarios and the end of the unloading activity for inbound scenarios)

Explanation Tool for Scheduling

With this app, you can analyze your scheduling results to better understand how the system has scheduled your objects.

Prerequisites

- You have the authorization to work with the explanation tool (authorization object T_EXP_USR).
- You have set user parameter /SCMTMS/EXP to X to allow the logging of scheduling data.
- The explanation tool for scheduling uses a Gantt chart for visualization. The system uses the colors defined in the default layout for the Gantt chart. If you want to use other colors that are defined in another layout, you need to select the **VSS Expl Tool Layout** checkbox for this other layout. For more information, see Customizing for Transportation Management under **Basic Functions > Gantt Chart > Define Layouts for Gantt Chart**.

For more information about prerequisites for scheduling, see [Scheduling](#).

Features

Overview Page

On this page, all explanation logs are displayed in a table with basic information, such as the following:

- Planning profile
- Scheduling settings
- Planning horizon
- Scheduling direction
- Scheduling strategy
- Scheduling status, that is, whether the scheduling run was successful or not
- Action that called scheduling, for example freight unit building

You can filter the content using different search criteria such as the planning profile and search for a specific explanation log. You can also directly navigate to a planning profile or specific scheduling settings.

By selecting an explanation log for a successful scheduling run, you can navigate to the detail page.

Detail Page

On this page, detailed scheduling results are displayed both in a table view on the left side and a Gantt chart on the right side.

You can do the following:

- Iterate through the individual **solution steps**

For example, you can display the results after the system has scheduled the activities as early as possible or you can display the final results.

- Switch between **activity groups** to check the results per activity

An activity group is a chain of activities that need to be scheduled and do not have a relationship with any other activity outside of this chain. For each activity group, the system performs scheduling separately. You can iterate through the

solution steps for each activity group.

- Switch between different views:
 - Activity
 - Truck resource or truck type
 - Handling resource
- Display different solution steps for different views
- Visualize relationships between activities in the Gantt chart
- Visualize constraints such as downtimes and non-working times in the Gantt chart
- Display the current start and end time of each activity as well as the allowed time period in which the activity can be scheduled

Activities

To analyze scheduling results, open the [Explain Scheduling Results](#) app from the launchpad.

Related Information

[Gantt Chart for Transportation Planning](#)

Transportation Requirement Routing and Scheduling

Transportation requirement routing and scheduling (TRS) is a foundation module that supports various functions, such as route determination, transportation duration determination, and scheduling of transportation requirements.

You can use TRS for the following:

- [Determination of Transportation Durations for BPS-Based Scheduling](#)

Routing for a given request from source to destination location is performed based on master data, such as transportation lanes and default routes. In Transportation Management (TM), durations are determined based on the routing results.

In order-based sales order scheduling, you can use this function to determine the durations of load, travel, and unload activities. Based on these durations, scheduling is then performed in Business Process Scheduling (BPS). For more information, see [Integration of Sales Document Scheduling](#).

- [Routing and Scheduling of Transportation Requirements](#)

Routing for a given request from source to destination location is performed based on master data, such as transportation lanes and default routes. Based on the routing results and a reference date, forward or backward scheduling is performed.

In order-based sales order scheduling, you can use this function to delegate scheduling of load, travel, and unload activities to TM using BPS. For more information, see [Integration of Sales Document Scheduling](#).

You can use the TRS test cockpit to simulate TRS. For more information, see [Test Cockpit for Transportation Requirement Routing and Scheduling](#).

Determination of Transportation Durations for BPS-Based Scheduling

Transportation requirement routing and scheduling (TRS) enables you to determine the transportation durations of your requirements in an order-based scheduling scenario. Based on this information, scheduling is performed in Business Process Scheduling (BPS), a flexible configurable scheduling solution in advanced Available-to-Promise (aATP).

Prerequisites

- You have defined a transportation duration determination profile. In this profile, you can indicate one or several routing and determination methods and the sequence in which the system is to perform them. You can also indicate a planning profile that has scheduling settings assigned to it if you want to use the fixed loading durations and durations for prepare and finalize that you have defined there. To create a profile, open the [Create Transportation Duration Determination Profile](#) app from the launchpad.

i Note

- If the system determines durations using one method, these durations are not changed by any subsequent method that you indicated.
- The system uses the default means of transport for determining the loading duration.
- The system doesn't use the planning horizon from the planning profile but a default horizon of 60 days.
- If you want to use this function for order-based sales order scheduling, you have set up the integration of sales order scheduling. For more information, see [Integration of Sales Document Scheduling](#).

Use

In TRS, travel durations as well as loading and unloading durations are determined based on the routing and determination methods that you selected in your transportation duration determination profile. For more information, see [Routing and Determination Methods](#).

For the determination, the scheduling settings of the assigned planning profile are used.

The system can use the means of transport and vehicle type determined by the corresponding determination method as the basis for determination. If no means of transport or vehicle type has been determined, the default means of transport is used as the basis for determination.

Activities

When you create a sales order in Sales and Distribution (SD), scheduling takes place automatically in BPS based on your settings when you enter a material and press ENTER. Based on the transportation durations determined in TRS, BPS calculates the delivery date. For more information about sales order scheduling based on TM durations, see [Integration of Sales Document Scheduling](#).

You can simulate this function in the TRS test cockpit. For more information, see [Test Cockpit for Transportation Requirement Routing and Scheduling](#).

Related Information

[Business Process Scheduling \(BPS\)](#)

Routing and Scheduling of Transportation Requirements

For a given request from a source to a destination location and a reference date, routing and scheduling are performed.

Prerequisites

- You have created a requirement routing profile. To do so, open the [Create Requirement Routing Profile](#) app from the launchpad. You have specified the following:
 - A planning profile
 - Routing and determination methods (see [Routing and Determination Methods](#))

Note that the system doesn't use the planning horizon from the planning profile but a default horizon of 60 days.

- If you want the system to consider fixed loading and unloading durations, you have defined them in the scheduling settings of the planning profile that you have assigned to your requirement routing profile. To do so, open the [Edit Scheduling Settings](#) app from the launchpad.
- If you want the system to consider factory calendars, shift sequences, and driving time and working hours constraints, you have defined them in the capacity availability settings of the planning profile that you have assigned to your requirement routing profile. To do so, open the [Edit Capacity Availability Settings](#) from the launchpad.
- If you want to use this function for order-based sales order scheduling, you have set up the integration of sales order scheduling. For more information, see [Integration of Sales Document Scheduling](#).

Use

Routing tries to determine a possible route from a source to a destination location for a transportation requirement. Scheduling determines dates and times for loading, traveling, and unloading. Scheduling considers a reference date as requested delivery date in backward scheduling or as requested loading date in forward scheduling.

The system determines loading and unloading durations based on the scheduling settings of the assigned planning profile. Scheduling considers factory calendars, shift sequences, and driving time and working hours constraints based on the capacity availability settings of the assigned planning profile. The system can use the means of transport and vehicle type determined by the corresponding determination method as the basis for determination and scheduling. If no means of transport or vehicle type has been determined, the default means of transport is used as the basis for determination.

Optionally, scheduling can try to find the next possible schedule departure for stages that have been assigned to a schedule during routing.

i Note

Embedded scheduling (VSS_EMBED) is supported only.

Activities

When you create a sales order in Sales and Distribution (SD), scheduling takes place automatically in Transportation Management (TM) based on your settings when you enter a material and press ENTER. For more information about sales order scheduling based on TM scheduling, see [Integration of Sales Document Scheduling](#)

You can simulate this function in the TRS test cockpit. For more information, see [Test Cockpit for Transportation Requirement Routing and Scheduling](#).

Routing and Determination Methods

Routing and determination methods can be used in transportation requirement routing and scheduling (TRS) processes to determine routing for requests from a source location to a destination location.

In the transportation duration determination profile and the requirement routing profile, you can define a sequence of multiple routing and determination methods. The system processes the methods one after the other. If a method determines a result, this result is not changed anymore by the succeeding methods.

The following methods are available:

- Means of transport determination (see [Means of Transport and Vehicle Type Determination](#))

Based on business rules, the system searches for a means of transport, a vehicle type, or both per stage. If it finds any, it copies it into the result.

Means of transport and vehicle type determination can be performed before and after any other method. It can be performed several times to improve the results of a preceding routing method, but not two times after each other.

❖ Example

- You can have the system determine stages using network path determination or default route determination first and then determine the means of transport or vehicle type per stage.
- You can have the system determine the means of transport or vehicle type first and then determine distances using straight-line determination or lane, distance, and duration determination.
- You can have the system determine the means of transport or vehicle type first and then determine stages using network path determination or default route determination.

- Default route determination (see [Default Route](#))

The system searches for default routes. If it finds any, it copies the routing information and the travel durations into the result. If it finds a default route without a duration in any of the stages, subsequent methods can determine the duration for these stages.

- Lane, distance, and duration determination (LDDD) (see [Lane, Distance, and Duration Determination \(LDDD\)](#))

The system searches for durations using the lane, distance and duration determination. Note that the standard configuration of LDDD first checks transportation lanes and then uses straight-line determination as fallback.

- Straight-line determination

The system determines straight-line distances and durations based on the speed of the default means of transport.

- Network path determination

The system tries to determine a path from source to destination location based on the transportation network.

To do so, it determines transshipment locations as well as schedules taking into account parameters that you have indicated in the requirement routing profile, such as the search depth for these locations or schedules.

You can indicate whether the system tries to determine the path with the shortest duration or distance.

i Note

- If you indicated that the path with the shortest duration is to be determined, the system adds an expected waiting time until the next possible schedule departure for a schedule.

- The system doesn't evaluate default routes.
- The system doesn't evaluate the capacity of schedules.
- You can simulate network path determination using the test cockpit for network path determination. For more information, see [Test Cockpit for Network Path Determination](#).

A Business Add-In (BAI) is available to change the process settings and the results of the routing and determination methods. For more information, see Customizing for [Transportation Management](#) under [Business Add-Ins \(BAIs\) for Transportation Management](#) > [Planning](#) > [BAI: Adjustment of TRS Methods](#).

In the following section, you find various examples of how these routing and determination methods are applied.

Let's assume you have indicated in your TRS profile that the following three routing and determination methods are to be executed in the following sequence:

- Default route determination
- Means of transport and vehicle type determination
- Network path determination
- Means of transport and vehicle type determination
- Lane, distance and duration determination

Example 1:

There is a requirement from A to C1.

There is a default route from A via B to C1 with no means of transport defined for any stage, in which the stage from A to B has a duration of one 1 day and the stage from B to C1 has a duration of one hour.

The system determines the default route and the stages are enhanced with the means of transport and the vehicle types. The system neither changes the routing during network path determination nor the durations during lane, distance and duration determination anymore.

Example 2:

There is a requirement from A to C2.

There is a default route from A via B to a zone that includes C2, in which the stage from A to B has a duration of one day and no means of transport defined. The stage from B to the zone has no duration defined (that means, the duration is set to zero) but has a means of transport defined.

The system determines the default route and only the first stage A to B is enhanced with the means of transport and vehicle type. The system doesn't change the routing during network path determination anymore but adds a duration for the stage from B to C2 during lane, distance and duration determination since the duration was still missing.

Example 3:

There is a requirement from A to C3.

There is no default route. There is no direct lane between A and C3 but B is a transshipment location between both and there are lanes from A to B and B to C.

The system doesn't determine a default route. The means of transport determination determines a means of transport and vehicle type for the stage between A and C3. Then, network path determination determines the possible path from A to C3 via B and already determines the durations. (Assumption: The corresponding lanes have the same means of transport as the stage between A and C3). The new stages are enhanced with the same means of transport. Lane, distance and duration determination doesn't need to add a duration.

Example 4:

There is a requirement from A to C4.

There is neither a default route nor a network connection between the locations.

The system neither determines a default route nor a network path. However, during lane, distance and duration determination, after an unsuccessful search for a lane, the system adds a distance and a duration based on the straight-line determination (fallback). The system enhances the stage with the means of transport and the vehicle type.

Test Cockpit for Transportation Requirement Routing and Scheduling

The test cockpit for transportation requirement routing and scheduling (TRS test cockpit) allows you to simulate either transportation duration determination or requirement routing and scheduling. You can then analyze the result.

Prerequisites

For more information, see [Determination of Transportation Durations for BPS-Based Scheduling](#) and [Routing and Scheduling of Transportation Requirements](#).

Features

Input

On the initial screen, you first need to select a TRS process category to indicate whether the system is to simulate transportation duration determination or requirement routing and scheduling.

Based on the TRS process category, you can indicate the following:

- TRS profile
- Scheduling direction
- Whether the system is allowed to parallelize
- Whether intermediate results are to be displayed as well
- Time zone and location-specific time zone
- Unit of measure for distances

i Note

The time zone, location-specific time zone, and unit of measure for distances are copied from your [Display Settings](#) by default. You can overwrite them here.

Further, you can indicate up to three request items which are defined by the following:

- A document ID (optional)

The document ID is used to group request items. It doesn't have any reference to business documents.

- The source and destination locations and their location types
- The source and destination countries/regions and regions
- The material freight group, transportation group, service level (sales), and transportation mode
- A reference date and time
- A time zone, which is applied to the reference date and time

Output

The result screen contains the following areas:

- **General Input**

In this area, you find the input data that you entered on the initial screen. You can change this information.

- **Item Input**

In this area, you can change your input. For example, you can change the source location of one request item or change the number of request items to be considered.

- **Model Assignment**

The system assigns each request item to an entity called "model". Each item needs to be assigned to a model but a model can have several items assigned to it. In this area, you should be able to find each of your items in one of the models.

- **Result Overview**

In this area, you find an overview of the result per model. If you have selected the **Show Intermediate Results** checkbox, the system displays the intermediate result for a routing or determination method after the system executed this method. By selecting **Show Details**, you can display details for each model in the **Result Details** area. By selecting **Show Network Path Determination**, you can switch to the test cockpit for network path determination. For more information, see [Test Cockpit for Network Path Determination](#).

- **Result Details**

In this area, you can display details for each model.

If you have selected **Transportation Duration Determination** as the TRS process category, you can see loading durations or stages that the system added, for example.

If you have selected **Requirement Routing and Scheduling** as the TRS process category, you can see the following information, for example:

- Loading start date/time
- Loading end date/time
- Departure date/time
- Arrival date/time
- Means of transport
- Vehicle type

- **Messages**

Activities

To start the TRS test cockpit (program /SCMTMS/TEST_TRS_ENGINE), on the SAP Easy Access screen choose ► **Transportation Management** ► **Administration** ► **Transportation Requirement Routing and Scheduling** ► **TRS - Test Cockpit** ►.

Test Cockpit for Network Path Determination

The test cockpit for network path determination allows you to simulate the determination of network paths. You can then analyze the result.

Prerequisites

If you want the system to consider transshipment locations, you have defined transshipment location assignments. To do so, on the SAP Easy Access screen, choose ► **Logistics** ► **Transportation Management** ► **Master Data** ► **Transportation Network** ► **Location** ► **Define Transshipment Location Assignments** ►.

For more information about other prerequisites, see [Determination of Transportation Durations for BPS-Based Scheduling](#) and [Routing and Scheduling of Transportation Requirements](#).

Features

Input

On the initial screen, you can indicate the following information:

- Routing option
- Search depth for transshipment locations
- Search depth for schedules
- Expected waiting time of schedule

Alternatively, you can select a transportation requirement routing and scheduling (TRS) profile. By doing so, the above fields are filled automatically as defined in the selected profile.

For more information, see [Routing and Determination Methods](#) and the system documentation.

Further, you can indicate up to three path requests which are defined by the following:

- The source and destination location
- The means of transport
- A reference date and time
- A time zone

Output

The result screen contains the following areas:

- **General Settings**

In this area, you find the input data that you entered on the initial screen. You can change this information.

- **Path Requests**

In this area, you can change your input. For example, you can change the source location of one path request or change the number of requests to be considered.

- **Locations**

In this area, the system displays all locations from your network that are relevant for your requests. This network includes both the source and destination locations from your requests and the transshipment locations that connect these source locations with the destination locations.

- **Connections**

In this area, the system displays all transportation lanes and schedules that are available for the connections between these locations. In addition, two buttons are available to display a detailed calculation of the expected waiting time for schedule connections and to convert durations from hours to seconds and vice versa.

- **Path Results**

In this area, the system displays results for each path request. Each path request can consist of multiple connections. Depending on your routing option, the result can either be the path with the shortest distance or the path with the shortest duration.

Activities

To start the test cockpit (program /SCMTMS/TEST_NPD_ENGINE), on the SAP Easy Access screen choose ► **Transportation Management** ► **Administration** ► **Transportation Requirement Routing and Scheduling** ► **Test Cockpit for Network Path Determination** ►.

Additional Planning Options

Planning offers various other planning options. For example, you can assign drivers to your freight orders or you can use package units, trailers, or multi-resources.

Planning Based on Equipment Types

If you regularly subcontract your transportation demands, you are usually not interested in physical resources, such as the actual truck with a plate number on your yard, but rather in resource types, which you can order from your carriers.

For such scenarios, we recommend that you define equipment types. That means, you don't need to define and assign resources anymore. You just need to specify or assign equipment types. The system treats equipment types similarly to unlimited multiresources.

Prerequisites

- You have defined equipment types (that is, vehicle types and container types) in Customizing under ► **Master Data** ► **Resources** ► **General Settings** ► **Define Equipment Groups and Equipment Types** ►

For more information, see [Equipment Groups and Equipment Types](#).

i Note

For these vehicle types and container types, you must **not** select the **Restricted Use in Planning** checkbox.

- Optionally, you can use Business Add-In (BAI) /SCMTMS/FU_STAGE_MODIFY to determine the preferred equipment type and group for freight unit stages.

For more information, see Customizing for Transportation Management under ►**Business Add-Ins (BAIs) for Transportation Management** ► **Planning** ► **BAI: Determination of Freight Unit Stages** ►. For more information about using preferred equipment types and groups, see [Preferred Equipment Groups and Types](#).

- Optionally, in the **Capacity Selection Settings** of your planning profile, you have indicated your equipment types on the **Vehicle Types** tab page and the **Container Types** tab page. For more information, see [Planning Profile](#).
- Optionally, in the **Capacity Availability Settings** of your planning profile, you have restricted the availability of your equipment types. For more information, see [Planning Profile](#).
- Optionally, in the **Constraints and Costs Settings** of your planning profile, you have indicated costs for your equipment types. For more information, see [Constraints and Costs Settings](#).
- If you want to display your objects grouped by equipment type in the hierarchies of the transportation cockpit, you have selected the standard grouping attribute **Equipment Type** in Customizing.

For more information, see Customizing for **Transportation Management** under ►**Planning** ► **General Settings** ► **Define Hierarchical Views for Business Documents** ►.

- If you work with incompatibilities, you have specified that your incompatibilities apply to equipment types. You do so in the **Validity** section of the **Edit Incompatibilities** app. Note that this field is only displayed for incompatibilities that involve vehicle resources or vehicle types.

Features

In the transportation cockpit, you can work with equipment types in the same way as you work with resources. For example, you can do the following:

- You can select a vehicle type and several freight unit stages and then select **Assign Selected Items**. The system creates a road freight order.
- You can select a trailer type and then select **Create Capacity Document**. The system creates a trailer unit.
- You can select a freight unit stage and a truck type and a trailer type and then select **Assign Selected Items**. The system creates a road freight order and a trailer unit.
- You can select a road freight order and then select **Remove Truck Type** to remove the truck type.

Drag and drop is also supported.

Further, the following planning actions are supported for equipment types:

- Scheduling
- Load planning and load consolidation

Load planning considers combination vehicle types similarly to combination resources.

- Gantt chart
- Vehicle scheduling and routing (VSR) optimization
- Generation of transportation proposals

- Creation of capacity documents based on groups with load consolidation

i Note

- By default, equipment types are considered as being always available. You can restrict their availability by indicating a shift sequence and a factory calendar in the [Capacity Availability Settings](#) of your planning profile (see "Prerequisites").
- Equipment types can't be displayed on the geographical map.

The following equipment types are supported:

- Vehicle type
 - Truck type
 - Trailer type
 - Locomotive type
 - Railcar type
- Container type

In the transportation cockpit, the most specific term is used on the user interface where possible. If this is not possible, the generic term "equipment type" or "vehicle type" is used.

Preferred Equipment Groups and Types

When you assign freight unit stages to capacity documents or create capacity documents based on freight unit stages, you can use preferred equipment groups and types to automatically create local items such as container items in capacity documents.

Prerequisites

- You have implemented the Business Add-In (BAI) [Determination of Freight Unit Stages](#) (/SCMTMS/FU_STAGE_MODIFY). For more information, see Customizing for Transportation Management under [Business Add-Ins \(BAIs\) for Transportation Management](#) > [Planning](#) > [BAI: Determination of Freight Unit Stages](#).
- You have defined your preferred equipment groups and types as master data in the system. For more information, see Customizing for Transportation Management under [Master Data](#) > [Resources](#) > [General Settings](#) > [Define Equipment Groups and Equipment Types](#).
- In the manual planning settings of your planning profile, you have selected [Create TU as Freight Document Item](#) in the [Create TU Item or TU Document](#) field.

Note that if you have selected [Create TU Document](#) in this field for other purposes, the system does not consider the preferred equipment group and type of a freight unit stage when you assign it to a freight document. Transportation unit documents (TU documents) can't be created automatically by the system when you assign freight unit stages to freight orders or freight bookings. You can, however, first assign the freight unit stage to a TU document and then assign this document to the freight document (in separate steps).

Features

Determination of the Preferred Equipment Group and Type

After freight units have been created during freight unit building or have been updated, the system determines the preferred equipment groups and types using the **Determination of Freight Unit Stages** BAdI. The BAdI considers the following information as an input:

- Material freight group of each requirement item
- Material of each requirement item
- Transportation mode of the requirement stages
- Source and destination location of the requirement stages
- Requested and acceptable time windows

You can enhance and adjust this input.

The result of the determination is displayed in the fields **Preferred Equipment Group** and **Preferred Equipment Type** in the freight unit stage. The fields are not displayed by default, but you can select them in the UI settings.

You can manually change the preferred equipment groups and types in the capacity document item.

Creation of Capacity Document Items Using the Transportation Cockpit

When you assign freight unit stages to a capacity document in the transportation cockpit or when you create a capacity document based on freight unit stages, the system automatically creates local items in the capacity document based on the preferred equipment groups and types that the system has determined using the BAdI. Examples:

- When you create a road freight order for a freight unit stage with a preferred equipment group and type, this road freight order is based on the preferred equipment group and type.
- When you create a road or rail freight order or an ocean freight booking for a freight unit stage that has a container group and type defined as a preferred equipment group and type, the system automatically creates a local container item with this container group and type in the new freight order or ocean freight booking.
- When you create a rail freight order for a freight unit stage that has a railcar group and type defined as a preferred equipment group and type, the system automatically creates a local railcar item with this railcar group and type in the new rail freight order.

To ensure that the system considers the preferred equipment group and type, please note the following restrictions. If you ignore these restrictions, the system displays a warning message:

- If several stages of the same freight unit are planned together, the preferred equipment groups and types of these stages must be the same.
- The preferred equipment groups and types must be compatible with the document types. If you have, for example, specified a railcar equipment group and type for the freight unit stage as preferred equipment group and type and you want to assign the stage to an ocean freight booking, this equipment group and type don't match and are therefore ignored by the system.
- If an equipment group and type have already been specified for the capacity document to which you assign a freight unit stage, the system doesn't consider the preferred equipment group and type of this freight unit stage.
- Values of the preferred equipment group and type that have been specified for the freight unit stages must be predefined as equipment groups or types in the system (see **Prerequisites**). Furthermore, equipment groups and types with restricted use in planning are not considered by the system.
- If the system creates a new local item in a capacity document, it does not check the capacity of the corresponding equipment type. This can lead to the equipment type being used above its capacity. However, the utilization of the item is

displayed so that you can easily identify whether the capacity is exceeded.

Creation of Capacity Document Items Using the Capacity Document Apps

The feature described above for manual planning in the transportation cockpit can also be used in the different apps for creating and editing capacity documents, such as the **Edit Ocean Freight Booking** app. For example, when you assign a freight unit stage to an ocean freight booking in the **Edit Ocean Freight Booking** app, the system also considers the preferred equipment group and type that has been determined for this freight unit stage. For example, it can then create a local container item in the ocean freight booking if the prerequisites are met.

Related Information

[Manual Planning with Local Transportation Unit Items](#)

Means of Transport and Vehicle Type Determination

Means of transport and vehicle types are determined based on business rules. They are considered during transportation requirement routing and scheduling (TRS).

Prerequisites

- You have defined a condition of condition type **Determine Means of Transport/Veh. Types** (/SCMTMS/MTR_VT_DET). For more information, see [Definition of a Condition for MTr/VT Determination](#).
- You have activated the means of transport and vehicle type determination and indicated your condition in your planning profile. For more information, see [Planning Profile](#).

Use

The system considers input criteria such as source and destination location data, service level (sales), payer and ship-to party to determine the appropriate means of transport and optionally, the vehicle type.

This determination is performed for each requested stage for which no means of transport is defined yet. If no vehicle type is defined, the determination is also performed to determine a vehicle type that is compatible with the existing means of transport. Otherwise, the determination is skipped.

The determined means of transport and vehicle type are copied into the stage.

The system can determine one or several means of transport, one or several vehicle types, or both.

A Business Add-In (BAI) is available to change the input data and the results. For more information, see Customizing for **Transportation Management** under **Business Add-Ins (BAIs for Transportation Management) > Planning > BAI: Adjustment of Means of Transport Determination**.

Related Information

[Transportation Requirement Routing and Scheduling](#)

Use of Package Units in Planning

You can create, edit, and display package units.

Prerequisites

- You have configured the relevant settings in Customizing so that you can configure your standard layout for the use of package units in the layout settings for the transportation cockpit. Several hierarchy types are predefined by default for the planning of package units.

In addition, you can define your own hierarchy types in Customizing for **Transportation Management** under **►Planning ►General Settings ►Define Hierarchical Views for Business Documents ►**. For more information, see [Use of Hierarchical Views](#).

- You have defined a transportation unit type in Customizing with the transportation unit category **Package**. The transportation unit type PKG is predefined by default for package units. For more information, see Customizing for **Transportation Management** under **►Planning ►Transportation Unit ►Define Transportation Unit Types ►**.

Use

Manual Creation of Package Units

- You choose **Package Unit** to create a new package unit. The system proposes the business document type that you entered for package units in your planning profile. You can overwrite this.
- You can also create package units based on freight unit stages. To do this, select one or more freight unit stages and then choose **►Create Capacity Document ►Create Package Unit ►**. The system creates a new package unit. If you want to create a package unit for each freight unit stage, choose **►Create Multiple Capacity Documents ►Create Multiple Package Units ►**.
- You can assign freight unit stages to existing package units. To do this, select one or more freight unit stages and a package unit and then choose **Assign Selected Items**. Alternatively, you can use drag and drop.

i Note

You can use default routes during the manual creation of package units. For more information, see [Default Route](#).

Manual Creation of Capacity Documents from Package Units

- You can create capacity documents based on package unit stages. To do this, for example, select several package unit stages and a truck and then choose **Assign Selected Items**. The system creates a new road freight order. Alternatively, you can use drag and drop or choose **Create Capacity Document**.
- You can assign package unit stages to existing capacity documents. To do this, for example, select several package unit stages and a road freight order and then choose **Assign Selected Items**. Alternatively, you can use drag and drop.

Automatic Creation of Package Units

- You can create package units based on unified package building for freight units. To do this, choose **Create PUs Based on UPB** in the freight unit stage list or hierarchy. First the system creates packages for the freight units and then it creates package units for these packages.

In doing so, it includes first all freight units for which you have selected **Exclusive Package for Customer**. This means that the package units only receive products for a specific customer.

It then groups the remaining freight units that can be consolidated into soft and hard location groups according to your setting in the unified package building profile. Then it creates packages for these freight units. As the final step, the system checks the utilization threshold value that you have specified in the planning profile for the location groups. If the utilization of the package exceeds the threshold value, the system creates a package unit for this package. If a freight unit is contained

in both a soft and a hard location group, the system first checks the threshold value for the soft location group and then the threshold value for the hard location group.

If you have also specified a utilization threshold value for business partners, this is ignored (that is, the location groups take priority).

If you have specified a utilization threshold value for business partners, the system groups the remaining freight units that can be consolidated based on the business partner and creates packages. The system checks the utilization threshold value that you have specified in the planning profile for the business partner. If the utilization of the package exceeds the threshold value, the system creates a package unit for this package.

Note that if several packages are created for one group during unified package building, only one of these packages needs to have a utilization above the threshold for the system to be able to create a package unit for all these packages. This applies to all the threshold values mentioned above.

In the planning profile, you can define a maximum number of packages up to which the system creates package units. If this value is exceeded, the system does not create any package units. This parameter is valid for all the above steps.

Package units that were created with this function are taken into account and planned in the next VSR optimization run, but the packing information is not changed. Freight units that cannot be planned in package units using this function are planned by the VSR optimization run as before.

As a prerequisite for this function, you must have entered the following in the planning profile in the settings for unified package building:

- A unified package building profile
- A default package unit type and, optionally, a default rule for the creation of package units

This rule controls whether the system is to create one package unit for each package item (standard setting) or one package unit for all package items.

In addition, you can enter a package unit type for package units with only one destination location and a corresponding rule.

- The prerequisite for considering the maximum number of packages per package unit is that you have selected the value **Create One PU for all Package Items** for one of the two package unit types.

For more information, see [Planning Profile](#).

The sample planning strategy **CEPU_ONLY** is provided for creating package units on the basis of unified package building for freight units. It contains the planning method **VSR_CEP**. You can include this method in your own planning strategies according to your needs.

You can also create package units in parallel processes. To do this, you have to define a parallel processing profile and assign it to your planning profile (in the subprofile **Settings for Unified Package Building** under **Unified Package Building > Parallel Processing Profile**). For more information, see [Definition of Parallel Processing Profiles](#). You can also execute this function in the background (see [Creation of Package Units Based on Unified Package Building for Freight Units](#)).

i Note

- In the default package unit type, you must have selected **Linear with Distribution** as the sequence type of stages. In the package unit type for package units with only one destination location, you must have selected **Defined and Linear** as the sequence type of stages. In addition, you must have specified in the **Assgmt of Pred.Docs** (Assignment of Predecessor Documents) field that freight units have to be assigned completely to a package unit.
- If you have specified in the default package unit type that freight units have to be assigned completely to a package unit (**Assgmt of Pred.Docs** field ("Assignment of Predecessor Documents")) and the system has then

had to distribute a freight unit to several package units during the creation of package units, it splits this freight unit. If you undo the creation of the package units, these freight units remain split.

- You cannot change the packaging hierarchy in these package units manually. Nor can you manually add or remove freight units to or from these package units.
- If you assign two of these package units containing product quantities from the same freight unit to different freight orders, the system splits the freight unit.
- If you want to undo the creation of package units, you must reverse the created package units. Alternatively, you can use the **Undo** pushbutton.

Automatic Planning of Package Units

- VSR optimization (VSR = vehicle scheduling and routing) including scheduling takes package units into account in the same way as container units, that is, only as requirement documents and not as capacity documents. This means that VSR optimization can plan, in other words assign, package units and container units in freight documents and other transportation units. It cannot, however, assign freight units to a package unit or container unit. This is possible in manual planning only (see above).
- VSR optimization including scheduling takes into account the pickup and delivery windows for freight units assigned to a package unit. However, if you have defined times at a given stop of a package unit, VSR optimization takes these times into account and ignores the time windows of all assigned freight units at this stop. The same is true for container units.
- Scheduling ignores the loading and unloading times of freight units assigned to package units. Only the loading and unloading times of package units assigned to freight documents or transportation units are taken into account.

i Note

You can use default routes during the automatic creation of package units. For more information, see [Use of Default Routes in VSR Optimization](#).

Editing and Displaying Package Units

- You can display package units in a tabular and graphical format.
- You can display and edit package units on the map, in the same way as container units and trailer units. For more information, see [Interactive Planning on the Map](#).

Activities

You create and edit package units in the transportation cockpit. Open the **Transportation Cockpit** app on the launchpad.

Related Information

[Package Unit](#)

[Creating a Package Unit](#)

[Planning with Distribution Stops](#)

Planning with Distribution Stops

You use package units with distribution stops to map the transportation of mixed pallets in delivery tours, for example. The goods are then unloaded directly from the truck one after the other at different locations.

Prerequisites

- You have defined a transportation unit type in Customizing that allows the use of distribution stops. For more information about this and about distribution stops, see [Package Unit](#).
- To display distribution stops in the transportation cockpit in your hierarchy for package units or package unit stages, in Customizing you must select the **Distribution Stops** hierarchy level and set the **Show Level** (Show Hierarchy Level) checkbox. For more information, see Customizing for Transportation Management under ► **Planning** ► **General Settings** ► **Define Hierarchical Views for Business Documents** ►.

Note that this setting may lead to increased runtime. To avoid this, you can alternatively specify in your page layout that you want to display distribution stops (detail area for demand). For more information, see [Page Layouts](#).

Display of Distribution Stops

In the transportation cockpit, you can display distribution stops in the hierarchies for package units and package unit stages. If you have configured your page layout accordingly, you can display the list of distribution stops for a package unit stage (hierarchy for package units or package unit stages) using **Show Details** (**Actions** columns).

Manual Planning

You can plan package units with distribution stops in manual planning like any other package unit. This means that you can, for example, drag and drop a distribution stop to assign it to another freight order, another package unit, or another resource (truck, trailer, or container). Alternatively, you can assign it using the **Assign Selected Items** pushbutton.

i Note

- Note that if you have created a package unit from a freight order (see [Transportation Planning with Mixed Packages and Deconsolidation](#)), you can only assign all distribution stops together to another freight order or resource. This means that the corresponding stage of the package unit is assigned to the other freight order.
- You cannot process package units with distributed freight units in manual planning in the transportation stage for the delivery tour. However, you can process them in the pre-carriage transportation stage.

Planning on the Map

You can display and plan package units with distribution stops on the map.

i Note

For example, if you drag and drop the stage for a distribution stop to another freight order all the other stages for further distribution stops are also assigned to the other freight order.

Automatic Planning and Scheduling

Package units with distribution stops are included in automatic planning (VSR optimization). Package units are included in scheduling only if they are assigned to another document (such as a freight order).

You can display the package unit in the explanation tool.

During scheduling, the quantities of freight units or package units (if the package unit is assigned to a container unit) are taken into account at all distribution stops to determine the variable unloading times. At all other stops, the quantities of the package

unit are taken into account to determine the variable unloading times.

i Note

- When the variable loading times are calculated for a freight order and a partial quantity of a distributed freight unit is assigned to a distribution stop of a package unit, the system includes only the partial quantity assigned to this package unit in the calculation.
- VSR optimization cannot create, change, or delete package units with distributed freight units.

Related Information

[Package Unit](#)

[Creating a Package Unit](#)

Transportation Planning with Mixed Packages and Deconsolidation

In Transportation Management (TM), you can transport orders from different customers together across several transportation stages and deconsolidate them on the last transportation stage in a “delivery tour”. With the help of mixed packages, you can optimize resource utilization and so reduce the freight costs.

You can use package units in TM to consolidate packages for different customers. Once the system has created packages using the unified package building function, you can group them together in package units and assign them to several freight orders. To optimize the utilization of the packages, a freight unit can be assigned to several package units. You do not have to split the freight unit provided the different package units to which a freight unit is assigned are transported with the same freight document for each stage. For more information, see [Package Unit](#).

Below is an example of how SAP TM supports you in the planning process. You first plan the delivery tour(s) and then the linear transportation stages of your route.

Prerequisites

- You have made all the required settings for unified package building. For more information, see [Unified Package Building in TM](#).

More specifically, you have entered a unified package building profile in the planning profile that is assigned to the document type or that you use to call the transportation cockpit. If you want to determine and display the exact position of the products in the package, you have to activate the **Optimizer-Based Mixed Package Building** function in this profile. For more information, see [Optimizer-Based Mixed Package Building](#).

- You have defined a transportation unit type for packages. Here you have made the following settings:
 - You have activated unified package building for this document type.
 - You have specified that, in addition to linear, consecutive transportation stages, your route contains a delivery tour for the deconsolidation of packages. To do this, select the value **Linear with Distribution** in the **Seq. Type of Stages** (Sequence Type of Stages) field.
 - You have specified that a freight unit can be assigned to multiple packages. To do this, select the value **Partially or Completely** in the **Assgmt of Pred.Docs** (Assignment of Predecessor Documents) field.

For more information, see Customizing for [Transportation Management](#) under [Freight Order Management](#) [Transportation Unit](#) [Define Transportation Unit Types](#).

- In Customizing for the freight order type, you have defined a default package unit type that contains the settings described above. For more information, see Customizing for [Transportation Management](#) under [Freight Order Management](#) [Freight Order](#) [Define Freight Order Types](#).
- In Customizing for the freight order type, you have indicated in the **Default Package Unit Creation Rule** field whether the system is to create one package unit per package or one package unit for all selected packages.
- You have created a freight unit building rule (FUB rule). If you want to use the VSR optimizer later to create freight orders and a rough estimation of the number of packages is to be taken into account, enter a unified package building profile on the **Advanced Settings** tab page of the FUB rule. Then, select the **Estimate Package Count Only** option in the **Unified Package Building Result** field.

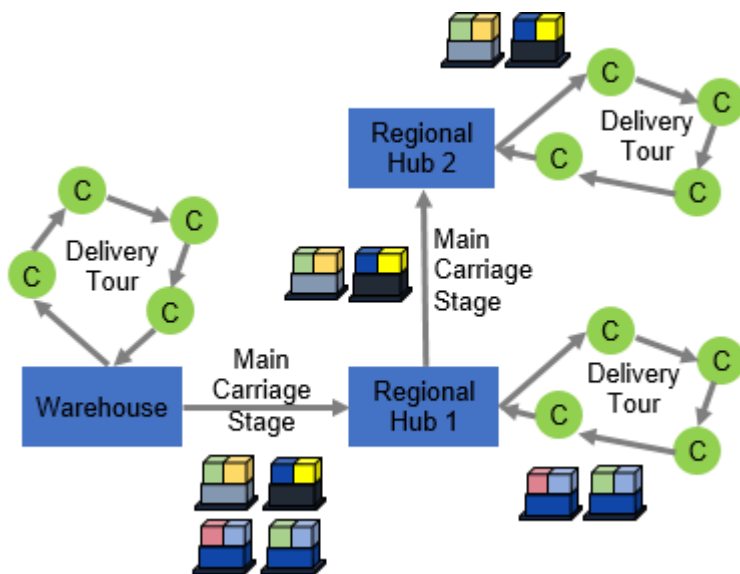
For more information, see [Freight Unit Building Rule](#).

- If you want to use the VSR optimizer or manual planning in the planning process described here, you can define planning strategies to combine the process steps and have them performed automatically by the system. The following planning strategies are available:
 - VSRI_CPB for manual planning with unified package building based on capacity documents
 - VSRI_PBLP for manual planning with unified package building based on capacity documents and load planning
 - CPB_ONLY for unified package building based on capacity documents
 - VSR_CPB for VSR optimization with unified package building based on capacity documents
 - VSR_CPBALP for VSR optimization with unified package building based on capacity documents and load planning

For more information, see [Planning Strategies](#).

Process

The following figure provides an overview of an example process:



Planning Process with Mixed Packages and Deconsolidation

Planning of Delivery Tours

1. Creating freight units

You have the system automatically create freight units based on your order documents (such as forwarding orders or sales orders) and using the freight unit building rule you defined. For more information, see [Creation and Editing of Freight Units](#).

2. Creating a freight order for the delivery tour

You create a freight order for the unplanned stages of the freight units that belong to your delivery tour. You can also use the VSR optimizer to do this. If you have specified this in your freight unit building rule, the system takes into account the number of packages estimated when the freight units were created. The optimizer also takes into account the maximum number of packages for each truck as well as weight and volume restrictions.

For more information, see [Creation and Editing of Freight Orders](#) and [VSR Optimization](#).

3. Unified package building

You have the system create packages for the items in the freight order. These packages are then available locally in the freight order. For more information, see [Unified Package Building in TM](#).

4. Load planning

You execute a load planning for these packages in the freight order. For more information, see [Load Planning](#).

5. Finalize the packaging

After you have checked the results of unified package building and made any necessary manual adjustments, you set the load plan status of the packages to "Finalized" for the freight order. You do this at header level in the freight orders under ► **Load Plan Status (Packaging)** ► **Set All Top-Level Packages to Finalized** ►. Alternatively, you can also select the vehicle resource on the **Items** tab page in the **Load Plan** hierarchy view and select the **Set All Top-Level Packages to Finalized** option or select individual packages and select the **Set to Completed** option.

After the status has been set, the packages are fixed for further planning. This means that you can only change the packages if you reset the status.

6. Creating package units

If you want to be able to use and plan these packages in other freight documents as well, you must create package units. You do this at header level in the freight order under ► **Follow Up** ► **Create Package Units** ►. Alternatively, as described under step 5, you can also select individual packages and select the **Create PUs** option. However, note the following: You must always select complete freight units. In other words, all items in the freight units selected with the packages must be included in the selected packages of this freight order. The system supports you with warning messages and input help.

The following applies to package units created by the system:

- The system creates the package units in such a way that a freight unit can be assigned to several package units. You can display the package units to which a freight unit is assigned in the stage details of a freight unit or in the transportation cockpit on the [Assignment to Package Units](#) tab page.
- The system determines the type of package unit created and the way in which package units are built from packages (1:1 or 1:n) from the Customizing settings for the freight order type and the planning profiles used.
- The package units contain both the linear transportation stages and a special stage with the distribution stops of the delivery tour. These stops are displayed as substops, but their sequence is arbitrary. The transportation stages of the freight orders and freight units remain unchanged.
- The package units also contain information about the exact position of products in a package.
- The package units that were created from a freight order cannot be changed. If you want to make any changes you have to cancel the package units (see step 7).

For more information, see [Package Unit](#) and [Creating a Package Unit](#).

7. Canceling package units (optional)

You can cancel the package units created by the system in step 6. You do this at header level or for individual packages by selecting the **Cancel Package Units and Merge** option. The system then integrates these packages back into the item hierarchy of the freight order as local packages.

However, the package units must not have already been assigned to other freight orders. Moreover, the same prerequisites apply to the selection of package units as described above for the creation of package units.

For more information about how the load plan status changes after the process steps described above, see [Load Plan Status Changes](#).

Planning of Main Carriage Stage

1. Creating a freight order for the main carriage stage of the package units

You manually create one or more freight orders for the main carriage stage of your package units or have these created by the VSR optimizer.

2. Load planning

You execute a load planning for the package units in the freight orders.

3. Finalizing the load plan

After you have made any necessary manual adjustments, you set the load plan status of the load planning to "Finalized" for the freight orders. You do this at header level in the freight orders under ► **Load Plan Status (Load Planning)** ► **Set Load Plan to Finalized** ►.

4. Splitting the freight units

The package units created by the system can contain freight units that are assigned to several package units. If the different package units to which a freight unit is assigned are transported with different freight orders in the main carriage, the system splits the freight unit accordingly and distributes it among the freight documents. Note that you must have set the load plan status (load planning) to "Finalized".

Related Information

[Use of Package Units in Planning](#)

[Planning with Distribution Stops](#)

[Use of Normalized Quantities and Capacities](#)

Load Plan Status Changes

Many of the steps of the "Transportation Planning with Mixed Packages and Deconsolidation" process have an effect on the load plan status. The following sections describe how the load plan status changes after each process step. Note however that they only refer to the first part of the process ("Planning of Delivery Tours").

Initial situation

Let's assume the freight order created in this process has three freight units assigned to it (FU1, FU2, and FU3).

On item level, for example in the load plan hierarchy, the load plan status looks like this:

	Load Plan Item Status (Cargo) (Description)	Load Plan Status (Capacity) (Description)
1. Vehicle Resource	Not Relevant	Not Planned

1.1 Product 1 (FU 1)	Not Planned	Not Relevant
1.2 Product 2 (FU 2)	Not Planned	Not Relevant
1.3 Product 3 (FU 3)	Not Planned	Not Relevant

Note that the status changes after each step are marked in bold below.

Package building

After you have performed package building for all items of your freight order, the load plan status looks like this:

	Load Plan Item Status (Cargo) (Description)	Load Plan Status (Capacity) (Description)
1. Vehicle Resource	Not Relevant	Not Planned
2. Package (local)	Not Planned	Planned
2.1 Product 1 (FU 1)	Planned	Not Relevant
3. Package (local)	Not Planned	Planned
3.1 Product 2 (FU 2)	Planned	Not Relevant
4. Package (local)	Not Planned	Planned
4.1 Product 3 (FU 3)	Planned	Not Relevant

Load planning

After you have performed load planning for all items of your freight order, the load plan status looks like this:

	Load Plan Item Status (Cargo) (Description)	Load Plan Status (Capacity) (Description)
1. Vehicle Resource	Not Relevant	Planned
2. Package (local)	Planned	Planned
2.1 Product 1 (FU 1)	Planned	Not Relevant
3. Package (local)	Planned	Planned
3.1 Product 2 (FU 2)	Planned	Not Relevant
4. Package (local)	Planned	Planned
4.1 Product 3 (FU 3)	Planned	Not Relevant

Finalizing packaging

After you have finalized packaging for all items of your freight order, the load plan status looks like this:

	Load Plan Item Status (Cargo) (Description)	Load Plan Status (Capacity) (Description)
1. Vehicle Resource	Not Relevant	Planned
2. Package (local)	Planned	Finalized

2.1 Product 1 (FU 1)	Finalized	Not Relevant
3. Package (local)	Planned	Finalized
3.1 Product 2 (FU 2)	Finalized	Not Relevant
4. Package (local)	Planned	Finalized
4.1 Product 3 (FU 3)	Finalized	Not Relevant

Note that the requirement assignments of local packages with status value **Finalized** are automatically fixed.

Creating package units (PUs)

After you have created package units for the freight order, the load plan status looks like this:

	Load Plan Item Status (Cargo) (Description)	Load Plan Status (Capacity) (Description)
1. Vehicle Resource	Not Relevant	Planned
2. Package (PU 1)	Planned	Not Relevant
2.1 Product 1 (FU 1)	Not Relevant	Not Relevant
3. Package (PU 2)	Planned	Not Relevant
3.1 Product 2 (FU 2)	Not Relevant	Not Relevant
4. Package (PU 3)	Planned	Not Relevant
4.1 Product 3 (FU 3)	Not Relevant	Not Relevant

Related Information

[Transportation Planning with Mixed Packages and Deconsolidation](#)

Use of Normalized Quantities and Capacities

If you are using different packaging materials for planning in a road scenario, you can use normalized load consumption quantities and capacities in Transportation Management (TM) to optimize the use of the loading space. If, for example, your packaging material consists of pallet cages and europallets, you can convert and plan these into europallets or loading meters. The capacity of a truck can also be defined as a normalized capacity.

Prerequisites

- To activate the use of normalized quantities (NLQ) and additional normalized quantities (ALQ), you have indicated in your business document type whether the relevant unit of measure for the NLQ or ALQ is determined based on a condition (with condition type /SCMTMS/NLQ_UOM) or on the default unit of measure for the NLQ or ALQ that you have specified in Customizing for your freight unit type, transportation unit type, and freight order type. We recommend that you use consistent unit of measures within one scenario.

For more information, see Customizing for **Transportation Management** under the following Customizing activities:

- ▮ **Planning** ▸ **Freight Unit** ▸ **Define Freight Unit Types** ▮
- ▮ **Freight Order Management** ▸ **Transportation Unit** ▸ **Define Transportation Unit Types** ▮

- ► [Freight Order Management](#) ► [Freight Order](#) ► [Define Freight Order Types](#) ►

- In the Customizing activity [Define UoM Relevance for NLQ and ALQ](#), you have optionally specified whether a unit of measure is relevant for NLQs or ALQs.

For more information, see Customizing for [Transportation Management](#) under ► [Basic Functions](#) ► [Normalized Quantities \(NLQ\) and Additional Normalized Quantities \(ALQ\)](#) ► [Define UoM Relevance for NLQ and ALQ](#) ►.

If you specify a unit of measure, the system offers it for selection in the corresponding user interfaces. If you don't specify a UoM in this Customizing activity, the system offers dimensionless UoMs.

- You have indicated a unit of measure for NLQ and ALQ in the [Define Transportation Cockpit Settings](#) app. The system needs this to be able to display NLQs on the map as well as for trucks, trailers, and containers in the transportation cockpit.

For more information, see [Planning Profile](#).

- Some of the values related to the normalized quantity can be displayed in a series of lists and hierarchies in the transportation cockpit using the corresponding field catalogs. For more information, see [Field Configuration for the Transportation Cockpit](#).
- If you are working with schedules, you must also enter a default unit of measure for the NLQ in the schedule type. The NLQ is copied from the schedule departure into the freight order only if this unit of measure corresponds to the default unit of measure in the freight order.

For more information, see Customizing for [Transportation Management](#) under ► [Master Data](#) ► [Transportation Network](#) ► [Schedule](#) ► [Define Schedule Types](#) ►.

- You have defined an alternative quantity for the unit of measure of the normalized quantity in your material master. To do so, on the SAP Easy Access screen, choose ► [Logistics](#) ► [Transportation Management](#) ► [Master Data](#) ► [Material](#) ► [Create](#) ► or [Change](#). Alternatively, you have specified an individual NLQ per product or reference product and unit of measure in the [Define Normalized Quantity](#) app.

For more information, see [Determination of Normalized Load Consumption Quantities](#).

- If you want a conversion into an additional normalized unit of measure, you need to define a conversion factor between the unit of measure of the NLQ and the unit of measure of the ALQ in the Customizing activity [Define Conversion Between NLQ and ALQ UoM](#).

For more information, see Customizing for [Transportation Management](#) under ► [Basic Functions](#) ► [Normalized Quantities \(NLQ\) and Additional Normalized Quantities \(ALQ\)](#) ► [Define Conversion Between NLQ and ALQ UoM](#) ►.

- You have entered the capacity for the NLQ in the default unit of measure in the master data for the resources you are using.

For more information, see [Definition of Capacity](#).

- You have specified the number of units and alternative number of units as well as the NLQ UoM and ALQ UoM for your equipment types in the Customizing activity [Define Equipment Groups and Equipment Types](#).

For more information, see Customizing for [Transportation Management](#) under ► [Master Data](#) ► [Resources](#) ► [General Settings](#) ► [Define Equipment Groups and Equipment Types](#) ►.

- If you want an overview of the packaging material used in a document, you must activate the option [Aggregate Load Consumption Quantities](#) in Customizing for the freight unit type, transportation unit type, and freight order type. For more information, see the Customizing activities mentioned above.

To display a compressed view in the lists and hierarchies of the transportation cockpit, you need to add the corresponding fields to your lists and hierarchies using field catalogs. For more information, see [Field Configuration for the Transportation Cockpit](#).

To display a compressed view in the Gantt chart, you must assign the corresponding fields to the specific lists. For more information, see Customizing for [Transportation Management](#) under ► [Basic Functions](#) ► [Gantt Chart](#) ► [Define Field Lists](#).

and Label Schemes for Gantt Chart .

- To display the utilization of the capacity in relation to the NLQ and to display the remaining capacity in the packaging material overview, you must define the method for calculating these values in the **NLQ Utilization Rule** field in Customizing for the freight order type.

For more information, see Customizing for **Transportation Management** under **Freight Order Management**  **Freight Order**  **Define Freight Order Types** .

- If you want to use the NLQ and ALQ as split criteria in freight unit building, you have defined the NLQ or ALQ or both as planning quantity in your freight unit building rule.

For more information, see [Creation and Editing of Freight Units](#).

Features

Determination of Normalized Quantity

The NLQ is used to allow the comparison of different packaging materials that are planned within the same scenario. The different units of measure of the packaging material are converted to a standard unit of measure to facilitate comparison.

The system determines NLQs in the following way:

First, the system takes into account the NLQs that you have defined individually for the product and UoM in the **Define Normalized Quantity** app. It then takes into account the NLQs that you have defined individually for the reference product in this app.

i Note

The system iteratively moves up in the reference product hierarchy until it finds a value.

The system then takes into account the full package definition of the product master data that you have defined in the transactions for creating or changing materials. It divides 1 by the defined full package quantity. For example, 1 EPL divided by 100 pieces results in 0.01 EPL per piece of the product. It then takes into account the full package definition of the reference product master data in one of these apps.

i Note

The system iteratively moves up in the reference product hierarchy until it finds a value.

If the system can determine a value in one of these steps, this is the result and subsequent steps are not evaluated anymore. If the system can't determine a value in any of the steps, it returns value 0.

The system also determines the load consumption quantity (LCQ) for the items based on the normalized quantity and the conversion factor that you defined in the master data of the packaging material.

The values determined by the system are displayed in the item details. The values for the entire freight unit document, transportation unit document, or freight order document are also aggregated. The aggregated values for the normalized quantity are displayed on the **General Data** tab page and the values for the LCQ are displayed on the **Packaging Material Overview** tab page (see below).

For more information, see [Determination of Normalized Load Consumption Quantities](#).

Determination of Additional Normalized Quantity

The system determines ALQs by multiplying the value for the NLQs with the conversion factor that you have defined in the Customizing activity [Define Conversion Between NLQ and ALQ UoM](#). If you haven't defined an appropriate conversion factor, the value for ALQs remains empty.

While the system determines the NLQs on product item level and then on document level, it determines ALQs on document level only.

For example, you can specify in Customizing that 1 europallet corresponds to 0.4 loading meters. If you have entered 20 europallets in the document, 8 loading meters are displayed as the additional normalized quantity of the document.

Packaging Material Overview

You can use the packaging material overview to get an overview of the packaging material contained in a document. If you have configured this in Customizing, a tab page appears in freight units, transportation units, and freight orders showing the packaged and unpackaged LCQ as well as the number of unpackaged freight items for a document.

In the case of freight units, the system displays the values determined during freight unit building in the [Unpackaged Load Consumption Quantity](#) field. The actual values that were calculated during planning and unified package building are displayed in freight order documents in the [Packaged Load Consumption Quantity](#) field.

The remaining capacity of the freight order document is also displayed. Depending on your Customizing settings, the system also takes into account the remaining capacity of the individual packages.

The system displays a compressed view of this information in your personal object worklist for freight orders. This is also displayed in the lists and hierarchies of the transportation cockpit and in the Gantt chart if you have completed the steps mentioned in the prerequisites.

Normalized Loading Space Capacity and Calculation of Maximum Utilization

If you have specified the number of units and alternative number of units as well as the NLQ UoM and ALQ UoM for your resources (in the transaction for creating resources in the SAP Easy Access menu) or your equipment types (in the Customizing activity [Define Equipment Groups and Equipment Types](#)), the system displays the utilization of the resources or equipment types with regard to the normalized quantity in the header data of the freight order. You can also specify this normalized loading space capacity for trailers. The utilization is then displayed separately for the truck and trailer and together for both resources.

The system calculates the maximum utilization, taking into account the most critical transportation stage and the most critical unit of measure (mass, volume, or normalized quantity).

Check Normalized Quantities in Documents

You can check whether the normalized quantities in your documents are current. You can cancel documents with obsolete normalized quantities. For more information, see [Check Normalized Quantities in Business Documents](#).

Example

The following example illustrates the calculation of an NLQ, taking into account the data that you have defined in the material and resource master data:

- Two different packaging materials are used in a road scenario. The first packaging material is a europallet (unit of measure is PA1). The second packaging material is defined in a unit of measure that is half the size of the europallet (PA2).
- You can now define the normalized unit of measure PAN in the master data of the packaging material as follows:
 - $PAN = PA1 * 1$
 - $PAN = PA2 * 2$

- If a resource, for example, has a capacity of 33 PAN (as the NLQ), different combinations of PA1 and PA2 can be checked so that the total capacity is not exceeded. For example, 23 PA1 and 20 PA2 would fit, but also 33 PA1 or 66 PA2.

NLQ Determination Based on Documents

Use

You can determine a value for the normalized load consumption quantity (normalized quantity, NLQ) per product and unit of measure (UoM) based on historical consolidation data. This value indicates how much space (footprint) a single product consumes in a top-level package (that is, the outer packaging).

The system determines this value in the following way:

1. For all products contained in the selected set of road freight orders, the system reads the NLQs, which are currently stored in the [Define Normalized Quantity](#) app.
2. The system determines NLQs for all products based on the packaging information from the selected road freight orders.

i Note

The system only takes into account selected road freight orders that were created after the last change of the NLQ in the [Define Normalized Quantity](#) app. If you have indicated a weighting factor of 100% on the selection screen, the system considers all selected road freight orders without any time restriction.

3. The system calculates the average NLQs taking into account the maximum tolerance for outliers that you have indicated on the selection screen.
4. The system calculates the new NLQ for these products according to the following formula:

$$\text{New NLQ} = \text{Weighting Factor} \times \text{Average NLQ from Road Freight Orders} + (1 - \text{Weighting Factor}) \times \text{Current NLQ}$$

5. Finally, the system saves the new NLQs in the [Define Normalized Quantity](#) app, taking into account the threshold that you have indicated on the selection screen.

For more information about the determination of NLQs, see [Determination of Normalized Load Consumption Quantities](#).

Prerequisites

- Road freight orders with NLQs exist in the system.
- You have created a selection profile.

For more information about prerequisites for package building, see [Package Building](#).

Features

On the selection screen, you indicate the following:

- A selection profile
- A UoM for the normalized quantity
- The level on which normalized quantities are to be determined (on product level or reference product level, or both)

We recommend that products belonging to the same reference product have the same dimension.

- A weighting factor for real data

This factor specifies to what extent the system uses the average NLQ determined from existing road freight orders when calculating the new NLQ for a product.

- A maximum tolerance for outliers

This tolerance defines the maximum percentage for NLQs that the system considers when averaging the NLQs from existing road freight orders.

- A threshold for saving the newly calculated NLQs

With this threshold, you can prevent saving NLQs with insignificant changes.

When you select **Continue**, the NLQ determination screen is displayed:

- On the left, the selected road freight orders are displayed. By selecting one or a small number of road freight orders and then selecting **Determine Products**, you can restrict the number of products for which you calculate the NLQs.
- On the right, the normalized quantities for the determined products are displayed. With **Calculate Normalized Quantity**, you can calculate the new NLQs.

The system saves the new NLQs in the **Normalized Quantity** field in the **Define Normalized Quantity** app. If this app already contains entries for combinations of product number and UoM for the normalized quantity, these entries are overwritten.

Activities

Open the **Determine Normalized Quantity Based on Documents** app from the launchpad. Alternatively, you can execute this function in the background. For more information, see [Determine NLQs Based on Documents](#).

We recommend that you check your business documents for outdated NLQs in regular intervals. For more information, see [Check Normalized Quantities in Business Documents](#).

Related Information

[Defining Package Type Assignments](#)

Check Normalized Quantities in Business Documents

You can check whether the normalized load consumption quantities (normalized quantities, NLQs) in your business documents are up to date and, if not, cancel these business documents.

Use

If the normalized quantities in the **Define Normalized Quantity** app have changed (for example, because you have changed them manually or have executed the report for determining normalized quantities based on randomized data), this change is not reflected in existing business documents, such as freight orders or freight units.

Planning with business documents that are based on outdated normalized quantities might lead to unexpected results, such as over-utilization or under-utilization. This function helps you to analyze such results, which would otherwise be very difficult - if not impossible.

Features

You need to select the following:

- A document category (that is, road freight orders, container units, trailer units, package units, consignment orders, or freight units)
- One or several business documents using a selection profile and additional selection criteria, such as the document ID or the document type

Moreover, you can define a threshold for canceling business documents. By doing so, you can prevent cancellation of business documents in case of insignificant NLQ changes.

You can specify that documents with empty NLQs are to be determined and displayed.

The system returns all business documents that fulfill the following criteria:

- Documents with changed normalized quantities taking into account the threshold that you have defined
- Documents with empty NLQs if you have selected the corresponding checkbox

You can select one or several of these business documents and cancel them. Note that you can only cancel road freight orders and unplanned requirement documents (that is, freight units, package units, container units, and trailer units) with this function.

If you want to cancel planned requirement documents, you first need to unassign them from the assigned capacity document from within the function. Then you need to restart the function, select the corresponding document category and run the function again to cancel them.

i Note

- If you cancel a road freight order, the status of the assigned requirement documents changes from **Planned** to **Unplanned**.
- If the unplanned requirement documents were created by automatic freight unit building, the cancellation triggers another freight unit building run that takes into account the most up-to-date normalized quantities.
- If you have several levels of assignments, check and cancel the business documents in the sequence of their assignment.

For example, you have a freight unit, which is assigned to a package unit, which is assigned to a container unit, which is assigned to a trailer unit, which is assigned to a road freight order. In this case, first cancel the road freight order, then the trailer unit, then the container unit, then the package unit, and finally the freight unit.

- This function is based on the standard cancellation function. This means that business documents with other statuses, such as **In Execution**, can't be canceled.

Activities

Open the [Check NLQs and Cancel Documents](#) app from the launchpad.

Alternatively, you can execute the function in the background [Check NLQs and Cancel Documents](#).

Related Information

[Use of Normalized Quantities and Capacities](#)

Use of Consignment Orders in Planning

You can use consignment orders in planning.

Prerequisites

- If you want to display and plan consignment orders on the geographical map and in the Gantt chart, you must have made the settings for the geographical map and the Gantt chart. For more information, see [Map Functions](#) and [Gantt Chart for Transportation Planning](#).
- If you want to work with hierarchies for consignment orders, you must have defined a hierarchical view in Customizing for Transportation Management under **►Planning ► General Settings ► Define Hierarchical Views for Business Documents ►**. For more information, see [Use of Hierarchical Views](#).

Use

The following manual planning operations are possible:

- You can assign one or several consignment orders to a road freight order.

i Note

The **Enable Settlement** checkbox must be selected only in the related freight order type or consignment order.

- You can assign one or several consignment orders to a truck. The system creates a road freight order.
- You can assign a freight unit stage to a consignment order.
- You can create a road freight order from within a consignment order.
- You can create a consignment order from a freight unit stage or package unit stage.
- You can create a transportation unit, such as a container unit, for one or more consignment orders.
- You can assign a consignment order to a transportation unit.
- You can create packages for consignment orders.

i Note

- If you create packages for a road freight order to which two consignment orders are assigned, the system creates two separate packages for the two consignment orders.
- The system packs items in one consignment order that belong to different external delivery notes into different packages.

- Consignment orders are taken into account in load planning and load consolidation.
- You can display a hierarchy for a consignment order or display consignment orders as part of a freight order hierarchy.

During manual planning, the system considers incompatibilities between freight units at consignment order level. This means that you can make sure that dangerous goods and non-dangerous goods are not planned in the same consignment order, for example. Similarly, VSR optimization cannot plan consignment orders with incompatible content.

Scheduling considers consignment orders in the same way as container units or package units. This means that the system considers the times of the assigned freight units if no requested and acceptable dates/times are defined for loading or unloading in the consignment order.

You can also display and plan consignment orders on the geographical map and in the Gantt chart.

Consignment orders are also taken into account in VSR optimization.

i Note

It is possible that the system assigns consignment orders with different carriers to a freight order during manual planning or VSR optimization. In this case, make sure that you edit the freight order manually.

Related Information

[Consignment Order](#)

[Creation and Editing of a Consignment Order](#)

Use of Drivers in Planning

Use

You can assign drivers to existing freight orders and vehicle resources. You can also create freight orders to which the system assigns drivers as soon as they are created.

This function is available only for road freight orders and trucks.

Prerequisites

- In your master data, you have created drivers as business partners with the role **Driver** and you have defined absences and non-working times for each driver. To define drivers, open the **Define Business Partner** app on the launchpad. For more information, see [Definition of Business Partners](#).
- In Customizing, you have created a freight order type with the relevant settings for the use of drivers. For example, you have made the following settings:
 - You have specified whether you want to assign a single driver or a driver team (consisting of two drivers) to your freight orders. If you want to create freight orders with assigned drivers in the transportation cockpit in the road freight order hierarchy or in the road freight order list, you have to enter the number of required drivers.

i Note

If you want to change the number of required drivers in the road freight order hierarchy or in the road freight order list, you have to select the **Settings Can Be Changed in Freight Order** checkbox.

- You have specified whether you want to assign your drivers for entire freight orders or for the individual transportation stages of a freight order.

For more information, see Customizing for **Transportation Management** under **Freight Order Management** > **Freight Order** > **Define Freight Order Types**.

- If you want to use the list of drivers in the transportation cockpit, you have to define a page layout in which you have made the following settings for the driver in the resources area:

- If you want to create freight documents from the list of drivers in the transportation cockpit, you have to select the **Create Freight Document** checkbox.
- If you want to add additional drivers to your list of drivers in the transportation cockpit, you have to select the **Insert** checkbox.
- If you want to use context determination in the transportation cockpit to display other orders within a specific planning horizon that are assigned to a driver, you have to select the **Context Determination** checkbox. For more information, see [Context Determination](#).

If you want to use the **Driver Assignment** tab page in the details of your freight orders in the transportation cockpit, you have to define a page layout in which you have selected the driver assignment in the order details area.

To define page layouts, open the **Define Page Layouts for Transportation Cockpit** app on the launchpad.

- If you want to use the driver as a grouping attribute in the freight order hierarchy, you have to define a hierarchy type in Customizing with the following settings:
 - Transportation mode category: **Road**
 - Consumer: **Transportation Cockpit: Freight Orders/Freight Bookings**
 - Grouping rule: **Standard Grouping Attribute**
 - Standard grouping attribute: **Driver**

For more information, see Customizing for **Transportation Management** under **►Planning►General Settings►Define Hierarchical Views for Business Documents►** and [Use of Hierarchical Views](#).

- Optionally, in the capacity availability settings, you have restricted the availability of your drivers (see [Planning Profile](#)). You can also indicate in these settings if the unavailability of your drivers is to be considered as an absence or a break. Breaks can be within the freight order, that is, the freight order can overlap with the driver's break. Absences cannot lie within the freight order, that is, the freight order cannot overlap with the driver's absence (this may lead to a split of the freight order).

i Note

- Absences of drivers, downtimes of vehicles, and validity periods of vehicles and drivers that you have defined in the master data are always considered as downtimes independent from the settings that you make in the capacity availability settings.
- The unavailability of your drivers is considered only during scheduling and VSR optimization (for the default driver; VSR = vehicle scheduling and routing).

Features

You can assign one or more drivers to an existing freight order or to the individual transportation stages of an existing freight order. You can also create a freight order to which you can directly assign one or more drivers. In a driver team, the individual drivers take turns to drive. The person who is not driving sleeps. You can also assign a driver to several existing freight orders.

i Note

If you assign the default driver of a specific vehicle resource to an existing freight order that has not yet been assigned a vehicle resource, the system automatically assigns this freight order to the vehicle resource to which this driver is assigned as the default driver. If a different vehicle resource has already been assigned to the freight order, this resource is kept. For more information, see [Driver](#).

i Note

If you have specified one driver per freight order as the number of required drivers in Customizing and a driver is already assigned to your freight order in the system, when you assign a different driver to the freight order the system replaces the driver originally assigned and assigns the new driver to the freight order.

Activities

You can assign one or more drivers to an existing freight order in the transportation cockpit as follows:

- Drag and drop

You can drag and drop drivers from your list of drivers in the transportation cockpit to the freight order being planned in your road freight order hierarchy or in your road freight order list. If you have specified in your freight order type that you want to assign your drivers per transportation stage, you can also drag and drop drivers in the road freight order hierarchy to the individual transportation stages.

- Manual creation and change

You can enter or change drivers manually in the **Driver** column in your road freight order hierarchy or in your road freight order list. If you have specified in your freight order type that you want to assign drivers per transportation stage, you can enter or change drivers for individual transportation stages in the **Driver** column in your road freight order hierarchy.

If you have activated driver assignment in the order details area in your layout for the transportation cockpit, you can select a freight order in the road freight order hierarchy or in the road freight order list and manually enter or change a driver in the details for this freight order.

- Dialog box

You can select a freight order in the road freight order hierarchy or in the road freight order list and choose the **Assign Driver** pushbutton. A dialog box appears in which you can enter drivers. You can decide in the dialog box whether to replace the currently assigned drivers with new drivers or whether to add drivers to the existing drivers.

You can create a freight order in the transportation cockpit to which you can directly assign one or more drivers as follows:

- You can select drivers in the list of drivers and choose the **Create Freight Document** pushbutton. A dialog box appears in which you can enter additional data. The system creates a freight document for each driver that you selected in the list of drivers.
- In the road freight order hierarchy or in the road freight order list, you can choose the **Road Freight Order** pushbutton. A dialog box appears in which you can directly assign drivers to the freight order as soon as they are created.
- You can select freight units in the freight unit list and trucks on the **Truck** tab page in the resources area and choose the **Optimizer Planning** pushbutton. The system generates freight orders to which the default driver of the corresponding resource is automatically assigned. The system considers the driver's non-working times and absences according to your capacity availability settings.
- In the **Freight Orders/Freight Bookings** list, you can choose **►New ► Road Freight Order ►**. A dialog box appears in which you can directly assign drivers to the freight order as soon as they are created.
- You can drag and drop drivers from your list of drivers to one or more freight units in the freight unit list. The system then automatically creates a freight order with the selected freight units that is assigned the selected driver with the vehicle resource to which it is assigned as the default driver. Alternatively, you can choose the **Assign Selected Items** pushbutton.

i Note

This function is available only for drivers who are assigned to a vehicle resource as the default driver.

Alternatively, you can assign drivers in the Gantt chart. For more information, see [Gantt Chart for Transportation Planning](#).

In the freight order hierarchy, you can use the standard grouping attribute **Driver** to display all freight orders that are assigned to a driver. You can then continue to plan. For example, you can assign all these freight orders to a truck resource or vice versa. Alternatively, you can display all freight orders for a driver on the geographical map.

If you want to display information about a second driver in the road freight order list, you must select the corresponding fields in your field catalog. For more information, see [Field Configuration for the Transportation Cockpit](#).

More Information

For more information about the use of drivers in freight order management, see [Use of Drivers in Freight Order Management](#).

Use of Multiresources

Use

You can use multiresources, for example, if an unknown number of vehicle resources of a means of transport are available, but you only want to make the settings for these vehicle resources once. You can define multiresources for vehicle resources and container resources.

Restrictions

- In the case of schedules, the system takes multiresources into account for the capacity only.
- We recommend that you assign a means of transport for which you have selected the **Multiresource** checkbox to **one** vehicle resource only.
- Container resources are **not** taken into account in VSR optimization and for the generation of transportation proposals.

Prerequisites

- For vehicle resources, you have selected the **Multiresource** checkbox in Customizing for your means of transport and have optionally specified the number of individual resources (**Number of Individual Resources** field).

For more information, see Customizing for **Transportation Management** under **►Master Data ► Resources ► Define Means of Transport. ►**

By selecting the **Lock Multiresource** checkbox for the means of transport, you can prevent multiple planners from using the multiresource in parallel in multiple sessions. Using the multiresource in multiple sessions can, for example, result in the maximum number of individual resources being violated.

- You have defined a vehicle resource or container resource. To do so, on the **SAP Easy Access** screen, choose **►Transportation Management ► Master Data ► Resource ► Resource ►**. You have assigned a means of transport to your vehicle resource. You have selected the **Multiresource** checkbox for your container resource.
- In the optimizer settings, you have optionally specified how the system is to react if the number of parallel freight orders exceeds the number of individual resources of the multiresource. To do this, open the **Create Optimizer Settings** app on the launchpad.

Features

Vehicle Resources

During VSR optimization, the system assumes that as many vehicle resources exist as you have specified in Customizing in the **Number of Individual Resources** field. In accordance with this number, the system can plan parallel freight orders for this vehicle resource.

If you have not specified a value in Customizing in the **Number of Individual Resources** field, the system assumes that an unlimited number of vehicle resources are available. Depending on the settings that you have made in Customizing and in the optimizer settings, the number of individual resources of a multiresource can be violated.

During manual planning for vehicle resources and container resources, the system checks whether the number of individual resources of a multiresource is violated. The system only performs this check if the method **VSR_CHECK** is contained in your planning controller strategy or if you have selected the pushbutton for checking. The system checks all resources belonging to the business documents for which you have called the check. During the check, the system interprets the number of individual resources as maximum. If this number is violated, the system generates an error message.

When it executes the report **/SCMTMS/VSR_MULTIRES_CHECK**, the system checks whether the number of individual resources of a multiresource is violated. During the check, the system interprets the number of individual resources as maximum. If this number is violated, the system generates an error message or an alert.

Only individual resources of vehicle resources and container resources are displayed on the geographical map. Multiresources cannot be displayed.

Use of Trailers

Use

You can create, edit, and display trailer units. You can also calculate charges based on trailers.

Prerequisites

You have configured the relevant settings in Customizing so that you can configure your standard layout for the use of trailers in the layout settings for the transportation cockpit. Hierarchy type **TUTRL** is predefined by default for the planning of trailer units. In addition, you can define your own hierarchy types. For more information, see Customizing for **Transportation Management** under **►Planning ►General Settings ►Define Hierarchical Views for Business Documents ►**.

You have defined a transportation unit type in Customizing. Transportation unit type **TRTL** is predefined by default for trailer units. For more information, see Customizing for **Transportation Management** under **►Planning ►Transportation Unit ►Define Transportation Unit Types ►**.

Features

Creation of Trailer Units

- You use drag and drop to assign a freight unit to a trailer. The system creates a trailer unit for this trailer. You can continue planning it by, for example, assigning further freight units.

To couple and uncouple the trailer unit to vehicle resources, you use drag and drop to assign this trailer unit to the freight order of the selected vehicle resource. The system checks whether the coupling and uncoupling locations exist in both documents. If there are several coupling and uncoupling locations, you can choose one.

In an alternative scenario, your trailer is not yet assigned to a vehicle resource, but you already want to plan the trailer. We recommend that you use this procedure, for example, if you have a trailer in the yard and you want to load freight units onto it. However, at the time of planning, it is not yet clear or it is not important with which vehicle resource the trailer is to be

moved or which route the vehicle resource is to take. In this case, you create a trailer unit as described above. You later assign the vehicle resource to this trailer unit using drag and drop. The system creates a new freight order for the vehicle resource and couples the trailer automatically.

- You use drag and drop to assign a trailer to an existing freight order. The system creates a new trailer unit and couples the trailer automatically to the vehicle resource of the existing freight order.
- You choose **New Trailer Unit**. You assign freight units and trailers to this trailer unit in any sequence you prefer. The system then carries out consistency checks, such as for the following attributes:
 - Means of transport
 - Transportation mode
 - Carrier

You can create an empty trailer unit and assign a trailer to it in this way, for example. However, you assign the freight units later on. We recommend that you use this procedure, for example, if you know that the trailer will be picked up the next day but, at the time of planning, it is not yet clear which goods it is to transport.

- You select a trailer and choose **Create Trailer Unit**. You assign a document type, a source and destination location, and departure and arrival dates to this trailer unit.

Editing and Displaying Trailer Units

- You can define separate hierarchies for trailer units. For more information, see [Use of Hierarchical Views](#) and [Use of Profile and Layout Sets](#).
- You can remove the assignment of a trailer to a vehicle resource by selecting both documents and choosing **Remove Trailer**.

i Note

If this pushbutton is not displayed, check your layout.

- If you add an additional location for a trailer unit that is already coupled to a vehicle resource, the system automatically adds it to the freight order for the vehicle resource, since the location sequence for the vehicle resource and the coupled trailer must be identical.

Charge Calculation Based on Trailers

You consolidate freight units from multiple forwarding orders into a trailer unit and charge the customer based on a standard rate, the weight of the trailer, or any other criteria. You can maintain a charge line in the forwarding order for the trailer using the resolution base **Resource**. You maintain rates for different types of trailers using the **Means of Transport** calculation base.

Activities

You create trailer units in the transportation cockpit. To call the transportation cockpit, open the **Transportation Cockpit** app on the launchpad.

More Information

[Trailer Unit](#)

[Creation and Editing of Transportation Units](#)

[Example: Trailer Swap](#)

Use of Combination Resources

A combination resource is the fixed amalgamation of one vehicle resource and one or more passive vehicle resources that are coupled in a fixed sequence and are always used together. An example of a combination resource is a tractor with two semi-trailers.

Use

Combination resources are an example of vehicle combinations and are different from vehicle-resource combinations. For more information about the difference between combination resources and vehicle-resource combinations, see [Vehicle Resource](#).

The use of combination resources in planning is especially recommended if you do not have your own fleet. In this case, you can perform VSR optimization on the level of the combination resource. This means that VSR optimization does not have to deal with the coupling and uncoupling of the individual components of the combination resource. The trailers are important for load planning, however. This means that the individual components of the combination resource and their attributes are considered in load planning.

Prerequisites

- You have defined combination resources in the resource master data. For more information, see [Definition of Combination Resources](#).
- You have also made the following planning-relevant settings in the resource master data:
 - In the **Capacity** view, you have defined the planning-relevant capacities for the combination resource (quantities and weights). These values are important for VSR optimization.

You have also defined additional capacities for the components of the combination resource (length, width, and height of the loading space).
 - In the **Phys. Properties** (Physical Properties) view, you have manually defined the tare weight and the maximum gross weight not only for the components but also for the combination resource because the system does not automatically aggregate the values of the individual components.
 - In the **Load Optimization** view, you have defined axles, empty weight and maximum weight as well as dimensions for the components of your combination resource. These values are important for load planning and pre-checks in VSR optimization. For more information, see [Master Data in Load Planning](#).
- You have defined incompatibilities in the incompatibility settings of your planning profile (optional). VSR optimization considers incompatibilities for the combination resource, whereas load planning considers them for the individual components.
- If you want simple load planning checks to be performed for your combination resource during VSR optimization, you must select the value **Partially** in the **Consider Load Planning** field in the optimizer settings of the planning profile. For more information, see the F1 Help for this field.

Features

Single Documents and Transportation Cockpit

In the road freight order, combination resources are marked as such by a corresponding indicator.

The trailers of a combination resource are displayed as local items in the road freight order. However, information about coupling and uncoupling is displayed on the **Overview** tab page because trucks and trailers always travel together.

In the transportation cockpit, the resource list shows only the combination resource. You can see the individual components in the load plan hierarchy, for example.

You cannot plan the components of a combination resource individually, that is, outside of the combination resource. If you want to do this, you must copy the combination resource, meaning that you have to define it twice. However, you can assign the individual components to several, different, combination resources.

You also cannot change combination resources in planning. For example, you cannot remove or add a trailer. Similarly, you cannot change the sequence of the trailers.

If you have your own fleet and want to dynamically couple or uncouple a tractor and trailer, we recommend that you work with vehicle-resource combinations. VSR optimization can dynamically combine the individual components for vehicle-resource combinations. This is based on the prerequisite that you have defined means-of-transport combinations in Customizing under ► [Master Data](#) ► [Resources](#) ► [Means of Transport and Compartment](#) ► [Define Means-of-Transport Combination](#) ►.

Load Planning

Combination resources are considered in load planning. The individual components are also displayed here.

During load planning, the system considers incompatibilities between freight units and incompatibilities between freight units and the truck and its trailers.

Load consolidation is not supported for combination resources.

Related Information

[Load Planning](#)

Integration of Real-Time Execution Data

In the transportation cockpit, you can display real-time execution data, such as delays or accidents, for freight documents.

Use

Real-time execution data helps you to identify delays during the execution of your transports at an early stage and to adapt the transportation plan if necessary.

Prerequisites

- You are working with the process controller. Predefined services and the following standard strategies are available for this function:
 - TES_D_TM - [Real-Time Exec. Data: Delay Determination](#)
 - TES_R_TM - [Real-Time Exec. Data: Resource Pos. Deter.](#)

In the standard system, delays for a specific freight document stage are calculated based on the estimated time of arrival (ETA). The system derives the ETA of the next stop from the estimated event date of the [Delay](#) event, which is stored in the freight order on the [Execution](#) tab page. If you want to have the ETA determined by an external provider, you must define your own strategy.

The resource position is sent by default for each event to Transportation Management and is considered in TM as the last reported position of the resource. If you want to have the resource position determined by an external provider, you must also define your own strategy.

There is no standard strategy for the determination of events. You need to define your own strategy.

i Note

Delays and events can be determined only for freight documents that have the status **In Execution**.

For more information about the process controller, see [Process Controller Configuration](#).

- You have made the following settings in the planning profile:
 - You have entered the standard strategies or your own strategies.
 - Optional: You have entered a threshold value starting from which a delay is to be classified as critical and is to be displayed on the geographical map and in the Gantt chart.
 - Optional: You have specified how long the real-time execution data is to be kept in the cache.
- Optional: You have defined in Customizing how real-time data is to be displayed on the geographical map and in the Gantt chart. For more information, see Customizing for Transportation Management under:
 - [Basic Functions](#) > [Geographical Map](#) > [Define Layouts for Geographical Map](#) >
 - [Basic Functions](#) > [Gantt Chart](#) > [Define Layouts for Gantt Chart](#) >

Features

On the **geographical map**, you can determine delays by choosing **Show Delays** in the context menu. All freight document stages with a delay are marked in a special color. The quick info and the label provide more details about the delay.

You can remove all real-time execution data from the geographical map by choosing **Remove Execution Data** in the context menu.

For an individual freight document, you can display the reason for the delay by choosing **Show Events**. The system can determine and display the following types of events:

- Traffic events, such as accidents or road works
- Weather events
- Other events

For an individual freight document, you can also determine and display delays by choosing **Show Delays** in the context menu.

In the **Gantt chart**, delayed freight documents are marked with a warning icon. The quick info provides more details about the delay.

Greenhouse Gas Emissions

Transportation Management (TM) allows you to calculate greenhouse gas (GHG) emissions during manual planning and vehicle scheduling and routing (VSR) optimization.

GHG emissions cover a range of gases, such as carbon dioxide (CO₂) and methane (CH₄). These GHG emissions are measured in CO₂ equivalents. On the user interfaces, GHG emissions are simply referred to as CO₂ emissions.

Use

When you assign freight unit stages to a road freight order (FO) manually or by selecting **Assign Selected Items**, the CO₂ emissions of the road FO are calculated automatically based on the settings that you made in Customizing for your vehicle types. The same holds true when you use VSR optimization. In this case, the CO₂ emissions costs that you defined in your constraints and costs settings are taken into account in addition.

The calculated CO₂ emissions with their unit of measure are displayed in the road FO list. The road FO hierarchy (both in the business document and in the transportation cockpit) provides the CO₂ emissions with their unit of measure on document and on stage level. Using the aggregation row, you can easily identify the total CO₂ emissions of all road FOs in the transportation cockpit.

In the truck list and the equipment types list, you can see the CO₂ emissions that you have defined in Customizing for your empty and fully loaded vehicles. Further, the CO₂ emissions per loaded weight are displayed. This value indicates the increase in CO₂ emissions per distance if you add one unit of load, such as one ton or kg, to the vehicle. In other words, it describes the gradient of the linear curve between the CO₂ emissions per distance of the empty vehicle and the fully loaded vehicle. The formula used is:

CO₂ emissions of the fully loaded vehicle minus CO₂ emissions of the empty vehicle divided by the capacity, that is the maximum allowed weight of the load

Note that the fields in the truck list and equipment types list are not displayed by default but need to be added using personalization.

You can display the CO₂ emissions for road FOs on the geographical map. They can be included in the tooltip or as a label. In the Gantt chart, you can also display the CO₂ emissions for road freight orders in the corresponding list and as a tooltip and a label in the chart panel.

Prerequisites

- You have indicated CO₂ emissions for your vehicle types. You can indicate CO₂ emissions both for empty and fully loaded trucks and specify a weight and distance unit of measure. Further, you have indicated a payload weight and a corresponding unit of measure. For more information, see Customizing for **Transportation Management** under ► **Master Data** ► **Resources** ► **General Settings** ► **Define Equipment Groups and Equipment Types** ►.
- To define the priority of CO₂ emissions for VSR optimization, you have optionally defined CO₂ emissions costs per unit of measure in your constraints and costs settings using the **Edit Constraints and Costs Settings** app.
- If you want to display the CO₂ emissions as tooltip and label in the chart panel of the Gantt chart, you have made the required settings in Customizing.

For more information, see [Gantt Chart for Transportation Planning](#).

- If you want to display the CO₂ emissions as tooltip and label in the geographical map, you have made the required settings in Customizing.

For more information, see [Map Functions](#).

Related Information

[Constraints and Costs Settings](#)

[Customizing for CO₂ Emissions Calculation](#)

[Calculation of CO₂ Emissions](#)

Calculation of Distances Traveled With or Without Load

Transportation Management (TM) provides visibility on loaded and empty transportation stages.

Prerequisites

- You have activated the calculation of distances in Customizing for **Transportation Management** under **Basic Functions** > **General Settings** > **Define General Settings for TM**.
- In the field catalogs for the lists and hierarchies for road and rail freight orders as well as trailer, container, and railcar units and the lists for trailer, container, and railcar unit stages, you have selected the following fields and indicated whether they are to be displayed by default or using personalization:
 - **Distance (Empty)**
 - **Distance (Loaded)**

For more information, see [Field Configuration for the Transportation Cockpit](#).

- You have added the following fields to the Gantt chart for road and rail freight orders as well as trailer, container, and railcar units:
 - **Total Distance**
 - **Total Distance UoM**
 - **Distance (Empty)**
 - **Distance (Loaded)**

For more information, see Customizing for **Transportation Management** under **Basic Functions** > **Gantt Chart** > **Define Field Lists and Label Schemes for Gantt Chart**.

Use

TM provides visibility regarding distances that are traveled either with or without load and whether the corresponding transportation stages are empty or not. This information is available in the corresponding lists and hierarchies of the transportation cockpit, the Gantt chart, and the single document user interfaces for road and rail freight orders as well as trailer, container, and railcar units.

Based on this information, you are able to do settlement, for example, or measure the efficiency of your fleet by calculating the ratio of the distance traveled with load and the total distance.

Further, TM provides dedicated key performance indicators (KPI).

The following assumptions are made when it comes to whether a vehicle resource or container is considered empty:

- A container is empty if it doesn't carry any product items or package items.
- A trailer is empty if it doesn't carry any product items, package items, or container.
- A truck is empty if it doesn't carry any product items, package items, or container and doesn't have a trailer attached to it. However, a combination resource is considered as empty even though it has a trailer attached to it.
- A truck transporting an empty container or having an empty trailer attached to it across a stage is considered as follows:
 - From the truck's viewpoint, the stage is not empty.
 - From the container's and the trailer's viewpoint, the stage is empty.

Related Information

[Key Performance Indicators in the Transportation Cockpit](#)

Customer-Specific Functions

Use

You can use this function to access planning data and to model your own planning processes, such as the following:

- Visually represent freight orders and freight units
- Remove forwarding orders from planning
- Add freight units to freight orders

Prerequisites

- You have defined at least one strategy of your own and at least one method of your own, and have assigned this method to your strategy. For more information about the process controller, see Customizing for [Transportation Management](#) under [▮▮Planning▮▮General Settings▮▮Configure Process Controller for Planning▮▮](#).
- You have defined a customer-specific function and have assigned your strategy to this function. For more information, see Customizing for [Transportation Management](#) under [▮▮Planning▮▮General Settings▮▮Define Customer-Specific Functions for Planning▮▮](#).

Features

You model your planning processes using strategies that consist of one or more methods. Each method represents an individual planning step. The process controller is the framework used for defining strategies and methods.

Once you have defined customer-specific functions in Customizing and assigned a strategy to each of them, these functions are automatically available for selection on the user interface for transportation planning. When you choose a function, the system automatically performs the corresponding planning steps.

For this, the process controller passes the planning request object /SCMTMS/CL_CUS_REQUEST or a customer-specific object derived from it, on through the methods. You can enhance your customer-specific object with your own data.

You can specify a request object class in Customizing. This class must be derived from the /SCMTMS/CL_CUS_REQUEST class. If you do not specify a class, the system uses the /SCMTMS/CL_CUS_REQUEST class. For more information, see Customizing for [Transportation Management](#) under [▮▮Planning▮▮General Settings▮▮Define Customer-Specific Functions for Planning▮▮](#).

Activities

To perform transportation planning using customer-specific functions, open the [Transportation Cockpit](#) app on the launchpad and then start planning. Use the corresponding pushbutton to choose the customer-specific function you want to perform.

More Information

[Process Controller Configuration](#)

Parallel Planning

Use

In manual planning and in VSR optimization, multiple planners can work simultaneously. If there are conflicts with dates or times, the system issues a message.

Features

If several planners access the same object in manual planning or in VSR optimization, such as a resource or a schedule, the system issues a message that the object is locked. For multiresources, you can specify whether the multiresource is to be locked for the means of transport.

If several planners perform separate pre-carriage, main carriage, or on-carriage planning and unplanned stages exist (for example, because you specified that only the route is to be copied when transportation proposals are generated), VSR optimization creates new stages and tries to adjust the dates and times of the stages to the ones determined in the transportation proposal. If a planner plans a stage (such as the main carriage) manually and a planned freight order or freight booking exists, the system copies the dates and times to the predecessor and successor stage. It considers them a hard constraint (acceptable values). If inconsistencies occur, for example, because acceptable dates overlap, the system issues a message. If two planners plan a freight unit at the same time, for example, pre-carriage and main carriage, and conflicts arise, the system also issues a message. For time conflicts, the system sets the appropriate planning status for the freight unit stage (see [Explanation of Statuses of Business Documents](#)).

→ Recommendation

We recommend that you first plan stages that are critical in terms of date and time, and then the remaining stages.

Definition of Carriers

Transportation Management (TM) offers various options for defining or determining carriers for business documents.

Planning offers the following options:

- Carrier determination

The system determines a unique carrier based on a set of simple rules that you define in the carrier determination profile.

Carrier determination is supported for freight units, freight orders, and consignment orders.

For more information, see [Carrier Determination](#).

- Carrier selection

The system assigns the carrier with the lowest costs based on specific constraints to a given freight document. Carrier selection can be performed manually or automatically. It takes into account your carrier selection settings.

Carrier selection creates a ranking list of all available and relevant carriers based on decision criteria. You can then manipulate the ranking list according to your needs. Finally, the carrier that is ranked first can be assigned to the freight document automatically or manually or the result can be used in a subsequent tendering process.

Carrier selection is supported for freight orders and freight bookings.

For more information, see [Carrier Selection](#).

In addition, the following options are available in TM:

- Manual definition

You can manually enter a carrier in the transportation cockpit for freight units, freight orders, freight bookings, and consignment orders. In addition, you can manually enter the carrier for freight units, freight documents, and consignment

orders on the respective user interfaces.

- Direct shipment options

Direct shipment options can be determined automatically for freight units during creation or can be generated manually for a freight unit on the user interface based on the freight unit type. Direct shipment options are generated for each carrier/service level combination based on a freight agreement. The costs for the direct shipment options come from charge management. Direct shipment options take into account your settings in the carrier selection settings.

Direct shipment options are supported for freight units.

For more information, see [Determination of Direct Shipment Options](#).

- Carrier propagation

The carrier is propagated from a predecessor document, such as the delivery, into the first TM document, between TM documents, and from the top-level TM document into the predecessor document.

- Using the carrier from the schedule

When you create a freight document based on a schedule, the carrier of the schedule is copied into the freight document.

For more information, see [Schedule](#).

- Default route

A default route can be applied to requirement documents (freight units or transportation units) and impose a carrier constraint per stage of the default route.

For more information, see [Default Route](#).

- Freight tendering

The freight tendering process can be used to tender freight orders to one or more carriers.

Tendering is performed based on a tendering profile. In this profile, you define the individual steps of the tendering process. For example, you can specify the tendering type, such as peer-to-peer or broadcast tendering, the tendering process based on request for quotation or direct, the relative price limit, or the maximum response time.

When a suitable and valid carrier is found and made responsible for execution, the respective carrier is stored on freight order root level.

For more information, see [Freight Tendering](#).

- External planning

In an external planning scenario, freight units or consignment orders are sent to an external transportation management system and freight orders including carrier information are sent back. For more information, see [External Planning](#).

- SAP Business Network for Logistics

SAP Business Network for Logistics is a cloud-based platform designed to optimize logistics processes by connecting business partners, enabling real-time collaboration, and providing comprehensive visibility into the supply chain. Through integration of Transportation Management with SAP Business Network Freight Collaboration, the solution serves as a hub for all communication between the carrier and shipper and carrier information is sent back to TM.

For more information, see [SAP Business Network for Logistics](#).

Carrier Determination

Carrier determination lets you automatically determine a carrier for freight units, consignment orders, and freight orders based on simple rules.

Prerequisites

- You have defined a carrier determination profile.

In your carrier determination profile, you define determination rules. A determination rule consists of a number of document attributes that are used to determine a unique carrier, such as gross weight, source and destination location, or transportation mode. Note that some of these fields need to be made visible using personalization. If data is available in hidden fields, a corresponding warning is displayed in a dedicated field of the affected determination rule.

There are checks in place that prevent duplicate entries and unreachable determination rules.

You can download your carrier determination profile as a spreadsheet or upload a spreadsheet to your carrier determination profile.

Finally, you can simulate carrier determination either by a specific business document or by document attributes, such as source and destination location, gross weight, or transportation group. You can see which carrier the system would choose based on which rule.

To define carrier determination profiles, open the [Create Carrier Determination Profile](#) app from the launchpad.

- You have activated carrier determination by assigning your carrier determination profile to your business document type in Customizing. For more information, see Customizing for [Transportation Management](#) under:
 - [Freight Order Management](#) > [Freight Order](#) > [Define Freight Order Types](#) >
 - [Freight Order Management](#) > [Consignment Order](#) > [Define Consignment Order Types](#) >
 - [Planning](#) > [Freight Unit](#) > [Define Freight Unit Types](#) >
- If you want to display information messages about which carrier was determined and when a carrier was changed in the application log, you have defined message settings in Customizing. For more information, see Customizing for [Transportation Management](#) under [Basic Functions](#) > [User Interface](#) > [Define Message Settings](#) >.

Features

Carrier determination is performed automatically after creation or update of a business document (that is, a freight unit, consignment order, or freight order). The system first evaluates whether the document is relevant for triggering carrier determination. Based on the assigned carrier determination profile, it checks whether relevant data was changed on root, item, or stop level. In a second step, the system determines or redetermines the carrier if necessary. To do this, it evaluates the determination rules of the assigned carrier determination profile from top to bottom according to the specified processing sequence. It applies the first matching determination rule and assigns the specified carrier to the business document. If no carrier is defined in this rule, it is also a valid result. If none of the rules match, the system takes the carrier from the predecessor document.

You can manually change the carrier in the business document. Note, however, that a manually changed carrier won't be changed by carrier determination anymore. Similarly, a carrier determined by carrier selection won't be changed by carrier determination anymore.

The Business Add-In (BAI) [Enhancements for Carrier Determination](#) (/SCMTMS/EX_TSPD) allows you to do the following:

- Adjust the carrier determination profile to be used for a given business document, for example, to activate or deactivate carrier determination for certain business documents

- Adjust the decision on whether a redetermination is needed, for example, to react to other changes in a custom determination
- Adjust the determined carrier, for example, to create a custom determination

In the application log, you can see the processing relevance of business documents. Further, it shows the carrier determination results, including the determination rule that was used. Finally, the log shows adjustments made by the BAdI.

i Note

- Carrier determination is performed on document level only.
- Only the first and the last stop are evaluated.

Carrier Selection

Use

You can use carrier selection to assign a suitable carrier to your subcontractable business documents (for example, freight orders or freight bookings) either manually or automatically. The aim is to find a carrier with the lowest costs under consideration of constraints.

Prerequisites

Basic Settings

You have defined transportation lanes in the master data and, in doing so, you have defined means of transport and carriers, including carrier profiles (optional).

Optimization Options

You use one or more of the following optimization options:

- Strategies (only relevant to automatic carrier selection)
 - Costs
 - Priority

For more information, see [Strategies](#).

- Continuous Move

For more information, see [Continuous Move](#).

- Business share (only relevant to automatic carrier selection)

For more information, see [Business Share](#).

- Check against transportation allocations (relevant to both automatic carrier selection and manual carrier selection).

For more information, see [Check Against Transportation Allocations](#).

Detailed Settings

- In Customizing, you have specified that your business document types (for example, freight order types) are relevant to subcontracting. For more information see, for example, for freight orders, Customizing for [Transportation Management](#) under [Freight Order Management](#) > [Freight Order](#) > [Define Freight Order Types](#).
- You have made further settings in the carrier selection settings. In the carrier selection settings, you define, for example, the process in which carrier selection is to be executed:
 - Regular carrier selection
 - Tendering (see [Freight Tendering](#))
 - Determination of direct shipment options (see [Determination of Direct Shipment Options](#))

To make the carrier selection settings, open the [Create Carrier Selection Settings](#) app on the launchpad.

- If you want to consider external freight charges during carrier selection, you have to specify in the settings for carrier selection that you want charges from charge management to be considered. In this case, you must configure charge management accordingly (see [Charge Management and Service Product Catalogs](#)).

Features

Manual Carrier Selection

In manual carrier selection, you manually assign the required carrier to your business documents. If you have configured the check against transportation allocations, the system takes into account and checks transportation capacities (transportation allocations) that you have defined for the individual carriers. If, during the allocation, the system finds relevant transportation allocations without capacity or transportation allocations are violated by the allocation, warning messages appear.

Automatic Carrier Selection

A separate optimization is available for the automatic carrier selection. It takes into account the selected optimization options when determining the most cost-effective carrier for all business documents that you have selected. If none of the carriers are available, the system does not assign a carrier to the relevant business documents.

Note that automatic carrier selection does not support consignment orders.

Determining Available Carriers

In this case, the system determines all available carriers. If you have specified in the carrier selection settings that transportation allocations are not to be considered, even those carriers without any free capacity are displayed.

Business Documents with Multiple Stops

In the case of business documents with multiple stops, the system tries to find a direct transportation lane between the source and destination locations and uses it as the starting point for the determination of available carriers.

❖ Example

A freight order goes from A via B and C to D. In the carrier selection settings, you have selected the checkbox for overall carrier availability.

The system determines the most concrete transportation lane (using the same technique as for strategy determination) for each stage (A to B, B to C, and C to D). The result is a list of all possible carriers for each stage. The system also uses strategy determination to determine the strategy for the main carriage (direct connection from A to D). The result is a list of possible carriers. The system now creates the intersection from all these lists. The result is no carriers, one carrier, or multiple carriers that can serve all the stages including the main stage. The consideration of stages and stops is used therefore as a filter.

The system first checks which carriers cover all the stages of the freight order (A to B, B to C, and C to D). In a second step, the system chooses the carrier from those found in the first step with a direct transportation lane between A and D.

If you have not selected the checkbox for overall carrier availability in the carrier selection settings, the system uses strategy determination to determine the strategy for the main stage only.

SAP delivers the standard planning strategy TSPS_DEF for carrier selection. You can specify this strategy in the carrier selection settings. SAP also delivers the standard planning strategy VSR_1STEP. For more information, see [One-Step Planning](#) and [Planning Strategies](#).

Activities

- To call carrier selection, open the **Carrier Selection** app on the launchpad.

First, a user interface is displayed where you can define profile and layout sets. For more information, see [Use of Profile and Layout Sets](#).

- To perform carrier selection in the background, schedule report /SCMTMS/TSPS_OPT_BGD. For more information, see [Selection of Carriers](#).

More Information

[Full Map Display](#)

Strategies

Use

Strategies are optimization options for the carrier selection. The system takes into account strategies during the automatic carrier selection. You define strategies either in the transportation lane or in carrier selection settings. In the first case, the system considers the appropriate transportation lane for each freight order. This allows it to consider different settings in one run.

Features

You can choose one of the following strategies or a combination of these strategies:

Priority

The system determines the carrier on the basis of a priority that you have defined. This means that the system chooses the carrier with the highest priority first from the transportation lane, taking into account the various restrictions.

Costs

The system determines the carrier on the basis of the lowest total costs. You can take into account either internal costs or charges calculated in charge management:

- Internal costs

Transportation costs and carrier costs that you define for your carriers:

- You define transportation costs per business document, for example, freight order (EA), weight, or volume.

- You define carrier costs per kilometer or mile.

You can define internal costs in the transportation lane (see [Transportation Lane](#)) and, in the case that no internal costs are defined in the specific transportation lane, you can define them in the carrier profile (see [Carrier Profile](#)).

i Note

The costs used here are not real costs. They are simply internal costs that are used for weighting.

- Charges Calculated in Charge Management

You have to create a charge calculation that is set up in charge management. For more information, see [Charge Management and Service Product Catalogs](#).

For more information about how the system takes into account strategies during the carrier selection, see [Strategy Determination](#).

Activities

1. Choose the required strategy for the means of transport in the transportation lane or in the carrier selection settings.

To choose the required strategy in the transportation lane, open the [Define Transportation Lane](#) app on the launchpad.

To choose the required strategy in the carrier selection settings, open the [Create Carrier Selection Settings](#) app on the launchpad.

2. If you want to use priorities, define the required priorities for the individual carriers in the transportation lane.
3. If you want to use internal costs, define transportation costs and carrier costs for the individual carriers either in the transportation lane or in the carrier profile.

Strategy Determination

The system tries to find the transportation lane that, out of all the transportation lanes that are relevant for the carrier selection, most closely corresponds to the business document (for example, freight order) . For more information, see [Transportation Lane](#).

❖ Example

You have defined the following transportation lanes:

- Transportation lane between source location A and destination location B
- Transportation lane between source location A and destination transportation zone B' (contains destination location B)
- Location Transportation Lane at source location A

The freight order is to go from A to B. In this case, the system chooses the transportation lane between source location A and destination location B. This is the most concrete transportation lane, since it runs between two specified locations. The transportation lane between source location A and destination transportation zone B' is less concrete since a transportation zone can contain multiple locations. The location transportation lane at A is also less concrete since a transportation lane between two locations is more concrete than a location transportation lane. Using the strategy that you have defined for this transportation lane, the system determines the carrier.

The system takes into account the carriers in the transportation lane only. If you have a freight order with multiple stops, you can use these stops as an additional filter. For more information, see [Carrier Selection](#).

Continuous Move

Use

Continuous moves are optimization options for carrier selection. The system takes into account continuous moves during automatic carrier selection only.

Prerequisites

- You have set the **Continuous Move** checkbox in the carrier profile of the suppliers. You define a carrier profile on the SAP Easy Access screen under **Logistics > Transportation Management > Master Data > Transportation Network > Define Carrier Profile**. For more information see [Carrier Profile](#).
- You have made the settings relevant to the continuous move in the carrier selection settings or in the transportation lanes. To define the settings for carrier selection, open the **Create Carrier Selection Settings** app on the launchpad. For more information, see [Transportation Lane](#).

Features

As part of automatic carrier selection, the system tries to combine subcontractable business documents (for example, freight orders) by forming continuous moves. The aim is to reduce the costs of the business documents by assigning additional business documents to a carrier. The following types of continuous moves are available:

- Simple continuous move: The carrier accepts one or more additional business documents and, after executing the first business document, directly executes the following business documents one after the other.
- Round trip: A round trip is a simple continuous move for which the following additional prerequisites must be met: The source location of the first business document and the destination location of the second business document are identical.

A round trip can only consist of exactly two business documents.

i Note

The system cannot take into account complex round trips where multiple source locations (pickup) or destination locations (delivery) are served.

The system can combine two or more business documents in a continuous move if the following conditions are met:

- The relevant carrier is available for both business documents.
- The arrival date/time of the first business document lies in the departure window of the second business document.
- The departure date/time of the second business document lies in the arrival window of the first business document.

The system performs a check of incompatibilities for the business documents if you have activated this function in the settings for carrier selection.

Furthermore, the distance between the destination location of the first business document and the source location of the second business document must not exceed the smaller maximum distance that you have defined for the carrier in the transportation lanes.

The system checks the type of continuous move. In the case of a round trip, the system checks if the destination location of the first business document and the source location of the second business document are identical, for example. In the case of a simple continuous move, the system can combine multiple business documents without the vehicle having to return to the source location.

Activities

1. Choose the continuous move type for the means of transport in the **Contin. Move Type** field of the transportation lane.
2. Define the following values, as required, for the individual carriers:
 - Arrival and departure windows
 - Maximum distance
 - Discount

Cost Calculation for Continuous Moves

The system first determines the costs per business document (for example, freight order). Here it considers the strategy and origin of the carrier costs that you defined in the carrier selection settings. The system then determines the discount per business document. When doing so, it takes into account the settings that you have made in the transportation lane for the carrier and the means of transport. The system deducts this discount from the costs determined previously for each business document.

❖ Example

You have defined the following data:

Carrier A is to receive a discount of 10% on the transportation lane A->B for means of transport 0001.

Carrier A is to receive a discount of 20% on the transportation lane B->C.

There are two freight orders (FRT1 for A->B and FRT2 for B->C). The costs for FRT1 are 1000, the costs for FRT2 are 2000. The optimizer has determined that both freight orders are suitable for a continuous move.

The optimizer now calculates costs of 900 (1000 minus 10%) for FRT1 and 1600 (2000 minus 20%) for FRT2. If the optimizer were to assign the continuous move to carrier A, the total costs would therefore be 2500. The optimizer compares these costs with carrier B, for example, who offers lower costs for each freight order (for example, 950 for FRT1 and 1800 for FRT2). If the optimizer were to assign the continuous move to carrier B, the total costs for the continuous move would then be 2750. Therefore, the best solution is to assign both freight orders as a continuous move to carrier A.

Recalculation of Costs from Charge Management

If you specified in the carrier selection settings that the system is to perform a recalculation of costs from charge management for continuous moves, the system considers the discounts for continuous moves for the business documents from charge management in the recalculation, however, only for the carriers that offer continuous moves. The system saves the result in the ranking list. For more information, see [Charge Management and Service Product Catalogs](#).

Business Share

Use

Business shares are optimization options for the carrier selection. The system takes into account business shares during the automatic carrier selection.

Prerequisites

You have defined business shares. To do this, open the [Create Business Shares](#) app on the launchpad.

Features

For each business share, you can define tolerances as well as penalty costs that the optimizer calculates when an excess or shortfall of the target value plus or minus these tolerances occurs. However, the penalty costs are not real costs, but instead are used for weighting purposes only.

→ Recommendation

It only makes sense to use business shares if you have defined penalty costs. We recommend that you also always define tolerances to give the optimizer a certain amount of leeway.

The optimizer takes into account the following when assigning the carriers in accordance with the defined business shares:

- Transportation zone hierarchies
- Hierarchical relationships between means of transport
- Historical business shares
- Documents to which a carrier has not yet been assigned

In the tendering profile, you can define whether the system, when it is checking the target value of a business share or allocation, includes only the business documents already assigned to the relevant business share or allocation or whether it also includes the corresponding freight requests during carrier selection. For more information, see the system documentation.

❖ Example

Carrier A is to receive 50% of all freight orders that are transported from source location DC_FRANKFURT with any means of transport of the transportation mode [Street](#) in June 2010.

More Information

[Creation of Business Shares and Transportation Allocations](#)

Creation of Business Shares and Transportation Allocations

Use

You can use this function to create business shares and transportation allocations. The system uses business shares and transportation allocations during carrier selection or for capacity management in the air cargo scenario, for example.

Prerequisites

You have configured the transportation allocation settings in Customizing – for example, you have activated the use of transportation allocations and business shares. For more information, see Customizing for [Transportation Management](#) under [► Planning ► General Settings ► Define Transportation Allocation Settings ►](#).

Features

Creation of Trade Lanes

You need trade lanes both for business shares and transportation allocations. A trade lane comprises the following:

- Geography, that is, a combination of zones or locations (see [Use of Transportation Orientations](#))
- Means of transport or transportation mode (optional)

❖ Example

Examples of trade lanes include:

- All transports from Hamburg to New York
- All transports within Germany
- All transports that leave Bavaria

You can use trade lanes to define geographical data with different degrees of detail. Furthermore, you can work with placeholders. Each business share and each transportation allocation is only valid for exactly one trade lane.

i Note

- Trade lanes do not represent connections between objects in the transportation network, for example, locations or zones. They are a context-independent classification of any transportation-related activities.
- Trade lanes can form a hierarchy. This means that more generic trade lanes can contain more specific trade lanes (for example, **Within Germany Using Means of Transport Truck** contains **Along the Way from Frankfurt to Hamburg with Means of Transport Refrigerated Truck** and **Within Bavaria with Means of Transport Truck for Liquids**).
- Trade lanes may overlap, for example, **From Germany Using Airplane** and **Within Europe Using Airplane**.
- Trade lane hierarchies contain the following:
 - Transportation zone hierarchies, for example, **From Germany** contains **From Bavaria**, and this contains **From Plant_MÜNCHEN**
 - Hierarchical relationships between means of transport, for example, **Within Germany Using Means of Transport Truck** contains **Within Germany Using Means of Transport Refrigerated Truck**
 - Means of transport/transportation mode relationship, for example, **Within Germany Using Transportation Mode Street** contains **Within Germany Using Means of Transport Refrigerated Truck**

Creation of Business Shares

When you create a business share, you define the following:

- Creation options

You can specify, for example, whether a business share is to refer to a transportation lane, that is, is only to be created if the corresponding transportation lane already exists. Additionally, you can specify whether time buckets can be overwritten.
- Trade lane (see above)
- Business share details

Here you define, among other things, tolerances and penalty costs that are incurred when an excess or shortfall of these tolerances occurs. You also define the unit for your target value.

- Target value

Here you define the actual business shares, for example, 40 % for carrier A and 60 % for carrier B.

You can use this user interface for mass creation. You can create, for example, a daily time bucket for a whole year.

Creation of Transportation Allocations

When you create a transportation allocation, you define the following:

- Creation options

You can specify, for example, whether a transportation allocation is to refer to a transportation lane and whether time buckets may be overwritten.

- Trade lane (see above)
- Scope of transportation allocation

Here you enter, for example, the planning period (yearly, weekly, and so on). You are free to choose the start of the planning period. For example, you can define a bucket from Monday 14:00 until Tuesday 14:00.

- Buckets

Here you can define, for example, a sub-bucket distribution. If a carrier can only deliver, for example, in the morning, you can define two sub-buckets (**Morning** and **Afternoon**) for the bucket **Daily**. As the percentage of the assigned business documents (for example, freight orders), you define 100% for **Morning** and 0% for **Afternoon**.

In addition, you can specify the minimum and maximum product allocation quantity.

Furthermore, you can create multiple transportation allocations by defining specific parameter combinations, for which you can define specific creation options. You can specify, for example, whether a transportation allocation is to refer to a transportation lane, that is, is only to be created if the corresponding transportation lane already exists. Additionally, you can specify whether time buckets can be overwritten.

Activities

To create business shares, open the **Create Business Shares** app on the launchpad.

To create carrier selection allocations, open the **Create Transportation Allocation** app on the launchpad.

To create flight schedule-based allocations, open the **Edit Schedule** app on the launchpad. You then choose the "Create Transportation Allocation" pushbutton in your master flight schedule. You can access and edit the transportation allocation from your personal object worklist (POWL). To do this, open the **Transportation Allocations** app on the launchpad.

To create air freight agreement allocations, open the **Edit Freight Agreement** app on the launchpad. You can now create a transportation allocation at item level in your freight agreement, assuming you have previously defined a corresponding trade lane.

Additionally, you can upload allocations from an external data source to Transportation Management (TM) by executing report /SCMTMS/TAL_IMPORT. If there are changes in existing buckets of allocations, you can adapt these buckets with this report, too. You call the report from the SAP Easy Access screen under **►Transportation Management ► Administration ► Background Processing ► Allocation Upload ►**. Alternatively, you can upload allocations in the **Allocations** worklist and in the allocation UI.

More Information

Check Against Transportation Allocations

Use

The check against transportation allocations is an optimization option that you can use as part of both the manual and automatic carrier selection.

In the check against transportation allocations, you assign transportation capacities (transportation allocations) to carriers for a certain period. You can define the following:

- The maximum number of business documents (for example, freight orders) for a carrier per transportation lane or free transportation lane for a certain period (day, week, month, quarter, or year), validity, and a certain means of transport. You define this by entering quantity unit EA for the transportation allocation.
- An allocation unit and quantity unit for each transportation allocation
- The minimum number of business documents that defines the number of agreed business documents per period (day, week, or month) and validity.

Prerequisites

- You have made the settings for transportation allocations in Customizing, for example, you have activated the use of transportation allocations. For more information, see Customizing for [Transportation Management](#) under [►Planning►General Settings►Define Transportation Allocation Settings►](#).
- You have defined carrier selection allocations. To do this, open the [Create Transportation Allocation](#) app on the launchpad.
- In the carrier selection settings, you have defined whether you want to use transportation allocations or whether the system is to consider the settings for the transportation lane. In the second case, you must have marked the [Use Transp. Allocations](#) checkbox for your means of transport.

Features

The system checks all relevant transportation allocations during the check against transportation allocations. This means that you can define restrictions.

❖ Example

You have defined the following capacities (transportation allocations):

- Transportation allocation belonging to a freight order from source location A to destination location B (maximum quantity = 3 EA). This is the most specific transportation allocation since the freight order runs between two specific locations (direction [Along](#)).
- Transportation allocation that belongs to a freight order from source transportation zone A' and destination transportation zone B' (maximum quantity = 10 EA)

- Transportation allocation for source location A (maximum quantity = 10 EA). This is the least specific transportation allocation.

The system can only assign capacities if capacities are free in all of the transportation allocations (you can override this in manual planning). In this case, the system can assign, at the most, three business documents between A and B even if a greater capacity is available for the transportation allocation between the source and destination transportation zones, and at the source location. The reason for this assignment is that the system takes into account the lowest available capacity.

The system also takes into account transportation zone hierarchies and hierarchies between means of transport. You define hierarchies between means of transport by specifying higher-level means of transport when defining means of transport. This means that you can also use hierarchical relationships to define restrictions, since the system takes into account the transportation allocations in all higher-level means of transport.

More Information

[Transportation Allocation](#)

[Creation of Business Shares and Transportation Allocations](#)

Transportation Allocation

Definition

You can store information about transportation allocations of carriers in geographical areas with specific means of transport or about specific departures of a schedule. A transportation allocation defines constraints on the minimum or maximum quantity available within a certain bucket and stores the assigned quantities.

Transportation Management distinguishes between three transportation allocation types:

- Air freight agreement allocation

Stores information about transportation allocations of carriers that act as constraints for air freight agreements. You can create air freight agreement allocations directly from an air freight agreement.

- Carrier selection allocation

Stores information about transportation allocations of carriers in a specific trade lane. The system uses carrier selection allocations when optimizing the carrier selection.

- Flight schedule-based allocation

Stores information about transportation allocations of carriers at schedule departure level in a master flight schedule. You can create flight schedule-based allocations directly from a master flight schedule.

If changes are made to the departure times and arrival times of the schedule departures in the corresponding master flight schedule, and if these no longer match the bucket-from date and bucket-to date of a period in your flight schedule-based allocation, the system changes the status of the referenced data for the corresponding period from **Data Is Up-to-Date** to **Data Is Not Up-to-Date**.

If the bucket-from date and the bucket-to date of your period are no longer current, you can update them directly in the period. To do so, choose **Update from Schedule**.

If the schedule to which you refer from your flight schedule-based allocation creates new departures, you can add a period to your allocation that contains the dates and times of the new departures of the corresponding schedule. To do so, choose **►Add Bucket► From New Sched. Dep. ►**.

You can use transaction **WUF** to display a list of schedules and schedule-based allocations that reference a specific schedule.

i Note

If you have defined multiple buckets for a schedule departure and execute report **/SCMTMS/MP_SCHED_CREATE_TOR** to generate air freight bookings, the system creates a separate air freight booking for each bucket.

You can define transportation allocation types in Customizing. For more information, see Customizing for **Transportation Management** under **►►Planning ► General Settings ► Define Transportation Allocation Types ►**.

For more information about creating transportation allocations, see [Creation of Business Shares and Transportation Allocations](#).

i Note

You can enter multiple capacity units and units of measure in parallel for all transportation allocation types. Carrier selection, however, takes just one unit into account for each transportation allocation. You therefore need to specify for carrier selection allocations in Customizing which unit is to be taken into account during carrier selection.

For example, you can specify a maximum weight of 500 tons and a maximum volume of 45 cubic meters for a transportation allocation. If you have already planned 500 tons and 40 cubic meters of this transportation allocation, you cannot use this allocation further. If you have defined cubic meters as the relevant unit for carrier selection, however, carrier selection ignores the weight that has been fully used up and determines that five cubic meters of capacity are still available.

When you create transportation allocation types, you can specify whether a particular transportation allocation type is to use attributes or carrier selection.

The following attributes can be used, for example:

- Shipping type, such as unit load device

Each shipping type is assigned to a transportation mode.

- Contract basis

You can create just one freight booking for each shipping type/contract basis combination.

- Periodicity

Depending on the transportation allocation type, you can specify various periodicities, such as **daily**, **monthly**, or **departure of a schedule**.

- Handling code

You can define buckets with handling codes for a transportation allocation. Within the validity period for a transportation allocation, you can define a number of buckets with different handling codes in parallel. For more information about handling codes, see [Handling Codes](#).

❖ Example

You create a flight schedule-based allocation from a master flight schedule for which you define two buckets. One bucket has the handling code **Upper Deck** and the other period has the handling code **Lower Deck**. If you now generate an air freight booking with the handling code **Upper Deck** from this master flight schedule, the system assigns the air freight booking to the bucket with the corresponding handling code automatically.

You can create a transportation allocation with the handling code **Dangerous Goods** in the same way. Any air freight bookings that you post to this transportation allocation must also have **Dangerous Goods** as the handling code.

Integration

Business documents (for example, freight orders) do not have a direct link (cross-business-object association) to the transportation allocation. However, if you change business document details that are relevant for the transportation allocation process, the system transfers the corresponding information to the transportation allocations. Relevant changes can be, for example:

- The assignment of a carrier to a business document
- The cancellation of a business document that has a carrier assigned to it
- Changes to the quantities

When you assign a carrier to a business document, the system books existing transportation allocations if the following data concurs in the business document and the existing transportation allocations:

- Source and destination
- Means of transport
- Carrier
- Requested dates (The system does not consider accepted dates.) In the case of overlapping periods, start and end dates are required.

i Note

If you create a booking for a specific schedule departure, the system also posts the relevant bucket of the transportation allocation assigned to the schedule, assuming you have previously made this assignment.

The system stores references to business documents in allocated capacities.

More Information

[Creation of Business Shares and Transportation Allocations](#)

[Check Against Transportation Allocations](#)

Creation of Business Shares and Transportation Allocations

Use

You can use this function to create business shares and transportation allocations. The system uses business shares and transportation allocations during carrier selection or for capacity management in the air cargo scenario, for example.

Prerequisites

You have configured the transportation allocation settings in Customizing – for example, you have activated the use of transportation allocations and business shares. For more information, see Customizing for [Transportation Management](#) under [Planning](#) > [General Settings](#) > [Define Transportation Allocation Settings](#) .

Features

Creation of Trade Lanes

You need trade lanes both for business shares and transportation allocations. A trade lane comprises the following:

- Geography, that is, a combination of zones or locations (see [Use of Transportation Orientations](#))
- Means of transport or transportation mode (optional)

❖ Example

Examples of trade lanes include:

- All transports from Hamburg to New York
- All transports within Germany
- All transports that leave Bavaria

You can use trade lanes to define geographical data with different degrees of detail. Furthermore, you can work with placeholders. Each business share and each transportation allocation is only valid for exactly one trade lane.

i Note

- Trade lanes do not represent connections between objects in the transportation network, for example, locations or zones. They are a context-independent classification of any transportation-related activities.
- Trade lanes can form a hierarchy. This means that more generic trade lanes can contain more specific trade lanes (for example, [Within Germany Using Means of Transport Truck](#) contains [Along the Way from Frankfurt to Hamburg with Means of Transport Refrigerated Truck](#) and [Within Bavaria with Means of Transport Truck for Liquids](#)).
- Trade lanes may overlap, for example, [From Germany Using Airplane](#) and [Within Europe Using Airplane](#).
- Trade lane hierarchies contain the following:
 - Transportation zone hierarchies, for example, [From Germany](#) contains [From Bavaria](#), and this contains [From Plant_MÜNCHEN](#)
 - Hierarchical relationships between means of transport, for example, [Within Germany Using Means of Transport Truck](#) contains [Within Germany Using Means of Transport Refrigerated Truck](#)
 - Means of transport/transportation mode relationship, for example, [Within Germany Using Transportation Mode Street](#) contains [Within Germany Using Means of Transport Refrigerated Truck](#)

Creation of Business Shares

When you create a business share, you define the following:

- Creation options

You can specify, for example, whether a business share is to refer to a transportation lane, that is, is only to be created if the corresponding transportation lane already exists. Additionally, you can specify whether time buckets can be overwritten.

- Trade lane (see above)
- Business share details

Here you define, among other things, tolerances and penalty costs that are incurred when an excess or shortfall of these tolerances occurs. You also define the unit for your target value.

- Target value

Here you define the actual business shares, for example, 40 % for carrier A and 60 % for carrier B.

You can use this user interface for mass creation. You can create, for example, a daily time bucket for a whole year.

Creation of Transportation Allocations

When you create a transportation allocation, you define the following:

- Creation options

You can specify, for example, whether a transportation allocation is to refer to a transportation lane and whether time buckets may be overwritten.

- Trade lane (see above)
- Scope of transportation allocation

Here you enter, for example, the planning period (yearly, weekly, and so on). You are free to choose the start of the planning period. For example, you can define a bucket from Monday 14:00 until Tuesday 14:00.

- Buckets

Here you can define, for example, a sub-bucket distribution. If a carrier can only deliver, for example, in the morning, you can define two sub-buckets (**Morning** and **Afternoon**) for the bucket **Daily**. As the percentage of the assigned business documents (for example, freight orders), you define 100% for **Morning** and 0% for **Afternoon**.

In addition, you can specify the minimum and maximum product allocation quantity.

Furthermore, you can create multiple transportation allocations by defining specific parameter combinations, for which you can define specific creation options. You can specify, for example, whether a transportation allocation is to refer to a transportation lane, that is, is only to be created if the corresponding transportation lane already exists. Additionally, you can specify whether time buckets can be overwritten.

Activities

To create business shares, open the **Create Business Shares** app on the launchpad.

To create carrier selection allocations, open the **Create Transportation Allocation** app on the launchpad.

To create flight schedule-based allocations, open the **Edit Schedule** app on the launchpad. You then choose the "Create Transportation Allocation" pushbutton in your master flight schedule. You can access and edit the transportation allocation from your personal object worklist (POWL). To do this, open the **Transportation Allocations** app on the launchpad.

To create air freight agreement allocations, open the **Edit Freight Agreement** app on the launchpad. You can now create a transportation allocation at item level in your freight agreement, assuming you have previously defined a corresponding trade lane.

Additionally, you can upload allocations from an external data source to Transportation Management (TM) by executing report /SCMTMS/TAL_IMPORT. If there are changes in existing buckets of allocations, you can adapt these buckets with this report, too. You call the report from the SAP Easy Access screen under **Transportation Management > Administration > Background Processing > Allocation Upload**. Alternatively, you can upload allocations in the **Allocations** worklist and in the allocation UI.

More Information

[Carrier Selection](#)

Use of Transportation Orientations

Use

You can use this function to create the transportation orientation for a trade lane. For more information about trade lanes, see [Creation of Business Shares and Transportation Allocations](#).

Features

The following transportation orientations are available (the parameters that are relevant for each orientation are in brackets):

- **Along** (parameter: Source **and** Destination; parameter type: Location or Zone)

Both the source and the destination are restricted. If the parameter type is **Zone**, the corresponding location of the relevant business document (for example, freight order) must be within the transportation zone defined. If the parameter type is **Location**, the location of the relevant business document must correspond to this parameter.

❖ Example

From PL_FRANKFURT To ITALY

If the source location is PL_FRANKFURT and the destination location belongs to transportation zone ITALY, each object is considered as context (and is therefore relevant).

- **Within** (parameter: Source; parameter type: Zone)

Both the source and the destination are restricted. Both locations of the relevant business document (for example, freight order) must be within the defined transportation zone.

❖ Example

Within GERMANY

If both the source location and the destination location belong to transportation zone GERMANY, each object is considered as context (and is therefore relevant).

- **From** (parameter: Source; parameter type: Zone or Location)

If the source parameter type is **Zone**, the source location of the relevant business document (for example, freight order) must be in the defined transportation zone, and the destination location must be outside of the transportation zone. If the source parameter type is **Location**, the source location of the relevant business document must correspond to this parameter. The destination is not restricted.

❖ Example

From GERMANY

If the source location belongs to transportation zone GERMANY and the destination location is outside of this transportation zone, each object is considered as context (and is therefore relevant).

- **Outbound** (parameter: Source; parameter type: Zone)

Only the source is restricted. The source location of the relevant business document (for example, freight order) must be within the defined transportation zone. The destination is not restricted and it can be within or outside of the transportation zone.

❖ Example

Outbound GERMANY

If the source location belongs to transportation zone GERMANY, each object is considered as context (and is therefore relevant).

- **To** (parameter: Destination; parameter type: Zone or Location)

If the destination parameter type is **Zone**, the destination location of the relevant business document (for example, freight order) must be in the defined transportation zone, and the source location must be outside of the transportation zone. If the destination parameter type is **Location**, the destination location of the relevant business document must correspond to this parameter. The source is not restricted.

❖ Example

To GERMANY

If the destination location belongs to transportation zone GERMANY and the source location is outside of this transportation zone, each object is considered as context (and is therefore relevant).

- **Inbound** (parameter: Destination; parameter type: Zone)

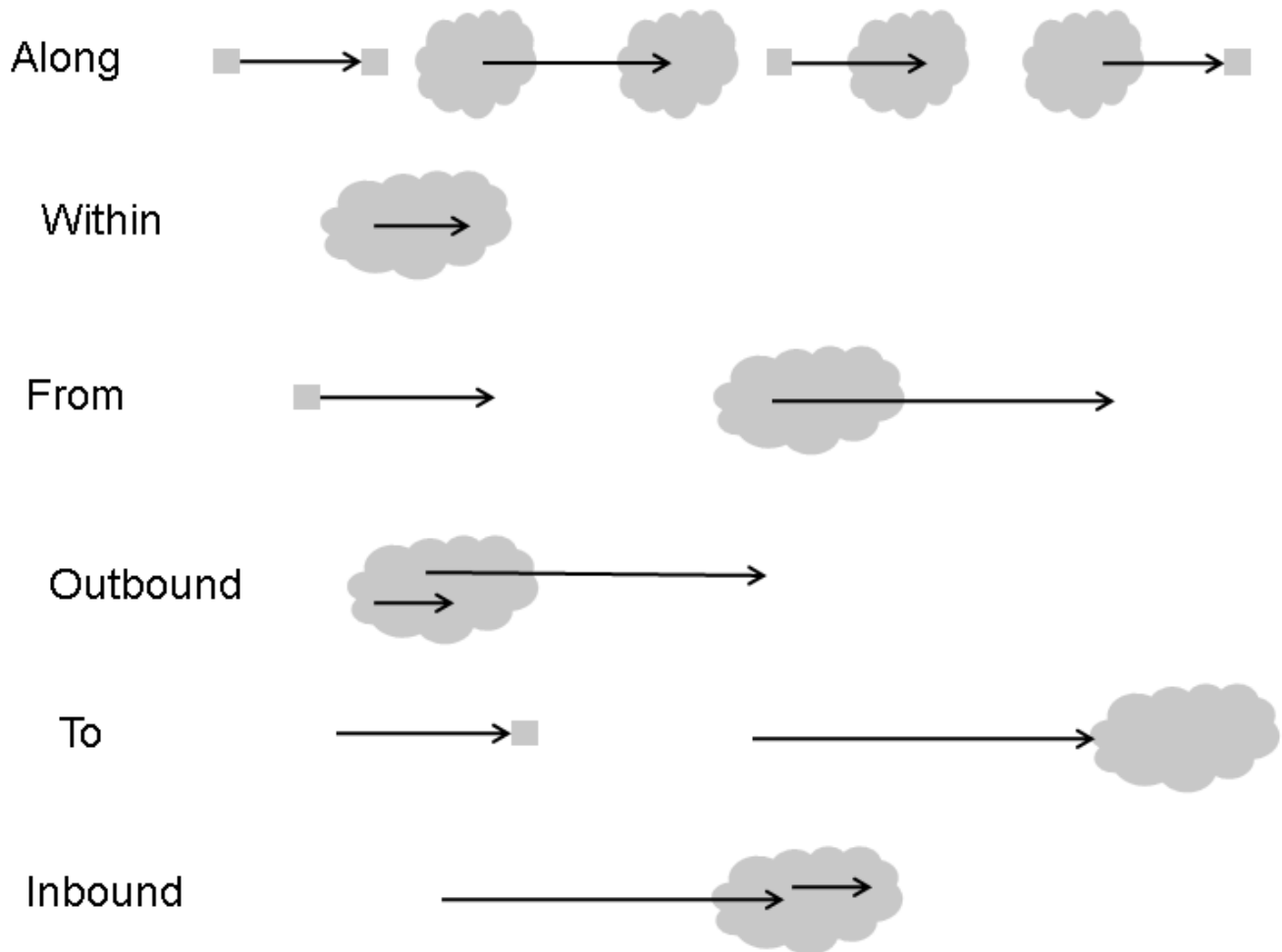
Only the destination is restricted. The destination location of the relevant business document (for example, freight order) must be within the defined transportation zone. The source is not restricted and it can be within or outside of the transportation zone.

❖ Example

Inbound GERMANY

If the destination location belongs to transportation zone GERMANY, each object is considered as context (and is therefore relevant).

The following graphic shows the transportation orientations described:



Transportation Orientations

Optimization as Part of Carrier Selection

Optimization as part of the carrier selection determines the assignment of carriers with the lowest costs to business documents (for example, freight orders).

It aims to achieve the following goals:

- Reduction in total costs (discounts for continuous moves)
- Consideration of all capacity restrictions for means of transport and carriers (minimum and maximum capacities are hard constraints)
- Consideration of business shares (soft constraints; from the transportation lane)
- Incompatibilities
- Carrier Activity

The total costs represent the weighted total of the following individual costs:

- Non-delivery costs for business documents that the optimizer for carrier selection cannot assign
- Costs for the assignment of carriers to business documents
- Penalty costs for violating business shares

- Discounts for continuous moves

The total of all the discounts for continuous moves is subtracted from these total costs.

The optimizer for carrier selection takes into account transportation zone hierarchies and hierarchical relationships between means of transport. You define hierarchies between means of transport by specifying higher-level means of transport when defining means of transport.

The optimization problem is solved by mixed-integer linear programming.

Determination of Direct Shipment Options

Use

You can use this function to have direct shipment options for each carrier/service level combination determined when generating freight units. In this case, the freight units are not consolidated, but rather assigned directly to the carrier (LTL or FTL). This carrier/service level combination is also then shown as the LTL price, even if it is an FTL case without consolidation.

The VSR optimizer considers the most cost-effective LTL price to be the reference cost. If the VSR optimizer finds a consolidated solution with even lower costs during the comparison, it uses this one. If the LTL price is lower, you can convert the freight unit to a freight order automatically, manually, or by using a background report.

Alternatively, you can use this function for scenarios in which a carrier transports products from a shipping point to several consignees. One example of this is the parcel shipment process. The system can automatically determine the direct shipment options for freight units and then assign the freight units to a suitable freight order that meets the requirements in terms of the source location, carrier, and pick-up date. For more information, see [Parcel Shipment](#).

Constraints:

- The LTL price is determined and saved for the entire transportation chain of the freight unit, and not on the level of the stage. For this reason, this function does not support scenarios in which an LTL decision is made for one stage and FTL is used for a second stage. For such scenarios, a separate FTL freight order must be generated and the LTL price is not transferred to the VSR optimizer.
- LTL prices are determined for a freight unit. If a consolidated LTL price is different from the sum of the FTL prices for a freight unit, it is not taken into account. It is therefore important for you to define and use the appropriate freight unit building rules.

i Note

The costs for the direct shipment come from charge management.

However, since the costs in VSR optimization do not usually correspond to the actual costs, but rather only represent an option for controlling the result of the optimization run, you have to select the costs for the direct shipment in such a way that they are comparable to the costs used by VSR optimization.

Prerequisites

- You have defined the carrier selection settings and specified there that they are to be used to determine the direct shipment options. To define the carrier selection settings, open the [Create Carrier Selection Settings](#) app on the launchpad.
- In Customizing for your freight unit type, you have made the following settings:

- You have assigned your freight unit type to the carrier selection settings previously defined.
- You have optionally assigned a strategy for determining carrier selection settings to your freight unit type. The system first evaluates the condition (if available). If this condition does not return a result or if no condition exists, the system uses the carrier selection settings you assigned to your freight unit type.
- You have assigned a strategy for determining direct shipment options to your freight unit type. You use this strategy to specify that the system is to continue processing the freight units automatically once it has determined the direct shipment options.
- In the **DSO Result Rule** field for your freight unit type, you have specified how the system is to continue processing freight units for which it has determined direct shipment options. The following options are available:
 - The system converts the freight units into freight orders for direct shipments.
 - The system assigns the freight units to a freight order that meets the requirements in terms of source location, carrier, and pick-up date. If the system cannot find a suitable freight order, it creates a new one and copies the freight unit data and the direct shipment option to this order.
- You have assigned a freight order type (directly or using a condition) to your freight unit type.
- If you want the system to automatically recalculate the direct shipment options when relevant changes are made to the freight unit, in the **Type DirectShipOpt.** field (type of direct shipment options), choose the corresponding option.

For more information, see Customizing for **Transportation Management** under **►►Planning►Freight Unit►Define Freight Unit Types ►**

- In the costs settings for the planning profile, you have defined a currency that is to be used when direct shipment options are determined. Open the **Create Constraints and Costs Settings** app on the launchpad.

Features

The direct shipment options are determined based on process controller strategies **DSO_DEF** or **DSO_RESULT** as described in the following section. You can replace these strategies with your own strategies.

1. Execution of Carrier Selection

The system identifies a list of direct shipment options and carrier service products for each combination of carrier and service level. When doing so, it takes into account transportation lanes and freight agreements. The system considers incompatibilities based on your carrier selection settings.

i Note

To restrict your results even further, you can define VSR incompatibility settings for incompatibilities between freight units and means of transport. In the carrier selection settings, you can specify that you want to use these settings for determining direct shipment options. Since the vehicle resource represents the means of transportation during the incompatibility determination (see [Incompatibilities](#)), you have to create a dummy vehicle resource and define an incompatibility for it with the incompatibility type **FU-Vehicle Resource**.

2. Determination of Transit Durations

The system determines transit durations based on the duration determination type defined in the freight agreement of the corresponding carrier. The following duration determination types are supported:

- Duration determination based on conditions or freight agreements

Transit duration and calendar are either determined based on a condition (if a condition has been entered in the freight agreement), or taken directly from the freight agreement or the freight agreement type.

- Consideration of freight unit dates only

In this case, the transit duration results from the pickup and delivery dates at the transportation stops of the freight unit.

- Duration determination with external strategy

Transit duration and calendar are not determined and are part of an external scheduling (see next step). The name of the external strategy (see [External Strategies](#)) is also contained in the freight agreement or the freight agreement type.

The result is a transit duration and calendar that are assigned to the freight unit.

3. Scheduling

The system performs scheduling based on the transit duration and calendar. The result is the earliest pickup date and the latest delivery date. The scheduling calculation depends on the duration determination type:

- Duration determination based on conditions or freight agreements

Backward scheduling is performed starting from the requested delivery date of the destination stop. Here the system considers the calendar of the source and destination location and the transit calendar.

- Consideration of freight unit dates only

Pickup and delivery dates correspond to the requested dates of the source and destination locations (transportation stops of the freight unit).

- Duration determination with external strategy

The external strategy defined in the freight agreement or the freight agreement type is called for the execution of an external scheduling. If an external strategy has not been entered, the process controller strategy DSO_EXTSCH is executed. It contains the method DSO_EXTSCH. You can replace it with your own method.

- Any direct shipment options that have a pickup date that occurs before the pickup date of the freight unit are removed from the list of direct shipment options by the system. You can replace this step with your own implementations.

5. Selection of the most cost-effective direct shipment option

The system chooses the most cost-effective direct shipment option by selecting it. You can replace this step with your own implementations.

6. Postprocessing

Activities

On the user interface for freight units, you can display the direct shipment options that have been identified. Open the [Freight Unit](#) app on the launchpad.

Depending on your Customizing settings, you can trigger the identification of direct shipment options manually from this user interface. You can also select direct shipment options here and convert freight units into freight orders for direct shipment or assign them to suitable freight orders.

Alternatively, you can schedule the conversion or assignment of freight units to run in the background. For more information, see [Processing of FUs for Direct Shipment](#).

i Note

In your personal object worklist for road freight orders and in the transportation cockpit, you can display the total of all selected DSO amounts for all freight units that are planned as part of a freight order. This amount specifies how high the costs would be

if each freight unit were to be transported individually. The difference between the total amount and the amount calculated in the freight order indicates the potential savings in the case of consolidation.

More Information

[Process Controller Configuration](#)

[Agreement](#)

External Planning

In an external planning scenario, the shipper sends transportation requirements in the form of freight units (FU) or consignment orders (CO) to the carrier, who is responsible for planning the transportation of goods and sends the planning results in the form of freight orders (FO) or bookings (FB) back to the shipper.

The external planning process requires the use of the following APIs for communication between shipper and carrier:

- `TransportationOrderGenericRequest_Out`
- `TransportationOrderGenericRequest_In`

The APIs enable a shipper to communicate their transportation requirements to the carrier. Conversely, the carrier can communicate transportation planning results, event notifications, updates, and cancellations to the shipper.

For an exhaustive list of elements that these APIs can process, see SAP Note [3420340](#) .

For technical documentation about the services, see: [APIs for Delivery and Transportation](#).

Prerequisites

General Settings

- You have configured the Post Processing Framework (PPF) in Customizing for **Cross-Application Components** under **Processes and Tools for Enterprise Applications** > **Reusable Objects and Functions for BOPF Environment** > **PPF Adapter for Output Management** > **Maintain PPF Settings** . For example, in application `/SCMTMS/TRANSPORTATION`, you activated the action definition **TOR_TPNOR_CO (Replication of Consignment Order)** of action profile `/SCMTMS/TOR`.

The output profiles and scheduling conditions control if and when a message is sent.

- You have assigned the output profile `/SCMTMS/TOR` to the container unit type you use in Customizing for **Transportation Management** under **Freight Order Management** > **Transportation Unit** > **Define Transportation Unit Types** .
- You have defined an EDI communication profile in Customizing for **Transportation Management** under **Integration** > **Enterprise Services** > **Define EDI Communication Profile** .

Road-Specific Settings

- You have defined the necessary freight order types in Customizing for **Transportation Management** under **Freight Order Management** > **Freight Order** > **Freight Order Types** .
- Optional: To allow FU splits in the receiving system, you have enabled FU split in Customizing for **Transportation Management** under **Integration** > **Enterprise Services** > **Define EDI Communication Profile** . Note that you also need to assign your EDI communication profile to the business partner (carrier).

- You have defined the necessary freight booking types in Customizing for **Transportation Management** under ► **Freight Order Management** ► **Freight Booking** ► **Freight Booking Types** ►.

Air-Specific Settings

- You have defined a consignment order type with transportation mode **Air** in Customizing for **Transportation Management** under ► **Freight Order Management** ► **Consignment Orders** ► **Define Consignment Order Types** ►. You define the transportation mode under **Basic Settings**.
- You have assigned the output profile **/SCMTMS/TOR** to the consignment order type in Customizing for **Transportation Management** under ► ► **Freight Order Management** ► **Consignment Order** ► **Define Consignment Order Types** ►.

! Restriction

The following are not currently supported:

- External planning with stage splits
- External planning of transportation units
- Only freight orders with the sequence type of stages **Defined and Linear** are supported

Features

SAP Business Network for Logistics

SAP Business Network for Logistics is a cloud-based platform designed to optimize logistics processes by connecting business partners, enabling real-time collaboration, and providing comprehensive visibility into the supply chain. Through integration of Transportation Management (TM) with SAP Business Network Freight Collaboration and SAP Business Network Global Track and Trace, the solution serves as a hub for all communication between the carrier and shipper.

For more information about SAP Business Network for Logistics, see [SAP Business Network for Logistics](#).

Outbound Communication

The transportation requirement is sent out to the carrier in the form of freight units (FU) or consignment orders (CO) with the aim of requesting a transportation plan.

For this purpose, the shipper uses the `TransportationOrderGenericRequest_Out` API to communicate the transportation request to the carrier.

Features

The `TransportationOrderGenericRequest_Out` API is used to communicate transportation demands to an external system.

Message Content

Outbound messages contain the relevant attributes of a business document, that is, a FU or CO, including the complete item hierarchy or structure, references to other business documents, and party information.

Transportation Order Request: XML Message Example

This section describes the different parts of an XML message and gives explanations for the most important nodes in a message. The first section contains the example message in its full form. The different parts of the message are broken down and described in the sections that follow.

Transportation Order Request: XML Example of Transportation Demand

The following is an example of a full transportation order request:

Sample Code

```
<n0:TransportationOrderGenericRequest xmlns:n0="http://sap.com/xi/SAPGlobal20/Global" xmlns:prx=
  <MessageHeader>
    <CreationDateTime>2025-04-10T12:08:01.929Z</CreationDateTime>
    <SenderBusinessSystemID>System_ID</SenderBusinessSystemID>
    <RecipientParty>
      <InternalID>RECIPIENT_ID</InternalID>
    </RecipientParty>
    <BusinessScope>
      <TypeCode>3</TypeCode>
      <ID>EXTERNAL_PLANNING</ID>
    </BusinessScope>
  </MessageHeader>
  <TransportationOrder actionCode="04" changeOrdinalNumberValue="2" completeTransmissionIndicato
    <TypeCode>S420</TypeCode>
    <CategoryCode>FU</CategoryCode>
    <DangerousGoodsIndicator>false</DangerousGoodsIndicator>
    <TotalDistanceMeasure unitCode="KMT">200.0</TotalDistanceMeasure>
    <TotalNetDuration>PT96H</TotalNetDuration>
    <ShipperParty>
      <InternalID>SHIPPER_ID</InternalID>
      <Address>
        <OrganisationFormattedName>ORGANIZATION_NAME</OrganisationFormattedName>
        <PhysicalAddress>
          <CountryCode>COUNTRY_CODE</CountryCode>
          <RegionCode>REGION_CODE</RegionCode>
          <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
          <CityName>CITY</CityName>
          <POBoxIndicator>false</POBoxIndicator>
          <StreetName>STREET</StreetName>
          <HouseID>HOUSE_NUMBER</HouseID>
        </PhysicalAddress>
        <Communication>
          <CorrespondenceLanguageCode>LANGUAGE</CorrespondenceLanguageCode>
        </Communication>
      </Address>
    </ShipperParty>
    <ConsigneeParty>
      <InternalID>CONSIGNEE_ID</InternalID>
      <Address>
        <OrganisationFormattedName>Consignee Name</OrganisationFormattedName>
        <PhysicalAddress>
          <CountryCode>COUNTRY_CODE</CountryCode>
          <RegionCode>REGION_CODE</RegionCode>
          <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
          <CityName>CITY</CityName>
          <POBoxIndicator>false</POBoxIndicator>
          <StreetName>STREET</StreetName>
          <HouseID>HOUSE_NUMBER</HouseID>
        </PhysicalAddress>
        <Communication>
          <CorrespondenceLanguageCode>LANGUAGE</CorrespondenceLanguageCode>
        </Communication>
      </Address>
```

```

</ConsigneeParty>
<CarrierParty>
  <InternalID>CARRIER_ID</InternalID>
  <Address>
    <OrganisationFormattedName>Carrier Name</OrganisationFormattedName>
    <PhysicalAddress>
      <CountryCode>COUNTRY_CODE</CountryCode>
      <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
      <CityName>CITY</CityName>
      <POBoxIndicator>>false</POBoxIndicator>
    </PhysicalAddress>
    <Communication>
      <CorrespondenceLanguageCode>LANGUAGE</CorrespondenceLanguageCode>
    </Communication>
  </Address>
</CarrierParty>
<AdditionalParty>
  <Party>
    <InternalID>INTERNAL_ID</InternalID>
    <Address>
      <OrganisationFormattedName>Organisation Name</OrganisationFormattedName>
      <PhysicalAddress>
        <CountryCode>COUNTRY_CODE</CountryCode>
        <RegionCode>REGION_CODE</RegionCode>
        <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
        <CityName>CITY</CityName>
        <POBoxIndicator>>false</POBoxIndicator>
        <StreetName>STREET</StreetName>
        <HouseID>HOUSE_NUMBER</HouseID>
      </PhysicalAddress>
      <Communication>
        <CorrespondenceLanguageCode>LANGUAGE</CorrespondenceLanguageCode>
      </Communication>
    </Address>
  </Party>
  <RoleCode>1</RoleCode>
</AdditionalParty>
<TransportationStage>
  <ID>10</ID>
  <OrdinalNumberValue>1</OrdinalNumberValue>
  <StatisticalIndicator>>false</StatisticalIndicator>
  <ExternalPlanningRelevanceIndicator>>true</ExternalPlanningRelevanceIndicator>
  <PayerCompanyID>PAYER_ID</PayerCompanyID>
  <DistanceMeasure unitCode="KMT">200.0</DistanceMeasure>
  <NetDuration>PT96H</NetDuration>
  <RequestedUnloadingDateTimePeriod>
    <StartDateTime>2025-04-09T22:00:00Z</StartDateTime>
    <EndDateTime>2025-04-09T22:00:00Z</EndDateTime>
  </RequestedUnloadingDateTimePeriod>
  <AcceptableUnloadingDateTimePeriod>
    <EndDateTime>2025-04-10T21:59:59Z</EndDateTime>
  </AcceptableUnloadingDateTimePeriod>
  <MainLoadingLocation>
    <InternalID>LOADING_LOCATION_ID</InternalID>
    <Address>
      <OrganisationFormattedName>Organisation Name</OrganisationFormattedName>
      <PhysicalAddress>
        <CountryCode>COUNTRY_CODE</CountryCode>
        <RegionCode>REGION_CODE</RegionCode>
        <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
        <CityName>CITY</CityName>
        <POBoxIndicator>>false</POBoxIndicator>
        <StreetName>STREET</StreetName>
      </PhysicalAddress>
    </Address>
  </MainLoadingLocation>
  <MainLoadingStopID>30</MainLoadingStopID>
  <MainUnloadingLocation>
    <InternalID>UNLOADING_LOCATION_ID</InternalID>
    <Address>
      <OrganisationFormattedName>Organisation Name</OrganisationFormattedName>
      <PhysicalAddress>
        <CountryCode>COUNTRY_CODE</CountryCode>
        <RegionCode>REGION_CODE</RegionCode>

```

```

        <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
        <CityName>CITY</CityName>
        <POBoxIndicator>>false</POBoxIndicator>
        <StreetName>STREET</StreetName>
        <HouseID>HOUSE_NUMBER</HouseID>
    </PhysicalAddress>
</Address>
</MainUnloadingLocation>
<MainUnloadingStopID>20</MainUnloadingStopID>
<ItemAssignment>
    <ItemID>10</ItemID>
</ItemAssignment>
<ItemAssignment>
    <ItemID>1000000</ItemID>
</ItemAssignment>
</TransportationStage>
<DeliveryTerms>
    <Incoterms>
        <ClassificationCode>DDP</ClassificationCode>
        <TransferLocationName>LOCATION_NAME</TransferLocationName>
    </Incoterms>
</DeliveryTerms>
<DeliveryInformation>
    <PlanningStatusCode>1</PlanningStatusCode>
    <GoodsMovementStatusCode>2</GoodsMovementStatusCode>
    <DistributionStatusCode>1</DistributionStatusCode>
</DeliveryInformation>
<BaseBusinessTransactionDocumentReference>
    <ID>FREIGHT_UNIT_ID</ID>
    <TypeCode>1122</TypeCode>
    <VersionID>9</VersionID>
</BaseBusinessTransactionDocumentReference>
<BusinessTransactionDocumentReference>
    <BusinessTransactionDocumentReference>
        <ID>OUTBOUND_DELIVERY_ORDER_ID</ID>
        <TypeCode>496</TypeCode>
    </BusinessTransactionDocumentReference>
    <Date>2025-04-10</Date>
</BusinessTransactionDocumentReference>
<BusinessTransactionDocumentReference>
    <BusinessTransactionDocumentReference>
        <ID>SALES_ORDER_ID</ID>
        <TypeCode>114</TypeCode>
    </BusinessTransactionDocumentReference>
    <Date>2025-04-10</Date>
</BusinessTransactionDocumentReference>
<BusinessTransactionDocumentReference>
    <BusinessTransactionDocumentReference>
        <ID>OUTBOUND_DELIVERY</ID>
        <TypeCode>73</TypeCode>
    </BusinessTransactionDocumentReference>
    <Date>2025-04-10</Date>
</BusinessTransactionDocumentReference>
<Item>
    <ID>1000000</ID>
    <CategoryCode>FUR</CategoryCode>
    <GrossWeightTransportationQuantity unitCode="KGM">50.0</GrossWeightTransportationQuantity>
    <GrossVolumeTransportationQuantity unitCode="MTQ">0.18</GrossVolumeTransportationQuantity>
    <PiecesTransportationQuantity unitCode="EA">5.0</PiecesTransportationQuantity>
    <OuterVolumeTransportationQuantity unitCode="MTQ">0.18</OuterVolumeTransportationQuantity>
    <DeliveryNetAmount currencyCode="EUR">0.0</DeliveryNetAmount>
    <ShipFromLocation>
        <InternalID>SOURCE_LOCATION_ID</InternalID>
        <Address>
            <OrganisationFormattedName>Organization Name</OrganisationFormattedName>
            <PhysicalAddress>
                <CountryCode>COUNTRY_CODE</CountryCode>
                <RegionCode>02</RegionCode>
                <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
                <CityName>CITY</CityName>
                <POBoxIndicator>>false</POBoxIndicator>
                <StreetName>STREET</StreetName>
            </PhysicalAddress>
        </Address>
    </ShipFromLocation>

```



```

</ShipFromLocation>
<ShipToLocation>
  <InternalID>DESTINATION_LOCATION_ID</InternalID>
  <Address>
    <OrganisationFormattedName>Organization Name</OrganisationFormattedName>
    <PhysicalAddress>
      <CountryCode>COUNTRY_CODE</CountryCode>
      <RegionCode>REGION_CODE</RegionCode>
      <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
      <CityName>CITY</CityName>
      <POBoxIndicator>>false</POBoxIndicator>
      <StreetName>STREET</StreetName>
      <HouseID>HOUSE_NUMBER</HouseID>
    </PhysicalAddress>
  </Address>
</ShipToLocation>
<DeliveryTerms>
  <Incoterms>
    <ClassificationCode>DDP</ClassificationCode>
    <TransferLocationName>LOCATION_NAME</TransferLocationName>
  </Incoterms>
</DeliveryTerms>
<BaseBusinessTransactionDocumentReference>
  <ID>FREIGHT_UNIT_ID</ID>
  <TypeCode>1122</TypeCode>
  <VersionID>9</VersionID>
  <ItemID>1000000</ItemID>
</BaseBusinessTransactionDocumentReference>
<BusinessTransactionDocumentReference>
  <BusinessTransactionDocumentReference>
    <ID>OUTBOUD_DELIVERY_ORDER</ID>
    <TypeCode>496</TypeCode>
  </BusinessTransactionDocumentReference>
</BusinessTransactionDocumentReference>
<BusinessTransactionDocumentReference>
  <BusinessTransactionDocumentReference>
    <ID>SALES_ORDER</ID>
    <TypeCode>114</TypeCode>
  </BusinessTransactionDocumentReference>
</BusinessTransactionDocumentReference>
</Item>
</TransportationOrder>
</n0:TransportationOrderGenericRequest>

```

Transportation Order Request: XML Snippet for <MessageHeader>

This XML message's message header specifies the information an EDI message from a shipper to a carrier must include so the freight unit, freight order, or freight booking can be created in the carrier's system. The <BusinessScope>'s <TypeCode> and <ID> are mandatory in an external planning scenario.

<MessageHeader>

Sample Code

```

<MessageHeader>
  <CreationDateTime>2025-04-10T12:08:01.929Z</CreationDateTime>
  <SenderBusinessSystemID>System_ID</SenderBusinessSystemID>
  <RecipientParty>
    <InternalID>CARRIER_ID</InternalID>
  </RecipientParty>
  <BusinessScope>
    <TypeCode>3</TypeCode>
    <ID>EXTERNAL_PLANNING</ID>
  </BusinessScope>
</MessageHeader>

```


Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<CreationDateTime>	Message creation UTC timestamp (Coordinated Universal Time)
<RecipientParty><InternalID>	The carrier to whom the transportation request is sent
<BusinessScope>	The external planning scenario is specified by value 3 and EXTERNAL_PLANNING .

Transportation Order Request: XML Snippet for <TransportationOrder>

This XML snippet specifies the header level information an EDI message from a shipper to a carrier must include so the freight unit, freight order, or freight booking can be created in the carrier’s system.

<TransportationOrder>

☰, Sample Code

```
<TransportationOrder actionCode="04" changeOrdinalNumberValue="2" completeTransmissionIndicator=
  <TypeCode>S420</TypeCode>
  <CategoryCode>FU</CategoryCode>
  <DangerousGoodsIndicator>false</DangerousGoodsIndicator>
  <TotalDistanceMeasure unitCode="KMT">200.0</TotalDistanceMeasure>
  <TotalNetDuration>PT96H</TotalNetDuration>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<TransportationOrder actionCode	A coded representation of an instruction to the recipient of a message specifying how to process a transmitted document 04: Save (here: update)
<TransportationOrder changeOrdinalNumberValue	A number indicating the position of a change in a sequence of changes that have been made to the business document being transmitted For outbound messages derived from the same business document, the system ensures that the changeOrdinalNumberValue attribute is populated with increasing attribute values. 2: The sequence number of the message
<TransportationOrder completeTransmissionIndicator	true: Full image create or update message false: Delta message
<TransportationOrder replicateIndicator	true: Replication on basis of common master data; otherwise false
<TransportationOrder><TypeCode>	A coded representation of a document type that applies to a transportation order of the category freight unit, freight order, or

Technical Name	Comment
	freight booking S420: Freight unit type
<TransportationOrder><CategoryCode>	A coded representation of a document category that applies to a transportation order; for example, categories freight unit, freight order, or freight booking FU: Transportation order category – Freight unit
<DangerousGoodsIndicator>	An indicator that specifies whether or not any of the items in a transportation order represent or contain dangerous goods false: No dangerous goods contained; otherwise true i Note For PS&S only

<ShipperParty>

This XML snippet contains information about the shipper's organization name and address as well as language used in communication.

 Sample Code

```
<ShipperParty>
  <InternalID>SHIPPER_ID</InternalID>
  <Address>
    <OrganisationFormattedName>ORGANIZATION_NAME</OrganisationFormattedName>
    <PhysicalAddress>
      <CountryCode>COUNTRY_CODE</CountryCode>
      <RegionCode>REGION_CODE</RegionCode>
      <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
      <CityName>CITY</CityName>
      <POBoxIndicator>>false</POBoxIndicator>
      <StreetName>STREET</StreetName>
      <HouseID>HOUSE_NUMBER</HouseID>
    </PhysicalAddress>
    <Communication>
      <CorrespondenceLanguageCode>de</CorrespondenceLanguageCode>
    </Communication>
  </Address>
</ShipperParty>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<ShipperParty>	A natural or legal person, organization, organizational unit, or group that is involved in a transportation order in the role of a shipper party Either an InternalID or a StandardID must be present. If a StandardID is present, its schemeAgencyID attribute must contain one of the following values: 3 IATA IATA code of a party

Technical Name	Comment
	9 GS1 GLN number of a party 16 D&B Dun & Bradstreet number of a party 182 NMFTA Standard Carrier Alpha Code (SCAC) If the <code>schemeAgencyID</code> attribute specifies the value 3, the <code>schemeID</code> attribute can be used to distinguish between the following usages: "" or not specified IATA Airline Code "PIMA Code" PIMA Code of a party "SITA/ARINC Code" SITA/ARINC Code of a party
InternalID	SHIPPER_ID : Unique ID of the shipper party involved in a transportation order
<CorrespondenceLanguageCode>	LANGUAGE_CODE : A coded representation of the language of communication; for example, en – English

<ConsigneeParty>

This XML snippet contains information about the consignee or send-to party. It includes details such as organization name, physical address, and language of correspondence.

☰ Sample Code

```
<ConsigneeParty>
  <InternalID>CONSIGNEE_ID</InternalID>
  <Address>
    <OrganisationFormattedName>Consignee Name</OrganisationFormattedName>
    <PhysicalAddress>
      <CountryCode>COUNTRY_CODE</CountryCode>
      <RegionCode>10</RegionCode>
      <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
      <CityName>CITY</CityName>
      <POBoxIndicator>>false</POBoxIndicator>
      <StreetName>STREET</StreetName>
      <HouseID>HOUSE_NUMBER</HouseID>
    </PhysicalAddress>
    <Communication>
      <CorrespondenceLanguageCode>LANGUAGE_CODE</CorrespondenceLanguageCode>
    </Communication>
  </Address>
</ConsigneeParty>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<ConsigneeParty>	A natural or legal person, organization, organizational unit, or group that is involved in a transportation order in the role of a consignee party. Either an <code>InternalID</code> or a <code>StandardID</code> must be present. If a <code>StandardID</code> is present, its <code>schemeAgencyID</code> attribute must

Technical Name	Comment
	contain one of the following values: • 3 IATA IATA code of a party • 9 GS1 GLN number of a party • 16 D&B Dun & Bradstreet number of a party • 182 NMFTA Standard Carrier Alpha Code (SCAC) If the <code>schemeAgencyID</code> attribute specifies the value 3, the <code>schemeID</code> attribute can be used to distinguish between the following usages: • "" or not specified IATA Airline Code • "PIMA Code" PIMA Code of a party • "SITA/ARINC Code" SITA/ARINC Code of a party
<InternalID>	CONSIGNEE_ID : Unique ID of the shipper party involved in a transportation order
<CorrespondenceLanguageCode>	LANGUAGE_CODE : A coded representation of the language of communication; for example, en – English

<CarrierParty>

This XML snippet contains information about the carrier responsible for the transportation plan. It includes details such as organization name, physical address, and language of correspondence.

 Sample Code

```
<CarrierParty>
  <InternalID>CARRIER_ID</InternalID>
  <Address>
    <OrganisationFormattedName>Carrier Name</OrganisationFormattedName>
    <PhysicalAddress>
      <CountryCode>COUNTRY_CODE</CountryCode>
      <StreetPostalCode>69126</StreetPostalCode>
      <CityName>CITY</CityName>
      <POBoxIndicator>>false</POBoxIndicator>
    </PhysicalAddress>
    <Communication>
      <CorrespondenceLanguageCode>de</CorrespondenceLanguageCode>
    </Communication>
  </Address>
</CarrierParty>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<CarrierParty>	A natural or legal person, organization, organizational unit, or group that is involved in a transportation order in the role of a carrier party. Either an <code>InternalID</code> or a <code>StandardID</code> must be present. If a <code>StandardID</code> is present, its <code>schemeAgencyID</code> attribute must

Technical Name	Comment
	contain one of the following values: • 3 IATA IATA code of a party • 9 GS1 GLN number of a party • 16 D&B Dun & Bradstreet number of a party • 182 NMFTA Standard Carrier Alpha Code (SCAC) If the <code>schemeAgencyID</code> attribute specifies the value 3, the <code>schemeID</code> attribute can be used to distinguish between the following usages: • "" or not specified IATA Airline Code • "PIMA Code" PIMA Code of a party • "SITA/ARINC Code" SITA/ARINC Code of a party
<InternalID>	CARRIER_ID : Unique ID of the carrier party involved in a transportation order
<CorrespondenceLanguageCode>	LANGUAGE_CODE : A coded representation of the language of communication; for example, en – English

<AdditionalParty>

This XML snippet contains information about an additional party involved in the transportation process. It includes details such as organization name, physical address, and language of correspondence.

 Sample Code

```
<AdditionalParty>
  <Party>
    <InternalID>ADDITIONAL_PARTY_ID</InternalID>
    <Address>
      <OrganisationFormattedName>Organization Name</OrganisationFormattedName>
      <PhysicalAddress>
        <CountryCode>COUNTRY_CODE</CountryCode>
        <RegionCode>10</RegionCode>
        <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
        <CityName>CITY</CityName>
        <POBoxIndicator>>false</POBoxIndicator>
        <StreetName>STREET</StreetName>
        <HouseID>HOUSE_NUMBER</HouseID>
      </PhysicalAddress>
      <Communication>
        <CorrespondenceLanguageCode>LANGUAGE_CODE</CorrespondenceLanguageCode>
      </Communication>
    </Address>
  </Party>
  <RoleCode>1</RoleCode>
</AdditionalParty>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<AdditionalParty>	A natural or legal person, organization, organizational unit, or group that is involved in a transportation order in a party role other than shipper, consignee, ordering party, or carrier party.
<Party>	A party that is involved in a transportation order. Either an <code>InternalID</code> or a <code>StandardID</code> must be present. If a <code>StandardID</code> is present, its <code>schemeAgencyID</code> attribute must contain one of the following values: • 3 IATA IATA code of a party • 9 GS1 GLN number of a party • 16 D&B Dun & Bradstreet number of a party • 182 NMFTA Standard Carrier Alpha Code (SCAC) If the <code>schemeAgencyID</code> attribute specifies the value 3, the <code>schemeID</code> attribute can be used to distinguish between the following usages: • "" or not specified IATA Airline Code • "PIMA Code" PIMA Code of a party • "SITA/ARINC Code" SITA/ARINC Code of a party
<InternalID>	ADDITIONAL_PARTY_ID: Unique ID of the additional party involved in a transportation order
<CorrespondenceLanguageCode>	LANGUAGE_CODE: A coded representation of the language of communication; for example, en – English
<RoleCode>	A coded representation of the role of a party in a specific transportation order item. Typically one of the following code values can occur: • 10 Bill to Party • TM009 Communication Party • TM010 Executing Carrier • TM013 Issuing Carrier's Agent Other party role codes could be specified for informational purposes, or to cover requirements that arise from special scenarios.

<TransportationStage>

This XML snippet contains information about stages in a transportation order. This includes requested times, loading and unloading location, and physical addresses of sending and receiving parties.

☰ Sample Code

```

<TransportationStage>
  <ID>10</ID>
  <OrdinalNumberValue>1</OrdinalNumberValue>
  <StatisticalIndicator>>false</StatisticalIndicator>

```

```

<ExternalPlanningRelevanceIndicator>true</ExternalPlanningRelevanceIndicator>
<PayerCompanyID>PAYER_ID</PayerCompanyID>
<DistanceMeasure unitCode="KMT">200.0</DistanceMeasure>
<NetDuration>PT96H</NetDuration>
<RequestedUnloadingDateTimePeriod>
  <StartDateTime>2025-04-09T22:00:00Z</StartDateTime>
  <EndDateTime>2025-04-09T22:00:00Z</EndDateTime>
</RequestedUnloadingDateTimePeriod>
<AcceptableUnloadingDateTimePeriod>
  <EndDateTime>2025-04-10T21:59:59Z</EndDateTime>
</AcceptableUnloadingDateTimePeriod>
<MainLoadingLocation>
  <InternalID>LOADING_LOCATION_ID</InternalID>
  <Address>
    <OrganisationFormattedName>Organization Name</OrganisationFormattedName>
    <PhysicalAddress>
      <CountryCode>COUNTRY_CODE</CountryCode>
      <RegionCode>REGION_CODE</RegionCode>
      <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
      <CityName>CITY</CityName>
      <POBoxIndicator>>false</POBoxIndicator>
      <StreetName>STREET</StreetName>
    </PhysicalAddress>
  </Address>
</MainLoadingLocation>
<MainLoadingStopID>30</MainLoadingStopID>
<MainUnloadingLocation>
  <InternalID>UNLOADING_LOCATION_ID</InternalID>
  <Address>
    <OrganisationFormattedName>Organization Name</OrganisationFormattedName>
    <PhysicalAddress>
      <CountryCode>COUNTRY_CODE</CountryCode>
      <RegionCode>REGION_CODE</RegionCode>
      <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
      <CityName>CITY</CityName>
      <POBoxIndicator>>false</POBoxIndicator>
      <StreetName>STREET</StreetName>
      <HouseID>HOUSE_NUMBER</HouseID>
    </PhysicalAddress>
  </Address>
</MainUnloadingLocation>
<MainUnloadingStopID>20</MainUnloadingStopID>
<ItemAssignment>
  <ItemID>10</ItemID>
</ItemAssignment>
<ItemAssignment>
  <ItemID>1000000</ItemID>
</ItemAssignment>
</TransportationStage>

```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<TransportationStage>	A collection of information about a section of the transportation chain represented by a transportation order
<ID>	10: An identifier of a transportation stage
<OrdinalNumberValue>	1: An ordinal number to indicate the position of a transportation stage in a sequence of transportation stages
<StatisticalIndicator>	false: An indicator that specifies whether or not a transportation stage is statistical (not planning relevant)

Technical Name	Comment
<ExternalPlanningRelevanceIndicator>	true : An indicator that specifies whether or not a transportation stage is relevant for external planning
<NetDuration>	The net duration of a transportation stage PT96H : 96 hours For more information about the format, see ISO 8601.
<RequestedUnloadingDateTimePeriod>	A period during which the goods must be unloaded, that is, unloading the goods earlier or later is unacceptable
<AcceptableUnloadingDateTimePeriod>	A period during which the goods must be unloaded, that is, unloading the goods earlier or later is unacceptable
<MainLoadingLocation>	A series of lines containing the ID and physical address of the location where the goods are to be picked up At least one of the elements MainLoadingLocation , MainLoadingIATALocationText , or MainLoadingUNLOCODELocationText must be specified. If the element MainLoadingLocation is specified, either an InternalID , a StandardID , or an Address must be present, otherwise the entire element must be omitted.
<MainLoadingStopID>	30 : An identifier of the transportation stop of the main loading location.
<MainUnloadingLocation>	A series of lines containing the ID and physical address of the location where the goods are to be delivered
<ItemAssignment>	10 : An assignment of a transportation order item to a transportation stage of a transportation order.

<DeliveryTerms>

This XML snippet contains information about the agreed Incoterms.

 Sample Code

```
<DeliveryTerms>
  <Incoterms>
    <ClassificationCode>DDP</ClassificationCode>
    <TransferLocationName>LOCATION_NAME</TransferLocationName>
  </Incoterms>
</DeliveryTerms>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<Incoterms>	The conventional contract formulations for delivery terms that correspond to the rules defined by the International Chamber of Commerce (ICC) Typically one of the following code values can occur:

Technical Name	Comment
	Rules for any mode or modes of transport - EXW Ex Works - FCA Free Carrier - CPT Carriage Paid To - CIP Carriage and Insurance Paid To - DAP Delivered at Place - DPU Delivered at Place Unloaded - DDP Delivered Duty Paid Rules for sea and inland waterway transport - FAS Free Alongside Ship - FOB Free On Board - CFR Cost and Freight - CIF Cost Insurance and Freight

<DeliveryInformation>

This XML snippet contains information about a delivery. It's only included in messages detailing freight units, and is not available in freight orders or freight bookings.

 Sample Code

```
<DeliveryInformation>
  <PlanningStatusCode>1</PlanningStatusCode>
  <GoodsMovementStatusCode>2</GoodsMovementStatusCode>
  <DistributionStatusCode>1</DistributionStatusCode>
</DeliveryInformation>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<DeliveryInformation>	A collection of information about the delivery for a freight unit
<DeliveryInformation><PlanningStatusCode>	A coded representation of the planning status of a delivery 1 Not relevant 2 Not yet processed 3 Partially processed 4 Completely processed
<DeliveryInformation><GoodsMovementStatusCode>	A coded representation of goods movement status of a delivery 1 Not relevant

Technical Name	Comment
	• 2 Not yet processed • 3 Partially processed • 4 Completely processed
<DeliveryInformation><DistributionStatusCode>	A coded representation of the distribution status of a delivery • 1 Not relevant • 2 Relevant • 3 Distributed

<BaseBusinessTransactionDocumentReference>

This XML snippet contains information about the document at hand. It includes ID, type code, and version.

 Sample Code

```
<BaseBusinessTransactionDocumentReference>
  <ID>FREIGHT_UNIT_ID</ID>
  <TypeCode>1122</TypeCode>
  <VersionID>9</VersionID>
</BaseBusinessTransactionDocumentReference>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<BaseBusinessTransactionDocumentReference>	This document's unique ID
<BaseBusinessTransactionDocumentReference> <TypeCode>	Document type code used for all transportation order request documents 1122: transportation order
<BaseBusinessTransactionDocumentReference> <VersionID>	9: An identifier for a version of a business document

<BusinessTransactionDocumentReference>

This XML snippet contains information about other predecessor documents related to the transportation order.

 Sample Code

```
<BusinessTransactionDocumentReference>
  <BusinessTransactionDocumentReference>
    <ID>OUTBOUND_DELIVERY_ORDER</ID>
    <TypeCode>496</TypeCode>
  </BusinessTransactionDocumentReference>
  <Date>2025-04-10</Date>
</BusinessTransactionDocumentReference>
<BusinessTransactionDocumentReference>
  <BusinessTransactionDocumentReference>
    <ID>SALES_ORDER</ID>
    <TypeCode>114</TypeCode>
```

```
</BusinessTransactionDocumentReference>
<Date>2025-04-10</Date>
</BusinessTransactionDocumentReference>
<BusinessTransactionDocumentReference>
  <BusinessTransactionDocumentReference>
    <ID>OUTBOUND_DELIVERY</ID>
    <TypeCode>73</TypeCode>
  </BusinessTransactionDocumentReference>
  <Date>2025-04-10</Date>
</BusinessTransactionDocumentReference>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<BusinessTransactionDocumentReference>	A reference of another business transaction document, for example, a predecessor document of the items of a transportation order
<BusinessTransactionDocumentReference> <BusinessTransactionDocumentReference><TypeCode>	A unique reference to another business document or business document item Typically one of the following code values can occur: • 496 Outbound Delivery Order • 114 Sales Order • 73 Outbound Delivery

<Item>

This XML snippet contains information about the items in an order. This includes such information as unique ID, related transaction documents, and ship-from and ship-to locations.

☰ Sample Code

```
<Item>
  <ID>1000000</ID>
  <CategoryCode>FUR</CategoryCode>
  <GrossWeightTransportationQuantity unitCode="KGM">50.0</GrossWeightTransportationQuantity>
  <GrossVolumeTransportationQuantity unitCode="MTQ">0.18</GrossVolumeTransportationQuantity>
  <PiecesTransportationQuantity unitCode="EA">5.0</PiecesTransportationQuantity>
  <OuterVolumeTransportationQuantity unitCode="MTQ">0.18</OuterVolumeTransportationQuantity>
  <DeliveryNetAmount currencyCode="EUR">0.0</DeliveryNetAmount>
  <ShipFromLocation>
    <InternalID>SOURCE_LOCATION_ID</InternalID>
    <Address>
      <OrganisationFormattedName>Organization Name</OrganisationFormattedName>
      <PhysicalAddress>
        <CountryCode>COUNTRY_CODE</CountryCode>
        <RegionCode>02</RegionCode>
        <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
        <CityName>CITY</CityName>
        <POBoxIndicator>>false</POBoxIndicator>
        <StreetName>STREET</StreetName>
      </PhysicalAddress>
    </Address>
  </ShipFromLocation>
  <ShipToLocation>
    <InternalID>DESTINATION_LOCATION_ID</InternalID>
    <Address>
      <OrganisationFormattedName>Organization Name</OrganisationFormattedName>
```

```

    <PhysicalAddress>
      <CountryCode>COUNTRY_CODE</CountryCode>
      <RegionCode>REGION_CODE</RegionCode>
      <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
      <CityName>CITY</CityName>
      <POBoxIndicator>>false</POBoxIndicator>
      <StreetName>STREET</StreetName>
      <HouseID>HOUSE_NUMBER</HouseID>
    </PhysicalAddress>
  </Address>
</ShipToLocation>
<DeliveryTerms>
  <Incoterms>
    <ClassificationCode>DDP</ClassificationCode>
    <TransferLocationName>Location Name</TransferLocationName>
  </Incoterms>
</DeliveryTerms>
<BaseBusinessTransactionDocumentReference>
  <ID>FREIGHT_UNIT_ID</ID>
  <TypeCode>1122</TypeCode>
  <VersionID>9</VersionID>
  <ItemID>1000000</ItemID>
</BaseBusinessTransactionDocumentReference>
<BusinessTransactionDocumentReference>
  <BusinessTransactionDocumentReference>
    <ID>OUTBOUND_DELIVERY_ORDER</ID>
    <TypeCode>496</TypeCode>
  </BusinessTransactionDocumentReference>
</BusinessTransactionDocumentReference>
<BusinessTransactionDocumentReference>
  <BusinessTransactionDocumentReference>
    <ID>SALES_ORDER</ID>
    <TypeCode>114</TypeCode>
  </BusinessTransactionDocumentReference>
</BusinessTransactionDocumentReference>
</Item>

```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<Item><ID>	1000000 : An identifier for a transportation order item
<Item><CategoryCode>	A coded representation of the category of a transportation order item FUR : Freight unit root item; the top most item of a freight unit
<Item><GrossWeightTransportationQuantity unitCode="KGM">	The gross weight of the goods represented by a transportation order item
<Item><GrossVolumeTransportationQuantity unitCode="MTQ">	The gross volume of the goods represented by a transportation order item
<Item><PiecesTransportationQuantity unitCode="EA">	The quantity of the goods represented by a transportation order item, measured in a dimensionless unit of measurement, such as pieces or canisters
<Item><OuterVolumeTransportationQuantity unitCode="MTQ">	The outer volume of the goods represented by a transportation order item
<Item><DeliveryNetAmount currencyCode="EUR">	The net value of the goods to be delivered that are represented by a transportation order item

Technical Name	Comment
<BusinessTransactionDocumentReference><TypeCode>	A unique reference to another business document or business document item Typically one of the following code values can occur: • 496 Outbound Delivery Order • 114 Sales Order • 73 Outbound Delivery • 1122 Transportation Order (generic type code)

Inbound Communication

The carrier plans the transportation based on the shipper's request and sends a message with a freight order (FO) or booking (FB).

The shipper system receives the message using the `TransportationOrderGenericRequest_In` API. With the message, a FO or FB is created in the shipper system, and the corresponding FUs and consignment orders (CO) are linked and updated according to the transportation plan. When a carrier sends an update, the capacity document created on the basis of the carrier's transportation plan is updated in the shipper's system.

→ Remember

- Currently, web services are only available for road FOs as well as ocean and air freight bookings (FB).
- Web service `TransportationOrderGenericRequest_In` can use referenced requirement documents such as the FU or delivery document to build up its stage sequence.
- To reference FUs, you can use the freight unit ID and don't need to specify all FU item IDs in the message.

Features

The shipper receives `TransportationOrderGenericRequest_In` with a transportation plan and uses it to update their business documents in Transportation Management (TM).

The inbound message can take one of two formats: full image or delta message. While a full image message contains all cargo item details, the delta message contains the information for a single cargo item so that a carrier must send a separate message for every cargo item.

Message Content

With `TransportationOrderGenericRequest_In`, the shipper receives confirmation that the transportation requirements have been planned. If the capacity document in the message has not yet been created, the system triggers its creation in the receiving system. The message must have category code B0 or F0 based on the scenario, and business scope ID `EXTERNAL_PLANNING`. Additionally, depending on the purpose (create, update, or event notification) of the message, the message can contain information about the transportation stages, items, and more. To identify the relevant FUs or COs, the receiving system requires that one of the following references be included in the inbound message:

- FU ID & FU Item ID

- FU ID (simplified FU reference without Item ID)
- CO ID (air freight booking only)
- Predecessor business document ID; for example, outbound delivery reference

The `completeTransmissionIndicator` in the message can be used to indicate a full image or delta message. A full image message contains the complete document including all the items, while a delta message may only contain a subset of the items, for example, in case of status updates.

Consolidation Variants

In ocean, as well as air freight scenarios, your freight booking's consolidation settings in Customizing determine whether consolidation locations are included in the transportation stages of the freight booking. These settings, defined by the shipper, are binding in relation to the inbound messages as well. For more information, see Customizing for **Transportation Management** under [Freight Order Management](#) > [Freight Booking](#) > [Define Freight Booking Types](#).

Choose the topic of interest below to learn more about the processes and messages.



Road

The carrier receives a transportation request from a shipper, organizes the transportation plan, and sends the results back to the shipper as a road freight order (FO).

The carrier sends an EDI message to communicate the transportation plan, and the shipper receives the message via the `TransportationOrderGenericRequest_In` API, which contains the transportation demands assigned to the FO and serves to update the FUs or consignment orders (CO) in the shipper's system. This includes, for example, information about the driver, the vehicle, transportation stages, and license plate.

Transportation Planning Results and Splits

It can happen that during transportation planning, FUs or the items in FUs that make up the transportation requirement need to be transported in different vehicles. As a result, the FU can be split in the receiving system when the carrier sends a message with the transportation result or an update. The split can be based on the item quantity or based on which handling units were assigned to the FO.

i Note

FU splits via EDI message are only supported for road FO scenarios.

Similarly, delivery splits occur when multiple freight units referencing the same outbound delivery exist and not all of them are planned onto the same road freight order. The carrier has the option of referring to the remaining transportation demand either by using the original delivery reference or the new split delivery reference, which is sent out automatically to the carrier after the split occurs in the shipper's system.

i Note

In cases involving splits, the following apply:

- Handling units in a message to the shipper only require top level package and not full item hierarchy
- The carrier can use the original or newly split delivery as reference
- Splits of packed products are not supported
- FUs can be split in outbound scenarios only
- FUs referenced must have been created in the shipper's system

Road: XML Message Example

This section describes the different parts of an XML sample and gives explanations for the most important nodes in a message.

i Note

You should not use other web services to update the freight order during the transportation planning phase because a subsequent freight order update from the external TM system with web service `TransportationOrderGenericRequest_In` could overwrite these updates.

Instead, necessary updates to the freight order must be performed in the external TM system first and then sent with web service `TransportationOrderGenericRequest_In`.

XML Example of Freight Order Information

This XML example specifies the information an EDI message from a carrier to a shipper must include so a road freight order can be created in the shipper's system. The breakdown of the `<TransportationOrder>` and `<Items>` segments are outlined in separate sections.

Sample Code

```
<n0:TransportationOrderGenericRequest xmlns:n0="http://sap.com/xi/SAPGlobal20/Global" xmlns:prx=
  <MessageHeader>
    <CreationDateTime>2021-07-27T10:36:32.732Z</CreationDateTime>
    <SenderBusinessSystemID>SYSTEM_ID</SenderBusinessSystemID>
    <SenderParty>
      <InternalID>CARRIER_ID</InternalID>
    </SenderParty>
    <BusinessScope>
      <TypeCode>3</TypeCode>
      <ID>EXTERNAL_PLANNING</ID>
    </BusinessScope>
  </MessageHeader>
  <TransportationOrder actionCode="04" completeTransmissionIndicator="true" replicateIndicator="
    <TypeCode>SM42</TypeCode>
    <CategoryCode>T0</CategoryCode>
    <CarrierParty>
      <InternalID>CARRIER_ID</InternalID>
    </CarrierParty>
    <Driver>
      <Party>
        <InternalID>DRIVER_ID</InternalID>
      </Party>
      <TransportationStage>
        <ID>10</ID>
      </TransportationStage>
    </Driver>
  </Driver>
```

```

<Name>Driver Name</Name>
<Party>
  <Address>
    <PersonName>
      <GivenName>Given Name</GivenName>
      <FamilyName>Surname</FamilyName>
    </PersonName>
    <PhysicalAddress>
      <CountryCode>COUNTRY_CODE</CountryCode>
      <RegionCode>REGION_CODE</RegionCode>
      <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
      <CityName>CITY</CityName>
      <POBoxIndicator>>false</POBoxIndicator>
      <StreetName>STREET</StreetName>
      <HouseID>HOUSE_NUMBER</HouseID>
    </PhysicalAddress>
    <Communication>
      <Telephone>
        <Number>
          <AreaID>AREA_CODE</AreaID>
          <SubscriberID>PREFIX</SubscriberID>
          <ExtensionID>UNIQUE_NUMBER</ExtensionID>
          <CountryCode>COUNTRY_CODE</CountryCode>
        </Number>
        <SMSEnabledIndicator>>false</SMSEnabledIndicator>
        <NumberDefaultIndicator>>true</NumberDefaultIndicator>
        <NumberUsageDenialIndicator>>false</NumberUsageDenialIndicator>
      </Telephone>
      <MobilePhone>
        <Number>
          <AreaID>ID</AreaID>
          <SubscriberID>ID</SubscriberID>
          <CountryCode>COUNTRY_CODE</CountryCode>
        </Number>
        <SMSEnabledIndicator>>false</SMSEnabledIndicator>
        <NumberDefaultIndicator>>true</NumberDefaultIndicator>
        <NumberUsageDenialIndicator>>false</NumberUsageDenialIndicator>
      </MobilePhone>
      <Email>
        <URI>EMAIL_ADDRESS</URI>
        <URIDefaultIndicator/>
        <URIUsageDenialIndicator>>false</URIUsageDenialIndicator>
      </Email>
    </Communication>
  </Address>
</Party>
<TransportationStage>
  <ID>10</ID>
</TransportationStage>
</Driver>
<TransportationStage>
  <ID>10</ID>
  <MainLoadingLocation>
    <InternalID>LOCATION_ID</InternalID>
  </MainLoadingLocation>
  <MainLoadingStopID>10</MainLoadingStopID>
  <MainUnloadingLocation>
    <InternalID>LOCATION_ID_2</InternalID>
  </MainUnloadingLocation>
  <MainUnloadingStopID>20</MainUnloadingStopID>
  <ExpectedDepartureDateTime>2025-04-09T22:00:00Z</ExpectedDepartureDateTime>
  <ExpectedArrivalDateTime>2025-04-12T22:00:00Z</ExpectedArrivalDateTime>
  <ItemAssignment>
    <ItemID>10</ItemID>
  </ItemAssignment>
  <ItemAssignment>
    <ItemID>1000000</ItemID>
  </ItemAssignment>
</TransportationStage>
<BaseBusinessTransactionDocumentReference>
  <ID>FREIGHT_ORDER_ID</ID>
  <TypeCode>3003</TypeCode>
</BaseBusinessTransactionDocumentReference>
<Item>

```



```

    <ID>1000000</ID>
    <CategoryCode>AVR</CategoryCode>
    <VehicleResource>
      <TransportMeansID>LICENSE_PLATE</TransportMeansID>
      <TransportMeansHomeCountryCode>COUNTRY_CODE</TransportMeansHomeCountryCode>
    </VehicleResource>
  </Item>
  <Item>
    <ID>10</ID>
    <CategoryCode>PRD</CategoryCode>
    <HierarchyRelationship>
      <ParentItemID>1000000</ParentItemID>
    </HierarchyRelationship>
    <BusinessTransactionDocumentReference>
      <BusinessTransactionDocumentReference>
        <ID>DELIVERY_NUMBER</ID>
        <TypeCode>T75</TypeCode>
      </BusinessTransactionDocumentReference>
    </BusinessTransactionDocumentReference>
  </Item>
  <Item>
    <ID>20</ID>
    <CategoryCode>PRD</CategoryCode>
    <HierarchyRelationship>
      <ParentItemID>1000000</ParentItemID>
    </HierarchyRelationship>
    <BusinessTransactionDocumentReference>
      <BusinessTransactionDocumentReference>
        <ID>DELIVERY_NUMBER</ID>
        <TypeCode>T75</TypeCode>
      </BusinessTransactionDocumentReference>
    </BusinessTransactionDocumentReference>
  </Item>
</TransportationOrder>
</n0:TransportationOrderGenericRequest>

```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<MessageHeader><CreationDateTime>	Message creation UTC timestamp (Coordinated Universal Time)
<MessageHeader><SenderParty><InternalID>	CARRIER_ID : A carrier's unique ID
<MessageHeader><BusinessScope>	The external planning scenario is specified by value 3 and EXTERNAL_PLANNING

XML Snippet for <TransportationOrder>

This XML snippet specifies the information for the transportation order segment of an EDI message for a road freight order. It needs to be included when creating or updating the road freight order. The <Driver> and <TransportationStage> segments are documented separately.

<TransportationOrder> General

Sample Code

```

<TransportationOrder actionCode="04" completeTransmissionIndicator="true" replicateIndicator="fa
  <TypeCode>SM42</TypeCode>
  <CategoryCode>T0</CategoryCode>
  <CarrierParty>
    <InternalID>CARRIER_ID</InternalID>

```

```
</CarrierParty>
[...]
```

```
<BaseBusinessTransactionDocumentReference>
  <ID>FREIGHT_ORDER_ID</ID>
  <TypeCode>TYPE_CODE</TypeCode>
</BaseBusinessTransactionDocumentReference>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<TransportationOrder actionCode	A coded representation of an instruction to the recipient of a message specifying how to process a transmitted document 04: Save (here: Update)
<TransportationOrder replicateIndicator	true: Replication on basis of common master data; otherwise false
<TransportationOrder completeTransmissionIndicator	An indicator that specifies whether the message is a full or delta image false: Delta message true: Full image create or update message
<TypeCode>	A coded representation of the type of a transportation order; for example, SM42 If not provided in the message, the default type specific to the TOR category.
<CategoryCode>	A coded representation of the category of a transportation order. Currently the following code values are supported: • BO Freight Booking • SO Service Order • TO Freight Order
<CarrierParty>(optional)	CARRIER_ID: A unique identifier of a contracted carrier party
<BaseBusinessTransactionDocumentReference> <TypeCode> (optional)	TYPE_CODE: For example, Type Code 3003 = Bordero, or T67 = reference number of carrier.

<Driver>

☰ Sample Code

```
<Driver>
  <Party>
    <InternalID>DRIVER_ID</InternalID>
  </Party>
  <TransportationStage>
    <ID>10</ID>
  </TransportationStage>
</Driver>
<Driver>
  <Name>Driver Name</Name>
  <Party>
    <Address>
```

```

    <PersonName>
      <GivenName>Given Name</GivenName>
      <FamilyName>Surname</FamilyName>
    </PersonName>
    <PhysicalAddress>
      <CountryCode>COUNTRY_CODE</CountryCode>
      <RegionCode>REGION_CODE</RegionCode>
      <StreetPostalCode>POSTAL_CODE</StreetPostalCode>
      <CityName>CITY</CityName>
      <POBoxIndicator>>false</POBoxIndicator>
      <StreetName>STREET</StreetName>
      <HouseID>HOUSE_NUMBER</HouseID>
    </PhysicalAddress>
    <Communication>
      <Telephone>
        <Number>
          <AreaID>ID</AreaID>
          <SubscriberID>ID</SubscriberID>
          <ExtensionID>ID</ExtensionID>
          <CountryCode>COUNTRY_CODE</CountryCode>
        </Number>
        <SMSEnabledIndicator>>false</SMSEnabledIndicator>
        <NumberDefaultIndicator>>true</NumberDefaultIndicator>
        <NumberUsageDenialIndicator>>false</NumberUsageDenialIndicator>
      </Telephone>
      <MobilePhone>
        <Number>
          <AreaID>0175</AreaID>
          <SubscriberID>42045321</SubscriberID>
          <CountryCode>DE</CountryCode>
        </Number>
        <SMSEnabledIndicator>>false</SMSEnabledIndicator>
        <NumberDefaultIndicator>>true</NumberDefaultIndicator>
        <NumberUsageDenialIndicator>>false</NumberUsageDenialIndicator>
      </MobilePhone>
      <Email>
        <URI>EMAIL_ADDRESS</URI>
        <URIDefaultIndicator/>
        <URIUsageDenialIndicator>>false</URIUsageDenialIndicator>
      </Email>
    </Communication>
  </Address>
</Party>
<TransportationStage>
  <ID>10</ID>
</TransportationStage>
</Driver>

```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<Driver><Party><InternalID>	DRIVER_ID: An internal ID of a driver to be assigned
<Driver><TransportationStage><ID>	10: An identifier of a transportation stage
<Driver><Name>	The name of an ad hoc driver

<TransportationStage>

Sample Code

```

<TransportationStage>
  <ID>10</ID>
  <MainLoadingLocation>
    <InternalID>LOCATION_ID</InternalID>
  </MainLoadingLocation>
</TransportationStage>

```

```
</MainLoadingLocation>
<MainLoadingStopID>10</MainLoadingStopID>
<MainUnloadingLocation>
  <InternalID>LOCATION_ID_2</InternalID>
</MainUnloadingLocation>
<MainUnloadingStopID>20</MainUnloadingStopID>
<ExpectedDepartureDateTime>2025-04-09T22:00:00Z</ExpectedDepartureDateTime>
<ExpectedArrivalDateTime>2025-04-12T22:00:00Z</ExpectedArrivalDateTime>
<ItemAssignment>
  <ItemID>10</ItemID>
</ItemAssignment>
<ItemAssignment>
  <ItemID>10000000</ItemID>
</ItemAssignment>
</TransportationStage>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<TransportationStage>	A collection of information about a section of the transportation chain represented by a transportation order
<TransportationStage><ID>	A unique numeric identifier of a transportation stage
<TransportationStage><MainLoadingLocation> <InternalID>	A physical place to which a transportation stage of a transportation order refers as a main loading location At least one of the elements MainLoadingLocation, MainLoadingIATALocationText, and MainLoadingUNLOCODELocationText must be specified. If the complex element MainLoadingLocation is specified, either an InternalID, a StandardID, or an Address must be present, otherwise the entire element must be omitted.
<TransportationStage><MainLoadingStopID>	An identifier of the transportation stop of the main loading location i Note This field is mandatory if the stage segment is filled in the message
<TransportationStage><MainUnloadingLocation> <InternalID>	A physical place to which an unloading transportation stage of a transportation order refers as a main unloading location Either an InternalID or a StandardID must be present. If a StandardID is present, the supported values of the schemeAgencyID attribute depend on the configuration of the receiving system. Leaving out the MainUnloadingLocation element is only allowed if the location is either uniquely identified by one of the elements MainUnloadingIATALocationText or MainUnloadingUNLOCODELocationText, being specified for the same instance of the TransportationStage node, or else if the location is identical to one that is specified in the MainLoadingLocation element (or related elements) of the successor stage. For example, the MainUnloadingLocation of a pre-carriage stage is not required because the system assumes

Technical Name	Comment
	that it is identical to the MainLoadingLocation of the succeeding main carriage stage.
<TransportationStage><MainUnloadingStopID>	An identifier of the transportation stop of the main unloading location i Note This field is mandatory
<TransportationStage><ItemAssignment><ItemID>	An assignment of a transportation order item to a transportation stage of a transportation order and its identifier

XML Snippet for <Item>

This XML snippet specifies the information for the item segment of an EDI message for a road freight order. It gives information about the item hierarchy for a freight order below the main item (AVR).

Sample Code

```

<Item>
  <ID>1000000</ID>
  <CategoryCode>AVR</CategoryCode>
  <VehicleResource>
    <TransportMeansID>LICENSE_PLATE</TransportMeansID>
    <TransportMeansHomeCountryCode>COUNTRY_CODE</TransportMeansHomeCountryCode>
  </VehicleResource>
</Item>
<Item>
  <ID>10</ID>
  <CategoryCode>PRD</CategoryCode>
  <HierarchyRelationship>
    <ParentItemID>1000000</ParentItemID>
  </HierarchyRelationship>
  <BusinessTransactionDocumentReference>
    <BusinessTransactionDocumentReference>
      <ID>DELIVERY_NUMBER</ID>
      <TypeCode>T75</TypeCode>
    </BusinessTransactionDocumentReference>
  </BusinessTransactionDocumentReference>
</Item>
<Item>
  <ID>20</ID>
  <CategoryCode>PRD</CategoryCode>
  <HierarchyRelationship>
    <ParentItemID>1000000</ParentItemID>
  </HierarchyRelationship>
  <BusinessTransactionDocumentReference>
    <BusinessTransactionDocumentReference>
      <ID>DELIVERY_NUMBER</ID>
      <TypeCode>T75</TypeCode>
    </BusinessTransactionDocumentReference>
  </BusinessTransactionDocumentReference>
</Item>
</TransportationOrder>
</n0:TransportationOrderGenericRequest>

```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<Item><CategoryCode>	A coded representation of the category of a transportation order item AVR: Active vehicle resource Typically one of the following code values can occur: • AVR Active Vehicle Resource • PRD Product • PKG Package • BOK Booking • FUR Freight Unit Root • TUR Transportation Unit Resource • SHP Shipment
<Item><VehicleResource><TransportMeansID>	LICENSE_PLATE: An identifier for a means of transport. Usually, a vehicle's license plate.
<Item><VehicleResource> <TransportMeansHomeCountryCode>	COUNTRY_CODE: A coded representation of the home country of a means of transport; for example, de – Germany
<Item><BusinessTransactionDocumentReference><ID>	DELIVERY_NUMBER: A unique identifier for a referenced business document In a simple freight unit reference, the ItemID can be omitted. The system relies on a representative product item to identify the freight unit and assign it to the freight order. i Note With a delivery reference, item ID and type code are optional.
<Item><BusinessTransactionDocumentReference> <TypeCode>	A coded representation of the type of business document that is referenced T75: The receiving system can resolve a variety of references. T75 is a generic type code used for delivery documents and customer returns deliveries. Other available type codes used in business scope EXTERNAL_PLANNING are: • 58 Inbound Delivery • 73 Delivery Document • 700 Customer Returns Delivery • 1122 Transportation Order for the Freight Unit

Road: XML Example of Departure Update Message

This section describes the different parts of an XML sample and gives explanations for the most important nodes in a message.

Update of Transportation Event

This XML example specifies the information an EDI message from a carrier to a shipper must include so a road freight order can be updated with transportation event information in the shipper's system. In this example, the <InternalID> of the location serves as stop identifier for the event reporting.

Sample Code

```
<glob:TransportationOrderGenericRequest>
  <MessageHeader>
    <CreationDateTime>2020-05-25T18:50:10.927Z</CreationDateTime>
    <SenderBusinessSystemID>SYSTEM_ID</SenderBusinessSystemID>
    <SenderParty>
      <InternalID>CARRIER_ID</InternalID>
    </SenderParty>
  </MessageHeader>
  <TransportationOrder actionCode="04" replicateIndicator="false" completeTransmissionIndicator=
    <BaseBusinessTransactionDocumentReference>
      <ID>FREIGHT_ORDER_ID</ID>
      <TypeCode>3003</TypeCode>
      <VersionID>2</VersionID>
    </BaseBusinessTransactionDocumentReference>
    <CategoryCode>T0</CategoryCode>
    <TransportationEvent>
      <EventTypeCode>ARRIVAL</EventTypeCode>
      <Location>
        <InternalID>LOCATION_ID</InternalID>
      </Location>
    </TransportationEvent>
    <TransportationEvent>
      <EventTypeCode>UNLOADING_BEGIN</EventTypeCode>
      <Location>
        <InternalID>LOCATION_ID</InternalID>
      </Location>
    </TransportationEvent>
    <TransportationEvent>
      <EventTypeCode>UNLOADING_END</EventTypeCode>
      <Location>
        <InternalID>LOCATION_ID</InternalID>
      </Location>
    </TransportationEvent>
  </TransportationOrder>
</glob:TransportationOrderGenericRequest>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<CreationDateTime>	Message creation UTC timestamp
<SenderParty><InternalID>	CARRIER_ID : A carrier's unique ID
<TransportationOrder actionCode	A coded representation of an instruction to the recipient of a message specifying how to process a transmitted document 04 : Save (here: update)
<TransportationOrder replicateIndicator	true : Replication on basis of common master data; otherwise false
<TransportationOrder completeTransmissionIndicator	An indicator that specifies whether the message is a full or delta image false : Delta message

Technical Name	Comment
	true: Full image create or update message
<BaseBusinessTransactionDocumentReference>	A unique self-reference to the transportation order
<ID>	External freight order ID
<TypeCode>	A coded representation of the type of a transportation order For example, type code 3003 = Bordero, or T67 = reference number of carrier.
<VersionID>	1: An identifier for a version of this business document
<CategoryCode>	A coded representation of the category of a transportation order. Currently the following code values are supported: • BO Freight Booking • SO Service Order • TO Freight Order
<TransportationEvent><EventTypeCode>	A coded representation of the type of a transportation event; for example, DEPARTURE
<TransportationEvent><Location><InternalID>	A location's unique ID

Ocean

The carrier has received a transportation request from a shipper in the form of freight units (FU), organizes a transportation plan, and sends a freight booking (FB) containing the transportation demands back to the shipper.

The carrier can send the shipper the transportation plan or a freight booking update using EDI communication. For example, the carrier can send confirmation of the departure from the port of loading, updates on the expected time of arrival, house bill of lading, or updates on changes to the port of discharge. The system supports either updates for FB with a full image message or for individual container units within a FB with a delta image message. The shipper receives the message via the **TransportationOrderGenericRequest_In** API, which contains the transportation demands assigned to the FB and serves to update the FUs in the shipper's system. This includes, for example, information about the vessel, transportation stages, and the carrier's reference number.

A process for left-behind goods is also in place. For more information, see [Cargo Left Behind at Port/Airport](#).

Related Information

[Ocean Freight Bookings for Shippers \(Outbound\)](#)

Ocean: Example of Create Message

This section describes the different segments of a sample XML message used to assign freight units to an ocean freight booking. It show structure and content samples for the EDI messages.

Example of Create Message

This XML snippet specifies the information an EDI message must contain so that the ocean freight booking can be created in the shipper's system.

 Sample Code

```
<n0:TransportationOrderGenericRequest xmlns:n0="http://sap.com/xi/SAPGlobal20/Global">
  <MessageHeader>
    <CreationDateTime>2024-02-12T21:10:00.000Z</CreationDateTime>
    <SenderParty>
      <InternalID>S42_CARR01</InternalID>
    </SenderParty>
    <BusinessScope>
      <TypeCode>3</TypeCode>
      <ID>EXTERNAL_PLANNING</ID>
    </BusinessScope>
  </MessageHeader>
  <TransportationOrder actionCode="04" replicateIndicator="false" completeTransmissionIndicato
    <TypeCode>S42L</TypeCode>
    <CategoryCode>B0</CategoryCode>
    <BaseBusinessTransactionDocumentReference>
      <ID>FREIGHT_BOOKING_ID</ID>
      <TypeCode>T67</TypeCode>
    </BaseBusinessTransactionDocumentReference>
    <TransportationTerms>
      <TransportModeCode listID="8067" listAgencyID="6">1</TransportModeCode>
    </TransportationTerms>
    <TransportationStage>
      <CategoryCode>2</CategoryCode>
      <MainLoadingLocation>
        <InternalID>LOADING_LOCATION_ID</InternalID>
      </MainLoadingLocation>
      <MainUnloadingLocation>
        <InternalID>UNLOADING_LOCATION_ID</InternalID>
      </MainUnloadingLocation>
      <JourneyID>JOURNEY_ID</JourneyID>
      <TransportMeansID>VESSEL_NAME</TransportMeansID>
      <TransportMeansStandardID schemeAgencyID="54">VESSEL_ID</TransportMeansStandardID>
    </TransportationStage>
    <Item>
      <ID>1000000</ID>
      <CategoryCode>B0K</CategoryCode>
    </Item>
    <Item>
      <ID>10</ID>
      <CategoryCode>PRD</CategoryCode>
      <HierarchyRelationship>
        <ParentItemID>1000000</ParentItemID>
      </HierarchyRelationship>
      <PiecesTransportationQuantity unitCode="EA">10.0</PiecesTransportationQuantity>
      <BaseBusinessTransactionDocumentReference>
        <ID>FREIGHT_UNIT_ID</ID>
        <TypeCode>1122</TypeCode>
        <ItemID>0000000010</ItemID>
      </BaseBusinessTransactionDocumentReference>
    </Item>
  </TransportationOrder>
</n0:TransportationOrderGenericRequest>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<CreationDateTime>	Message creation UTC timestamp
<BusinessScope>	The external planning scenario is specified by value 3 and EXTERNAL_PLANNING .

Technical Name	Comment
<SenderParty>/<CarrierParty> <StandardID schemeAgencyID="..."	Agency issuing Party Standard IDs such as GS1 (9), IATA (3), D&B (16). For values see UNECE DE 3055.
<TransportationOrder actionCode	A coded representation of an instruction to the recipient of a message specifying how to process a transmitted document 04: Save (here: Update)
<TransportationOrder replicateIndicator	true: Replication on basis of common master data; otherwise false
<TransportationOrder completeTransmissionIndicator	An indicator that specifies whether the message is a full or delta image false: Delta message true: Full image create or update message
<TransportationOrder><TypeCode>	A coded representation of a document type that applies to a transportation order of the category freight unit, freight order, or freight booking S42L: Freight booking type
<TransportationOrder><CategoryCode>	BO: Transportation order type category – Freight booking
<TransportationEvent><EventTypeCode>	DEPARTURE: Departure event
<BaseBusinessTransactionDocumentReference><ID>	External freight booking ID
<BaseBusinessTransactionDocumentReference><TypeCode>	T67: Reference number of carrier
<TransportationTerms>	A collection of transportation conditions and agreements that apply when transporting the goods that are represented by a transportation order, and when providing the necessary services and activities for this. At least one of the optional elements of the TransportationTerms node must be present, otherwise the entire node must be omitted.
<TransportationTerms><TransportModeCode listID="8067"	A coded representation of the mode of transport that is to be used for the main carriage stage(s) of a transportation order.
<TransportationTerms><TransportModeCode listAgencyID="6">	A coded representation of the mode of transport that is to be used for the main carriage stage(s) of a transportation order. The code value refers to the configurable code list for this data type (Customizing).
<TransportationStage><CategoryCode>	A coded representation of the category of a transportation stage. 2: Main carriage
<TransportationStage><MainLoadingLocation><InternalID>	A physical place to which a transportation stage of a transportation order refers as a main loading location At least one of the elements MainLoadingLocation , MainLoadingIATALocationText , and MainLoadingUNLOCODELocationText must be specified. If the complex element MainLoadingLocation is specified, either

Technical Name	Comment
	an InternalID, a StandardID, or an Address must be present, otherwise the entire element must be omitted.
<TransportationStage><MainUnloadingLocation> <InternalID>	A physical place to which an unloading transportation stage of a transportation order refers as a main unloading location Either an InternalID or a StandardID must be present. If a StandardID is present, the supported values of the schemeAgencyID attribute depend on the configuration of the receiving system. Leaving out the MainUnloadingLocation element is only allowed if the location is either uniquely identified by one of the elements MainUnloadingIATALocationText or MainUnloadingUNLOCODELocationText, being specified for the same instance of the TransportationStage node, or else if the location is identical to one that is specified in the MainLoadingLocation element (or related elements) of the successor stage. For example, the MainUnloadingLocation of a pre-carriage stage is not required because the system assumes that it is identical to the MainLoadingLocation of the succeeding main carriage stage.
<TransportationStage><TransportMeansID>	The name of a means of transport
<TransportMeansStandardID schemeAgencyID="54">	A standardized identifier for a means of transport, whereby the identification scheme used is controlled by an agency from the code list DE 3055; for example, 54 : IMO (International Maritime Organization)
Item><CategoryCode>	A coded representation of the category of a transportation order item BOK: Booking Typically one of the following code values can occur: • AVR Active Vehicle Resource • PRD Product • PKG Package • BOK Booking • FUR Freight Unit Root • TUR Transportation Unit Resource • SHP Shipment
<BaseBusinessTransactionDocumentReference><ID>	Freight unit ID as sent upfront to carrier
<BaseBusinessTransactionDocumentReference> <TypeCode>	Document type code used for all transportation order request documents 1122: transportation order

Ocean: XML Example of Departure Status Update

This section describes XML sample snippets for an ocean freight booking and includes the assignment of container units to the ocean freight booking. They show structure and content samples for the EDI messages.

XML Example of Departure Status Update

This XML snippet specifies the information for the transportation order segment of an EDI message so that the ocean freight booking can be created or updated (delta message) in the shipper’s system. The samples for **Stages** and **Items** are outlined in a separate section.

Sample Code

```
<n0:TransportationOrderGenericRequest xmlns:n0="http://sap.com/xi/SAPGlobal20/Global">
  <MessageHeader>
    <CreationDateTime>2024-01-22T21:10:00.000Z</CreationDateTime>
    <SenderParty>
      <InternalID>CARRIER_ID</InternalID>
    </SenderParty>
    <BusinessScope>
      <TypeCode>3</TypeCode>
      <ID>EXTERNAL_PLANNING</ID>
    </BusinessScope>
  </MessageHeader>
  <TransportationOrder actionCode="04" replicateIndicator="false" completeTransmissionIndicator=
    <TypeCode>S42L</TypeCode>
    <CategoryCode>B0</CategoryCode>
    <TransportationEvent>
      <EventTypeCode>DEPARTURE</EventTypeCode>
      <ActualDateTime>2024-01-22T21:00:00.000Z</ActualDateTime>
      <UNLOCODELocationText>UNLOCODE_LOCATION</UNLOCODELocationText>
    </TransportationEvent>
    <TransportationStage>
      <CategoryCode>2</CategoryCode>
      <MainLoadingUNLOCODELocationText>UNLOCODE_LOADING_PORT</MainLoadingUNLOCODELocationText>
      <MainUnloadingUNLOCODELocationText>UNLOCODE_UNLOADING_PORT</MainUnloadingUNLOCODELocationT
      <JourneyID>VOYAGE_ID</JourneyID>
      <TransportMeansStandardID schemeAgencyID="54">IMO_VESSEL_NUMBER</TransportMeansStandardID>
    </TransportationStage>
    <BaseBusinessTransactionDocumentReference>
      <ID>BOOKING_ID</ID>
      <TypeCode>1122</TypeCode>
    </BaseBusinessTransactionDocumentReference>
    <Item>
      <ID>1000000</ID>
      <CategoryCode>B0K</CategoryCode>
    </Item>
    <Item>
      <ID>1000010</ID>
      <HierarchyRelationship>
        <ParentItemID>1000000</ParentItemID>
      </HierarchyRelationship>
      <CategoryCode>TUR</CategoryCode>
      <TransportationUnitResource>
        <ID>CONTAINER_NUMBER</ID>
      </TransportationUnitResource>
    </Item>
  </TransportationOrder>
</n0:TransportationOrderGenericRequest>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<CreationDateTime>	Message creation UTC timestamp

Technical Name	Comment
<BusinessScope>	The external planning scenario is specified by value 3 and EXTERNAL_PLANNING .
<SenderParty>/<CarrierParty> <StandardID schemeAgencyID="..."	Agency issuing Party Standard IDs such as GS1 (9), IATA (3), D&B (16). For values see UNECE DE 3055.
<TransportationOrder actionCode	A coded representation of an instruction to the recipient of a message specifying how to process a transmitted document 04 : Save (here: Update)
<TransportationOrder replicateIndicator	true : Replication on basis of common master data; otherwise false
<TransportationOrder completeTransmissionIndicator	An indicator that specifies whether the message is a full or delta image false : Delta message true : Full image create or update message
<TransportationOrder><TypeCode>	Freight booking type, for example S42L
<TransportationOrder><CategoryCode>	BO : Transportation order type category – Freight booking
<TransportationEvent><EventTypeCode>	DEPARTURE : Departure event
<BaseBusinessTransactionDocumentReference><ID>	External freight booking ID
<BaseBusinessTransactionDocumentReference> <TypeCode>	T67 : Reference number of carrier

XML Snippet for Voyage and IMO Vessel Number <TransportationStage>

This XML snippet specifies the information for the transportation stage segment of an ocean freight booking. It can to be included when creating or updating an ocean freight booking.

Sample Code

```
<TransportationStage>
  <CategoryCode>2</CategoryCode>
  <MainLoadingUNLOCODELocationText>UNLOCODE_LOADING_PORT</MainLoadingUNLOCODELocationText>
  <MainUnloadingUNLOCODELocationText>UNLOCODE_UNLOADING_PORT</MainUnloadingUNLOCODELocationText>
  <JourneyID>VOYAGE_ID</JourneyID>
  <TransportMeansStandardID schemeAgencyID="54">IMO_VESSEL_NUMBER</TransportMeansStandardID>
</TransportationStage>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<TransporationStage><CategoryCode>	A coded representation of the category of a transportation stage. 2 : Main carriage → Tip

Technical Name	Comment
	In multi-port booking scenarios, it is possible to have multiple main carriage stages sent by the carrier in the inbound message.
<MainLoadingUNLOCODELocationText>	A text string that represents a location referenced by a transportation stage as a main loading location Only a string listed in the LOCODE column of the corresponding country-specific list provided by UNECE may occur as a value of a UNLOCODELocationText element; for example, DE AAH – Aachen. For more information, see UN/LOCODE Code List by Country and Territory .
<MainUnloadingUNLOCODELocationText>	A text string that represents a location referenced by a transportation stage as an main unloading location Only a string listed in the LOCODE column of the corresponding country-specific list provided by UNECE may occur as a value of a UNLOCODELocationText element; for example, DE AAH – Aachen. For more information, see UN/LOCODE Code List by Country and Territory .
<TransportationStage><JourneyID> (optional)	An identifier of a journey
<TransportMeansStandardID (optional) schemeAgencyID	54: IMO (International Maritime Organization) from the code list UN/EDIFACT DE 3055; also specifies the IMO vessel number

XML Snippet for <Item>

This XML snippet specifies the information for the item segment of an EDI message of an ocean freight booking. It describes how to assign container units to an ocean freight booking.

Sample Code

```
<Item>
  <ID>1000000</ID>
  <CategoryCode>BOK</CategoryCode>
</Item>
<Item>
  <ID>1000010</ID>
  <HierarchyRelationship>
    <ParentItemID>1000000</ParentItemID>
  </HierarchyRelationship>
  <CategoryCode>TUR</CategoryCode>
  <TransportationUnitResource>
    <ID>CONTAINER_NUMBER</ID>
  </TransportationUnitResource>
</Item>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<Item><CategoryCode>	BOK: Main item of a freight booking

Technical Name	Comment
	TUR: Transportation unit resource – container item
<TransportationUnitResource><ID>	Container registration number

Ocean: XML Example of ETA Update Message

XML Example of ETA Update (on Event Level) <TransportationEvent>

This XML snippet specifies information for updating the expected date and time of arrival.

Sample Code

```
<n0:TransportationOrderGenericRequest xmlns:n0="http://sap.com/xi/SAPGlobal20/Global">
  <MessageHeader>
    <CreationDateTime>2024-02-12T21:10:00.000Z</CreationDateTime>
    <SenderParty>
      <StandardID schemeAgencyID="9">SENDER_ID</StandardID>
    </SenderParty>
    <BusinessScope>
      <ID>EXTERNAL_PLANNING</ID>
    </BusinessScope>
  </MessageHeader>
  <TransportationOrder actionCode="04" replicateIndicator="false" completeTransmissionIndicator=
    <CategoryCode>B0</CategoryCode>
    <BaseBusinessTransactionDocumentReference>
      <ID>CARRIER_REFERENCE_NUMBER</ID>
      <TypeCode>T67</TypeCode>
    </BaseBusinessTransactionDocumentReference>
    <TransportationEvent>
      <EventTypeCode>EXPECTED_ARRIVAL</EventTypeCode>
      <UNLOCODELocationText>UN_LOCODE_UNLOADING_PORT</UNLOCODELocationText>
      <EstimatedDateTime>2024-05-18T11:10:15.000Z</EstimatedDateTime>
    </TransportationEvent>
  </TransportationOrder>
</n0:TransportationOrderGenericRequest>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<CreationDateTime>	Message creation UTC timestamp
<SenderParty>/<CarrierParty> <StandardID schemeAgencyID	Agency issuing Party Standard IDs such as GS1 (9), IATA (3), D&B (16). For values see UNECE DE 3055.
<TransportationOrder actionCode	04: Save (= Create or Update)
<TransportationOrder replicateIndicator	true: Replication on basis of common master data; otherwise false
completeTransmissionIndicator	false: Delta message true: Full image create or update message
<TransportationOrder><CategoryCode>	Transportation order category BO: Freight booking

Technical Name	Comment
<TransportationEvent><TypeCode>	EXPECTED_ARRIVAL: Expected arrival event
<TransportationEvent><EstimatedDateTime>	Estimated event timestamp in UTC
<BaseBusinessTransactionDocumentReference><ID>	CARRIER_REFERENCE_NUMBER: External freight booking ID
<BaseBusinessTransactionDocumentReference> <TypeCode>	T67: Reference number of carrier

Ocean: House Bill of Lading Update Message

XML Example of House Bill of Lading Update

This XML snippet specifies information for an update containing the house bill of lading.

Sample Code

```
<n0:TransportationOrderGenericRequest xmlns:n0="http://sap.com/xi/SAPGlobal20/Global">
  <MessageHeader>
    <CreationDateTime>2024-02-12T21:10:00.000Z</CreationDateTime>
    <SenderParty>
      <InternalID>CARRIER_ID</InternalID>
    </SenderParty>
  </MessageHeader>
  <TransportationOrder actionCode="04" replicateIndicator="false" completeTransmissionIndicator=
    <TypeCode>S42L</TypeCode>
    <CategoryCode>B0</CategoryCode>
    <BaseBusinessTransactionDocumentReference>
      <ID>FREIGHT_BOOKING_ID</ID>
      <TypeCode>1122</TypeCode>
    </BaseBusinessTransactionDocumentReference>
    <Item>
      <ID>1000010</ID>
      <CategoryCode>TUR</CategoryCode>
      <TransportationUnitResource>
        <ID>CONTAINER_REGISTRATION_NUMBER</ID>
      </TransportationUnitResource>
      <BusinessTransactionDocumentReference>
        <BusinessTransactionDocumentReference>
          <ID>HOUSE_BILL_OF_LADING</ID>
        </BusinessTransactionDocumentReference>
        <TransportationDocumentTypeCode>T51</TransportationDocumentTypeCode>
      </BusinessTransactionDocumentReference>
    </Item>
  </TransportationOrder>
</n0:TransportationOrderGenericRequest>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<MessageHeader><CreationDateTime>	Message creation UTC timestamp
<MessageHeader><SenderParty>	A carrier's unique ID
<TransportationOrder actionCode	04: Save (= Create or Update)

Technical Name	Comment
<TransportationOrder replicateIndicator	true: Replication on basis of common master data; otherwise false
<TransportationOrder completeTransmissionIndicator	false: Delta message true: Full image create or update message
<TransportationOrder><TypeCode>	A coded representation of a document type that applies to a transportation order of the category freight unit, freight order, or freight booking S42L: Freight booking type
<TransportationOrder><CategoryCode>	Transportation order category BO: Freight booking
<BaseBusinessTransactionDocumentReference> <TypeCode>	A coded representation of the type of this document 1122: Transportation Order (here: Freight Booking)
<Item><CategoryCode>	A coded representation of the category of a transportation order item TUR: Transportation Unit Resource Typically one of the following code values can occur: • AVR Active Vehicle Resource • PRD Product • PKG Package • BOK Booking • FUR Freight Unit Root • SHP Shipment
<Item><TransportationUnitResource>	A container's registration number
<Item><BusinessTransactionDocumentReference>	A unique identifier for a referenced business document
<Item><TransportationDocumentTypeCode>	A unique reference between a transportation order and another business transaction document; for example, a predecessor document T51: House Bill of Lading

Ocean: Change of Port of Discharge Update Message

XML Example of Change of Port of Discharge Update

This XML snippet specifies a change in the port of discharge. In this example, the message is sent as an update. The original port of discharge was New York and, with this update message, changed to Bremerhaven. The respective code can be found under the <TransportationStage> segment.

☰, **Sample Code**

```

<n0:TransportationOrderGenericRequest xmlns:n0="http://sap.com/xi/SAPGlobal20/Global">
  <MessageHeader>
    <CreationDateTime>2024-02-12T21:10:00.000Z</CreationDateTime>
    <SenderParty>
      <InternalID>S42_CARR01</InternalID>
    </SenderParty>
  </MessageHeader>
  <TransportationOrder actionCode="04" replicateIndicator="false" completeTransmissionIndicator=
    <TypeCode>S42L</TypeCode>
    <CategoryCode>BO</CategoryCode>
    <BaseBusinessTransactionDocumentReference>
      <ID>FREIGHT_BOOKING_ID</ID>
      <TypeCode>1122</TypeCode>
    </BaseBusinessTransactionDocumentReference>
    <TransportationStage>
      <ID>20</ID>
      <MainLoadingUNLOCODELocationText>DEHAM</MainLoadingUNLOCODELocationText>
      <MainUnloadingUNLOCODELocationText>DEBRV</MainUnloadingUNLOCODELocationText>
      <TransportMeansID>VESSEL_NAME</TransportMeansID>
      <ExpectedArrivalDateTime>2024-02-15T21:10:00.000Z</ExpectedArrivalDateTime>
    </TransportationStage>
    <Item>
      <ID>1000010</ID>
      <CategoryCode>TUR</CategoryCode>
      <TransportationUnitResource>
        <ID>CONTAINER_REGISTRATION_NUMBER</ID>
      </TransportationUnitResource>
    </Item>
  </TransportationOrder>
</n0:TransportationOrderGenericRequest>

```

Breakdown

Technical Name	Comment
<CreationDateTime>	Message creation UTC timestamp
<SenderParty>	A carrier's unique ID
<TransportationOrder actionCode	04: Save (= Create or Update)
<TransportationOrder replicateIndicator	true: Replication on basis of common master data; otherwise false
<TransportationOrder completeTransmissionIndicator	false: Delta message true: Full image create or update message
<TransportationOrder><CategoryCode>	Transportation order category BO: Freight booking
<TransportationEvent><TypeCode>	Freight booking type, for example, S42L
<BaseBusinessTransactionDocumentReference> <TypeCode>	A coded representation of the type of business document that is referenced. Here, this ocean freight booking. 1122: Transportation order for the freight unit
<TransportationStage> <MainLoadingUNLOCODELocationText>	A text string that represents a location referenced by a transportation stage as a main loading location Only a string listed in the LOCODE column of the corresponding country-specific list provided by UNECE may occur as a value of a UNLOCODELocationText element; for example, DE AAH – Aachen. For more information, see UN/LOCODE Code List by Country and Territory  .

Technical Name	Comment
<TransportationStage> <MainUnloadingUNLOCODELocationText>	A text string that represents a location referenced by a transportation stage as an main unloading location Only a string listed in the LOCODE column of the corresponding country-specific list provided by UNECE may occur as a value of a UNLOCODELocationText element; for example, DE AAH – Aachen. For more information, see UN/LOCODE Code List by Country and Territory  .
<Item><CategoryCode>	A coded representation of the category of a transportation order item TUR: Transportation Unit Resource Typically one of the following code values can occur: • AVR Active Vehicle Resource • PRD Product • PKG Package • BOK Booking • FUR Freight Unit Root • SHP Shipment
<Item><TransportationUnitResource>	A container's registration number

Air

The carrier has received a transportation request from a shipper in the form of freight units (FU) or consignment orders (CO), creates a transportation plan and sends the results back to the shipper.

The carrier sends the results of transportation planning based on FU or CO using the `TransportationOrderGenericRequest_In` API with business scope `EXTERNAL_PLANNING`. With this, the shipper's system creates the necessary freight bookings and assigns the transportation demands. The message includes information such as IATA airline and airport codes, flight number, and planned arrival date.

A process for left-behind goods is also in place. For more information, see [Cargo Left Behind at Port/Airport](#).

Related Information

[Air Freight Bookings for Shippers \(Outbound\)](#)

Air: XML Message Example

This section describes XML sample snippets for creating an air freight booking and includes assigning consignment orders to the air freight booking. The samples also include information for updating air freight bookings. They show structure and content samples for the EDI messages.

XML Example of General Booking Information

This XML example specifies the information an EDI message from a carrier to a shipper must include so an air freight booking can be created in the shipper's system. The samples for **Stages** and **Items** segments are outlined in a separate section.

≡ Sample Code

```
<n0:TransportationOrderGenericRequest xmlns:n0="http://sap.com/xi/SAPGlobal20/Global">
  <MessageHeader>
    <CreationDateTime>2024-02-12T21:10:00.000Z</CreationDateTime>
    <SenderParty>
      <StandardID schemeAgencyID="9">SENDER_ID</StandardID>
    </SenderParty>
    <BusinessScope>
      <TypeCode>3</TypeCode>
      <ID>EXTERNAL_PLANNING</ID>
    </BusinessScope>
  </MessageHeader>
  <TransportationOrder actionCode="04" replicateIndicator="false" completeTransmissionIndicator=
    <TypeCode>BAIR</TypeCode>
    <CategoryCode>B0</CategoryCode>
    <BaseBusinessTransactionDocumentReference>
      <ID>EXTERNAL_FREIGHT_BOOKING_ID</ID>
      <TypeCode>T67</TypeCode>
    </BaseBusinessTransactionDocumentReference>
    <CarrierParty>
      <StandardID schemeAgencyID="9">CARRIER_ID</StandardID>
    </CarrierParty>
    <TransportationStage>
      [...]
    </TransportationStage>
    <Item>
      [...]
    </Item>
    <TransportationCharge>
      [...]
    </TransportationCharge>
  </TransportationOrder>
</n0:TransportationOrderGenericRequest>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<CreationDateTime>	Message creation UTC timestamp (Coordinated Universal Time)
<BusinessScope>	The external planning scenario is specified by value 3 and EXTERNAL_PLANNING .
<SenderParty><StandardID schemeAgencyID	SENDER_ID : The sender's unique ID An agency which issues party standard IDs such as GS1 (9), IATA (3), D&B (16). For values, see UNECE DE 3055.
<TransportationOrder actionCode	04 : Save (either create or update)
<TransportationOrder replicateIndicator	true : Replication on basis of common master data; otherwise false
<TransportationOrder completeTransmissionIndicator	false : Delta message true : Full image create or update message
<TransportationOrder><TypeCode>	Freight booking type: for example, BAIR – Air freight booking
<TransportationOrder><CategoryCode>	Transportation order category BO – Freight booking

Technical Name	Comment
<BaseBusinessTransactionDocumentReference><ID>	External freight booking ID
<BaseBusinessTransactionDocumentReference> <TypeCode>	A coded representation of the type of business document that is referenced T67: Reference number of carrier
<CarrierParty><StandardID schemeAgencyID	CARRIER_ID: The carrier's unique ID An agency which issues party standard IDs such as GS1 (9), IATA (3), D&B (16). For values, see UNECE DE 3055.

XML Snippet for Flight Code and ETA Information

This XML snippet specifies the information for updating the expected departure and the expected time of arrival (ETA).

☰ Sample Code

```
<TransportationStage>
  <CategoryCode>2</CategoryCode>
  <MainLoadingIATALocationText>AIRPORT_CODE_OF_DEPARTURE</MainLoadingIATALocationText>
  <MainUnloadingIATALocationText>AIRPORT_CODE_OF_DESTINATION</MainUnloadingIATALocationText>
  <JourneyID>FLIGHT_CODE</JourneyID>
  <ExpectedDepartureDateTime>2024-04-17T21:10:00.000Z</ExpectedDepartureDateTime>
  <ExpectedArrivalDateTime>2024-05-17T21:10:00.000Z</ExpectedArrivalDateTime>
</TransportationStage>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above

Technical Name	Comment
<TransporationStage><CategoryCode>	A coded representation of the category of a transportation stage. 2: Main carriage
<TransportationStage> <MainLoadingIATALocationText>	A text string that represents an airport location referenced by a transportation stage as a loading location or that represents a group of airport locations situated near a particular city, in the notation that is defined by the International Air Transport Association (IATA). Only a string listed in the LOCODE column of the corresponding country-specific list provided by UNECE may occur as a value of a UNLOCODELocationText element; for example, DE AAH – Aachen. For more information, see UN/LOCODE Code List by Country and Territory .
<TransportationStage> <MainUnloadingIATALocationText>	A text string that represents an airport location referenced by a transportation stage as an unloading location or that represents a group of airport locations situated near a particular city, in the notation that is defined by the IATA. Only a string listed in the LOCODE column of the corresponding country-specific list provided by UNECE may occur as a value of a UNLOCODELocationText element; for example, DE AAH –

Technical Name	Comment
	Aachen. For more information, see UN/LOCODE Code List by Country and Territory .
<TransportationStage><JourneyID>	FLIGHT_CODE : what type of code; for example, LH8210

XML Snippet for <Item>

This XML snippet specifies the information for the item segment of an EDI message for an air freight booking. It gives information about the item hierarchy for a freight booking main item (BOK) with a SHP (consignment) item.

Sample Code

```
<Item>
  <ID>1000000</ID>
  <CategoryCode>BOK</CategoryCode>
</Item>
<Item>
  <ID>1000040</ID>
  <CategoryCode>SHP</CategoryCode>
  <HierarchyRelationship>
    <ParentItemID>1000000</ParentItemID>
  </HierarchyRelationship>
  <BaseBusinessTransactionDocumentReference>
    <ID>CONSIGNMENT_ORDER_ID</ID>
    <TypeCode>1122</TypeCode>
  </BaseBusinessTransactionDocumentReference>
  <BusinessTransactionDocumentReference>
    <BusinessTransactionDocumentReference>
      <ID>HAWB_NUMBER</ID>
    </BusinessTransactionDocumentReference>
    <TransportationDocumentTypeCode>T54</TransportationDocumentTypeCode>
  </BusinessTransactionDocumentReference>
</Item>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<Item><CategoryCode>	A coded representation of the category of a transportation order item BOK : Main item of a freight booking SHP : Assigned consignment order
<BaseBusinessTransactionDocumentReference><ID>	CONSIGNMENT_ORDER_ID : Consignment order ID as sent upfront to carrier
<BaseBusinessTransactionDocumentReference><TypeCode>	1122 : Generic type code for transportation order
<BusinessTransactionDocumentReference><ID>	A unique reference to another business document or to a business document item that is of significance within a freight booking process HAWB_NUMBER : House Air Waybill (HAWB) number

Technical Name	Comment
<BusinessTransactionDocumentReference> <TransportationDocumentTypeCode>	A unique reference between a transportation order and another business transaction document; for example, a predecessor document T54: House airway bill

XML Snippet for <TransportationCharge>

This XML snippet specifies the information for the transportation charge segment of an EDI message for an air freight booking. A possible scenario is that the carrier sends the charges information once the consignment orders have been assigned to flights.

Sample Code

```

<TransportationCharge>
  <ChargeItem>
    <Party>
      <StandardID schemeAgencyID="9">PARTY_ID</StandardID>
    </Party>
    <ChargeElement>
      <ID>CHARGE_ELEMENT_ID</ID>
      <OrdinalNumberValue>1</OrdinalNumberValue>
      <TypeCode>FB00</TypeCode>
      <PriceComponent>
        <DecimalValue>200</DecimalValue>
        <CurrencyCode>USD</CurrencyCode>
      </PriceComponent>
    </ChargeElement>
  </ChargeItem>
</TransportationCharge>

```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<TransportationCharge>	A collection of information that provides the hierarchy of charge items and charge elements that the charge calculation was based on.
<TransportationCharge><ChargeItem>	A collection of information that provides the hierarchy of charge items and charge elements that the charge calculation was based on.
<TransportationCharge><ChargeItem><Party> <StandardID schemeAgencyID	Agency issuing Party Standard IDs such as GS1 (9), IATA (3), D&B (16). For values see UNECE DE 3055.
<TransportationCharge><ChargeElement>	A collection of information that represents a charge element that was part of a charge calculation.
<TransportationCharge><ChargeElement><ID>	A unique identifier of a charge element.
<TransportationCharge><ChargeElement> <OrdinalNumberValue>	A number that specifies the position or order of a charge element relative to other charge elements. If not filled the charge elements will be created in a random order.

Technical Name	Comment
<TransportationCharge><ChargeElement><TypeCode>	A coded representation of the type of a charge element FB00: Basic rate
<TransportationCharge><ChargeElement> <PriceComponent>	The numerator of a fraction that is used for the calculation of the amount resulting from a charge element, and whose numerator and denominator are quantities, values, or dimensionless factors.

Air: XML Example of Departure Update Message

XML Snippet for Departure Update <TransportationEvent>

This XML snippet specifies how to send departure status information for an air freight booking based on events.

Sample Code

```
<n0:TransportationOrderGenericRequest xmlns:n0="http://sap.com/xi/SAPGlobal20/Global">
  <MessageHeader>
    <CreationDateTime>2024-02-12T21:10:00.000Z</CreationDateTime>
    <SenderParty>
      <StandardID schemeAgencyID="9">SENDER_ID</StandardID>
    </SenderParty>
    <BusinessScope>
      <ID>EXTERNAL_PLANNING</ID>
    </BusinessScope>
  </MessageHeader>
  <TransportationOrder actionCode="04" replicateIndicator="false" completeTransmissionIndicator=
    <CategoryCode>B0</CategoryCode>
    <BaseBusinessTransactionDocumentReference>
      <ID>CARRIER_REFERENCE_NUMBER</ID>
      <TypeCode>T67</TypeCode>
    </BaseBusinessTransactionDocumentReference>
    <TransportationEvent>
      <EventTypeCode>DEPARTURE</EventTypeCode>
      <IATALocationText>IATA_AIRPORT_CODE</IATALocationText>
      <ActualDateTime>2024-01-22T21:00:00.000Z</ActualDateTime>
    </TransportationEvent>
  </TransportationOrder>
</n0:TransportationOrderGenericRequest>
```

Breakdown

The table explains the main entries and abbreviations of the sample code above.

Technical Name	Comment
<SenderParty><StandardID schemeAgencyID	SENDER_ID: The sender's unique ID An agency which issues party standard IDs such as GS1 (9), IATA (3), D&B (16). For values, see UNECE DE 3055.
<TransportationOrder actionCode	04: Save (either create or update)
<TransportationOrder replicateIndicator	false: No replication on basis of common master data; otherwise, true
<TransportationOrder completeTransmissionIndicator	An indicator that specifies whether the message is a full or delta image

Technical Name	Comment
	false: Delta message true: Full image create or update message
<TransportationOrder><CategoryCode>	Transportation order category BO: Freight booking
<BaseBusinessTransactionDocumentReference><ID>	CARRIER_REFERENCE_NUMBER: A self-identifier for this business document
<BaseBusinessTransactionDocumentReference> <TypeCode>	A coded representation of the type of business document that is referenced T67: Reference number of carrier
<TransportationEvent><TypeCode>	A coded representation of the type of a transportation event like Departure, Arrival, Proof of Delivery DEPARTURE: Departure event
<TransportationEvent><IATALocationText>	A text string that represents an airport location in the notation that is defined by the International Air Transport Association (IATA) For the allowed values, see Airline and Airport Code Search ➡ .
<TransportationEvent><ActualDateTime>	Actual event timestamp in UTC